UNSW SYDNEY

Assimilating complex information

Author:
Pollock, Edwina Jane

Publication Date:
2000

DOI:
https://doi.org/10.26190/unsworks/19783

License:
https://creativecommons.org/licenses/by-nc-nd/3.0/au/
Link to license to see what you are allowed to do with this resource.

Downloaded from http://hdl.handle.net/1959.4/17813 in https://unsworks.unsw.edu.au on 2026-08-24

Assimilating Complex Information

Edwina Jane Pollock

Bachelor of Economics (Social Sciences) (Hons)

Thesis submitted to the University of New South Wales in fulfilment of the conditions for the degree of Doctor of Philosophy

School of Education

University of New South Wales

Australia

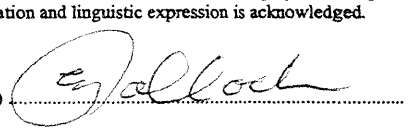
2000

Declaration

CERTIFICATE OF ORIGINALITY

I hereby declare that this submission is my own work and to the best of my knowledge it contains no materials previously published or written by another person, nor material which to a substantial extent has been accepted for the award of any other degree or diploma at UNSW or any other educational institution, except where due acknowledgement is made in the thesis. Any contribution made to the research by others, with whom I have worked at UNSW or elsewhere, is explicitly acknowledged in the thesis.

I also declare that the intellectual content of this thesis is the product of my own work, except to the extent that assistance from others in the project's design and conception or in style, presentation and linguistic expression is acknowledged.

(Signed) 

Acknowledgments

Many people have been instrumental in making the completion of this thesis a reality. Firstly, I would like to thank my supervisors. John Sweller has been unfailing in his guidance and expertise; thank-you also for your thoroughness and availability in providing me with answers to all my questions. Paul Chandler's support and strategic advice have been extremely useful in helping completion become reality. Thank-you both also for the opportunity to do this project.

The support, friendship and practical *thesis survival* advice provided by Graham Cooper and Sharon Tindall-Ford were invaluable. The management, trainers and apprentices of the Email training centre and BHP Ltd. need to be mentioned because without their participation there would have been no data to collect. Also, the support of Neil Trivett and Greg Hampton at the crucial end stage of the thesis was much appreciated.

Most of all I would like to thank my family for their love and encouragement: Mum, Dad and Ben have been constant supporters while the unconditional love and belief in me provided by husband Wade and our dog Rocket have allowed me to complete this project.

Table of Contents

Declaration ..... **ii**

Acknowledgements ..... **iii**

Table of Figures ..... **viii**

Table of Tables ..... **xi**

List of Appendices ..... **xiii**

Abstract ..... **xv**

Chapter 1 Human memory & problem solving **1**

1.1 The information processing view of memory 1

1.1.1 An overview of the information processing view of memory: the modal model 1

1.1.2 Sensory memory 3

1.1.3 Short term or working memory 5

1.1.4 Chunking 8

1.1.5 Long term memory 11

1.1.6 Beyond the modal model of memory 16

1.2 Schemas 18

1.2.1 Schemas: an overview 18

1.2.2 The features of schemas 21

1.2.3 Types of schemas 25

1.2.4 Schemas & remembering 29

1.2.5 The role of schemas in interpretation and comprehension 36

1.2.6 Schemas: a conclusion 41

1.3 Problem solving 42

1.3.1 Parts of a problem 42
1.3.2 Problem solving research in the second half of the twentieth century 46
1.4 Expert-novice differences 52
1.4.1 An introduction to expertise 52
1.4.2 Memory of problem states 54
1.4.3 Categorisation techniques 61
1.4.4 Problem solving strategies 75
1.4.5 Other differences between experts and novices in problem solving. 84
1.4.6 Summary of sections 1.3 – 1.4: Problem solving and expert-novice differences 86
1.5 Automation 88
1.5.1 Automatic and controlled processing 88
1.5.2 Automation and problem solving 92
1.6 Summary of Chapter 1 94

Chapter 2 Cognitive load theory 96

2.1 An Introduction to cognitive load theory 96
2.2 Sources of cognitive load 97
2.2.1 Intrinsic cognitive load and element interactivity 98
2.2.2 Extraneous cognitive load 102
2.2.3 Germane cognitive load 103
2.2.4 Total cognitive load 104
2.3 Instructional techniques generated by cognitive load theory 106
2.3.1 Goal free problems 106
2.3.2 Worked Examples 113
2.3.3 Split Attention Effect 123
2.3.4 Redundancy Effect 131
2.3.5 Modality Effect 137
2.4 Instruction and element interactivity 141

2.5 Cognitive Load Theory: a summary 146

Chapter 3 Pathways to meaningful learning 148

3.1 Types of learning 148
3.2 An alternative path to understanding 150

Chapter 4 Introduction to the empirical investigations 154

4.1 A summary of the research pertinent to this thesis 154
4.2 Learning complex information: a paradox 157
4.3 An isolated–interacting elements instructional technique 159
4.4 An introduction to the experiments conducted in this thesis 164

Chapter 5 Assimilating complex information: learning isolated elements then interacting elements 166

5.1 Experiment 1 166
5.2 Experiment 2 191

Chapter 6 Assimilating complex information: a replication 206

6.1 Introduction to Experiments 3 and 4 206
6.2 Experiment 3 207
6.3 Experiment 4 222

Chapter 7 Assimilating complex information: theoretical support 230

7.1 Introduction to Experiment 5 230
7.2 Experiment 5 233

Chapter 8 General discussion 252

8.1 The paradox of learning complex concepts 252
8.2 Summary of results 254
8.3 Theoretical and methodological issues arising from the research 257

8.3.1 Intrinsic cognitive load 257
8.3.2 Relationship to the redundancy effect 258
8.4 Application of the research & future directions 259
8.5 Concluding Comments 261

List of References **262**

Appendices **281**

Table of Figures

Chapter One

Figure 1.1: A basic model of human memory.....2

Figure 1.2: The sequence of pictures used by Baggett (1975) to tell the story of a haircut. 32

Figure 1.3: Some examples of the ambiguous figures, the figures' labels and subjects' recollections of these figures from Carmichael et al.'s (1932) research. 34

Figure 1.4: The text passage and the figure used to manipulate the presence or absence of context in an experiment from Bransford and Johnson (1972).. 39

Figure 1.5: A representation of the problem solving process (adapted from Gick, 1986). 45

Figure 1.6: The initial state of the tower of Hanoi problem. 48

Figure 1.7: An example of an typical novice and expert problem categorisation and explanation for this grouping (from Chi et al, 1981). 64

Figure 1.8: A depiction of an expert and a novice knowledge about the problem category label 'inclined planes' (from Chi et al., 1981).....70

Chapter Two

Figure 2.1: An algebra worked example. 114

Figure 2.2: A conventional geometry worked example 125

Figure 2.3: An integrated geometry worked example.....126

Figure 2.4: An example of redundant instructional material.....132

Figure 2.5: An example of some of the low element interactivity material and the high element interactivity material used in Tindall-Ford et al. (1997). ...144

Chapter Four

Figure 4.1: An example of interacting elements instructional material 160

Figure 4.2: An example of isolated elements instructional material. 162

Chapter Five

Figure 5.1: Relative instruction condition efficiency as a function of performance and mental effort ratings in Experiment 1. 182

Figure 5.2: Relative instruction condition efficiency as a function of performance and instruction time in Experiment 1. 187

Figure 5.3: Relative instruction condition efficiency as a function of performance and mental effort ratings in Experiment 2. 199

Figure 5.4: Relative instruction condition efficiency as a function of performance and instruction time in Experiment 2. 203

Chapter Six

Figure 6.1: Relative instruction condition efficiency as a function of performance and mental effort ratings in Experiment 3. 219

Figure 6.2: Relative instruction condition efficiency as a function of performance and mental effort ratings for Experiment 4. 228

Chapter Seven

Figure 7.1: Relative instruction condition efficiency as a function of performance and mental effort ratings for Experiment 5. 245

Figure 7.2: Relative instruction condition efficiency as a function of performance and instruction time for Experiment 5. 249

Table of Tables

Chapter Five

Table 5.1: Test Scores, Mental Load Ratings and Instruction Time (in seconds) for Experiment 1. 176

Table 5.2: Normalised (Z) scores and Condition Efficiency Scores for Performance and Mental Load for Experiment 1. 180

Table 5.3: Normalised (Z) scores and Condition Efficiency Scores for Performance and Instructional Time (in seconds) for Experiment 1. 185

Table 5.4: Test Scores, Mental Load Ratings and Instruction Time (in seconds) for Experiment 2. 194

Table 5.5: Normalised (Z) scores and Condition Efficiency Scores for Performance and Mental Load for Experiment 2. 197

Table 5.6: Normalised (Z) scores and Condition Efficiency Scores for Performance and Instructional Time (in seconds) for Experiment 2. 201

Chapter Six

Table 6.1: Test Scores and Mental Load Ratings for Experiment 3. 215

Table 6.2: Normalised (Z) scores and Condition Efficiency Scores for Performance and Mental Load for Experiment 3. 217

Table 6.3: Test Scores and Mental Load Ratings for Experiment 4. 224

Table 6.4: Normalised (Z) scores and Condition Efficiency Scores for Performance and Mental Load for Experiment 4. 226

Chapter Seven

Table 7.1: Test Score, Instruction Time (in seconds) and Mental Load Ratings for Experiment 5. 240

Table 7.2: Normalised (Z) scores and Condition Efficiency Scores for Performance and Mental Load for Experiment 5. 243

Table 7.3: Normalised (Z) scores and Condition Efficiency Scores for Performance and Instructional Time (in seconds) for Experiment 5. 247

List of Appendices

Appendix 1: Isolated elements instructional material used in Experiments 1 & 2.

Appendix 2: Interacting elements instructional material used in Experiments 1 & 2.

Appendix 3: An analysis of the element interactivity of the instructional material used in Experiments 1 & 2.

Appendix 4: Subjective mental load rating scale used in Experiments 1 & 2.

Appendix 5: Written test used in Experiments 1 & 2.

Appendix 6: An analysis of the element interactivity of the the instructional material used in Experiments 3 & 4.

Appendix 7: Isolated elements instructional material used in Experiments 3 & 4.

Appendix 8: Interacting elements instructional material used in Experiments 3 & 4.

Appendix 9: Diagram illistrating the oven baking process used in Experiments 3 & 4.

Appendix 10: Diagram depicting the oven electrical circuit used in Experiments 3 & 4.

Appendix 11: Subjective mental load rating scale used in Experiments 3 & 4.

Appendix 12: Written test used in Experiments 3 & 4.

Appendix 13: An analysis of the element interactivity of the the instructional material used in Experiment 5.

Appendix 14: Isolated elements instructional material used in Experiment 5.

Appendix 15: Interacting elements instructional material used in Experiment 5.

Appendix 16: Subjective mental load rating scale used in Experiment 5.

Appendix 17: Fault finding test used in Experiment 5.

Appendix 18: Pollock, E.; Chandler, P. & Sweller, J. (in press) Assimilating complex information. *Learning and Instruction.*

Abstract

Methods of instruction which are intended to facilitate understanding tend to incorporate all of the elements required for understanding in the instructions. Frequently, these types of instructions may overwhelm a learner's limited working memory and hinder learning. The five experiments reported in this thesis were generated by cognitive load theory. Cognitive load theory is based on the assumption that features of human cognitive architecture have important implications for how we learn and understand. The experiments examined an alternative approach to instructional design which had the goal of promoting schema construction in order to foster understanding. A two phase, isolated-interacting elements learning approach was developed which, in the first phase, artificially reduced the element interactivity of complex material by presenting the material as isolated elements of information that could be processed serially, rather than simultaneously, in working memory. In the second phase of instruction, all the information for understanding was presented. The control group's instruction was a more conventional approach to promoting understanding where all the information for understanding was presented in both Phase 1 and 2. Although a possible consequence of artificially reducing the element interactivity of material may be an initial decrease in a student's capacity for understanding, it was hypothesised that over the longer term it would lead to an increase in learners' understanding.

The results reported in this thesis provide powerful evidence that for certain groups of learners, information is better learnt through the isolated-interacting elements instructional method. Evidence is also provided in support of the proposal that schema construction is the mechanism underlying the success of the isolated-interacting elements instructional method. Experiments 1 and 3 demonstrated that when novice students are required to learn complex information, the isolated-interacting elements method of instruction was superior to the control method. In

contrast, no evidence was found in Experiments 2 and 4 to suggest a difference between these instructional techniques. These contrasting results can be explained by the different levels of domain knowledge possessed by the participants of each experiment. The results of Experiment 5 were similar to those of Experiments 1 and 3, namely that students performed superiorly if they had studied the isolated elements followed by interacting elements instruction. These results provide significant theoretical support for the claim that expertise is the key factor implicated in the beneficial effects associated with the isolated-interacting elements instructional technique since the participants in Experiment 5 had previously participated in Experiments 2 and 4. The difference in the outcomes of these experiments can be explained by the level of domain specific knowledge the participants of Experiment 5 had in each of the knowledge domains tested by the various experiments.

Chapter 1 Human memory & problem solving

1.1 The information processing view of memory

1.1.1 An overview of the information processing view of memory: the modal model

When developing the information processing model of memory, the cognitive school of psychology was strongly influenced by a computer metaphor. In the information processing model, the process of memory is seen to replicate the three stage process enacted upon data entered into a computer: encoding, storage and retrieval. Encoding is the process whereby information from the environment is transferred into the memory system; the information is then stored for a variable period before it is recovered or 'retrieved' back into consciousness.

During the 1960s, the field's understanding of memory was revolutionised by the proposal of a memory system that contained both a long and short term memory store. Examples of this style of model are those proposed by Atkinson and Shiffrin (1968) and Norman and Rumelhart (1970). Over the next thirty years the debate about the processes of memory have not subsided, and research in this area continues to flourish. Despite differences in the models that have been developed to explain the operation of memory, a strong consensus exists among researchers in the field over

some essential features of the memory system (Ericsson and Kintsch, 1995; Eysenck and Keane, 1990; Norman, 1970). For instance, modern conceptualisations of memory still maintain the basic division of memory into a long term and a short term store, despite their increased complexity of explanation. In fact, Shiffrin has argued that "(T)he concept of short term memory has virtually become a universal feature in theories of cognition" (1993: 196). Most modern theories can be viewed as refinements rather than rejections of the basic storage model of memory (Eysenck and Keane, 1990). Given this situation, the constraints and capabilities of the basic information processing model will be examined before looking at modern variations.

The model divides human memory into three interacting stores: sensory memory, working memory and long term memory. Figure 1.1 represents the modal model of memory. The arrows between the boxes indicate the flow of information. To summarise the figure, information from the environment is perceived by sensory registers, it then moves into short term memory were it is encoded and briefly stored. If attended to, information will move into long term memory where more permanent storage occurs. Information can be retrieved back from long term memory into consciousness or short term memory. Each section of the model will now be examined in more detail.

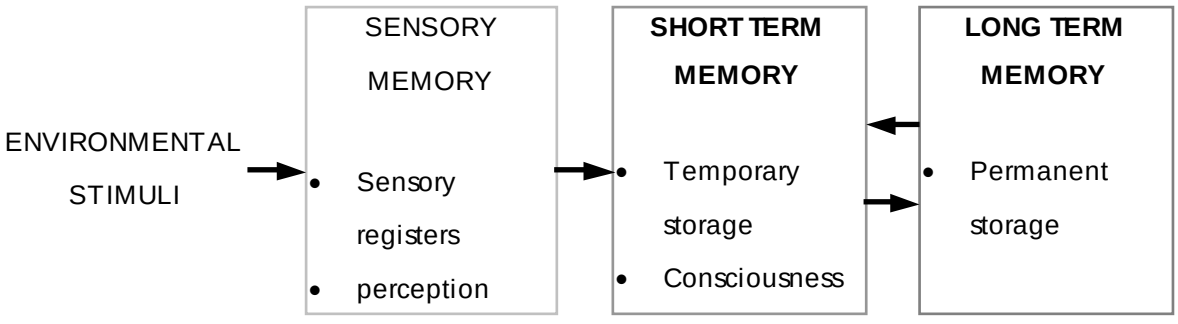


Figure 1.1: A basic model of human memory.

1.1.2 Sensory memory

Sensory memory is a holding system where information from the environment is able to enter into the human information processing system. Modality specific sensory registers exist in sensory memory and their function is to perceive information from the environment. These registers have the capacity to transform physical stimuli from the environment into physiological representations; in this form information is able to be temporarily retained in the memory system (Norman, 1970). The majority of research in this field has focused on the visual and the auditory sensory registers, although each of the senses theoretically does have this capability (Eysenck and Keane, 1990).

Although the sensory registers are capable of recording a vast number of stimuli at any one time, the system has severe limitations in terms of the duration it can retain information. Sperling's (1960) research on the visual register, using letter arrays, indicated that although most visual information was initially registered, the majority was lost in less than half a second. Sperling (1960) concluded that little recognition or assignment of meaning had occurred by this point; however, Butler (1974) disputed this contention. Butler's results suggested that the number of letters that could be remembered increased if the letters in the array resembled English words.

In a manner similar to Sperling (1960), Darwin, Turvey and Crowder (1972) investigated the capabilities of the auditory sensory register. Similarities between the function and process of visual and auditory registers were found to exist. Darwin et al.'s (1972) results indicated that as the time delay between the experimental stimuli and the instruction to recall the information increased, memory for auditory stimuli decreased. Auditory registers were found to have the ability to

retain stimuli for periods of 2-3 seconds (Darwin et al., 1972; Treisman, 1964), which is longer than visual stimuli can be held.

To summarise, the key function of the sensory memory system is its ability to retain some representation of stimuli after the physical stimuli no longer exists in the environment: hence the conceptualisation of sensory memory as a ‘holding tank’. Temporary retention of stimuli allows the transferral of information to other parts of the memory system and facilitates the process of meaning extraction from these stimuli within the memory system as a whole (Eysenck and Keane, 1990). Norman and Rumelhart (1970) contended that recognition and identification of meaning take place in the transferral of stimuli between sensory and short term memory because their findings suggested that it was the meaning rather than the physical characteristics of the stimuli that was present in short term memory.

1.1.3 Short term or working memory

Limitations are also integral to defining the operation of short term memory. Miller's (1956a) landmark paper identified the severe capacity limitations of short term memory. In fact, Miller asserted that our immediate memory system is limited to dealing with around only 5 to 9 items or chunks of information at any one time. In subsequent years, this estimate of short term memory capacity has been further reduced: Simon (1974) suggested working memory capacity was closer to five chunks; while Halford, Maybery & Bain (1986) and Sweller & Chandler (1994) proposed this number was reduced further if the information elements needed to be combined or processed concurrently.

Short-term memory also possesses limitations in the duration information can be retained. It can only be held for a few seconds unless rehearsal takes place; if attention from rehearsal is diverted, the information currently stored in short term memory becomes unavailable (Ericsson and Kintsch, 1995). Peterson and Peterson (1959) illustrated this phenomena by asking participants to recall a string of letters after a delay ranging from between three and eighteen seconds. During the delay period, participants completed an alternative task which hampered their ability to rehearse the letter string. The researchers observed a strong relationship between the amount recalled and the length of the delay before recall. As the period of delay increased there was a rapid deterioration in the participant's ability to recall the letters; in fact, after eighteen seconds only about ten percent of the original material was recalled.

1.1.3.1 Working Memory

“The concept of working memory has increasingly replaced the older concept of short-term memory” (Baddeley, 1992: 556). The first cognitivists viewed short term memory as a unitary system whose major role was temporary storage (Baddeley, 1992); thus short term memory was viewed as playing a relatively passive role in cognition. However, during the 1970s, evidence from the field of problem solving emerged indicating short term memory’s involvement in information processing rather than just information storage (Greeno, 1973; Simon, 1975). These findings fuelled an evolution of the concept of short term memory into ‘working memory’: a system of memory that processed or ‘worked on’ information as well as temporarily storing it. It was hypothesised that while working memory executed processes such as addition or comparing the premises of arguments, it also kept the information required by the processing accessible (Haberlandt, 1994). For example, a piece of text often refers to its subject by using a pronoun such as ‘it’ rather than repeating the noun. Working memory is involved in keeping information important for comprehension, such as the referent of ‘it‘ active, as well as executing the cognitive process of reading. The limitations of the short term store are still hypothesised to be critical to its method of functioning, despite the transformation in nomenclature. For example, the storage and processing functions of working memory directly compete for resources if necessary: Posner and Rossman (1965) demonstrated this characteristic by showing that as processing became more cognitively demanding, the ability to store information decreased.

In contrast to short term memory, working memory’s role in the cognition process was viewed as extremely active. The ‘active processor’ view has progressed in the field to the extent that it has been proposed that all cognitive operations are carried in the short term memory system (Schneider, 1993). Shiffrin questions the ability of the short term system to cope with all these functions “Is there room in the short term system for

all the necessary information and machinery?" (1993: 195). He has suggested instead that many of these operations are automatic, and therefore are not using short term capacity. Other theorists have taken the view that although processing and storage are important functions of working memory, they are certainly not the only functions. Baddeley (1992) and Schneider and Detweiler (1987) have contended that it is the coordination of memory resources that is the prime function of working memory.

Modern theories of memory also contend that the working memory system is divided into multiple components: each component has different capabilities to deal with certain types of information. An example of the multi component approach is Baddeley's research (1981, 1990, 1992). He divided working memory into three major parts: the central executive, the articulatory or phonological loop and the visual spatial sketch pad. It is proposed that the central executive has an overseeing role: it is the control centre of the system, it therefore selects and operates the appropriate cognitive processes. Additionally, the central executive has a storage function with the result that capacity limitations are applicable to this part of the system. Storage also occurs elsewhere in the system, the articulatory loop maintains verbal material through sub vocal rehearsal and the Visuo-Spatial Scratch Pad conserves imagery and spatial material through visualisation.

Whichever detailed model of working memory operations is accepted, the most important point about working memory for this thesis are the limitations of capacity and duration that apply. It is these limitations that have the most readily visible impact upon memory and learning.

1.1.4 Chunking

Although Miller (1956a) initially received much acclaim for placing a numerical value on short term memory capacity, Baddeley (1994) has contended, when assessing the impact of Miller's work for the centennial issue of Psychological Review, that his enduring legacy is the concept of chunking. A chunk is any stimulus that has been unitised through previous experience (Simon, 1986).

Miller (1956a) stated that short term memory was limited to dealing with 5-9 chunks of information at any one time. However he further proposed that the amount of information in a 'chunk' was not set: the amount was dependent upon the amount of previous knowledge a person had in the specific area. Thus, although the memory span is fixed by the number of chunks, the amount of information involved can increase by building larger chunks. Humans form chunks by grouping information according to patterns or meaning. As learning in an area increases so does the information that can be grouped into a chunk and therefore the amount of information the person can remember. For example, a chunk of information could be single letters, single words or whole phrases, yet native language speakers would have the ability to remember approximately 7 items from each of these categories. Thus while most English speaking people would remember around seven of these twelve letters K, B, A, N, E, Y, E, K, C, L, H, S; if the same letters were presented in another order, as English words, LAKE; BENCH; SKY, the same participants would have little difficulty remembering all twelve letters. Miller (1956b) used the analogy of a money purse and coins to explain chunking and the capacity limitations of short term memory: short term memory is a like a small purse that can only ever hold around seven coins. What can change however, is the denomination or value of each of the seven coins: they can be of low value (single letters or numbers) or of a much greater value (whole words concepts or relationships). There is some evidence however, that as the information contained

in a chunk increases, the number of chunks retained in memory decreases (Simon, 1974). The ability to chunk information is an essential part of learning; in fact, Newell (1990) contends it is *the* most essential process of learning.

A valid criticism, however, has been levelled at the concept of chunking: since there is no way to determine the size of a chunk independently of the measurement of memory span, the contention that the number of chunks retained is fixed, loses it's empirical validity. In fact, the definition of a chunk becomes quite circular since a chunk is the amount of material that short term memory has the capacity to hold five of (Simon, 1974).

Despite these criticisms, several individual's remarkable memory performances, based on chunking strategies, have been recorded. Ericsson, Chase and Faloon (1980) investigated the digit span ability of a person over the course of eighteen months. The participant, a university athlete, practised 3 to 5 hours a week over the period and improved the number of digits he could remember from seven to nearly eighty digits. The researchers investigated the strategies the participant had used: he had related the strings of random digits to running times. The participant was able therefore, to make meaningful units from the lists of numbers. As the list of numbers continued, he constantly recoded and regrouped numbers into a set hierarchical pattern, while rehearsing the most recently presented numbers. These results do not point to an improvement in the subject's working memory: when presented with strings of letters his performance was average. What had changed was the amount of a certain type of information the participant could include in a chunk.

Gobet and Simon (1996) explored the ability of the first author, a former Chess Master, to remember chess positions. Over 150 sessions his retention of pieces improved from 40 pieces to an average of 123 pieces. In 4 cases he was able to correctly place more than 160 pieces. Insight into the memory process was

provided by this relatively unique experimental methodology. Initially, the subject did not use any memory techniques so a baseline measure was recorded. He then developed the strategy of meaningfully associating set chess positions with the names of world champion chess players. Over the trials, the subject shortened the list of names to their first syllables in order to facilitate subvocal rehearsal of the list during the association phase.

Ericsson and Polson (1988) investigated another case of amazing memory ability. The memory skills of a waiter, who could remember up to twenty food orders at once without the use of external memory aids, were examined. The waiter's strategy involved regrouping the presented information categorically; for example, rather than remembering all the aspects of a meal in the order they were presented by each diner (as the control group of university students did), he grouped, for example, the whole table's orders for salad dressing into one category. Each category was then linked to a mnemonic strategy. The researchers probed the generalisability of the waiter's encoding strategy by changing the order information was presented, the type of information presented and the category structure of the information. It was found that the waiter used his original encoding strategy regardless of the type of information involved or the order information was presented in. As the categorical structure of the information was randomised and the information became less meaningful, the waiter's memory advantage was increasingly eliminated. In fact, when the information became the most random, the participant was recorded using a novice recall strategy. This example of skilled memory performance illustrates the role of prior knowledge in chunking: underlying the waiter's performance was a strategy to recode the given information into meaningful chunks, based on prior learning. Results from an analysis of the waiter's encoding and retrieval processes allowed the authors to reject the theory that information was chunked in short term memory; it was their conclusion that skilled memory performance involves the use of long term memory.

1.1.5 Long term memory

After being processed in working memory, information may be passed on to long term memory for storage. Where sensory memory and working memory are characterised by limitations, long term memory is characterised by a lack of restrictions in two critical respects. Firstly, long term memory has the capability to store information in a relatively permanent manner; and secondly, it seems to have a vast, virtually unlimited capacity to store information. Humans are generally unaware of the vast bulk of information they have stored in long term memory because this information does not impede upon conscious awareness until required. When certain information is required, it is retrieved back into consciousness from long term memory stores.

Storage in long term memory is much more durable than in the sensory or short term stores; yet, it remains true that a massive percentage of the information we encounter in our daily lives is quickly forgotten. Generally, it is accepted that storage in long term memory occurs only after storage in short term memory as was originally proposed in early cognitive models of memory such as Atkinson and Shiffrin's (1968) model. These researchers also considered storage to be a function of amount of time stimuli had been held in short term memory. On the other hand, Anderson (1983) has suggested storage in long term memory is a function of the amount of times information enters short term memory. Debate upon this issue still continues, however, research has been able to more clearly define factors that improve the probability of information recall.

One proposal for improving the probability of information being stored and subsequently recalled from long term memory is the encoding specificity hypothesis (Tulving and Osler, 1968 and Tulving, 1983). This states that if the conditions that were present at encoding are present at retrieval, memory will be

enhanced. The hypothesis has been shown to hold true for a large range of circumstances such as location, mood, presence of cues (Godden & Baddeley, 1980; Overton, 1985; Tulving, 1983). For example, information recall improves if cues provided at encoding are also present at retrieval. This point was strongly illustrated by Mantyla's (1986) research where participants studied lists of six hundred words and generated either one or three associated word cues to the target words. Previous research has shown that the average recall rate, without any cues at retrieval, would be about 5-7% of 600 words. At retrieval, subjects were shown the cue words that they had generated: in the three cue group, recall for the 600 word list averaged at 90%; while in the one cue group, recall stood at 62%.

Another theory explaining the influences upon long term memory recall is the 'levels of processing' theory by Craik and Lockhart (1972). Their argument suggested that when information was encoded, the more time spent thinking about the meaning of the information and its semantic relationship to previously learnt knowledge, the more likely information was to be recalled. Semantically driven processing was labelled 'deep' processing, while processing which focussed on superficial characteristics of the information, such as the number of words starting with capital letter, was labelled as shallow processing. Deeper processing increased the likelihood of information recall because it produced a stronger memory trace. The theory has been made more substantial in subsequent years through the inclusion of other factors which influence long term memory recall such as the elaboration of the processing and the distinctiveness of the information to be remembered.

Apart from the relative permanence of long term memory storage, the other remarkable feature of this system is its seemingly unlimited capacity to store information (Newell & Simon, 1972). Evidence of the storage potential of the system came from research in the field of problem solving, an area previously

regarded as having little relationship with long term memory. De Groot's (1966) results suggested that the only factor differentiating more and less expert chess players was their ability to accurately recall realistic chess board configurations. de Groot concluded chess expertise was developed through an ability to recognise most of the configurations encountered in a typical game, consequently allowing the player to plan the best move associated with each particular configuration. Thus from this finding, de Groot (1966) hypothesised that expertise was a function of the amount of prior knowledge of the subject area stored in long term memory. Simon and Gilmartin (1973) built a computer model to reflect this process and were able to estimate that a chess Grand master holds between 10 000 and 100 000 chess board configurations in their long term memory. Research in many other subject domains supports de Groot's hypothesis that long term memory plays a critical role in expertise (Egan & Schwartz, 1979; Jefferies, Turner, Polson, & Atwood, 1981; Sweller and Cooper, 1985).

1.1.5.1 Types of knowledge stored in long term memory

Long term memory has a vast storage capacity; in addition, it is able to store many different types of knowledge. Distinctions have been drawn between declarative and procedural knowledge; semantic and episodic knowledge and implicit and explicit knowledge. Anderson (1983) drew a distinction between declarative knowledge and procedural knowledge. Declarative knowledge is factual information such as 'blue and yellow are primary colours that when mixed together form the colour green'. Procedural knowledge is the knowledge that tells you how to perform certain activities; for example, what proportion of blue and yellow oil paint is required to mix an emerald shade of green. Tulving (1972) has made a distinction between episodic and semantic knowledge. Episodic knowledge is information that has some autobiographical component because it is associated

with a certain time or place in your life while semantic knowledge is general factual knowledge that has no binding ties. Thus, a person may have episodic knowledge of the city of Jakarta through memories of a holiday spent there, while their semantic knowledge may include the fact that Jakarta is the capital city of Indonesia. The final distinction in the type of knowledge stored by long term memory is implicit and explicit memory as proposed by researchers such as Jacoby and Witherspoon (1982). Explicit memory involves conscious recall of information such as recognising words on a list from a previous list. Implicit memory occurs when behaviour is affected by a previous experience despite a lack of conscious memory for that experience. For instance, Jacoby and Witherspoon (1982) concluded that participants spelt a homophonic word (such as read or reed) in the less common way because in a previous unrelated task the less common word had been referred to. The authors concluded that this previous exposure to the less common word had unconsciously biased the participant's subsequent free interpretation of the homophonic word.

1.1.5.2 How knowledge is stored in long term memory

Long term memory has a vast capacity to store many types of information relatively permanently. Many researchers have explored the mechanisms that allow this storage. It is generally held that storage in long term memory is associative, where items related in meaning are stored together (Ericsson and Kintsch, 1995). In addition, a metaphor is commonly drawn between knowledge storage and the chemistry of matter. The most basic units, atoms, combine together in various ways to form increasingly complex molecules of knowledge (Eysenck and Keane, 1990). Despite this commonality of approach, many theories have been proposed to explain how information storage is stored in long term memory.

One theory is of 'atom' like units called concepts. A concept is a mental structure representing a meaningful category of information such as *dog* or *colour*. The question of how knowledge is organised into concepts is also much debated. It has been proposed it is rule governed (Bruner, Goodnow and Austin, 1956), dictated by the similarity of the current example to a prototype of the concept in memory (Rosch, 1978) or verified by a probabilistic weighing up of defining and characteristic attributes of the concept (Tversky, 1977). In fact, it may be the case that categorisation of information into concepts is dependent upon all these mechanisms, altering in response to the nature and difficultly of the knowledge to be categorised (Medin, Wattenmaker, and Hampson, 1987).

Propositions are another way that knowledge can be represented. They represent the relationship between concepts (Anderson, 1985) such as *'the door is open'*. The concept of *door* has a relationship to the concept of *open* in this proposition. Productions are *if/then* statements and represent a manner in which procedural knowledge can be organised (Anderson, 1985). They state an action to be performed if a certain set of conditions exist, for example, *'IF the door is open and it starts to rain, THEN shut the door'.*

Schemata are another manner in which knowledge is hypothesised to be organised. Schema theory has become a central concept to cognitive psychology since the 1970s, although the concept was originally proposed much earlier by Bartlett in 1932. Schemas are complex cognitive constructs that store knowledge (Rumelhart and Ortony, 1977; Rumelhart, 1981) although it is proposed that they are also involved in the cognitive processes of encoding and retrieval of information (Rumelhart, 1984). Schema theory will be discussed in more depth in section 1.2 as it is central to this thesis.

1.1.6 Beyond the modal model of memory

Since the 1970s, ideas about memory have continued to evolve. The prominence of a *stage* like approach to memory is decreasing as memory researchers move towards network models of memory and consider connectionist models of cognitive processing. Network or connectionist models of memory aim to simulate the parallel processing humans seem to be able to undertake: in these models processing occurs along many dimensions at the same time; in contrast, the computer metaphor dictates serial processing (McCelland, Rumelhart and Hinton, 1986).

Despite these new directions in memory research, modern conceptualisations of memory still include the stages of memory as has been discussed in this thesis; for example, Anderson's ACT* network model of memory (1983) has used these distinctions. The model, however, emphasised the interrelations between the memory systems. Thus, as a search of memory is undertaken, activation spreads from working memory to units storing information related to the concept in long term memory.

The distinction between the long term and short term stores is further blurred by Ericsson and Kintsch's (1995) proposal of a long term working memory. This memory store has the processing characteristics of working memory but the stability of storage of long term memory. Long term working memory can account for expert performance which places large demands on working memory.

Despite recent advances in memory theory, for the purposes of this thesis it is useful to understand the different stages of the modal model of memory. The key features of each stage of memory: the limited capacity of working memory to process and store information in comparison to the relatively unlimited capacity of

long term memory are salient to this thesis as are the theories of chunking and schemas.

1.2 Schemas

1.2.1 Schemas: an overview

Human cognitive architecture circumvents the problem of an extremely limited capacity working memory by constructing schemas. A schema is a cognitive structure that organises elements of information categorically and stores them in long term memory (Thorndyke and Hayes-Roth, 1977; Chi, Glaser & Rees, 1982; Sweller, 1993). Schemas reduce the impact of working memory's limited capacity by shifting the burden of memory storage to the seemingly limitless storage reservoir of long term memory. A single schema contains an organised network of knowledge, ideas and processes related to a particular concept. Bartlett provided an early conceptualisation of schema theory that forms the foundation of modern work: " 'Schema' refers to an active organisation of past reactions, or of past experiences ... (these) operate, not simply as individual members coming one after another, but as a unitary mass" (1932/1995: 201).

Schemas represent and organise knowledge in long term memory according to the manner in which it is used since a schema represents knowledge from some or all of a stimulus domain (Howard, 1987). Schemas are involved in all stages of cognitive processing. They allow new information to be interpreted and encoded, stored, organised and importantly retrieved back into consciousness (working memory) in the event of the information being required. They also play a role in determining goals, allocating memory resources and more generally facilitate the flow of information through the information processing system (Rumelhart, 1980). In fact, Rumelhart has stated that "schemata are truly the building blocks of

cognition. They are the fundamental elements upon which all information processing depends" (1980: 33).

Schemas allow us to recognise information from different contexts and categorise it accordingly. Consequently, the infinite variety of information captured by our senses can be ignored, with the result that working memory capacity is not overburdened (Sweller and Chandler, 1994). For example, we have a schema for the letter ‘f’ which allows immediate recognition of it, irrespective of the countless range of formats the letter can be presented in - f, **f**, *f*, f, F, F, F, f, f, f. The marks on the page creating the letter *f* correspond to our previously acquired schema for this letter, allowing us to recognise and assign meaning to it; the recognition process does not tax working memory however, since the stimuli activates the vast amount of knowledge we possess about the letter *f* in long term memory. Thorndyke and Hayes-Roth (1977) have suggested that since schemas are an abstraction of knowledge from multiple and diverse contexts they therefore possess *structural integrity,* that is, they are independent of any particular instance of stimuli represented by the schema. The functionality of schemas is being able to use knowledge represented from one context in many contexts. In addition, the dynamic nature of schemas results in constantly updated knowledge; Sweller (1994) contended it is this ability to construct and modify schemas that underlies learning.

Schemas also reduce the processing burden on working memory because interacting elements that would otherwise overwhelm working memory can be incorporated in a schema that acts as a single element and imposes a minimal working memory load. For example, algebraic schemas tell us that $a/b = c$ can be transformed into $a = cb$. Both equations and their transformation can be processed easily in working memory by anyone who is competent at elementary algebra despite the large number of elements. (Elements include relations between symbols as well as the symbols themselves.) A higher-order schema tells us the

location of each symbol and the mathematical procedure required to transform the first equation into the second. The presence of schemas allows pieces of new information to be chunked together, thus, the otherwise difficult task of processing all the interacting elements in working memory becomes a relatively simple task because the elements are incorporated in a schema that itself acts as an element and can be processed readily in working memory.

The availability of a certain schema in memory depends on a combination of factors such as strength of the stored information, the degree of the match between the incoming information and the schema as well as the recency and frequency of previous activations (Thorndyke and Hayes-Roth, 1977). The details of each particular representation of a schema are tied to the general schema or representation of that knowledge through a 'context-preserving relation' (Thorndyke and Hayes-Roth, 1977). This relation dictates that in order to remember a specific detail from a specific occurrence of an event the general schema as well as the specific schema is recalled.

1.2.2 The features of schemas

Rumelhart and Ortony (1977) have identified some key features of schemas. Firstly, schemas are composed of variable slots; for example the schema of a car includes wheels, body, doors, seats, a steering wheel and a windscreen. These variables can be filled by a wide variety of values; for example a car body can vary from a small, sleek, enclosed sports car to a large bodied utility with a cabin and attached tray. Rumelhart (1980) used the analogy of a play to describe this structural characteristic of schemas: a play is composed of different roles played by a variety of actors just as schemas are composed of different variables that can be filled by a variety of data or situations. An example of this analogy is the schema for 'buy'. Rumelhart has suggested that our schema for 'buy' is composed of roles such as buyer, seller, merchandise and money; these roles can be filled by a wide range of people or situations without changing the essential nature of the schema. It is the case, however, that a prescribed order or relationship between schema's variables does exist; for example, in the 'buy' schema, buyers will always be buying from sellers regardless of any unique features of a specific 'buy' situation.

Secondly, Rumelhart and Ortony (1977) stated that schemas can embed within one another; for example, wheels form part of our schema for car, but this concept itself is formed by sub-schemas such as axle, hub, tyre, rim and so on. Thirdly, the level of abstraction of the knowledge represented by schemas can vary widely: a car schema is a concrete object, yet we have schemas for concepts such as 'compassion' or 'creative'. And finally, Rumelhart and Ortony (1977) suggested that schemas encompass all knowledge rather than just conceptual definition of objects or situations. Thus, a schema contains both factual knowledge as well as personal experience; additionally, a schema is flexible enough to allow for the enormous range of stimuli that exist within a category. Rumelhart (1980) has expanded upon this list of schema features by adding two points: he suggested firstly that schemas

are active processors; and secondly their function is to evaluate whether the activated schema can adequately account for the pattern of data under processing.

Information comprehension is one of the key functions of schemas. "To understand something is to select a schema that provides a plausible account of it, and thus allows us to assimilate it to something we know" (Howard, 1987: 39). When a schema is selected, it is said to be instantiated, that is, a specific instance of the schema is created. Schemas guide the organisation of incoming knowledge since information appropriate to the various variable slots of the schema are actively sought (Thorndyke and Hayes-Roth, 1977). The schema, when some or all of the variable slots are filled, allows comprehension of the stimuli to occur. Rumelhart (1980) viewed the interpretation and comprehension role of schemas as analogous to parsers as they are a device that sorts through the stimuli in order to verify if the information variables that constitute the schema are present. He also suggested schemas are like procedures because an active process occurs where the component parts of the schema are involved in determining whether the schema itself can account for the stimuli it is seeking to explain.

If information to fill various slots in the schema is unavailable, the presence of this feature is still able to be inferred because the slot is then filled by the default value of the schema. Thus, when passing a restaurant and witnessing food arriving at a table, one is able to understand this event because the presumption is made that the food has been ordered, by the diners, from a menu. The default value of the *eating out* schema is that food is ordered by diners. The default values of schemas aid understanding and reduce the information processing burden of working memory because information can be inferred from the generalised case. However, the presence of default values may lead to errors; for example, it may be the fact the these particular diners have chosen to partake of a *degustation* menu and thus the chef, rather than the diners was responsible for ordering the food. Rumelhart (1980) likened this characteristic of schemas to a theory: schemas, like theories,

attempt to describe the nature of things and allow humans to make predictions about unobserved events.

The knowledge contained in schemas is induced or abstracted from many experiences (Eysenck and Keane, 1990). Rumelhart and Norman (1978) and Rumelhart (1980) further proposed that schemas evolved in three ways. Firstly, *accretion* occurs when a new experience is recorded and added to the bank of information held within the schema. Secondly, *tuning* occurs when the information maintained in the schema is refined, elaborated or made more generally applicable. Finally, *restructuring* of schemas involves the formation of new schemas either through the copying and subsequent modification of an existing schema or through the repetition of a certain spatial-temporal configuration of stimuli. Thorndyke and Hayes-Roth (1977) investigated how memory was affected by repeatedly using the same schema to remember different instances of a type of information. Given the slot structure of a schema, repeated use of the schema, within a certain time period, entails many pieces of information of the same type in each slot. The results of this research indicated that as use of the single schema increased, memory for shared information was strengthened, increasing its accessibility in memory. However, the repeated use of the schema did cause interference in learning because the experiment's participants showed difficulty retaining the information that was unique to each of the contexts. Furthermore, the authors contended that as the discriminability between the learning contexts increased, the interference to learning was reduced. In fact, Thorndyke and Hayes-Roth (1977) were able to correctly predict the facilitative and inhibitory effect that repeated use of the same schema had on learning.

Therefore, the nature of schemas as described above allows them to play an active role in such processes as perception, understanding, remembering, learning and problem solving (Rumelhart, 1980). Moreover, through the knowledge that they hold, schemas ultimately guide our learning, actions and behaviour (Howard,1987).

1.2.3 Types of schemas

Mandler (1984) has suggested we possess three different types of schemas: schemas for scenes, schemas for events and schemas for stories. Scenes involve the physical or spatial relations of objects in space. For example, the schema for kitchens involves the relations among typical objects found in that room such as oven, sink and refrigerator, while a landscape includes hills that rise from the ground and rivers that flow down stream (Mandler, 1984).

Events involve a sequence of events in time such as catching the bus, which requires checking the time the desired bus leaves; getting to the bus stop; signalling the bus driver; boarding the bus; and paying the fare. Events could also be classified as scripts in the manner of Schank and Abelson (1977). Bower, Black and Turner (1979) investigated Schank and Abelson's proposal of a *restaurant eating* script by collecting a list actions and sub-events involved in this topic. They found considerable agreement in the actions that were nominated to be part of this event. Rabinowitz and Mandler (1983) examined how event schemas influenced memory and retention of information. Their experiment contained two groups: one group received an organised sequence of phrases describing the actions involved in various events (schema consistent information); for example, the phrases making up the "going skiing" event schema was as follows: go to mountain; put on down jacket; buy lift ticket; ski down slopes; drink hot chocolate. The other group received the same phrases, yet the order of the topics the phrases belonged to was random (information inconsistent to schema); for example, this group may receive a phrase from the "going skiing" event followed by a phrase from the "going to a baseball game" event schema and so on. The groups were compared for their memory of the phrases; memory for the phrases was significantly increased if the phrases had been presented in the schematically organised fashion.

Kintsch has proposed that a story schema is "a set of expectations about the structure of the story" (1977: 40). Story schemas aid in understanding of stories of cultural relevance because the story schema possesses slots for variables such as setting, characters, episodes of the story such as beginning or end as well as the goal of the story (Howard, 1987). Research has also indicated story schemas benefit learning in an educational setting (Yates & Chandler, 1991; 1994); for example, university level science students, when taught a schema for the construction of typical journal articles, increased their reading speed, conceptual understanding and ability to recall information from scientific papers (Samuels, Termyson, Sax, Mulcahy, Schermer and Hajovy, 1988). Similar results were recorded by Cook and Mayer (1988); they found recall and problem solving performance significantly increased after college science students were taught six common text structures underlying scientific text. These findings have been repeated with other forms of expository text (Pinchus, Geller and Stover, 1986). The successful results arising from teaching schemas for certain specific categories of stories seem similar to the concept of a 'problem' schema: a schema that contains the information on solution strategies for specific categories of problems (Hinsley, Hayes and Simon, 1977; Chi, Feltovich and Glaser, 1981; Gick and Holyoak, 1983). Indeed, Hinsley et al.'s (1977) research on the existence of schemas for types of algebra word problems actually viewed these problems as a type of 20th century story.

Evidence suggesting the existence of a problem schema is provided by the speed[1] and relative lack of information[2] by which experts are able to categorise problems (Hinsley et al., 1977; Chi et al., 1981). Hinsley et al. (1977) showed that even when a substantial part of the words in a problem were nonsense words, schema

[1] Lesgold, Rubinson, Feltovich, Glaser, Klopfer and Wang (1988) found that for expert radiologists faced with a diagnostic task, a problem type schema was triggered within the first 2 seconds of viewing an X-ray.
[2] Some subjects in Hinsley et al.'s (1977) experiment were able to categorise a problem as a certain type after only reading the first phrase of the problem statement (less than 18% of the total problem statement). This was also apparent in the verbal protocols taken from the expert in study 4 of Chi et al.'s (1981) work.

knowledge about a problem type was still used to aid comprehension and solution. In order to assess the knowledge associated with the problem type labels expert and novice physicists had given to categories of problems, Chi et al.'s (1981) subjects were given three minutes to talk about the category label, the problems included in that category and the solution methods involved in these problems. Subsequent analysis of these data revealed that for physics experts, problem category labels activated basic physics principles, procedural knowledge about how to apply these principles for solution as well as conditions under which they apply. This was not the case with more novice physicists. Expert subjects also had access to *IF - THEN* specific problem solution action rules that did not appear to be present in the novice's problem category knowledge. It seems likely, given these results that part of an expert's problem solving success is the presence of detailed problem schema, containing knowledge about the solution paths. Chi et al. (1981) hypothesised that for expert problem solvers in a domain, the problem statement leads to a partial cognitive representation of the problem, which activates a problem schema. The schema, with its various information slots, then guides further information gathering toward a solution path. Progressively, the schema slots require information to be derived from the information given in the problem statement rather than just this information itself. An algebraic representation of the problem can then be developed and the problem solution achieved.

Gick and Holyoak (1983) have further developed the idea of a *problem type* schema. These authors suggest specific problem schemas contain a network of information about a certain type of problem such as problem goals and constraints as well as procedural knowledge about the problem solution. For example, the problem schema for algebra tells us that $a/b = c$ is equivalent to $a = cb$ and $b = a/c$, it provides information about how to execute these transformations as well as the knowledge that a,b,c are symbols with the ability to be replaced by any numbers. Problem schemas also fill in information not provided as well as distinguishing

between information that is necessary or unnecessary for problem solution (Low and Over, 1992).

Other researchers have suggested a categorisation of schema types should also include schemas for persons (Hastie, 1981; Anderson, 1980). A general person schema has variable slots for things like motivations or personality traits and this is instantiated when a personal relationship is developed. The instantiated schema allows predictions about the behaviour of others to be made, resulting in the ability to judge if the person is acting out of character (Anderson, 1980). Additionally, the person schema can be instantiated to represent stereotypical groups of people such as librarians or surfers.

1.2.4 Schemas & remembering

Bartlett (1932) published a large series of experiments that pioneered research into the nature of remembering. Due to the predominance of Behaviourist thought during this period in psychology (Mayer, 1992), in addition to Bartlett's imprecisely controlled and measured experimental techniques (Kinsch, 1995), the work did not receive the attention it deserved. This situation changed after the 1970s when cognitive psychology focused on schema theory. Bartlett's (1932/1995) contribution to memory research cannot be understated (Kinsch, 1995): he was the first to identify in psychological terms[3] how a person's previous experience of the world affected their understanding and memory of new events. In fact, Bartlett took the view that "(R)emembering obviously involves determination by the past. The influence of schemata is influence by the past" (1932/1995: 202). Bartlett also pioneered the concept of memory as a reconstructive process, a notion that enjoys widespread support today.

Bartlett's most famous experiments used the *method of serial reproduction* and a native American folk tale, *The War of the Ghosts*. The method of serial reproduction is similar to the game of Chinese whispers: recollections of a story are passed along a chain of people. Bartlett got British University students to read a story and write out their recollection of the story after a delay. This version of the story was handed to the next person to read and so on. Bartlett looked at the changes to the story that resulted. He had deliberately chosen a piece of text that would be culturally unfamiliar to the participants in his study as cultural experiences and expectations form part of the schematic knowledge he was interested in exploring.

Bartlett found that the folk tale was transformed down the serial chain: it became shorter and more coherent, more ordinary, more rational and culturally palatable than the initial folk tale. For instance, supernatural elements of the tale such as the mention of ghosts and the *something black* that had come out of one of the warrior's mouth were quickly conventionalised then expunged altogether. To explain this Bartlett stated "any element of an imported culture which finds very little background in the culture to which it comes must fail to be assimilated" (1932/1995: 125). This proposal, that appropriate schemas must be available for information to be retained, underlies learning and schema theory today (Mayer, 1992). Bartlett also found details from the original story were transformed, *canoes* and *paddling* were conventionalised into the more familiar terms of *boats* and *rowing*. This example further clarifies the role of schemas in Bartlett's notion of reconstructive memory: memory does not store an exact duplicate of a stimuli, rather it retains the *gist* or central meaning of stimuli. When required, the retained information melds with a person's previous experience, that is, the schema to which the new learning was assimilated to; this schema then facilitates the reconstruction of the memory. Previous knowledge therefore, exerts influence upon the contents of our recollections (Mayer, 1992).

Memory for a stimuli is attached to a schema and it is the general knowledge and default values stored in the schema that help to reconstruct a detailed memory. Baggett (1975) has provided further evidence that it is the *gist* of the information that is encoded rather than the exact details. In this experiment, participants viewed a series of cartoon images which told a story; for example, Figure 1.2 tells the story of a long haired man going to the barber shop and then leaving the shop with short hair. The actual act of the haircut was not depicted, although this event

---

[3] The philosopher, Kant, in 1787, had proposed that humans possess innate structures for organising their perceptions of the world (Eysenck and Keane, 1990). His is the first recorded use of the word schema to describe this conceptual content. This historical perspective has influenced subsequent psychological

can be inferred from the sequence of images and in fact, this event forms the *gist* or explanatory meaning behind the images. In a subsequent recognition test, an image of the haircut actually occurring was included (see Figure 1.2). Immediately after the study, participants were able to recognise that they had not seen this image; at a post test a week later however, the majority claimed they had seen this image. These results suggest that over time, the distinction between the actual event and the inferences drawn at the point of encoding blur. The meaning behind the images was retained and when required, the *haircut* schema was reactivated and it was this schema that filled in the details of the memory and shaped what was recalled. Further evidence that memory retains a general impression of meaning rather than detailed accounts has been reported by Bransford and Franks (1971), Bransford, Barclay and Franks (1972) and Johnson, Bransford and Solomon (1973). Each of these studies used language rather than images as the basis for their research, adding to the robustness of this proposal.

---

researchers, such as Rumelhart (1980), in their choice of terminology.

Figure 1.2: The sequence of pictures used by Baggett (1975) to tell the story of a haircut. Note that the test picture was not included in the study sequence but can be inferred to have occurred between the 3rd and 4th pictures.

Bartlett further described the nature of remembering as “imaginative reconstruction” (1932/1995: 213): information from previous experience, held in schemas, helps to reconstruct our recollection of a stimuli from the *gist* retained in memory. The notion of memory as reconstruction has many strong correlates in the literature. An example of this is Carmichael, Hogan and Walter’s (1932) investigation of visual memory; in particular, the researchers examined participant’s memory for ambiguous figures. A set of figures were shown to two groups of subjects, differing only in the label attached to each figure. Figure 1.3 provides an example of some of the figures and labels used in the experiment; for example, the first figure in Figure 1.3 was labelled as *eyeglasses* for one group and as a *dumbbell* for the other group. When the figures were reproduced by the participants from memory, it was found that the label attached to each figure clearly influenced the manner in which it was recollected (see Figure 1.3). Thus, schemas influence how information is recalled; there is also much evidence to suggest that schemas greatly influence how information is interpreted, comprehended and therefore remembered.

Figure 1.3: Some examples of the ambiguous figures, the figures' labels and subjects' recollections of these figures from Carmichael et al.'s (1932) research.

Pichert and Anderson's (1977) research provided a strong illustration of the capacity of schemas to influence what information is attended to and therefore, what information is retained in memory. Students were asked to read a passage describing a house from the perspective of either a potential home buyer or a burglar. The researchers felt these instructions would activate different schemas hence influencing the information recalled. The results supported this hypothesis since the home buyer group remembered significantly more home information such as the number of bedrooms and the presence of a leaky roof while the burglar group recalled other details such as the number and locations of bikes and jewellery. Given the capacity limitations of working memory, the results of Pichert and Anderson (1977) point to another function of schemas: schemas act as a guide, directing attention to the most relevant information.

1.2.5 The role of schemas in interpretation and comprehension

Anderson, Reynolds, Schallert and Goetz (1977) investigated how schemas can affect the interpretation of information and hence memory and learning. The authors suggest that a piece of text has no *absolute* meaning because the meaning of the text will be influenced by the schemas that the person reading the text holds. The type of schemas and the complexity of those schemas are positively related to an individual's expertise in a given domain[4]. The authors further predicted that text which is ambiguous will be interpreted by the learner's higher order or more complex schemas. Ambiguous words and phrases can have several meanings: within a passage of text it is possible to deduce their meaning from the context of the surrounding sentence or sentences. In order to test these hypotheses, Anderson et al. (1977) constructed two passages of text which could have two sensible interpretations. One passage described a prisoner planning his escape, yet it could also be interpreted as a passage describing a wrestling match by those who understood the terminology and nuances of wrestling. The second passage described a group of friends playing cards but could also be understood as a group of friends playing musical instruments together. Two groups of participants were recruited: music experts and wrestling experts. The results, from both recall and recognition tests, supported the authors' hypotheses**.** Wrestling experts interpreted the first passage to be about wrestling, the second about cards; in contrast, the music experts interpreted the first passage to be about a prison escape and the second to be about music playing. Interestingly, the researchers also found that even when explicitly asked, less than twenty percent of all the participants had any awareness that the passage could be interpreted differently: thus generally, the activated schemas were so dominant that other possibilities

[4] See the next section of this thesis, section 1.4, for further details on expertise and schema development.

were not even entertained. Howard (1987) contended the difficulty in dislodging the interpretation of an activated schema is one of the few drawbacks schemas have for information processing. Overall however, these results elucidate the active role schemas play in the interpretation, understanding and learning of information.

Bransford and Johnson (1972) provided some interesting evidence about the influence of schemas upon learning and understanding. Their thesis stated that comprehension of text is achieved through the joint product of the input (sentences of the text) as well as prior knowledge. This hypothesis was tested by manipulating the prior knowledge of the participants, in regard to the context of the text. Figure 1.4 depicts the scene ruminated upon in the passage of text the participants of the experiment were asked to study (also see Figure 1.4). This figure therefore provided contextual information relevant for understanding the passage of text and thus the presence or absence of Figure 1.4 allowed prior knowledge to be manipulated. One group of participants, the *context before* group, received the figure before reading the passage; another, the *context after* group, received it after reading the passage; another group, the *partial context* group, received it in a partial form (the elements of the figure were rearranged in a manner that did not explain the text passage) and the final group of participants, the *no context* group, did not view the figure at all. In comparison to all the other groups, the *context before* group showed significantly better comprehension and recall of the text. This result was achieved despite all participants being familiar with the words and lexical structure used in the passage. The figure provided the *context before* group with a schema which allowed them to understand and remember the passage more effectively. The poor results of the *context after* group suggest that it is not merely contextual knowledge itself which improves performance but contextual knowledge at the time of learning that is vital for comprehension and thus recollection. Rumelhart (1980) suggested a failure of comprehension occurs when a schema has not been activated because either a person cannot locate or cannot construct an appropriate schema. This explanation seems valid when applied to

Bransford and Johnson's (1972) experiment. The *context after*, *partial context* and *no context* groups all struggled to understand and remember the text passage, yet the group that was provided with an appropriate schema, the *context before* group, were able to perform these activities.

If the materials popped, the sound wouldn't be able to carry since everything would be to far away from the correct floor. A closed window would also prevent the sound from carrying, since most buildings tend to be well insulated. Since the whole operation depends on a steady flow of electricity, a break in the middle of the wire would also cause problems. Of course, the fellow could shout, but the human voice is not loud enough to carry that far. An additional problem is that a string could break on the instrument. Then there could be no accompaniment to the message. It is clear the best solution would involve less distance. Then there would be fewer potential problems. With face to face contact, the least number of things could go wrong.

Figure 1.4: The text passage and the figure used to manipulate the presence or absence of context in an experiment from Bransford and Johnson (1972).

Bransford and Johnson (1972) replicated and extended upon the previous results using different textual passages. These passages described a series of commonplace procedures such as washing clothes or making and flying a kite; the action itself however, was not named within the passage. Context was manipulated through the presence or absence of a descriptive title to the text passages. Three experimental groups with different levels of prior knowledge were compared in this series of experiments: the *context before*, *context after* and *no context* groups. In a similar manner to the previous experiment, the *context before* group averaged significantly greater comprehension and recall scores than the other groups. These experiments illustrate the profound impact prior knowledge or context can have on learning: the activation of schematic knowledge aids comprehension of information at the time of learning and thus increases later recollection of it.

1.2.6 Schemas: a conclusion

Together, the studies discussed in this section reveal the enormous role schemas play in information processing. Humans are not passive learners; they are active learners, whose prior knowledge influences their perception, learning and recollection of knowledge. Schemas not only provide the structure for long term memory but form the basis of our intellectual capacity (de Groot, 1966; Sweller and Cooper, 1985). This notion will be explored further in the next section.

1.3 Problem solving

1.3.1 Parts of a problem

Despite differences in terminology, cognitive psychologists have identified the main parts of a problem to be:

- *Initial state* – the problem begins with certain givens such as information on values, objects or conditions;
- *Goal* – the desired state to be achieved upon successful completion of the problem;
- *Operators* – objects or concepts that can be manipulated to reach solution;
- *Constraints on operators* – include those factors in the problem that restrict the use of the operators in the problem (Mayer, 1992; Brunning, Schraw and Ronning, 1990).

A problem exists therefore when the initial state differs from the goal state. For example, with the problem:

'*you are required to get to Perth from Sydney within 12 hours*' .

The initial state of this problem is Sydney because this is where the problem solver begins from; the goal state is Perth, as this is the desired point of completion to this dilemma; possible operators include means of transport from Sydney to Perth such as plane, bus, car, bike, helicopter or boat; a constraint on the operators exists because there is a time limit of less than 12 hours allowed for travelling between the locations. Given the time constraint on this problem, the only possible solution

is to travel by air either by plane or helicopter. All of the parts of the problem, together with the possible paths to solution between the initial state and the goal state form the problem space (Newell and Simon, 1972).

1.3.1.1 Ill Defined and Well Defined Problems

The degree to which the parts of problem are defined varies greatly. Reitman (1965) has identified two variables to classify problems types: the degree to which the problem's initial state is defined and the degree to which the problem's goal state is defined. Thus, problems can be categorised into one of four states:

- well defined initial and goal state, for example*, 'x= y+3, y=4, find x*[5];
- ill defined initial and goal state, for example*, 'create something beautiful using useful materials'*[6];
- well defined initial state but ill defined goal state, for example, *'using an oven proof dish and a gas oven combine a tomato, an onion and garlic to make a good meal'*[7]*;*
- ill defined initial state with well defined goal state, for example, *'explain the factors most responsible for global warming'*[8]*.*

[5] Both the initial state (x= y+3, y=4) and the goal state (find x) are clearly defined, with only one possible interpretation.
[6] The initial state (useful materials) and the goal state (something beautiful) are both vague, subjective terms open to multiple interpretations.
[7] The initial state (an oven proof dish, a gas oven, a tomato, an onion and garlic) is clearly defined yet the goal state (a good meal) is vague and subjective.
[8] The goal state (global warming) is clearly defined yet the initial state (responsible factors) is vague and allows multiple interpretations.

Although simplistic, this categorisation does serve to highlight the real difference that exists between problem types.

Most information processing and expertise research has concentrated upon well defined problems and the manner in which humans attempt to solve them. In fact, most problem solving research as concentrated upon transformation problems (Greeno, 1978). Transformation problems have a well defined initial and goal state and a set of operators, constrained by certain rules, which can be manipulated to transform the problem through a set of intermediate problem states to reach solution (Mayer, 1992; Ellis and Hunt, 1990). A wide variety of problems can be classified as transformation problems such as the algebra example used to illustrate the well defined problem type category mentioned above, physics or chemistry problems as well as puzzle and maze type problems.

1.3.1.2 The processes of problem solving

Two important processes occur in problem solving: firstly, a representation of the problem space needs to be generated and secondly, the solution process, requiring a search through the problem space must be undertaken (Gick, 1986). A representation of these two stages of problem solving are presented in Figure 1.5. The problem solver must initially construct a representation of the problem; it is at this point that Gick (1986) has drawn the distinction between schema driven problem solving and other problem solving. If, during the representation phase the appropriate problem schema is activated, then, the correct solution strategy can be implemented. This would allow the solution to be reached unhindered. If no schema for the problem exists or it has failed to be activated, then the problem solver must search for a solution (perhaps using general problem solving strategies); if this search is successful, the task has been completed. If the solution

fails, the problem solver must retrace their steps: either by representing the problem in a different manner or by using a different solution. Schema driven problem solving and problem solving using general problem strategies will be discussed later in this section.

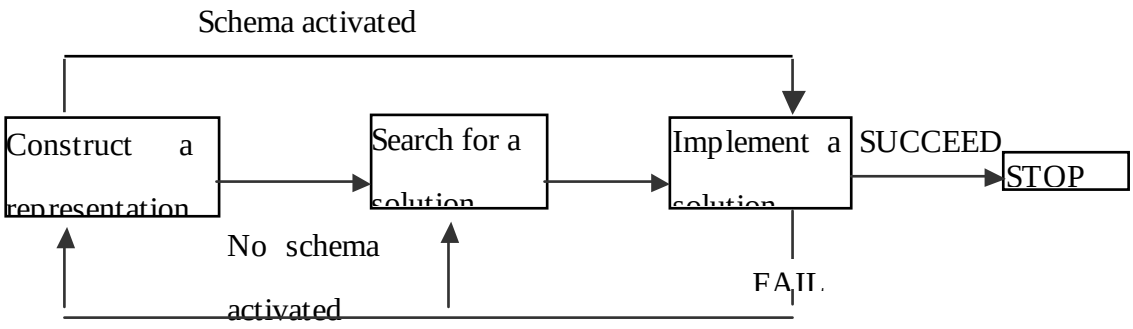


Figure 1.5: A representation of the problem solving process (adapted from Gick, 1986).

1.3.2 Problem solving research in the second half of the twentieth century

In the 1980's, the focus of problem solving research shifted from 'semantically impoverished' areas of knowledge to 'semantically rich' areas (Chi, Glaser and Rees, 1982): thus, the domain the stimulus problems of investigation have been drawn from has shifted towards knowledge filled domains such as chess, mathematics or physics. Before the mid 1970s, the favoured stimulus problem of investigation was artificial puzzle type problems (Gick, 1986). Problems of this type include transformation problems such as the water jug problem (Luchins, 1942), the Tower of Hanoi problem (Newell and Simon, 1972) and the Missionaries and Cannibals problem (Reed, Ernst and Banerji, 1974). These artificial problems were employed because they provided an opportunity for the investigation of problem solving skill development in areas to which subjects were unanimously naive.

1.3.2.1 The golden age of heuristics

The era before 1970 has been labelled as the *golden age of heuristics* (Perkins & Salomon, 1989). Heuristics are rules of thumb or general problem solving strategies such as breaking a problem into sub-problems; using diagrams to represent a problem; the generate-test strategy; working backwards to the solution or limiting the search space through such strategies as constructive search or analogy (Wessells, 1982; Ellis and Hunt, 1990). These strategies do not guarantee a solution but they often facilitate the problem solving process (Polya, 1957). It was commonly thought, before the 1970s, that problem solving was a general ability that was transferable across domains (Perkins & Salomon, 1989), hence general

problem solving strategies were embraced and promoted. Further support for this cause was contributed by the field of computer science, as work in Artificial Intelligence (AI) developed and was used to investigate problem solving; for example, the Logic Theorist, an early AI program, modelled human problem solving in the domain of symbolic logic. This program did not depend on the speed of its processing ability but upon its use of heuristic strategies (Newell, Simon and Shaw, 1965). In 1972, problem solving simulation improved with the development of the *General Problem Solver* (GPS) computer program; this program could solve problems in many domains using the general problem solving strategy of means end analysis (Newell and Simon, 1972).

1.3.2.2 Means End Analysis

Means end analysis, although just one of several general problem solving heuristics, is the most commonly used general problem solving strategy (Wessells, 1982); it aims to reduce the difference between the initial state and the goal state (Newell and Simon, 1972). The following steps apply:

- note the difference between the current state and the goal state;
- develop a subgoal to reduce this difference;
- select an operator to reduce this subgoal;
- repeat the process until the current state is equal to the goal state (Eysenck and Keane, 1990).

To illustrate the use of means end analysis, the solution steps for the Tower of Hanoi problem will be discussed. This problem involves three pegs and in its simplest format, three disks of different sizes (see Figure 1.6). The task involves moving the disks from peg A to peg C in the least number of moves. Two

constraints exist: only one disk can be moved at a time and a larger disk can never be placed on top of a smaller disk.

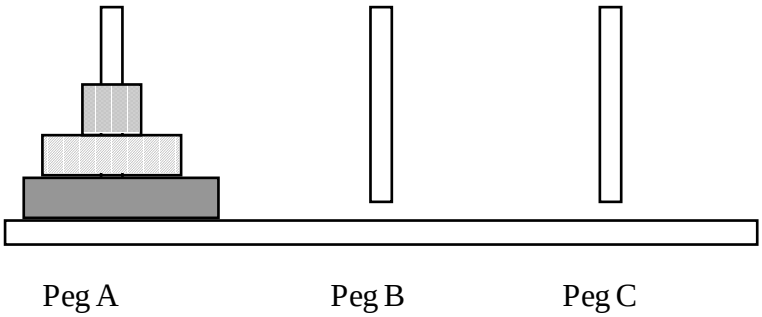


Figure 1.6: The initial state of the tower of Hanoi problem.

In order to solve this problem using means end analysis, the difference between the current state and the goal state must be examined. Initially the disks are on Peg A, while the goal is place them, in the same size order, on Peg C. Moving the smallest disk to Peg C could reduce this difference, and as this move does not violate the constraints on the operators this move is established as a subgoal and carried out. Differences still exist so the process continues until these differences are resolved. Researchers have found evidence not only for the existence of subgoals but also for a series of minor subgoals attached to a major subgoal (Egan and Greeno, 1974). For example, a major subgoal in the current problem would be to get the largest disk onto Peg C. The move described previously in this paragraph forms one of a sequence of subgoals designed to achieve this major subgoal. Placing the middle disk on to Peg B, and the smallest disk on top of that are also examples of the minor subgoals that may be attached to the major subgoal of moving the largest disk onto Peg C. After all three of these subgoals have been completed, the major subgoal can be achieved. Since only one subgoal can be achieved at a time in means end analysis, subgoals must be *stacked* for

future moves in the order that they are obtainable (Newell and Simon, 1972; Mayer, 1992).

Means end analysis is an efficient problem solving strategy but it will be contended later in this thesis that it may not be an efficient strategy for learning about problem structure and solution paths (Sweller, Mawer, Ward, 1983). Humans often interchange between several strategies when solving problems and this is where AI programs such as the GPS deviate from replicating human performance since they rigidly apply one strategy (Wessells, 1982). Furthermore, the application of general problem solving strategies may not necessarily be a conscious decision, it may merely be a label that can be applied post hoc by a researcher to the steps undertaken in the problem solving process by the problem solver (Gick, 1986).

More effective methods of problem solving are used as knowledge is acquired in a domain (Gick, 1986). For example, Anzai and Simon (1979) examined a single subject complete a five disk version of the Tower of Hanoi repeatedly; they found evidence for a shift from general problem solving strategies to domain specific strategies that were influenced by a knowledge of the problem, such as the sophisticated pyramid-subgoal strategy. Domain specific problem solving will be discussed in more detail presently.

1.3.2.3 A shift towards domain specific knowledge as the basis of problem solving ability

From the early 1970s, problem solving research shifted towards studying problem solving behaviour in semantically rich areas or knowledge-rich domains, that is, tasks that require hundreds perhaps thousands of hours of practise (Glaser & Chi,

1988). Part of this shift involved the realisation that learners' prior knowledge "rather than being a 'nuisance' factor to control, could itself be seen as a powerful independent variable in research design" (Yates and Chandler, 1991:131). Experts-novice differences in semantically rich domains were examined with the aim of understanding the nature and acquisition of expertise (Green and Gilhooly, 1992).

Findings in heuristic research and artificial intelligence also contributed to this shift in the research paradigm of problem solving. Firstly, further research in heuristics seemed to indicate that students taught Polya's heuristics for mathematical problem solving did not gain performance benefits:

> (t)hey didn't know what to do with the heuristics. They understood the heuristics in broad terms but didn't seem to understand the mathematics well enough to apply them in the rather complex and content sensitive way required (Perkins & Salomon, 1989: 19).

In addition, research indicated that heuristic instruction when taught in a very contextualised way (so that the general problem solving strategies were linked to the domain knowledge) yielded a significant performance gains (Schoenfeld, 1979; Schoenfeld & Herrmann, 1982). Secondly, AI programs such as the GPS, despite their initial success on multiple puzzle type problems, performed weakly in complex knowledge rich problem solving domains such as chess (Perkins & Salomon, 1989); in fact, it was found that humans could still outperform computers despite their enormous search capabilities (Glaser & Chi, 1988). In contrast, programs designed specifically for these knowledge areas proved to be successful (Rich, 1983).

Research into expertise suggested experts developed an enormous database of knowledge related to their domain which allowed them to rapidly recognise situations and patterns (de Groot, 1966; Chase & Simon; 1973a, b) as well as appropriate rules and solution paths (Perkins & Salomon, 1989). The implication of

this research was that knowledge was organised into specialised structures or schemas (Glaser & Chi, 1988). AI programs were developed mimicking this approach: this signalled a shift to a knowledge based approach in AI rather than the processing power approach followed previously (Posner, 1988). Programs such as ACT (Anderson, 1983) and SOAR (Laird, Newell & Rosenbloom, 1987) learnt from experience, *chunking* sequences of actions into procedures that achieved a goal. A database of chunks could then develop, creating, in time, a knowledge store that becomes the power behind the problem solving: thus, "cognition is replaced by recognition" (Gabrys, Weiner and Lesgold, 1993: 131). With programs such as SOAR, if the specific moves stored in its knowledge bank fail, it will resort to general problem solving strategies (Laird, Newell & Rosenbloom, 1987). The machine is therefore replicating the approach experts seem to take in a situation where they are asked to solve unfamiliar problems (Clement, 1991).

1.4 Expert-novice Differences

1.4.1 An introduction to expertise

The shift to the investigation of problem solving in semantically rich domains such as chess, maths, physics and computer programming also heralded a shift towards the study of experts and novices in these various domains. An expert possesses a large body of relevant knowledge and procedural problem solving skills. The proficiency of an expert, however, is limited to a defined domain (Green and Gilhooly, 1992; Glaser and Chi, 1988; Chi, Glaser, Rees, 1982). In contrast to an expert's performance, a novice lacks speed, accuracy and efficiency in problem solving. Expertise, however, is a relative concept: a chess grandmaster is likely to be more expert than an unranked but competent chess player, however this player is relatively more expert than a beginner player. In addition, the expert chess player is unlikely to show the same proficiency at chemistry or motor mechanics given the domain specific nature of expertise (Green and Gilhooly, 1992). The concept of problem solving as a general ability, capable of transference across domains has been overtaken by the multitude of evidence from many domains that suggests expertise is developed as domain specific knowledge is acquired.

It is now widely accepted in the field of psychology that expertise is heavily reliant on the acquisition and formation of a large and interconnected body of knowledge that has little or no application to other domains (Perkins & Salomon, 1989; Glaser & Chi, 1988). An expert's body of knowledge is hypothesised to exist in knowledge units or schemas (previously discussed in section 1.3 of this thesis); these schemas reside in long term memory. The expert's possession of a detailed and hierarchically organised network of knowledge stored in schemas (Reif, 1995;

Solway, Adelson and Erhlich, 1988; Larkin, McDermott, Simon and Simon, 1980a,b; de Groot, 1966; Chase and Simon, 1973a,b) can explain the robust and generalisable performance differences that have been found between expert and novice problem solvers (Glaser and Chi, 1988). The following sections of this thesis will look at each of the qualitative differences between experts and novices.

1.4.2 Memory of problem states

Early work to investigate the basis of expert performance centred on the domain of chess, since this game provided a model task environment for the investigation of problem solving (Charness, 1992). De Groot (1965, 1966) investigated the difference between expert and novice chess players through the collection of verbal '*think aloud*'' protocols as players choose moves during a real game. The players represented various skill levels such as grandmaster and master level, expert players and weekend or novice players. De Groot analysed many aspects of the player's style and thought processes in an attempt to uncover the basis of chess skill. For example, he studied the number of moves considered before a move was chosen, the degree to which the players searched the board and looked ahead for potential moves as well as their use of heuristics relevant to chess. Although the skill of the expert players was evident from their superior choice of moves[9], no quantitative differences were evident in the play and thought patterns of the differentially skilled players. In one investigation however, de Groot briefly exposed[10] the players to a chess board with 20-24 pieces configured to represent a point midway through a real game[11]. After they could no longer see the board the players were asked to reproduce it. Large performance differences on this task were evident between the skilled and less skilled players. Experts showed superior performance in terms of the number of pieces they could replace as well as the accuracy of this placement. Experts (Grandmasters and Masters) averaged over 90% accuracy, strong players around 50% and weekend players around 40%. These figures were replicated over many experiments. To summarise de Groot's findings, the differences between chess experts and novices are that chess experts

[9] As rated by independent experts.
[10] For five seconds only.
[11] Such as the twentieth move of a real chess master's game.

make superior moves and they have a superior memory of real chess board configurations.

Chase and Simon (1973a, 1973b) further explored the nature of chess expertise. A chess master, a strong A grade player and a beginner were tested on their ability to recall actual board configurations; the results were similar to de Groot's: the master's recall was superior to the strong player and the strong player was superior to the beginner. In fact, the master player took four trials to reproduce a configuration with 100% accuracy, the strong player took seven trials while the beginner player still had not achieved this level after 7 trials. However, in addition to remembering real chess board configurations, the players in these experiments were also asked to reproduce the pieces of randomly configured chess boards. If the results for memory of real board configurations reflected the superior general memory skills of chess experts, then the chess experts should excel on a memory task regardless of the meaningfulness or meaninglessness of chess configurations. However, the data collected challenged this view since no significant difference in the number of pieces recalled was found between the players of different skill levels. Players from all three skill levels were able to place about 4 chess pieces, a level that corresponded the beginner player's initial performance on the real board configuration memory task. The authors concluded that an expert's superior memory ability is limited to meaningful information concerning their domain; when a problem state is unfamiliar, an expert is reduced to a level of memory performance similar to a novice.

On the basis of their results and those of de Groot (1965, 1966), Chase and Simon contended (1973b) that memory is the determinant of chess skill. More specifically, an expert player's ability to retain and make sense of large patterns or chunks of chess pieces in memory underlay their ability to search for and choose good moves when playing chess. A chunk, in this instance was conceived as a group of chess pieces configured in a pattern that was encountered frequently when

playing. Therefore, a chess expert, when viewing a board configuration of a real game, was able to perceive and recognise these chunks allowing superior recall of the board configurations at a later interval. When faced with a randomly configured chess board, the skilled player's memory superiority vanished because they recognised few if any patterns on the board. The recognition of chunks is critical to skilled performance because skill involves the ability to search for and evaluate good moves. The ability to recognise more and larger patterns on a board allows an expert player to access information about chess moves that may be advantageous when a particular chunk or configuration of chunks are present on a board (Chase and Simon, 1973b). Chunks were discussed earlier in this thesis (section 1.1.4) and the notion raised by Chase and Simon in terms of chess, is similar to this earlier discussion. Chunking involves the meaningful grouping of information into a single entity; expert chess players in this case were grouping together several chess pieces into a meaningful single configuration.

Chase and Simon (1973a) further defined what constituted a chunk in chess through a perceptual experiment. Players, while able to view a board configuration, placed pieces on their own board to replicate this configuration. The researchers measured the length of each glance at the experimental board and how many pieces were placed on the subject's own board as a result. In addition, they delineated chunks of pieces by setting a 2 seconds as the length of time between the placement of pieces: that is, a piece was considered in the same chunk as the previous pieces if it was placed within a 2 second period. The chess master spent less time looking at the experimental board yet included more pieces in a single chunk. Moreover, the expert remembered more chunks per board. This evidence further suggested that an expert player was able to recognise the relationships among the chess pieces faster.

The achievement of expertise is not effortless. In fact, it is estimated that a chess master has spent between 10 000 - 20 000 hours looking at chess boards (Chase

& Simon, 1973a). A chess playing AI program, MAPP, developed by Simon and Gilmartin (1973) was able to accurately simulate the expert-novice related differences in recall found in human subjects[12]. MAPP modelled the process of expertise development in chess and thus made it possible to attempt an estimation of amount of chess chunks stored in long term memory; this figure is estimated at between 10 000 and 100 000 chess board configurations (Simon and Gilmartin, 1973). Given this range, the authors contended a skilled player would have built up a store of approximately 50 000 chess chunks (Simon and Gilmartin, 1973). This number is equivalent to the number of English words an average person would recognise.

Chase and Simon (1973a,b) originally conceived an expert's ability to group a stimuli into less chunks containing more information, as a working memory advantage. Evidence, also from the domain of chess, suggested the basis of the superior memory skill was long term memory. Charness (1976) found that recall of a chess board configuration was little affected by increasing the delay between the display of the stimulus and its recall to 30 seconds, even if this delay involved an interference task which used short term memory resources. In addition, Frey and Adesman (1976) showed that subjects could recall configurations from one of two boards they had studied, without prior knowledge of which one they would be asked to recall. Recall on this task was only affected in a minor way by increasing the display-recall delay to 30 seconds and requiring subjects to undertake an interference task during the delay, in comparison to a group that only had to remember a single board configuration.

Gobet and Simon (1996) have revised Chase and Simon's (1973a,b) original chunking hypothesis as the basis of chess expertise. In their multiple template theory, the author's postulated that some chunks evolve into complex long term

[12] such as de Groot's 1965, 1966 findings and Chase and Simon's 1973a,b findings

memory retrieval templates or schemas whose slot structure can be filled with various types of information such as the location of pieces (chunks), the opening move a particular configuration is likely to have come from and potential moves related to a position. Time taken to fill a slot in an existing template is rapid, as is the time taken to access a template for recall. In contrast however, the time taken to acquire these templates is lengthy.

The domain specific basis to expertise and the ability of experts to recall more information was illustrated clearly by Chi (1978). University students were compared to ten year old children on two memory tasks: digit span tasks and memory of chess board configurations. On the digit span task, traditionally accepted as a baseline measure of working memory capacity, the students could recall more numbers than the children. On the chess task however, this difference was reversed. The results can be explained by a difference in the expertise level of the two groups: the children were all experienced chess players, while none of the students played. The results of this experiment also clearly illustrate how information stored in long term memory manifests in working memory advantages.

The superior ability of experts to recall patterns of information has been shown in many domains apart from chess such as bridge (Charness, 1979); electrical circuitry (Egan and Schwartz, 1979); algebra (Sweller and Cooper, 1985); computer programming (Jeffries et al., 1981; McKeithen, Reitman, Reuter, and Hirtle, 1981; Adelson, 1981; Soloway, Adelson, & Ehrlich, 1988); x-ray interpretation (Lesgold, Rubinson, Feltovich, Glaser, Klopfer and Wang, 1988) and architectural plans (Akin, 1980). An expert's superior memory skills for real information from a domain as opposed to their novice equivalent performance for random information has also been replicated in other domains such as computer language recall (McKeithen et al., 1981); algebra (Sweller and Cooper, 1985) and electrical circuitry (Egan and Schwartz, 1979).

With the research in other domains suggesting the validity of the existence of an expert's database of domain specific information in long term memory, the simplicity of a chunk as defined by a time partition has also been challenged. Reitman's (1976) investigations into the game of GO indicated that a chunk could involve complex, overlapping patterns. Egan & Schwartz (1979) found that as expertise increased the patterns constituting chunks were increasingly abstract. Evidence was found indicating the existence of higher order or conceptual category chunks in architecture (Akin, 1980). Further evidence for the meaningfully organised categorisation of knowledge held in long term memory comes from McKeithen et al.'s (1981) study where expert programmers recalled important words from the ALGOL W programming language in meaningful clusters. In addition, Soloway et al. (1988) found expert programmers recall related programming steps as subroutines.

The findings of de Groot (1965, 1966) and Chase and Simon (1973 a,b) had wider implications for the field of psychology in terms of the long debated question over the principal basis of human intelligence: nature or nurture. While the view that problem solving skill was general in nature had fed into the argument that nature was the dominant factor, the evidence from chess clearly challenged this by showing that experts are made rather than born. Possession of domain specific knowledge was clearly more important than potentially inheritable global factors such as IQ (Yates & Chandler, 1991) or superior memory skill (Charness, 1992) and this type of knowledge and problem solving skills were acquired through thousands of hours of practise.

In conclusion, research into memory of problem states has suggested experts consciously or unconsciously chunk information into semantically meaningful units. Experts can hold more information in a single chunk leading to an effective increase in the amount of information an expert can hold in working memory within their domain. Yet it is not working memory capacity that defines expertise but the

large and highly organised body of knowledge stored in long term memory that allows experts to recognise these chunks or meaningful configurations of information in their domain. This ability leads to expert's superior recall of real problem states in their domain. Importantly, this ability also forms the basis of their problem solving prowess as information on successful moves is activated upon recognition of these chunks or patterns. Thus, expert's performance does not rely upon superior search skills or use of heuristics but upon the ability to recognise patterns of information. This ability is just one of the several differences that has been shown to exist between experts and novices. The literature also shows at least two other differences: categorisation and solution strategies. These differences will be discussed in the next sections of this thesis.

1.4.3 Categorisation techniques

Experts spend a great deal of time analysing a problem (Glaser and Chi, 1988). Evidence from verbal protocols suggests that experts initially spend time trying to represent and classify a problem whereas novices attempt to solve the problem unknowns immediately. A problem representation is a cognitive construct corresponding to the problem that is developed by the problem solver. Several groups of researchers have investigated how experts and novices represented problems and whether problem representation changes with expertise. A focus of this work was to investigate whether problem solvers categorised problems as a certain *type* of problem before they solved it. If this sequence of events was established, it would indicate that problem solvers have stored knowledge about different types of problems in their domain and furthermore, this knowledge could be a major influence upon problem representation and solution.

Hinsley et al. (1977) asked a group of high school and college algebra students to sort a variety of algebra problems according to their problem type. The participants were able to successfully complete this task. Importantly, there appeared to be clear agreement between participants about the categories of problem types. These categories tended to be indicative of the solution equations required such as *area* or *probability*. In a second experiment, the authors investigated the amount of information required from a problem for a categorisation decision to be made. The problem statement was segregated into parts such as noun phrases, dependent clauses or sentences. On average, categorisation occurred after less than 18% of the problem statement had been presented. The speed and relative lack of information required for a categorisation decision to be made seemed to indicate that a problem schema was invoked before problem representation and solution could have occurred, suggesting that the problem schema facilitates these processes.

Chi, Feltovich and Glaser (1981) investigated the hypothesis that problem representation differed according to the degree of expertise held by a problem solver. The researchers suggested that representation of a problem was due to the knowledge a problem solver had of a particular type of problem, thus the categorisation of a problem as a certain type may be a key difference in the organisation of knowledge between experts and novices.

In one of Chi et al.'s (1981) studies, expert physicists (doctoral students) and novice physicists (undergraduate students) were asked to sort 24 physics problems on the basis of their solution. No differences were found in the gross quantitative measures taken such as the number of categories or the ability to replicate the sorting pattern[13]. A significant difference did exist in the time taken to sort the problems, with the expert group taking longer. The most marked difference between the groups however, was the type of problems experts and novices categorised together. Using a cluster analysis to establish which problems were most frequently grouped together, the researchers observed that novice physicists tended to categorise problems by surface problem features such as objects referred to in the problem (i.e. a spring), literal physics terms mentioned in the problem (i.e. friction) or the physical configuration described by the problem (i.e. a block on an inclined plane). In contrast, the expert physicists seem to group problems together because of the underlying physic principle that would be used in solution such as Newton's second law, $F=ma$. The authors labelled the expert's categorisation as *deep* and the novices categorisation as *surface*. An example of both an expert and a novice problem category is shown in Figure 1.7. From the expert problem category[14] depicted in Figure 1.7, it is possible to see that the two problems grouped together have different surface features since one involves a

[13] This finding that the sorting strategies used by both groups could be replicated and the replication occurred within a faster time frame suggested the sorting strategies used by both groups were meaningful (Chi et al, 1981).

[14] 6 of the 8 experts in this experiment grouped these problems together (Chi et al, 1981).

spring while the other involves a block on an inclined plane. Yet, both these problems can be solved by the conservation of energy theorem and this is the label experts gave to this group. In contrast, the novices grouped the block on an inclined plane problem with another block on an inclined plane problem[15] (see Figure 1.7). Subsequent verbal protocols confirmed this interpretation as the novices labelled groups of problems with labels such as *blocks on inclined planes* and *problems that have something rotating*.

[15] 7 of the 8 novices in this experiment grouped these problems together (Chi et al, 1981).

Figure 1.7: An example of an typical novice and expert problem categorisation and explanation for this grouping (from Chi et al, 1981).

The expert and novice groups in Chi et al.'s (1981) experiment showed reasonable consistency across groups in terms of the categories they developed to differentiate problems. Importantly, there was also little overlap in the type of categories used by the different groups of participants (Chi et al., 1981). The differences in categorisation were also confirmed by a second study (Chi et al., 1981) and by further studies (Chi, Glaser and Rees, 1982). In the second study of Chi et al. (1981), problems were deliberately picked so that they contained a variety of surface features for each underlying solution law. Similar categorisation differences between the expert subject and the novice subject resulted: the novice student sorted the problems according to the surface features while the expert, a physics professor used physics principles. An intermediate subject, a physics graduate student, seemed to fall between these two since he used physics principle categories but misgrouped several problems. The authors suspected this occurred as a result of the problem statement miscuing the correct solution principle. In a related study, Hardiman, Dufresne and Mestre (1989) found that novices who made a greater attempt to use solution principles in their problem categorisation decisions were more capable at both problem solving and their ability to categorise problems. The authors concluded that as expertise develops, so does a student's tendency and ability to categorise on the basis of deep structures.

The difference in the principles by which experts and novices categorise problems has been replicated in several domains. For example, Schoenfeld and Herrmann's (1982) research in the domain of mathematics indicated that novices sorted on surface features of the problem such as objects in the problem like polynomials while the experts based their categories on deep solution principles. The effect has also been found with computer programmers categorising lines of code. Adelson (1981) reported that novice programmers organised lines of computer code according to the control word in the line of code such as IF or FOR whereas expert

programmers grouped code according to its membership of a particular program. Thus, the experts in this experiment seemed to be using semantics as the basis of their categorisation while the novices seemed to be using syntactics. This finding illustrates, as do the results from other domains, that novices and experts have a conceptual system organising their knowledge of a domain; it seems however, that as knowledge increases in a domain, the framework organising the knowledge changes to become more conceptually deep (Green and Gilhooly, 1992).

The way in which knowledge is organised may be according to the way in which it is used rather than according to deep principles or surface features of problems. Smith (1992) observed the categories into which genetics academics, genetics counsellors and genetics undergraduate students organised genetic problems. In this situation, both academics and counsellors can be seen as an expert group, however, their specialisation in this domain can roughly be seen to vary on a theory versus practical dimension. For genetics academic experts and genetics novices, Smith's findings resembled those of Chi et al. (1981): novices used superficial aspects of the problem, such as the wording of the problem or the type of organism involved to categorise the problems while experts organised them according to the deep principles of the discipline. Genetic counsellors however, seemed to use a mixture of both organisational strategies. Some deep principles such as the method of inheritance formed a key part of their organisational strategy, yet superficial problem variables such as the type of organism involved in the problem were also deemed important[16]. Smith suggested that the category groups of genetics counsellors reflects in the way they typically use knowledge from the domain in their professional lives. For genetics academics, their knowledge is organised around the concepts and principles forming their pedagogy; problem solving methodology is more important than correct answers and even then problems are only an adjunct to their role of promoting the learning

[16] Although the counsellors seem to focus on different superficial problem variables to the students (Smith, 1992).

of genetic concepts. For genetics counsellors however, problems are an integral part of their professional role where the answer to the genetic problem itself is a goal. Smith concluded that “genetic counsellors would be expected to focus on knowns and unknowns (in the problem) because of the manner in which they are required to apply their problem-solving skills in the course of their daily professional lives” (1992: 199).

Several other studies also question Chi and colleague’s (1981, 1982) interpretation of the organisation of problem solving knowledge. For instance, Sweller et al. (1983) tracked the development of expertise of novice problem solvers in a domain that had been artificially limited through the restriction of the number of equations that could be manipulated to achieve problem solution. In this study, it seemed that the participants did not organise knowledge according to the broad solution principles that Chi and her colleagues (1981, 1982) have suggested but rather on highly specific condition-action solution sequences. The authors hypothesised that problem schemas form a hierarchy ranging from broad principles down to specific solution procedures; expertise develops as schemas are acquired and it is Sweller et al.’s (1983) contention that the schemas that are most readily acquired are of the specific solution type. In addition, Wineburg (1991) offers an alternative proposal for the organisation of expert knowledge. He investigated the differences between experts and novices in the ill defined area of history problem solving. He did not find evidence of a specific schema for the specific battle that formed the content material of the experiment nor did he find evidence for a more general *American Revolutionary Battle* schema. Instead, Wineburg proposed that expertise lay in the ability of the historians to construct “a context specific schema tailored to a specific event” (1991: 83). Wineburg also contended the expert group did not show signs of automation. However, Wineburg’s observations did suggest that the expert historians were able to go beyond the face value of the information provided by the historical documents since they took into account the author’s motivation in writing the document and the circumstances that surrounded the generation of the

document. As a result, the expert group were judged as having a more sophisticated understanding and interpretation of the historical event. In comparison, the novice group seemed to view the texts merely as a vehicle for the provision of factual information. Wineburg interpreted this difference as a result of the ability to develop an elaborate schema of this event. However, another explanation of the experts' superior performance exists: the experts' performance could be driven by general and automatic *critical approach and analysis of information* schemas that, regardless of the specific content of a historical problem, guide the experts' approach to problems in this domain. This explains how they were able to develop a more sophisticated answer to the problems by the same set of documents, in comparison to the novice group. This interpretation also is aligned more closely to Chi and colleague's (1981, 1982) position since the critical analysis of information is an important principle underlying historical explanation.

1.4.3.1 Problems, schemas and knowledge

The knowledge contained in problem schemas was investigated by Chi et al. (1981). They hypothesised that the names experts and novices used to label their problem categories were representative of knowledge units or schemas. This schema knowledge was investigated by giving expert and novice physics subjects a variety of problem type category labels[17] and asking them to talk for three minutes on a topic, relating it to the problems involved in that category as well as the solution methods required for problems of this type. Obvious knowledge differences were evident in between the groups. Both expert and novice subjects could describe the variables and structural features involved in certain topics. However, the experts immediately focused upon the physics principles involved in

solving problems, solution procedures and conditions constraining the applicability of the physics principles. Figure 1.8 represents a network depiction of the knowledge verbalised by an expert and a novice participant on the category label 'inclined planes'. Although the novice participant mentioned the physics principle, it was not framed as an explicit solution procedure. In contrast, this knowledge formed the core feature of the expert subject's knowledge network and was greatly elaborated upon. The authors further analysed the verbal protocols into production rules. The majority of the expert's production rules could be seen as explicit solution action rules. This was not the case when the novices protocols were analysed; in fact, it was suggested that one novice student "knew what problem cues are relevant, but did not know what to do with them" (Chi et al., 1981: 18).

[17] The list was developed from the responses of the both expert and novice subjects in previous experiments (Chi et al, 1981).

EXPERT

NOVICE

Figure 1.8: A depiction of an expert and a novice knowledge about the problem category label 'inclined planes' (from Chi et al., 1981).

The finding that an expert's knowledge about a problem category is focussed around solutions was further verified by the verbal protocols collected from experts and novice students about their basic approach to solving specific problems (Chi et al., 1981). Experts interpreted the *basic approach* instruction as an opportunity to elaborate upon the major physics principle they would use to solve the problem and the steps they would use to do this. In contrast, novices "were unable to produce any abstracted solution method" (Chi et al., 1981: 20). Furthermore, when solving the problems, novices concentrated on surface features included in the problem statement such as the presence of the word *friction*. In comparison, experts identified conditions and states of the physical situation described in the problem. Some of these states including second order features, that is features derived from the problem givens, such as the presence of a *before and after* situation in the problem. The authors concluded that experts broader domain knowledge allowed them to understand more from a problem statement than just the words. They were able to understand the implications of the words for problem solution purposes (Chi et al., 1981).

Knowledge differences as a factor in expertise were also evident in the verbal protocols Chi et al. (1982) collected from experts and novices asked to summarise the contents of a textbook chapter they had read. Each group produced summaries of a similar length and both mentioned the three physics laws that formed the basis of the chapter, yet the physics experts clearly displayed a broader and deeper knowledge about the topics. The differences in knowledge seem to reflect the differences in categorisation of problems by problem solvers. For example, the findings of Chi et al. (1981) showed problem solvers of three different levels of expertise categorised problems differently. In addition, Schoenfeld and Herrmann's (1982) results illustrated that student's categorisation of problems changes as expertise in a specific domain is developed. These researchers asked novice mathematicians to categorise maths problems; the results were similar to the

results for novices of Chi and her colleagues (1981, 1982). The students were then split into two groups: one group spent 45 hours studying computer programming while the other spent 45 hours studying mathematical problem solving using problems similar to those used in the sorting task. After their training, the students were asked to recategorise the same set of maths problems. The results indicated that the only group to change their categorisation of the problems was the group who had undertaken mathematical training. This group's subsequent method of categorisation was classified as *deep* by the researchers since it was based around solution principles; it was also similar to the categories a group of mathematics professors (experts) had sorted the problems into.

Methodology other than problem categorisation has been used to probe the differences in the differential manner in which individuals at different levels of expertise organise their knowledge. Freyhof, Gruber and Ziegler (1992) analysed the differences in 3 level (superordinate, basic and elementary) hierarchical categories that chess experts (chess masters) and novices (Class B players) grouped the pieces of a real board configurations of chess into as they were able to view the board. At each level of the hierarchy, experts had more pieces in their categories than novices. This result increased as the typicality of the board position increased. In addition, the categories novices developed for the basic level and the superordinate level were very similar. In contrast, the experts grouped basic categories together to form superordinate categories: experts therefore, had the ability to meaningfully relate chess chunks[18]. These results lead to the conclusion that experts are able to find more connections between the pieces as well as perceiving relationships between the chunks. The authors concluded that "(T)he ability to perceive or to produce such (domain specific) connections between chunks distinguishes experts from average players" (1992: 36) and this is what also distinguishes the hierarchical organisation of knowledge in experts to novices.

1.4.3.2 Categorisation, problem representation and problem schemas

Several ideas on the organisation of knowledge arise as a result of work in the area of expert-novice differences in problem categorisation (Hinsley et al. 1977; Chi et al., 1981; Adelson, 1981; Chi et al. 1982; Schoenfeld and Herrmann, 1982). This area furthers research in chess by de Groot (1965, 1966) and Chase and Simon (1973a,b) since the categorisation of problems as types requires that meaningful patterns or problem states have to be recognised. Experts and novices tend to categorise problems differently: experts on the deep physics principles that would be used to solve the problems and novices on surface or literal features of the problem statement. Thus experts are able to recognise and categorise pattern types or problem states much more specifically. As a consequence, experts also represent problems in their domain at a deeper (solution based) level than novices (Glaser and Chi, 1988).

The labels given to categories of problems by experts and novices seem to represent a schema containing the problem solver's knowledge about that topic (Chi et al., 1981). Categorising a problem as a certain type activates the body of information relevant to that problem state. In experts, this knowledge seems to be related to the principles, strategies and actions that would be used to solve particular problem states. Novice problem solvers lack schemas for specific problems, instead, literal objects in a problem or problem terms form the basis for their schemas in a domain. These schemas contain some solution equations but a specific action rule solution processes, such as those held by expert in the domain, are absent. The activated problem schema guides the problem solving process, because the information slots of the schema direct the information collection

---

[18] The basic level categories described in Freyhof et al's (1992) work had many similarities to the chunks identified in Chase and Simon's (1973a,b) work.

process that allows the solution to the problem to be gained. The influence of a problem schema in the problem solving process was illustrated by Hinsley et al. (1977) in their final experiment. The researchers developed a problem that included information that suggested membership of two possible problem type categories, one of which was incorrect. The experiment's participants were equally divided in the category to which they allocated the problem. The problem category chosen however, was integral to the problem solver's subsequent treatment of the problem in terms of the information that was attended to, the information regarded as relevant and the type of errors made. This finding has been replicated in other domains. For instance, Lesgold, Rubinson, Feltovich, Glaser, Klopfer and Wang (1988) found that the early categorisation of a problem guided expert radiologists subsequent diagnostic search of an X-ray film.

The schema activated by the process of problem categorisation is self-regulating because if information cannot be extracted from the problem to fit its slots, the initial categorisation decision may be rejected. For expert problem solvers in a domain, the information slots of the schema require information given in the problem but also information to be derived from the problem: thus, the schema contains procedural knowledge about the information and manipulations required for problem solution. Problem solving is therefore both bottom up and top down: the initial choice of category activation is data driven, yet once this has occurred the schema is heavily involved in developing the problem solver's problem representation, their choice of solution path and the procedural knowledge that allows the solution to be achieved (Chi et al., 1981). It appears it is the presence of problem schemas containing both declarative and procedural knowledge about a problem's solution that allows an expert problem solver to work forward through a problem (Larkin et al., 1980a,b; Larkin, 1979; Simon and Simon, 1978) since the schemas guide the solution procedure undertaken. The difference in the solution strategies of experts and novices will be discussed in the following section.

1.4.4 Problem solving strategies

Several groups of researchers extended the findings that experts have the ability to recognise problem states in their domain (de Groot, 1965, 1966; Chase and Simon, 1973a,b) by investigating the strategies experts and novices use to solve problems. This work concentrated upon the problem solving behaviour of experts and novices within the domain of physics in areas such as kinematics and dynamics (Simon and Simon, 1978; Larkin, 1979; Larkin et al., 1980a,b; Larkin 1981). Methodology used in these investigations generally included the collection of written work sheets and verbal protocols from participants. These sources of evidence also formed the basis for the development of several computer programs which mimicked the strategies experts and novices used to solve problems (Simon and Simon, 1978; Larkin et al., 1980a). Two consistent findings have emerged from these studies of expert and novice problem solving. Firstly, experts solve problems faster than novices[19]. Secondly, novices tend to work backwards from the unknown problem goal to the quantities given in the problem while experts consistently worked forward from the information given in the problem to the desired problem goal (Simon and Simon, 1978; Larkin, 1979; Larkin et al., 1980a,b; Larkin 1981).

Generally, the starting point for novices when faced with physics problems, was the selection of an equation containing a problem goal. The second step involved searching the problem statement for the values of the other variables required by the chosen solution equation. If the value of these variables remained unknown, then they would be nominated as subgoals for the problem solver to identify in pursuit of the goal. The problem solver would then choose another equation that would provide values for some, or all, of the unknown variables in the previous

[19] In Larkin et al (1980a) the expert solved kinematic problems 4 times as fast as the novice.

equation. These backward steps leading away from the problem goal would continue until it was possible to solve the subgoals set by the problem solver, using the values contained in the problem statement. Once this had occurred, the novice was in a position to work forwards through the equation path they had established. The novice therefore, worked from the unknowns required by the problems towards the information given in the problem (Simon and Simon, 1978; Larkin, 1979; Larkin et al., 1980a,b; Larkin 1981). Larkin et al. contended "(n)ovices … seem to require goals and subgoals to direct their search (of the problem space)" (1980a: 1338).

The following problem in kinematics:

> *A car uniformly accelerates for 89 seconds. It travels 756 metres. What is the car's final velocity, V, if average velocity, v, = 0.5V and s = vt?*[20]*;*

contains two pieces of given information: *s*, the distance travelled and *t*, the length of time of acceleration. To solve this problem, a novice would typically note that *V,* the car's final velocity, is the problem goal and this quantity can be obtained by using the equation *v=0.5V*. Since the quantity of *v* is unknown but is necessary for the calculation of *V, v* becomes a subgoal:

STEP 1: *Goal V:* → *v= 0.5V;*

STEP 2: *v is unknown:* → *Subgoal v.*

This is a backward working step because the novice is moving backwards from the problem goal. The novice then looks for an equation containing *v* such as *s=vt*.

[20] This problem has been adapted from Sweller et al (1983).

Since s and t are given in the question, *v* can be calculated and the novice starts to work in a forward fashion:

STEP 3: *Subgoal v:* → *s=vt ;* → *756 = (v) 89;* *v = 8.5.*

STEP 4: *Goal V:* → *v=0.5V;* → *8.5= 0.5 (V); V = 17.*

Larkin et al. (1980b) claim the novice's strategy typifies a means end strategy. This general problem solving heuristic (described previously in section 1.3.2.1) establishes goals and subgoals in order to reduce the difference between the given state and the goal state. The management of goals and subgoals places a considerable burden on limited working memory, in some way accounting for the longer problem completion time required by novices (Larkin et al., 1980a).

In contrast to a novice, an expert would tend to start by using an equation that involved quantities given in the problem statement. He or she would work forwards from the given information to the unknown by systematically calculating new quantities from the given information that allowed the goal quantities to be defined (Simon and Simon, 1978; Larkin, 1979; Larkin et al., 1980a,b; Larkin 1981). In relation to the example problem outlined in the previous paragraph, an expert's problem solving strategy would typically differ from a novice's because the expert would choose an equation that uses quantities provided by the problem.

STEP 1: *s=vt;* → *756 = (v) 89;* *v = 8.5*

This is a forward working step because the expert has moved forwards from the problem givens. Using the value calculated from this equation, the expert would solve the problem for the unknown in less solution steps than the novice.

STEP 2: *v=0.5V;* → *8.5= 0.5 (V); V = 17.*

The expert group did not take the initial backwards step, although they did use identical equations to the novice group to solve the problems (Larkin, 1981).

Production models have been developed to replicate the different problem solving strategies of experts and novices (Simon and Simon, 1978; Larkin et al., 1980a). The key difference between the expert and novice problem solving models is that the expert model produced answers for any equations for which independent variables were given or established, while the novice model started by trying to solve the problem for the desired variable and if the action was not possible they created subgoals to find the value of unknown variables in the solution equation.

It has been observed that experts used a forward working strategy only on easy problems (Larkin et al., 1980a). However, the validity of the distinction between expert and novice solution strategies remains since although experts trialed several different schemas as they attempted to solve harder problems, when the correct schema was located, they then worked forwards to solution (Larkin, 1983). This finding relates to Sweller et al.'s (1983) contention that as expertise builds so does the number of schemas based around solutions to specific problem types rather than general solution principles. Part of an expert's strategy for solving problems involves attempts to locate an appropriate schema which can then direct the solution process.

The difference in strategy use by experts and novices is hypothesised to be evidence of schemas. The evidence has suggested that experts possess large quantities of domain specific knowledge stored in schemas. Part of an expert's schematic knowledge in a domain deals directly with typical problems as well as the solutions to these problems. Therefore, it is rarely necessary for experts, when faced with a typical problem from their domain, to go through the general search processes novices rely on to solve problems (refer back to Figure 1.5) (Gick,

1986). A problem schema also allows the expert to construct a detailed physical representation of the problem in memory (Simon and Simon, 1978). The problem representation holds the information given in the problem statement and so makes clear the steps to be taken to achieve the desired goal: thereby promoting a forward working strategy. Sweller et al. have summarised this position: ”(b)ecause the schema is based on problem givens, a forward working strategy is used” (1983: 640). As a result of their more rudimentary knowledge about a problem and it relations, a novice’s representation of the problem is much less detailed than an expert’s. A novice is therefore unable to work forwards and so uses means end analysis as a guide to solution.

Although the initial findings for the difference in expert and novice solution strategies were in the domain of physics, these findings have been replicated in other skilled domains such as medicine. For example, Patel and Groen (1986) observed that experts in the domain of cardiology making correct diagnoses used a forward reasoning strategy. In comparison, those who made unsuccessful diagnoses tended to use backward strategies. In addition, Sweller et al.’s (1983) use of a longitudinal methodology to investigate the issue of expert-novice differences in solution strategies is a different approach to the majority of research in this area. They observed that as a participant’s expertise developed (through problem solving practise), a significant shift towards a forward working strategy occurred. Moreover, the number of moves the participants required to achieve problem solution significantly decreased as expertise increased. This finding also provided evidence of the participants forward working as the initial backward steps are no longer required.

A variety of other evidence also points to the presence of problem solving schemas. For example, the speed with which experts solve problems in comparison to novices (Larkin et al., 1980a) suggests that experts have no need to search for potential solution paths; rather, they merely enact them. Also, experts

and novices seem to differ in the amount they explicitly mention a formula equation. Larkin et al. (1980a) observed that while their novice participant stated each equation she was about to use and then substituted values into the equation, their expert participant only mentioned the end result of the substitution. Similar results have been reported by Simon and Simon (1978) while Sweller et al. (1983) observed a significant decrease in this behaviour as their participant's expertise increased. It is proposed that experts have made the process of locating an appropriate equation, substituting independent variables and solving the equation for the dependent variables into a one step process (Larkin et al., 1980a). In comparison, for novices this process is still composed of several independent steps (Larkin et al., 1980a). This difference is explained as due to the expert's automation of key sections of the sequence (Larkin et al., 1980a) (automation is discussed in section 1.4).

The difference in experts and novices solution process behaviour can also be explained by the organisation of their stored knowledge. Larkin (1979) contended that experts had stored information about problem solution in large functional units while novices possessed just single, isolated units of knowledge. This hypothesis was based upon the observation that novices would produce equations as single units at a random rate while experts would generate equations in clusters. These findings about explicit reference to equations and the manner in which equations are clustered by experts and novices seemed to indicate that novices access individual equations and then go through a step by step process of solving each equation before then searching for an equation appropriate for the next step of solution. On the other hand, experts are able to access a body of knowledge that contains several equations, knowledge about the order the equations need to be solved in as well as procedural knowledge about how to solve equations. Therefore, the use of external memory aids by experts has become unnecessary. The knowledge contained in an expert's problem schema reduces the amount of working an expert requires to reach solution.

Evidence from other problem solving research has also found evidence for the active role problem schemas play in the solution process. Soloway, Adelson, & Ehrlich (1988) hypothesised expert computer programmers have knowledge of programming plans which are governed by the rules of programming discourse. They suggests that a programming plan is a schema of stereotypical type of program that guides a programmer's expectation of that particular program type. They designed programs that were typical and complied fully with the rules of programming discourse (schema conforming) as well as programs that would work yet violated the rules that typically govern programming conventions (schema non-conforming). The problem solving task required participants to fill in a blank line of code in different types of programs with the code that would best complete the program. The results of the experiment suggested when a program was atypical (schema non-conforming), expert programmers were reduced to a performance level similar to novices in terms of problem solving. Atypical problems forced the expert programmers to go back to bottom up novice strategies rather than relying on stored knowledge of typical programs. In addition, further evidence of schemas in the problem solving process was indicated by the finding that the majority of incorrect responses made when solving atypical program problems would have been correct responses to corresponding typical program (schema conforming) problems. These results provided further evidence that schemas guide problem solving responses.

Other research using non-standard problems to investigate the presence and role of problem schemas has also been fruitful. For instance, Hinsley et al. (1977) compared expert's performance in solving standard and non standard algebra word problems. The evidence suggested schemas were used when students solved standard problems since the students were able to solve standard problem in many less steps than non-standard problems. In results that have similarities to Soloway et al.'s (1988) findings, it seemed that when problems were non-standard,

processing of the problem statement occurred line by line. The author's have suggested this is a default strategy that is used when a problem is difficult to categorise as a type and thus suitable a schema is not available to aid solution. Hinsley et al. (1977) also found evidence for the use of problem schemas to solve nonsense problems. The participants in this experiment solved a problem more easily when they were able to categorise it as a certain type (because of the presence of an intact problem statement) despite the fact that the problem did not make sense because it included nonsense words. In fact, one participant of the experiment explained their approach to solving the nonsense problems this way:

> What I was trying to do all the time was to try and fit it (the problem) into some normal schema…..As soon as I could pop it into a frame…..it was much easier to deal with" (Hinsley et al., 1977: 101).

This finding is reminiscent of Adelson's (1981) work where expert programmers recalled lines of randomly presented code in program type groups suggesting their schemas for program types helped them to organise and thus remember the random information more effectively[21].

Some research has investigated how experts solve problems when a problem schema is not available. Clement's (1991) work with physics experts suggested that experts turn to general problem solving strategies when they are faced with unfamiliar, unusual problems within their domain, for which straightforward approaches have yielded no solution path. Strategies such as generating analogies to familiar objects and the examination of extreme cases (parameters set to zero or infinity) for certain variables as well as searching for misanalogies in the analogy developed for the problem were undertaken. Although the strategies utilised for problem solving were general, their use was not divorced from the subject's domain specific knowledge since their application was driven by the fundamental

laws of physics. Gick (1986) also argues that an expert's domain knowledge also allows him or her to successfully use strategies, including general problem solving strategies, that a novice uses infrequently or poorly.

To summarise, underlying experts' success at problem solving is their possession of large functional units of solution knowledge related to specific problems. This knowledge guides problem solving performance, making it forward working; fast; and less prone to errors. Therefore for novices to achieve problems solving success, they need to be taught problem solving knowledge such as appropriate equations, procedural knowledge about how to use them and the conditions under which certain solution procedures are applicable (Smith, 1992).

[21] Experts recall performance averaged 33% better than novices for the randomly presented information (Adelson, 1981).

1.4.5 Other differences between experts and novices in problem solving

Quantitative evidence has been found to support several other differences between expert and novice problem solvers. These findings relate back to the differences between experts and novices discussed earlier in this section. For instance, experts spent a more time analysing a problem qualitatively than novices (Chi et al., 1982). Part of this time is used to categorise the problem as a certain type, a process which aids in the problem solver's mental representation of the problem. Using their problem representation, experts also seemed to have a superior ability to infer relations about the problem and additionally think about possible constraints. For example, using ill-defined problems, Voss and Post (1988) found almost 25% of their expert participants elaborated on the initial state of the problem while only 1% novices did. The elaborations allowed the experts to identify possible constraints of the problem resulting in a reduction in the problem search space.

In addition, evidence has suggested that experts have stronger self-monitoring skills than novices since they are better at judging the difficulty of problems (Chi, Glaser & Rees, 1982). This ability appeared to be related to experts' superior ability to categorise problems as certain types. The ability to make a judgement on the difficulty of solving a problem also meant that experts are better at allocating their time during problem solving. Experts also seem more able at judging the relevance of problem information. In the domain of internal medicine, Groen & Patel (1988) found that experts make inferences about a problem from relevant information while novices base their inferences on less relevant material. The authors suggested that an expert's problem schema is responsible for filtering out irrelevant information. It also appears that an expert's ability to judge information relevance is integral to their successful decision making in the problem solving

process and thus successful problem solving. Shanteau (1992) concludes an expert's superior decision making during problem solving is not due to the amount of information an expert uses to make a decision but to the relevance of the information he or she uses.

1.4.6 Summary of sections 1.3-1.4: Problem solving and expert-novice differences

Humans face and solve a broad range of problems constantly throughout their lives. Psychological and educational research into problem solving has tended to focus on well defined problems. It was believed, before the 1970s, that problem solving skill was a general ability which was useful for solving problems across many domains. During this time in education, the teaching of heuristics or rules of thumb for problem solving was heavily promoted. From the mid 1970s, a multitude of evidence suggested that problem solving expertise was not a general ability. It was, in fact, domain specific.

Problem solving research shifted during this period into the semantically rich domains of chess, maths and science and the evidence suggested that the superior skill of a problem solver was restricted to their single domain. Subsequent research investigated the difference between experts and novices in specific domains. Several findings emerged: experts possess better memory skills for real problem states in their domain than novices; they are able to recognise and categorise problems on the basis of their solution rather than their surface features; and they work forward to solve problems, reducing the amount of steps they require to solve a problem in comparison to novice problem solvers. Cumulatively, this evidence points to the existence of large, functional, interconnected units of knowledge holding relevant declarative, strategic and procedural knowledge about problems and their solution. These units of knowledge are consistent with schema theory. Expertise can be understood to develop as an expert acquires a hierarchy of problem based schemas in their domain. Evidence suggests that expertise is also dependent upon the transfer of schematic knowledge from controlled to

automatic processing. Automatic processing will be discussed in the following section.

1.5 Automation

1.5.1 Automatic and controlled processing

The evidence presented in this thesis to this point has indicated the important role schemas play in learning, the development of expertise and problem solving prowess. Another factor has also been shown to play a critical role in these processes: the automation of skill. It seems that with extensive practise, certain processes become automatic, that is they demand few if any attentional and processing resources (Schneider and Shiffrin, 1977; Shiffrin and Schneider, 1977). Automation facilitates problem solving and the development of expertise because a problem solver can progressively solve larger and more difficult problems since basic problem solving skills became automatised. Attention can therefore be focused simultaneously on newer or more complex tasks, despite the limited capacity of the attentional resources and the working memory system (Shiffrin, 1988).

A distinction needs to be made at this point between automatic processes and controlled or attentional processes. This distinction has been supported by the history of psychological research into attention (Shiffrin, 1988); in fact, Shiffrin argues that “(m)ost researchers and theorists agree that a distinction between automatic and attentive processes is useful, necessary, and valid” (1988: 774). Controlled processes are any processes that require deliberate thought (Sweller, 1993); new skills therefore, tend to require controlled processing. Since they requires conscious attention, controlled processes use working memory capacity and so are constrained by the capacity limitations of this system (Schneider and Shiffrin, 1977; Shiffrin and Schneider, 1977). An example of a controlled process

would be trying to locate a particular street in an unfamiliar town, using a map. Conscious attention is paid to each street and landmark and these are matched to the map's representation. This task consumes all attentional resources. On the other hand, automatic processing is relatively effortless since it is largely unconscious. It is driven by long term memory and processing occurs in parallel and is not restricted by capacity limitations. An example of automatic processing is finding the street you have lived in for many years from the shops that you frequently visit. No attention is required to follow this route and in fact capacity is left over to carry out other tasks such as holding a conversation or completing some addition problems mentally to calculate the sum of your shopping purchases.

To reiterate, automatic processes typically develop only on highly practiced tasks (Detweiler and Schneider, 1991; Schneider and Shiffrin, 1977; Shiffrin and Schneider, 1977). Schneider and Shiffrin (1977) and Shiffrin and Schneider (1977) showed that an adult's ability to discriminate letters from numbers is automatic. In fact, on average 2100 trials were required for a participant to discriminate between letters and letters as fast as he or she could between letters and numbers. This result also demonstrates that with sufficient practise, automatic processing can be established with sufficient practice. Automatic processing is also much faster than controlled processing. Schneider and Shiffrin's (1977) results indicated that a group that developed automatic processing skills on a task needed approximately a fifth of the decision making time a group forced to use controlled processing required to achieve 95% accuracy on a visual search task. Evidence from other fields has also indicated the speed of experts using automatic processing. For example in medicine, novice radiologists require more time than experts to detect abnormalities in an X-ray (Christensen, Murry, Holland, Reynolds, Landy & Moore, 1981) and in fact, expert radiologists were able to detect abnormalities with 70% accuracy from X-rays they saw for only one fifth of one second (Kundel and Nodine, 1975). The author's conclusion in this case was that experts have automated recognition procedures. Wiedenbeck (1985) reached the same

conclusion about experienced FORTRAN computer programmers. In comparison to a group of much less experienced programmers[22], expert programmers were 25% faster and made 40% fewer errors when judging the grammatical correctness of lines of FORTRAN code, a basic programming skill.

Another critical difference between automatic and controlled processing is the ability of these processes to alter as circumstances change. Automatic processes are extremely inflexible since they are so highly practiced (Detweiler and Schneider, 1991); while controlled processes prove to be much more flexible in regard to changing circumstances. For example, Shiffrin and Schneider (1977) observed that if a participant was using automatic processes to search for a target in a visual search task, if circumstances such as the definition of a target or the type of stimuli used as distracter variables altered, nearly 1000 trials were required to regain the initial performance level of the participants. If participants had always used controlled processing because of the constantly varied nature of their search task, performance was effected to a much lesser degree and the search processes were able to be modified much more quickly.

Some debate has occurred over the nature and roles of automatic and controlled processing despite the wide acceptance of this distinction (Shiffrin, 1988). For example, Shiffrin, Dumais and Schneider (1981) argue against the often mooted belief that automation is a continuum. These authors contended that automatic and controlled states of processing are qualitatively different. Logan (1988a,b) proposed automation is a memory phenomenon rather than a processing phenomenon. He proposed as practise increases so does the domain specific knowledge stored about a stimulus. If a task is consistent, processing can rely on stored responses to a stimulus, thus processing becomes automatic. Logan summarised his position thus: “Automaticity is memory retrieval: Performance is

[22] Mean programming experience was 11 000 hours as opposed to 200 hours (Wiedenbeck, 1985).

automatic when it is based on a single-step direct-access retrieval of past solutions from memory" (1988a: 493). Although more explicit about the effects of practise on developing automation, Logan's contentions to not seem to differ dramatically from the commonly held beliefs about automation.

The controversy in the area of automation has even lead Shiffrin (1988) to revise his original theory of automation (Schneider and Shiffrin, 1977; Shiffrin and Schneider, 1977) to now include automatic processes and attentive mechanisms rather than controlled processing. In essence, the distinction is similar to Shiffrin and Schneider's earlier model since automatic processes are innate or learned; are not restricted by capacity limitations and can operate in parallel to other automatic or attentive processes without interfering in the execution of these processes. However, attentive processes are hindered by capacity limitations and interfere, leading to their use in a serial manner.

1.5.2 Automation and problem solving

Automatic processing plays a major role in skilled problem solving performance and the development of expertise. Kotovsky, Hayes and Simon (1985) developed 'isomorphs' of the Tower of Hanoi problem (discussed earlier in section 1.4.2.2.). An isomorph is a version of a problem that has different surface features but has the same underlying problem space in terms of the size of the problem space, the number of branches the sequence of operations and the minimum solution path length (Kotovsky, Hayes and Simon, 1985). Despite the underlying similarities in the structure of the isomorphic problems, large differences in the time taken to solve different versions of the problems had been observed. For instance, some isomorphs took up to 16 times as long to solve (Kotovsky, Hayes and Simon, 1985). As these differences cannot be attributed to the problem structure, the authors speculated they were due to the way the problem solver modelled or thought about the problem. The results of their experiments provided evidence for this contention. Various factors such as the ease of learning and applying problem rules; the plausibility of problem solving rules, given real world knowledge; and provision of a physical representation of a problem all were shown to affect the difficulty of a problem.

Evidence was also found suggesting several other elements of problem solving behaviour affected the difficulty of solving problem isomorphs. For example, the authors found that explicit training on problem solving rules, even if this training was removed from knowledge about the problem and its setting, facilitated subsequent problem solving. The researchers microanalysed the participants' problem solving moves and found that initially participants seem to make a substantial number of exploratory moves. Following this behaviour, the problem solver entered the final solution path and quickly and directly solved the goal. Kotovsky et al. (1985) postulated that these initial moves are rule familiarisation

and this behaviour actually leads to rule automation. Automation of the rules of the problem led to a reduction in working memory load imposed by the problem's variables such as information on the current state, the goal state, the rules and testing for move legality. As a result of this reduction in working memory load, the participant had the ability to plan ahead for moves and therefore the final solution could be reached relatively quickly. Automation also accounts for the improved problem solving performance that resulted from explicit rule training mentioned earlier in this paragraph. Kotovsky et al. summarised their findings:

> We have shown that the solution process depends on becoming expert at utilising the problem rules to make moves, that the automation of the rule using behaviour is a necessary precursor to planning behaviour, and that a small amount of planning capability enables a rapid solution" (1985: 290-291).

In a related finding, Cooper and Sweller (1987) in the domain of algebra, found that schema acquisition occurs before rule automation. Rule automation was an important step involved in problem solving transfer. Indeed without automation, problem solving is likely to be awkward and error prone (Sweller, 1993).

To summarise, the automation of skills and problem solving rules with practice frees up attentional and working memory capacity for the simultaneous consideration of other aspects of the problem solving process or the performance of other tasks. Automation increases the speed and accuracy of performance and is an essential element of learning and the development of skilled performance.

1.6 Summary of Chapter 1

To conclude, this chapter has discussed the fundamentals of human cognitive architecture and information processing theory. Working memory is where conscious processing occurs but is extremely limited in the amount of elements it can process at any time. Long term memory is seemingly unhampered with limitations and forms the basis of human intellectual prowess. Knowledge is stored in long term memory in cognitive constructs called schemas. Schemas develop through learning and hold vast amounts of information. They are hierarchically organised and are involved in all aspects of cognitive processing such as recognition, interpretation and problem solving.

Initially, research in problem solving postulated that problem solving was a general ability. Evidence has now emerged to strongly suggest that the differences between experts and novices is their domain specific knowledge about problems and problem solutions stored in schemas. Experts can recognise and retain more and larger problem states from their domain than novices, they represent and categorise problems on their underlying solution principles and they work forward to solution. All of these differences can be tied to an expert's development of domain specific schemas. Automation of problem solving skills is also critical to problems solving expertise as it frees processing capacity to allow for more complex tasks to be undertaken.

In summary, schema acquisition and automatic processing are the critical factors involved in human learning, understanding and expertise development. Automated schemas provide the structure for long term memory as well as allowing effortless processing that bypasses the limitations of working memory.

In Chapter 2, a variety of techniques involving the design of educational instruction will be discussed. These techniques have been developed as a result of cognitive load theory (Sweller, 1999). The basis of this theory is the enhancement of learning outcomes through the facilitation of the processes of schema acquisition and schema automation.

Chapter 2 Cognitive load theory

2.1 An introduction to cognitive load theory

Cognitive load theory (Sweller, 1988; 1989; 1993; 1994; 1999; Sweller and Chandler, 1991; 1994; Sweller, van Merrienboer and Paas, 1998; Chandler and Sweller, 1991; 1992, 1996) has its foundations in the cognitive architecture described in Chapter 1. It uses some aspects of human cognitive architecture and of the structure of information to provide instructional designs that facilitate understanding, learning and problem solving. The theory acknowledges the limitations of working memory and the important role long term memory has as the repository of intellectual skill. It asserts that the focus of instruction and, therefore, instructional design should be the facilitation of schema acquisition and the automation of those schemas. Cognitive load theory contends that the amount of mental load placed on working memory by an instructional task will be the critical factor in determining how effective learning has been. It further argues that many common instruction techniques fail to acknowledge the fundamental features of our cognitive architecture, and consequently tend to impose a heavy working memory load. High working memory load has the effect of impeding learning. In response to these issues, cognitive load theory has been used to generate a range of alternative methods of instruction designed to enhance learning by making make more efficient use of limited working memory. It is important to realise, however, that cognitive load theory is not concerned with any particular instructional design, rather its major concern is that the cognitive load imposed by instruction on working memory is minimal. In particular, if working memory is high

because of the instructional format, cognitive load theory advocates the use of alternative methods of instruction that reduce or ideally eliminate mental activities that are extraneous to learning. In fact, Sweller contends:

> an instructional design will be deficient to the extent that it requires learners to engage in cognitive processes that are irrelevant to schema acquisition and automation. Any such cognitive processes will require working memory resources that consequently are unavailable for schema acquisition and automation and to that extent, the instructional design will be inadequate (1999: 37).

2.2 Sources of cognitive load

Several factors are involved in learning instructions. Initially, Sweller (1993) and Sweller and Chandler (1994) identified two separate sources of cognitive load: intrinsic and extraneous cognitive load. The intrinsic cognitive load of instructional material is generated by the intellectual complexity of the instructional material itself (e.g. the difficulty involved in learning a particular physics theorem). The extraneous cognitive load is determined by how the information is presented (e.g. text format, diagrams or audio-visual). More recently, a third source of cognitive load has been identified: germane cognitive load. Germane cognitive load is related to the cognitive processing which is consciously directed toward the construction of schemas (Sweller, van Merrienboer and Paas, 1998). Intrinsic, extraneous and germane load together contribute to the total cognitive load involved in a learning task. The factors that affect intrinsic cognitive load will first be discussed.

2.2.1 Intrinsic cognitive load and element interactivity

Human cognitive architecture, as outlined in Chapter One, can be assumed to have evolved in order to handle some critical characteristics of information. Human cognitive architecture must mesh with the information structures it is required to process. All information is composed of elements. An element is information that can be processed by a particular learner as a single unit in working memory. Elements include relations between symbols as well as the symbols themselves. One aspect of the structure of information is the extent to which its various elements interact. While all information will impose some degree of intrinsic cognitive load, the degree to which information elements interact, rather than the number of information elements present, is the key factor responsible for the intrinsic cognitive load imposed (Sweller and Chandler, 1994; Sweller, 1993).

Some information can be understood and learned, individual element by individual element because the elements do not interact. Learning a new vocabulary provides an example. Each new vocabulary item can be learned without reference to any of the other items. For instance, the Italian word for *table* can be assimilated and learned independently of the word for *chair*. The information is, therefore, low in element interactivity. It imposes a low intrinsic cognitive load since there is little if any interaction between learning elements; as a result, only a limited number of elements need to be processed in working memory at a given time. It may be difficult to learn this type of information because there is a large number of words to be learnt, however, since this information can be processed in a serial fashion, the intrinsic load it imposes remains low.

In contrast, high element interactivity material consists of elements that cannot be understood in isolation because they interact. We can learn vocabulary items

individually but we cannot learn grammatical syntax without considering several vocabulary items and their relations: We can learn the names of various chemical elements in isolation but we cannot learn to balance a chemical equation without considering several elements and their relations simultaneously: We can learn the names and perhaps even the functions of individual electrical components one at a time but we cannot understand an electrical circuit consisting of those components without simultaneously considering several components and their relations. Understanding syntax, chemical equations or an electrical circuit are high element interactivity endeavours for learners with little experience in these areas. As a consequence, they impose a high, intrinsic cognitive load because many elements must be processed in working memory simultaneously. Difficult to understand material is difficult precisely because it is high in element interactivity and imposes a heavy load on working memory (Sweller and Chandler, 1994; Marcus, Cooper & Sweller, 1996; Sweller, 1994; see also Halford, Wilson & Phillips, 1998, for a similar formulation).

All information falls somewhere on a continuum of a small number of elements with low interactivity to many elements with a high degree of interactivity. It is possible to estimate the interactivity of material by counting the number of elements of information learners will be required to simultaneously process (Sweller and Chandler, 1994). This task, however, is made more difficult by learners' differing levels of prior experience; this issue will be discussed below (section 2.2.1.2).

2.2.1.1 Research on element interactivity

Marcus, et al. (1996) investigated the effect of learning information in which the information elements increasingly interacted. In this series of experiments, primary school students were required to join resistors and connectors together to solve

problems in basic electronics. The researchers devised problems in which the interactivity between a fixed number of information elements was varied. The problems required the participants to construct a single series connection, a multiple series connection and a parallel connection; the number of information elements requiring simultaneous processing in the parallel connection problem was substantially higher than either the single or the multiple series connection problem. The results indicated that as the interactivity of the problem was increased so was the time required for solution. In addition, the researchers found that instruction via diagram (a schema facilitating instructional technique) was of most benefit for the higher element interactivity problems. Analysis of dual task performance and subjective ratings of mental load also provided evidence to support the authors' conclusion as the degree of interactivity between the elements increased so did the cognitive burden placed on working memory.

In related research, Halford, Maybery and Bain (1986) and Maybery, Bain and Halford (1986) investigated the working memory load imposed by transitive inference problems such as:

X is smaller than Y; Y is smaller than Z: which is the smallest?

The researchers found that working memory load was reduced if the two premises were separated, however, in order to solve the problem, the premises must interact. The data indicated that working memory load was highest when the premises interacted and this was the point at which students had the most difficulty solving the problem.

2.2.1.2 Element interactivity and expertise

Our cognitive architecture handles the problem of an excessively high working memory load by constructing schemas[1]. An expert is someone who through extensive practice has acquired many schemas in a specific domain. The acquisition of a schema allows the interacting elements of information that would otherwise overwhelm the limited capacity of working memory to be incorporated. The schema then acts as a single element and imposes a minimal working memory load. Thus, relatively expert English speakers have a schema that tells them that "the cat sat on the mat" is syntactically correct but "cat the on sat mat the" is not. Algebraic schemas tell us that $a/b = c$ can be transformed into $a = cb$. Both equations and their transformation can be processed easily in working memory by anyone who is competent at elementary algebra despite the large number of elements. A higher-order schema tells us the location of each symbol and the mathematical procedure required to transform the first equation into the second. Once acquired, the otherwise difficult task of processing all the interacting elements in working memory becomes a relatively simple task because the elements are incorporated in a schema that itself acts as an element and can be processed readily in working memory.

The presence of a schema or a network of schemas, therefore, has an enormous impact upon the element interactivity and, hence, the intrinsic cognitive load imposed by instructional material since “what constitutes an element does not depend just on the nature of the material, it also depends on the expertise of the learner” (Tindall-Ford, Chandler and Sweller, 1997: 260). The number of elements and the degree to which they interact will vary greatly depending on the expertise of the students involved; for example, a competent mechanical engineer required

[23] For more information on schemas see section 1.2

to learn about the configuration and elements of a pneumatic circuit should be able to process all of the elements that create a standard pneumatic circuit as a few elements or even as a single entity. This processing can occur simultaneously in working memory because of the mechanical engineer's previously acquired schemas. The presence of schemas has shifted much of the processing burden to long term memory. In contrast, a relative novice (e.g., first year apprentice or trainee) required to learn about the same pneumatic circuit would only be able to hold and process a few of the circuit's elements simultaneously in working memory since he or she does not possess the requisite schemas and so is restricted to the limited processing capacity of working memory. Ausubel has concluded that "the single most important factor influencing learning is what the learner already knows" (1968, *vi*).

2.2.2 Extraneous cognitive load

While intrinsic cognitive load is generated by the learning material itself, extraneous cognitive load is generated by the instructional technique or the format of the instructional material. A variety of methods can be used to present or format instructions; examples of instructional techniques include written text, diagrams, formal lectures and multimedia computer presentations. The extraneous load imposed by each method of instruction varies. A central assumption of cognitive load theory is that the most prominent consideration when designing instructions should be the reduction of unnecessary extraneous working memory load (Sweller, 1993; 1994; 1999; Sweller and Chandler, 1994; Sweller, et al., 1998; Chandler and Sweller, 1996). Cognitive load theory has identified many commonly used instructional techniques that impose high extraneous load on learners. In response to these findings, cognitive load researchers have generated several alternative methods of instruction; these methods fulfil the same instructional needs as the conventionally used techniques yet have been designed to make more efficient

use of limited working memory. Evidence from measures such as subjective rating scales and dual task performance suggest that the alternative instructional techniques developed from cognitive load theory impose much less extraneous load and result in performance improvements (Cerpa, Chandler & Sweller, 1996; Chandler and Sweller, 1996; Chandler & Sweller, 1991; Cooper & Sweller, 1987; Jeung, Chandler and Sweller, 1997; Kalyuga, Chandler & Sweller, 1999; Kalyuga, Chandler & Sweller, 1998; Mousavi, Low and Sweller, 1995; Owen and Sweller, 1985; Sweller and Chandler, 1994; Sweller & Cooper, 1985; Sweller, Mawer and Ward, 1983; Tarmizi and Sweller, 1988; Ward and Sweller, 1990; Tindall-Ford et al., 1997). These alternative instructional techniques will discussed in section 2.3.

2.2.3 Germane cognitive load

Germane cognitive load involves ensuring the learner is focused on processes that are relevant to learning. This type of cognitive load should be taken into account by instructional designers if the total cognitive load associated with the learning task from other causes is not excessive. Learning benefits if the learner's attention is directed toward processes relevant to the construction of schemas, even if the consequence of achieving this effect is an increase in the total cognitive load imposed on working memory during learning (Sweller et al., 1998; Paas and van Merrienboer, 1994). Some examples of increasing the germane cognitive load of an instructional task are requiring learners to answer questions about an example or finish the solution to an incomplete example. Tasks such as these ensure the learner has properly processed the example before they are required to undertake some further learning activity such as solving related problems. In addition, increasing the interference imposed on the learner during training by presenting random rather than blocks of similar solution problems can increase germane cognitive load during training (van Merrienboer, Schuurman, de Croock & Paas, in press). Total cognitive load may be increased by including tasks such as those

described above, however, the tasks ensure that the learner's attention has been appropriately engaged and have been shown to benefit subsequent performance on similar and transfer problems (van Merrienboer, Schuurman, de Croock & Paas, in press; Sweller et al., 1998; Paas and van Merrienboer, 1994).

2.2.4 Total cognitive load

The total cognitive load imposed on a learner's working memory is a function of intrinsic, extraneous and germane load. If total cognitive load exceeds the limited capacity of working memory, schema acquisition and schema automation will be impeded. Several experiments have shown that sometimes, despite the use of an instructional technique that imposes a high extraneous load, learning remains unaffected. In this situation the intrinsic load generated by the instructional material itself is not sufficient to cause excessive total cognitive load.

When intrinsic load is high, research from a variety of domains indicates that the manner in which instructions are formatted is critical to the processes of schema acquisition and automation. Specifically, in this situation instructional techniques that reduce extraneous load or increase the processing power of working memory positively impact upon learning. When intrinsic load is low, the method of instruction used is much less relevant (Cerpa et al., 1996; Chandler & Sweller, 1996; Marcus et al., 1996; Sweller & Chandler, 1994; Tindall-Ford et al., 1997)[2]. Increasing the germane cognitive load imposed by a learning task is similarly detrimental for learning if total cognitive load from other sources is high; for example, an increase in germane cognitive load was found to only be of benefit to learning if the extraneous load imposed by the instructional material was reduced. Paas and van Merrienboer (1994) compared a group that studied geometry

[24] This research is discussed in greater detail in this chapter in section 2.8.

worked examples to a group that solved geometry problems under conditions of high and low variability of practice. The learning benefits associated with increasing germane cognitive load through the increase in practice variability (Jelsma and van Merrienboer, 1989) only materialised when extraneous loadwas low (the worked examples condition).

2.3 Instructional techniques generated by cognitive load theory

2.3.1 Goal free problems

2.3.1.1 Investigations of the effectiveness of means end analysis as a problem solving strategy

From a cognitive load perspective, the effectiveness of a problem solving strategy or an instructional technique is the degree to which this strategy or technique facilitates schema acquisition and automation. This requires that extraneous cognitive load is low and that a problem solver's attention is duly directed at features of a problem important for schema acquisition and automation (Sweller, 1999).

Means end analysis (see section 1.3.2.2) is a problem solving strategy commonly used by novices. Problem solvers using this strategy are always focussed on the goal since they are required to search for a legal move to reduce the difference between the current state of the problem and the goal state. Once a move has been undertaken, the problem solver can ignore the previous problem state and simply focus on reducing the difference between the new problem state and the goal (Gick, 1986). As a strategy, means end analysis is efficient at helping problem solvers obtain the problem goal since it leads to behaviours such a reduction in moves down dead end solution paths (Owen and Sweller, 1985); however, it has been suggested that "one can achieve a solution to a problem using means end analysis without ever remembering how that solution was achieved" (Gick, 1986:

109). These characteristics suggest that means end analysis is inefficient at facilitating schema acquisition because problem solvers focus only on the goal not on the sequence of problem states and their associated moves (Sweller et al., 1983; Owen and Sweller, 1985). In fact, Sweller, Mawer and Ward state that:

> The conditions that allow the goal to be obtained are not necessarily those that will facilitate learning. When learning, it may be important to consider how the problem state has been obtained rather than the difference between the current problem state and the goal" (1983: 648).

The well-developed problem solving schemas of experts contain explicit problem solution information (Chi et al, 1981). To develop schemas containing this type of information, the problem solver must be able to categorise problems as certain types and associate certain solution moves with these categories of problem types (Sweller, 1988). Means end analysis does not allow the problem solver to develop an expert-like problem schema because of the heavy processing demands it imposes on the problem solver. The problem solver is required to simultaneously maintain information about the current problem state, the goal state, any subgoals developed and the relations between these states. As well as maintaining this information, the problem solver is also required to manipulate this information and search for an operator to reduce the difference between these various states (Owen and Sweller, 1985). Given this working memory load and the previously discussed limitations of the working memory system[3], the ability of the novice problem solver to process information about problem states, their relationship to the problem solving operators and fundamental solution principles is severely impeded (Owen and Sweller, 1985; Sweller et al, 1983). Yet, for schema acquisition to occur, the problem solver's attention must be directed towards these

[25] The limitations of the working memory system were discussed in depth in section 1.1.3.

parts of a problem rather than at the goal (Pierce, Duncan, Gholson, Ray, Kamhi, 1993; Gick, 1986)

Early research into what has become known as cognitive load theory reflected these negative aspects of means end analysis. Using number transformation puzzle problems, Sweller, Mawer and Howe (1982) and Mawer and Sweller (1982) compared the performance of students using a means end strategy to students who used a schema based strategy that directed attention to previous successful moves in order to generate new ones. Sweller and colleagues found evidence that means end analysis impeded learning since participants using this strategy did not learn the solution rule or learn about problem structure despite often successfully achieving problem solution. This contrasted to the other group, which had developed a basic problem solution schema for these types of problems containing information about the problem structure, the problem states and solution paths.

2.3.1.2 The concept of a goal free problem

Means end analysis requires the problem solver to focus on the problem goal and reduce the difference between this state and their current state. A problem that contains no specific goal would circumvent the use of this strategy (Sweller et al., 1983). As a result, processing resources can then be directed toward the aspects of the problem beneficial for the development of schemas and automation: namely, problem types and their associated solution moves.

A typical problem includes a specific goal; for example:

> *In 18 seconds, a racing car can start from rest and travel 305.1 metres.*
> ***What speed will it reach?***

In contrast, a goal free problem does not nominate a specific problem goal; for example:

> *In 18 seconds, a racing car can start from rest and travel 305.1 metres.*
> ***Calculate the value of as many variables as you can.***
>
> (Adapted from Sweller et al., 1983: 643)

The format of a goal free problem, therefore, allows the problem solver to explore the problem space, attending to individual problem states and their associated move operators (Sweller, 1999) while simultaneously ensuring the cognitive load is kept to a minimum (Ayres, 1993). These conditions should facilitate schema acquisition and rule automation since a greater amount of cognitive resources are available for these learning activities (Sweller, 1999).

2.3.1.3 Investigations of goal free problems and learning

Many studies have investigated the effect of goal free problems upon learning. For instance, Sweller and Levine (1982) developed maze puzzle problems which through the manipulation of knowledge about the goal allowed them to compare a means end analysis group and a goal free group which was forced to generate moves from an analysis of their previous and current moves. A simple alternation rule (turn left, turn right) provided the solution to the maze, yet, the group who were aware of the goal had not learnt this rule after 300 moves. The goal free group learned this rule quickly indicating the presence of a schema.

Sweller et al. (1983) and Owens and Sweller (1985) expanded upon Sweller and Levine's research and demonstrated the significance of goal free problems for education. Using physics kinematics problems, Sweller et al. (1983) found that

goal free problems facilitated the use of a forward working or expert problem solving strategy (see section 1.3.3.4 for more information on forward working strategies). The students in the goal free group solved the problems in fewer moves and required a visual representation of the solution equation (problem solving operator) less often than the group solving conventional goal problems. These findings were replicated in the domain of circle geometry, even in a situation where the goal free group calculated variables that were identical to those thegoal group had to calculate in order to attain the goal (Experiment 4, Sweller et al., 1983). In the domain of trigonometry, Owen and Sweller (1985) compared the performance on a test of conventional problems of students trained on goal free problems to students trained on conventional goal problems. The results indicated that the goal free group solved more parts of the problem with fewer errors than the group that had solved conventional problems in training. In further experiments, the authors found firstly, that novice students developed greater knowledge of fundamental trigonometric principles if they had trained on goal free problems and secondly, that training on goal free problems improved ability to solve a transfer problem. In summary, the authors contended:

> over a very short acquisition period, no goal group subjects had developed superior cognitive representations of the trigonometric ratios and had acquired the schemas to apply them appropriately (Owen and Sweller, 1985: 278).

The finding that goal free problems benefit learning is robust since it has been replicated and generalised in many domains; for example, in the domain of geometry, Tarmizi and Sweller's (1988) results indicated that the group trained on goal free problems solved similar problems more rapidly and with fewer errors. In addition, verbal protocols indicated this group was more likely to use a forward working strategy than the group trained on conventional problems. Also in the domain of geometry, Lim, Dixon and Moore (1996) compared students trained using goal free problems to students trained using worked examples which

contained specific goals. Although hampered by a small sample size, the results of this longitudinal study indicated that goal free problems were associated with better test performance and that this performance benefit was maintained over time.

Goal free problems have also been associated with some other learning benefits; for instance, using two-stage problems that require the calculation of a subgoal before the goal can be achieved, Ayres (1993) further investigated the benefits of goal free problems for mathematics. A *stage effect* is commonly observed with two-stage problems; this occurs when the number of errors on the subgoalstage is greater than the number of errors on the goal stage. The stage effect occurs when students are forced to used means end analysis because the problem is unfamiliar. Ayres (1993) hypothesised that the stage effect would be reduced if goal free problems were used for training on two-stage problems since they eliminate the use of means end analysis. The results supported this hypothesis, as a stage effect was not observed in the goal free group; the students in this group also successfully completed more problems and negotiated more correct solution paths. Pierce et al. (1993) have identified another learning benefit of goal free problems. The researchers investigated analogical transfer using non-isomorphic target problems. In order to solve analogous problems, a problem solver had to notice the difference between the surface features of the problem in addition to the underlying difference in the problem space. To solve the problem successfully, the problem solvers had to change their solution procedure to account for these differences. Students in the goal free problem training condition, a condition that emphasised problem space exploration, were more able to shift their solution knowledge to the analogous problem. The authors regarded this as evidence of a higher quality problem schema.

Evidence supporting the contention that solving goal free problems impose less cognitive load than problem solving by means end analysis has also been

generated through computer modelling. Sweller (1988) developed a computer driven production model that worked forward or backward depending on the specificity of the problem goal. This system modelled a novice's behaviour on goal specific and non goal specific problems. It was designed to compare problem solving using means end analysis and the forward working problem solving strategy enforced by goal free problems. Cognitive load could be estimated from the number of production statements in the production system's memory. The model offered strong support for the reduced cognitive load imposed by goal free problems: while means end analysis on a goal specific problem required a minimum of 4 productions to be held simultaneously, the forward working strategy generated by a goal free problem required only a single production to be held at any one time. Sweller (1988) supported this computer model with empirical research that investigated students solving geometry problems while performing a secondary task. Secondary tasks are a commonly accepted measure of the mental load associated with the primary task (Fisk and Schnieder, 1984). This experiment showed that the group solving goal free problems performed significantly better on the secondary task: this result seemed to indicate that this task imposed a lower cognitive load.

2.3.1.4 Shortcomings

Although goal free problems have clear benefits for learning, their use is somewhat restricted by the nature of the domain. In some domains such as algebra, a problem where a goal is not present to establish boundaries upon the possible calculations can cause the creation of a virtually limitless problem space. In this case, the problems solver is subjected to "fruitless calculations" (Ayres, 1993: 381) and a specific goal is a necessary limiting device (Sweller, 1999).

2.3.2 Worked examples

This thesis has previously discussed the fact that means end analysis interferes with the acquisition of schemas and is therefore detrimental to learning. Goal free problems were developed as a direct response to this fact. The experimental results suggested that learning improvements resulted from this instructional technique which, through the format of the problem itself, did not allow the use of a means end strategy. If the success of goal free problems was indeed due to this factor, as cognitive load theorists claim, it follows that any instructional method which circumvents the use of means end analysis would also be beneficial for learning.

A worked example consists of a problem paired with the steps to its solution (see Figure 2.1 below for an example of a worked example). An instructional strategy that requires students to review worked examples eliminates the need to use means end analysis because, since the solution is provided, it is no longer necessary to search for an operator to reduce the difference between the current state and the goal state. In fact, the effect of presenting the problem solution is that the student focuses on individual problem states, the problem solving moves associated with them and the problem states resulting from these moves. Since this information is the type of information contained in a problem solving schema (Chi et al., 1981) it was hypothesised that reviewing worked examples would be beneficial for activities such as schema acquisition and automation (Sweller & Cooper, 1985; Cooper & Sweller, 1987). Moreover, these researchers contended that since worked examples deconstruct a problem solution into its parts, working memory load should be reduced which should further benefit learning. In this situation, the worked example is acting as an external memory aid, allowing the student to simply attend to a single problem state and its associated move. So despite their surface dissimilarity, goal free problems and worked examples share

many similar features when they are viewed from a cognitive load perspective: both instructional techniques reduce the use of means end analysis; both focus the attention of the learner onto problem states and their associated moves and both reduce the cognitive load associated with conventional problem solving techniques (Sweller, 1999).

Problem: Calculate x when $10x = z + 80$ and $z = 100$.

Solution:

$$10x = (100) + 80$$

$$10x = 180$$

$$\frac{10x}{10} = \frac{180}{10}$$

$$x = 18$$

Figure 2.1: An algebra worked example.

The effectiveness of studying worked examples for learning was investigated by Sweller and Cooper (1985). In this series of experiments, the researchers compared two groups of students with a limited knowledge of algebra on algebraic transformation problems such as:

$\frac{b(a+c)}{e} = d$ *Express 'a' in terms of the other variables.*

Exposure to worked examples was compared to the conventional strategy of simply solving many problems (an activity that is likely to require novices to use means end analysis). In the acquisition phase of the experiment, after a brief introduction to this area of algebra, both groups were given a set of eight problems

to complete which contained 4 pairs of similar problems. The problems included in the problem set were identical for both groups except that for the worked example group the first problem of each group was a worked example (thus the solution and the solution working was provided). In this group, the students were asked to study the worked example before solving the subsequent problem of the problem pair. In the conventional group, all eight problems from the problem set required solving. It is noteworthy that the researchers did not compare a *worked examples only* condition to a *problem solving only* condition. Although theoretically a *worked examples only* condition should provide the most benefits for learning, the researchers were concerned about the motivation of the students if they only studied worked examples. The condition used in the experiment required students to study a worked example and apply the same technique to a similar problem. The design of this condition, therefore, provided the students with the motivation to properly process the worked examples[4].

The results from Sweller and Cooper (1985) experiment indicated that the worked example group required significantly less time than the problem solving group to complete the acquisition phase. The acquisition phase was followed by a test phase; it consisted of problems that were similar in structure to the problems used in the acquisition phase. The worked example group completed the test problems significantly faster than the problem-solving group and they made significantly fewer errors. Thus, despite less time for learning, the students in the worked example group were able to solve similar problems faster and with less errors than a group that had more problem solving experience. These results seemed to indicate that studying worked examples does facilitate the acquisition of schemas.

[26] Paas and Van Merrienboer (1994) have reported the results of an experiment that compared a purely worked examples group to a conventional problem solving group. Results similar to other studies where worked examples are interspersed with problems were reported. The results indicated similar and transfer problem solving performance was enhanced in the worked examples group.

In the experiment outlined above, the test phase used similar problems to the acquisition phase. Sweller and Cooper (1985) also investigated the effect of worked examples on ability to solve dissimilar or transfer problems. Studying worked examples had no effect on the student's ability to solve transfer problems: the performance of the groups could not be differentiated.

Upon consideration of their previous research (Sweller and Cooper, 1985), Cooper and Sweller (1987) hypothesised that although the worked examples had facilitated schema acquisition as indicated by the results on similar test problems, several aspects of the experimental procedure had hindered the ability of the students to automate the problem solving schemas they had acquired. It was contended that the automation of problem solving schemas was a key process required for the transfer of knowledge since this resulted in a reduction in working memory load. Indeed, Sweller proposed that "(a)utomation is likely to be a necessary component of expertise. Understanding alone may be insufficient" (1999:77). Cooper and Sweller (1987) identified factors such as the relatively small number of worked examples and small amount of practice provided in the acquisition phase combined with the large number of problem sub-types and problem rules the students were required to learn about as factors from the previous experimental procedure which had impeded the development of automation. The authors hypothesised that by increasing the number of examples and by reducing the number of problem solving rules required for solution, improvements in schema automation and transfer problem performance would occur. When schemas are not automated, a problem solver still has a relatively high cognitive burden: although able to understand the problem, the problem solver still needs to retrieve the appropriate mechanisms of the schema and to consciously evaluate a problem solving move to ensure it is appropriate (Ward and Sweller, 1990). If a problem schema is automated, however, extra cognitive resources become available because resources are no longer required for schema retrieval. These extra cognitive resources are presumed to be significant since

they provide the student with the ability to explore the new problem space and to reflect on the differences between familiar problems and new, different problems (Sweller, 1999).

Cooper and Sweller (1987) investigated the performance of a group of students who studied worked examples (and solved a similar problem) in comparison to a group who only solved problems on transfer problems. The procedure of the experiment was similar to Sweller and Cooper (1985) except the changes to the experimental design outlined in the previous paragraph were incorporated: namely, the problems studied were less complex and more examples were provided. In terms of acquisition time, the results of this experiment replicated the findings of Cooper and Sweller (1985): the worked example group spent significantly less time acquiring the concepts. In comparison, however, no differences were found between the groups on similar problems yet large differences were found on transfer problem. The worked example group performed better on the transfer problems: in one experiment, while 75% of the worked example group solved the transfer problem, no students in the conventional problem solving group were able to solve it in the time available. In addition, the worked example group also made significantly fewer errors on the transfer problem than the problem-solving group.

The lack of difference on similar problems between the groups can be explained through both the worked example group and the conventional problem group having acquired appropriate schemas. This conclusion is supported by data indicating that both groups solved the problems swiftly and with few errors. While neither group had a schema for the transfer problem due to its structural differences from the training problems, the transfer problem results suggested that the worked example group had been able to automate their schemas. Although the novel problem forced both groups to use a means end strategy, the worked example group was able to solve the problem successfully because of the

processing advantage gained by the automation of several problem solving rules required in the problem's solution. In comparison, students in the conventional group, without the benefit of this automation, were overwhelmed by the joint load of a means end strategy and the need to search for appropriate rules (Sweller, 1999).

In another experiment, Cooper and Sweller (1987) compared a group of low and high ability students who studied worked examples or solved problems. In addition, the length of the acquisition phase of the experiment was also manipulated. The results supported the researchers' explanation of the effect of worked examples on transfer problem performance. In the short acquisition phase condition, performance of similar problems was improved by studying worked examples; however, in the longer acquisition phase, performance of transfer problems was improved by studying worked examples. The authors concluded that although schema acquisition could occur relatively quickly, rule automation occurred much more slowly. Automation increases a problem solver's success in situations of high memory load such as solving transfer problems. Practice, particularly if interspersed with the study of worked examples, facilitates the automation of schemas. Schemas must be acquired before they can be automated. Studying worked examples facilitates the process of schema acquisition leading to faster acquisition time as well as superior performance.

2.3.2.1 The effect of studying worked examples in real classrooms

While the experiments reported in the previous paragraphs indicated the beneficial results of studying worked examples in a laboratory experimental situation, the use of worked examples in a classroom situation has also been assessed. Ward and Sweller's (1990) research used physics problems as part of a high school teaching

program. After initial instruction on two topics, two classes taught by the same teacher received homework problems that consisted of a mixture of worked examples and conventional problems on one topic and conventional problems only on the other topic. The topic matched with the worked examples was counterbalanced across the classes. On the following day, the students sat a classroom test with questions on both topics. Significant class and topic interaction effects were found indicating that studying worked examples improved subsequent test performance. These effects were found for both similar and transfer problems.

The significance of these findings concerning worked examples lies not in an advocacy of their use since worked examples are commonly used in classrooms and in textbooks. The significance lies in Sweller and Cooper's (1985) and Cooper and Sweller's (1987) suggestion that the number of worked examples used to teach a concept be substantially increased and, indeed, that studying worked examples should be used as a replacement for problem solving. This suggestion has proved to be fruitful: test performance in many domains has been shown to improve through increasing the provision of worked examples. In addition, problem solving flexibility, as indicated by the results for transfer problems, has also shown improvement. A remarkable testimony to the efficiency of a worked examples instructional technique is the results of Zhu and Simon's (1987) study. This study found that a three year mathematics curriculum could be completed in two years if worked examples were studied as opposed to learning via a more traditional style of teaching that involved classroom lecturing and problem solving practice. In fact, learning was not only expedited in this study, the test results indicated that students in the worked example group actually performed slightly better than the conventional group.

The finding of superior performance as a result of learning from worked examples has been extended beyond the school classroom to include elderly learners. It was predicted that since working memory resources decline with age, a training

technique that maximises the learners' use of the remaining resources would be beneficial. As predicted, a group of elderly learners studying worked examples showed higher training efficiency than a group solving conventional problems (Van Gerven, Paas, van Merrienboer, Schmidt, in press).

2.3.2.2 Extending the worked examples effect

Despite the learning benefits associated with studying worked examples, student motivation is a key concern. Typically, worked examples are interspersed with practice to confront this problem, however, even in this situations there is no way to ensure that learners are dedicating appropriate attentional resources to study the worked example before solving the accompanying problem (Sweller et al., 1998). Completion problems provide some solution to this dilemma. A completion problem is a partially solved problem that requires the learner to *complete* the solution. While possessing the benefits of a worked example in decreasing the extraneous cognitive load imposed upon a problem solver, completion problems also solve the problem of motivating students and ensuring their attention is focused on the solution strategy provided. Students are motivated to study the example because they are required to solve it. They focus their attention on the relevant part of the task, since, in order to complete a part of a solution, they must study the pre and post problem states (Sweller et al., 1998). Paas (1992) investigated training strategies for transfer in the domain of statistics and compared students who solved conventional problems, students who solved completion problems and students who studied worked examples. Both the worked example group and the completion problem group performed significantly better than the conventional problem solving group, however, no performance differences could be differentiated between groups studying these different types of worked examples.

Student motivation and the use of worked examples in instruction to develop expertise has been further refined by Renkl, Atkinson & Maier (2000). These researchers investigated a dynamic model of integrating worked examples into instruction. This model used both complete worked examples, partially complete worked examples (completion problems) and pure problem solving. Initially, students studied complete examples. Gradually, the amount of problem solving students were required to undertake within the examples was increased. The final stage of the instructional procedure required the students to completely solve problems themselves. The results suggested this instructional procedureproduced greater near transfer performance than pairing a complete worked example with a problem to-be-solved.

The 'self-explanation effect', the learners' ability to explain the solution steps of a worked example to themselves, has been identified as a critical factor influencing learning outcomes associated with worked examples (Atkinson, Derry, Renkl & Wortham, in press; Renkl, 1999; Renkl, Stark, Gruber & Mandl, 1998; Renkl, 1997). In particular, effective self-explanation leads to an increase in performance on transfer problems. It is hypothesised this effect is achieved because self-explanation encourages reflection about the more general aspects of problem solving in a domain (Atkinson et al., in press; Renkl et al., 1998; Renkl, 1997). Effective self-explanation includes behaviours such as anticipatory reasoning of problem solving steps and identification of underlying problem-solving principles and the connection between problem goals and operators (Renkl, 1997). Moreover, through analysis of the qualitative aspects of learners' self-explanations, Renkl (1997) and Stark, Mandl, Gruber and Renkl (in press) were able to predict the type of performance gains achieved by the learners.

Instructional techniques aiming to facilitate self-explanation have been investigated; direct training or elicitation of self-explanation have been shown to be most successful (Stark et al., in press; Renkl, 1999). In fact, in Stark et al.'s (in

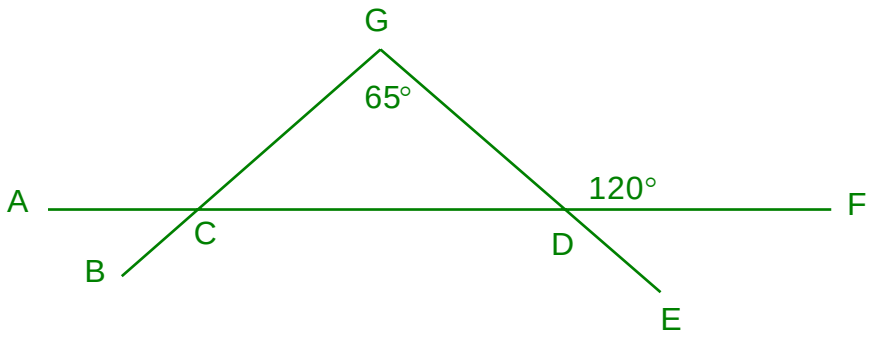
press) study, learners provided with this training produced twice as many elaborations as those who were not provided with the training; in addition, interms of *deep cognitive* elaborations, the learners provided with elaboration training produced twice as many of these elaborations, twice as often.

2.3.3 Split Attention Effect

The previous section examined the extensive body of research that suggests studying worked examples can benefit learning. It is the case, however, that sometimes learning is not improved through the study of worked examples. Tarmizi and Sweller (1988) investigated the usefulness of worked examples in the area of circle geometry. Due to the nature of this content, the worked examples consisted of two sources of information: a diagram and explanatory text. Both sources of information were necessary for the concepts being taught to be understood. Little performance differences could be detected between the groups: the worked example group spent just as long as the problem solving group in the acquisition phase (an extremely different result to Sweller and Cooper's (1985) and Cooper and Sweller's (1987) findings); they spent longer in the test phase and their performance on the test was equivalent. These results seemed to contradict the previous findings of Sweller and Cooper (1985) and Cooper and Sweller (1987) which had been conducted in the domains of algebra and physics. The difference between worked examples in the domains of algebra or physics and the domain of geometry was that learners were forced to split their attention between two sources of information which were both necessary for understanding. Although the geometry worked examples stopped problem solvers using the high cognitive load strategy of means end analysis, they required learners to process two physically separate sources of information which referred to each other. This requirement seemed to have dissipated the worked example's benefits for learning. Ward and Sweller concluded that "the critical factors are not whether worked examples are used, but rather whether the (instructional) techniques used appropriately direct attention and impose a relatively light cognitive load" (1990:4). Traditionally figured geometry worked examples and even physics problems that required students to split their attention between two sources of textual information

(the problem statement and the solution equations) did not seem to met this criterion (Tarmizi and Sweller, 1988; Ward and Sweller, 1990).

In order to comprehend the geometry worked example in Figure 2.2, the learner needs to understand both the diagram and the text below. The text does not make sense without referring to the diagram and the diagram does not make sense without referring to the text. The learners, therefore, are forced to split their attention between the sources of information and to undertake a *search and match* process whereby the diagram is *searched* and the corresponding text statements are *matched* to the relevant angles in the diagram. This task imposes a cognitive load; in addition, the learner is also required to hold all the *matched information* in working memory, imposing an additional cognitive burden. The result of the 'search and match' process is that the learner forms a mental picture in which the two separate sources of information become integrated. This task also uses limited working memory resources; as a consequence, less working memory resources are available for learning. According to cognitive load theory, an instructional format that imposes a cognitive load that is extraneous to the processes of schema acquisition and automation is less than ideal. It follows, therefore, that presenting information with separate sources of mutually referring information is less than ideal since this instructional design requires the learner to undertake many activities that are extraneous to learning.



Find Angle ACG.

Step 1: Angle GCF = Angle FDG – Angle EGB (external angles of a triangle equal the sum of the opposite internal angles)

Angle GCF = 120° - 65°

Angle GCF = 55°

Step 2: Angle GCF + Angle ACG = 180° (straight line)

55° + Angle ACG = 180°

Angle ACG = 125°

Figure 2.2: A conventional geometry worked example. An example formatted in this fashion requires the learner to split their attention between the diagram and the text.

The extraneous burden imposed by a split attention format can be eliminated if the process of integrating related information elements is done *before* learning commences. Mutually referring information can be physically integrated obviating the need for the learner to undertake this process. The worked example shown in Figure 2.3 presents the same information as the worked example in Figure 2.2, yet the two sources of information (text and diagram) have been physically integrated. The learner's attention can be focused on this single source of information rather than having to undertake the extraneous process of *matching* the related

information from the different sources. According to cognitive load theory, the worked example shown in Figure 2.3 will be of more benefit to learning since extra cognitive resources will be available for learning.

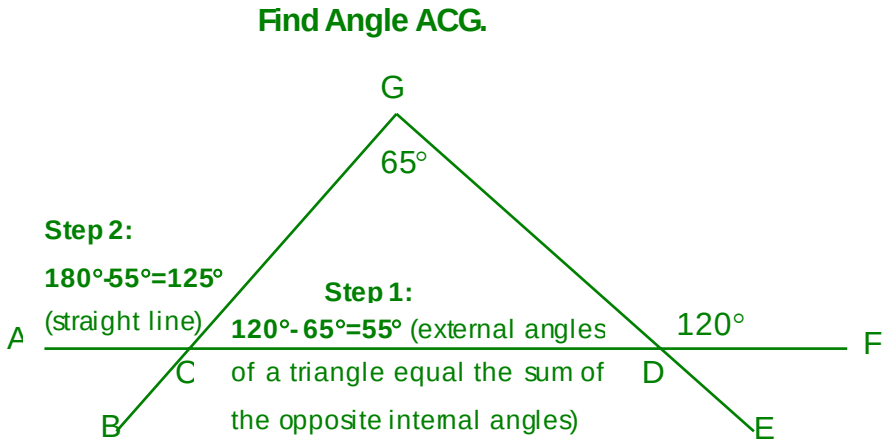


Figure 2.3: An integrated geometry worked example. The text of Figure 2.2 has been integrated into the diagram

Many studies support the notion that physical integration of split sources of information results in reduced cognitive load and leads to improvements in learning. Tarmizi and Sweller (1988) found that while split source geometry worked examples produced performance at a level of the problem solving only group, physically integrated worked examples produced superior performance. Split source information is not restricted to diagram and text. Ward and Sweller (1990) found that similar results to Tarmizi and Sweller with physics worked examples that involved two sources of textual information. If the example was restructured so that the two sources of information were physically integrated, the learning improvements previously associated with worked examples were reinstated (Ward

and Sweller, 1990). Physical integration of the multiple sources of information was achieved by integrating the solution equations into the physics problem statements (Ward and Sweller, 1990).

The benefits of physically integrated instructional material in comparison to split source instructional material is not restricted to worked examples; for example, Sweller, Chandler, Tierney and Cooper (1990) developed typical text book instructions for coordinate geometry that was formatted so that text was physically separate from diagrams. They also restructured the material so that the text was physically integrated into the diagram. Students in the integrated instruction group solved the subsequent test problems faster and more efficiently than the split source group. Similar results were recorded in other studies that instructed students on the technical area of numerically controlled machine programming. These results were recorded regardless of whether the split sources of information were text and diagram or text and text.

These results have been replicated and generalised to several other areas. In a long term study, Chandler and Sweller (1991) found that students who studied electrical engineering materials in a split source format did not reach the initial test performance level of students in the physically integrated instruction group despite 12 weeks of access to the instructional material. The researchers found similar results in the domain of biology where the integrated instruction group performed better than the split source group on the majority of test problems despite spending less time in the acquisition phase of the experiment.

The detrimental learning effects of presenting split source information has even been extended to the format of scientific journal articles. Traditional scientific journal articles separate information describing how an experiment is conducted; information describing the results and information describing the conclusions drawn into separate sections. Chandler and Sweller (1992) argued that the

traditional journal article format is a classic example of split attention since the reader is required to mentally integrate information from several sources in order to comprehend the concepts being presented; for example, a conclusion can only be understood in relation to the data recorded and the method under which this data was collected. A group of students read an article either traditionally formatted or restructured so that the method, results and conclusion of a procedure were presented simultaneously. Students then answered questions on details of the report. Students in the modified article group performed significantly better in the test despite requiring significantly less time to read the article.

A split attention effect is not limited to situations where text and diagrams or text and text refer to each other and are required to be mentally integrated. Another common instructional situation requires learners to split their attention between equipment and instructional material describing how the equipment works; for example, computer software is frequently taught through the provision of a manual that constantly refers to aspects of the computer or software. Often a learner's attention is split between not only the computer and the manual but the manual itself is composed of split source text and diagrams. This phenomena was investigated by Chandler and Sweller (1996) and Sweller and Chandler (1994) who found that a group who learned about several types of computer software without access to a computer (the group studied a manual that physically integrated text into diagrams of the computer screen and keyboard) performed significantly better on the subsequent test of computer skills than a group that studied a textual manual and used the computer to follow the steps suggested. Thus, despite having no ability to practice using the software, the modified manual group, learning under conditions of lowered cognitive load, developed superior computer software knowledge. Further support for a cognitive load explanation of these results was provided by Cerpa et al. (1996). The results of this series of studies suggested that it was not absence of the computer per se that produced the superior performance of the modified manual group in Chandler and Sweller's

(1996) and Sweller and Chandler's (1994) studies but the split attention due to having to mental integrate information from two sources. Cerpa et al. (1996) developed a self-contained computer based training package about spreadsheet software: instructions on how to use a spreadsheet were physically integrated into the software itself. This group was compared to a group using a conventional computer manual and the software package on a computer. It was predicted that the integrated computer based training group would outperform the conventional manual plus computer group because the learner's attention would be split between several sources of information in the latter group. The test results supported this hypothesis. In another study, self ratings of the mental effort required for learning were collected from the participants; these indicated that the integrated group required less mental effort to acquire a superior level of learning.

Interestingly, it has been suggested that split attention should be considered in innovative educational settings such as collaborative computer supported learning environments (Van Bruggen, Kirschner and Jochems, in press). Collaboration is achieved in this environment through discussion; discussion can be both synchronous or asynchronous. The authors suggest, however, that each of these discussion formats imposes extraneous cognitive load on the learner. Coherent asynchronous communication requires the learner to refer back to previous entries in the discussion list and possibly refer to external information sources creating a situation where the learners attention is split between several sources of information. Synchronous discussion makes the same demands while imposing additional external pace constraints on the learner. Van Bruggen et al. (in press) suggest software that allows viewpoints and 'rising above' notes may be useful tools to reduce split attention in this environment.

The split attention effect is general and robust. It has been shown to occur across many academic and technical domains using a variety of information sources that interact such as text and diagram, text and text and text and equipment.

Resolution of the split attention effect through integration of information was of particular significance for cognitive load theory as it illustrated the far reaching impact a theoretically based theory of instructional design could have. This design technique had relevance not only for classroom settings but for realistic training environments as well as instructional design in general.

2.3.4 Redundancy effect

Research into the effect of redundant information upon learning originally developed from an investigation by Chandler and Sweller (1991) to define the conditions under which physical integration to reduce split attention was useful. Redundancy also involves information from several different sources but, in contrast to split attention, the sources of information are self-contained and can be understood in isolation. An example of redundancy is text that replicates the information already contained in a diagram (see Figure 2.4). If the student is capable of gaining information about the concept from a single, highly intelligible source of information, the other source of information becomes redundant. Redundant information is detrimental to learning because it imposes an extraneous cognitive load. Cognitive resources are required to process the repeated information; as a consequence, less resources are available for the essential learning processes of schema acquisition and automation.

Figure 2.4: An example of redundant instructional material. The text in this figure provides the learner with the same information as the diagram (from Chandler and Sweller, 1991).

Using several realistic content areas such as blood flow diagrams from biology and electrical circuitry, Chandler and Sweller (1991) found that when students studied self-contained diagrams and text which repeated the information contained in the diagrams, learning was reduced and study times were often increased. This pattern of results was achieved regardless of whether redundant text was integrated or remained separate to the diagram. In a surprising result, Sweller and Chandler (1994) and Chandler and Sweller (1996) replicated the redundancy effect by finding the computer redundant when learning to use a computer program, if the program instruction manual was self contained (included diagrams of screen and keyboard). This effect was achieved only if the instructional material imposed a high cognitive load[5]. Cerpa et al. (1996) also investigated the effect of redundant information in computer software training. A group learning from a computer based training program containing physically integrated instructions was compared to a group which had access to the integrated computer based training as well as to a hard copy of these training materials and to a group studying a conventional (non-integrated) manual. Results from this research indicated that the group that studied the integrated software with no access to the hard copy materials (this group, therefore, had no access to redundant material) performed most successfully and also rated their mental effort as significantly less than the other groups on a subjective rating scale. The efficiency of the instructional conditions was assessed using the formula devised by Paas and van Merrienboer (1993) that standardises and then relates the mental effort and the test performance associated with a particular instructional condition. This procedure allows a comparison between instructional conditions to be drawn. It indicated that the performance efficiency of the group with no exposure to redundant material was superior to the redundancy group.

[27] The effect of learning material of varying intrinsic load will be discussed in section 1.4.

The redundancy effect is not restricted to visually presented material. Kalyuga et al. (1999) replicated the redundancy effect using multimedia computer based materials. Students given a visual diagram, auditory explanatory text and visual explanatory text which replicated the auditory text, reported increased mental load (assessed by subjective mental load scales); performed worse on both multiple choice and fault finding test questions and required a greater number of learning trials than a group which received auditory explanatory text only with the visual diagram. Performance efficiency scores indicated the instructional condition with no redundant material was a significantly more efficient method of instruction than the redundancy condition.

A learner's level of expertise in a domain is a major influence mediating the negative impact of redundant information upon learning. For example, Kalyuga et al. (1998) found the negative learning consequences of including redundant information altered depending on the level of expertise of the learner. Using material from electrical engineering, it was found that novice learners learnt best from a physically integrated instruction format that did not require them to split their attention between two sources of information. This instructional format integrated explanatory text into a self-explanatory electrical circuit diagram. Thus, the results indicated that novice students benefited from the presence of redundant material in their instruction. After intensive training in the principles of electrical circuitry, however, the same students performed best from an instructional format that included the self-explanatory diagram only. These results indicate that the text became redundant as domain knowledge increased. It was concluded that:

> (w)hile the theoretical distinction between sources of information that are or are not intelligible in isolation is clear cut, it is reasonable to hypothesise that decisions concerning the intelligibility of modules of information cannot be made without reference to the learners for whom the information is intended" (Kalyuga et al., 1998: 14).

Yeung, Jin and Sweller (1998) investigated the effects of expertise and the usefulness of redundant information in the domain of language comprehension and vocabulary learning. The findings of Kalyuga et al. (1998) were generalised to this area. As language comprehension ability increased, the participant's skill at comprehending a passage that included vocabulary definitions integrated into the text decreased. Thus, with increasing expertise, these definitions were redundant.

The experiments discussed in the previous paragraphs indicate that certain parts of instructional material, which were initially beneficial to learning, can become redundant and detrimental to learning as expertise increases. Schema acquisition can explain this finding (Kalyuga et al., 1998; Yeung et al., 1998). Explanatory text, despite merely repeating information available from another source, is useful to the novice learner as one source of information serves to explain another (see for example, the concept of the starter circuit in Kalyuga et al, 1998: 3-4). As schemas develop in the relevant knowledge domain, the inclusion of the explanation becomes unnecessary. In fact, the presence of this repetitive information impedes further learning since limited processing resources are expended upon redundant information (Kalyuga et al., 1998; Yeung et al., 1998). This process is extraneous to learning yet it causes less resources to be available for processes such as schema acquisition and automation that are critical to learning.

The benefits for learning caused by the removal of redundant information have been shown through test and study times performance (Kalyuga et al, 1999; Kalyuga et al, 1998; Yeung et al, 1998; Cerpa et al, 1996; Chandler and Sweller, 1991). In addition, measures of cognitive load support the contention that the presence of redundant information causes an increase in extraneous cognitive load. The cognitive load caused by redundant information has been assessed through such measures as subjective ratings of mental load (Kalyuga et al, 1999;

Kalyuga et al, 1998; Yeung et al, 1998; Cerpa et al, 1996 and secondary task performance (Chandler and Sweller, 1996).

There is a history of evidence supporting the detrimental effect of redundant information on learning. For example, Miller's (1937) investigation of children's reading suggested that pictures presented in conjunction with nouns hampered reading progress. Several more recent studies have replicated this result (e.g. Wu & Solman, 1994); in addition, the finding has been extended to passages of text rather than single words. Reder and Anderson (1980, 1982) investigated learning from whole text book chapters in comparison to chapter summaries. In a robust result, they reported that students learning from summaries showed superior retention of the main ideas of the text and this advantage was retained over time. Their investigations suggested this result was not caused by the reduced amount of time allotted to studying main points (Reder and Anderson, 1982) but rather was a consequence of the limited nature of human cognitive processing. It was suggested that processing capacity was used to learn details from the full text to the detriment of *main point* learning (Reder & Anderson, 1980). It was concluded from these results that the full text was redundant. Mayer, Bove, Bryman, Mars and Tapangco (1996) replicated Reder and Anderson's findings using a scientific text. The group that received the information summary performed significantly better than the group that received the full text. Performance was assessed by both free recall and problem solving measures. In fact, the performance of the summary group on the problem solving task was approximately double that of the full text group. These results were also used to support the hypothesis that the limitations of human cognitive architecture impede the ability to learn when a learner is required to process redundant information.

2.3.5 Modality effect

The cognitive load effects previously discussed in this thesis have developed from the assumption that working memory capacity is limited and, in order to maximise the processing resources devoted to learning, instructional techniques should aim to reduce the extraneous load they impose on learners. In contrast, the modality effect arises from the assumption that working memory capacity can be expanded. While the former assumption is based on the concept of working memory as a unitary entity, the latter assumption takes into account more recent theories of memory which suggest that working memory is composed of several partially independent processors that handle information from different sensory modalities such as auditory and visual (Baddeley, 1981, 1990, 1992; Paivio, 1990)[6]. Thus, according to this assumption, if instructional material uses both a diagrammatic or written format and an auditory format, the capacity of working memory to process it will be increased in comparison to its capacity to process the same information if it was presented in a solely visual or solely auditory format. Some of the evidence supporting this assumption includes the finding that the ability to carry several tasks at the same time is improved by the presentation of the tasks in different modalities (Allport, Antonis and Reynolds, 1972) and that recall is improved if presentation of stimuli is bi-modal (Frick, 1984).

Although evidence supports the notion that working memory contains processors for different modes of information, it does not suggest that these processors are completely separate. If this were true, bi-modal presentation of information would lead to increases in recall of 100%; while memory increases are apparent, they are not of this order of magnitude.

---

[28] A more detailed explanation of these theories is provided in section 1.2.2.1.

If the assumption that working memory contains partially independent information processors differentiated by information modality is held to be true, it follows that an instructional technique that takes these features of working memory into account could have benefits for learning. Thus, if the information to be learnt is presented through both auditory and visual channels, theoretically, the capacity of working memory to process the information should be increased; thus, learning will be more efficient. In addition, the use of such a dual mode instruction should provide an alternative technique for overcoming the extraneous cognitive load generated by the split attention effect. The extra processing resources available from a dual mode presentation should alleviate the extra cognitive load generated by split source materials that require a student to mentally integrate the sources of information.

Several studies have investigated such an instructional technique; for example, Mousavi et al. (1995) investigated the use of a dual mode instructional technique in the area of geometry. Geometry is a domain where traditionally, due to the nature of its content, instructional resources have often caused the learner to split their attention between a diagram and explanatory text. Using geometry worked examples, a group that received a visual diagram and an auditory explanation was compared to a group that received a visual diagram and a visual (written text) explanation and to a group that received a visual diagram, a visual explanation and the same explanation presented auditorily. The visual/auditory group consistently solved similar and transfer problems faster than the visual only group although no test score advantages were reported. The results of several other studies indicated that superiority of the dual mode group were unlikely to be due to factors such as students in the visual /visual group having to hold information from one source in working memory before attending to the other source or students finding listening easier than reading. In addition, the benefits of a dual mode instruction were achieved regardless of whether the format of the visual component of the information was in written or diagrammatic form.

The conditions under which dual mode instruction are most beneficial have been further refined. In areas of low visual search in geometry, Jeung et al. (1997) replicated the findings of Mousavi et al. (1995): dual mode of instruction was superior to visual only instruction. However, in conditions of high visual search, dual mode instruction was of no more benefit than visual only instruction. It was hypothesised that in this situation the benefits of dual modal instruction were alleviated by the extraneous cognitive load imposed by the task of coordinating the audio and visual information. When a visual marker in the form of electronic flashing was included in the instructional format to facilitate the information *search and match* process, the superiority of the dual mode instruction group was regained.

Tindall-Ford et al. (1997) further explored the benefits of dual mode instruction for ameliorating the cognitive burden of the split attention effect. Using high element interactivity content from the technical area of electrical engineering, Tindall-Ford et al. compared a group that received a visual diagram and auditory explanation (dual mode group), a group that received a conventional split source visual diagram and visual text explanation (split attention group) and a group that received a visual diagram and visual explanation that had been physically integrated (integrated group). The results indicated that both the dual mode instruction group and the integrated instruction group outperformed the split attention group in terms of both transfer test scores and practical test scores. The performance of the dual mode group and the integrated group could not be differentiated. These results indicate that the cognitive burden imposed by the split attention effect is solved equally well by instructional techniques that reduce extraneous load through physical integration or techniques that expand the capacity of working memory.

In a second study, the generality of a dual mode instruction was examined through the use of a text table rather than a diagram. In this experiment a group that received the visual information table and an auditory explanation was compared to a group that received the visual table and a visual (text) explanation. In addition, this experiment was significant because the cognitive load of the participants was directly assessed through subjective mental effort ratings. The dual mode group performed significantly better than the visual only group while rating their mental effort significantly lower. When the performance efficiency of the instructional conditions was compared, the dual mode condition was a significantly more efficient method of instruction then the visual only technique.

If the success of the dual mode instructional technique is due to the fact that working memory resources are expanded, as cognitive load theory contends, it follows that dual mode instruction will only be of benefit when cognitive load is high. Tindall-Ford et al. (1997) assessed this assumption by comparing dual mode instruction to visual only instruction on both low and high element interactivity materials. Both test performance and subjective mental load ratings supported this conclusion indicating that the use of two sensory modalities for instruction expands the capacity of working memory; the greater processing power that results is of most benefit in situations where the cognitive load imposed by the instructional material, the *intrinsic* load, is high enough to jeopardise working memory's limited processing ability. This study is discussed in greater detail in the following section (section 2.4).

2.4 Instruction and element interactivity

Despite the evidence discussed previously that suggests that presenting instructions using instructional techniques that impose a high extraneous cognitive load is detrimental to learning, further research indicates that this is only the case when the intrinsic cognitive load imposed by the instructional content itself is high. When this situation arises, the total cognitive load imposed upon the learner by the content and the method of instruction overburdens working memory and little if any learning results. For example, Sweller and Chandler (1994) found that when students were required to solve a problem where there was minimal interaction between the information elements (e.g. moving a cursor a set distance on the grid screen of a computer aided design package) no difference existed between a group learning from a conventional instructional manual and a computer, a group that learnt from a self-contained integrated manual and a group provided with the integrated manual as well as a computer. Thus, this experiment was designed to assess both the split attention and the redundancy effect. In contrast, when the participants were required to solve problems that imposed a higher intrinsic load (e.g. the steps required to draw a line between two absolute coordinate positions on the grid screen of a computer aided design package) the results indicated a significant performance difference. In these types of problems the integrated group performed better. The element interactivity of the material was estimated by assessing both the number of elements of information that that was required to be processed concurrently as well as the student's expertise in the domain. Large differences in the number of simultaneously processed elements existed between problems that were considered to be low in element interactivity and problems that were considered to be high in element interactivity.

Similar results were reported by Chandler and Sweller (1996) using the same experimental groups as Sweller and Chandler (1994). The self-contained manual

group performed better on written and practical tasks only when the material imposed a high intrinsic cognitive load. These researchers also measured the cognitive load imposed by the learning tasks by measuring the participant's performance on a secondary task. While low element interactivity material did not affect secondary task performance, significant differences were present when the participants solved problems based on high element interactivity material. In this case, the secondary task performance of students forced to learn under conditions of split attention or redundancy were significantly decreased in comparison to students where the instructional information did not impose an extraneous cognitive load. In addition the trend of the difference was such that while both the split attention and the redundancy groups' performance on the secondary task decreased dramatically, a minimal performance decrement was recorded for the integrated manual group. As the intrinsic cognitive load of the content material was standardised across all groups, this result provides strong support for the notion that it was extraneous load associated with the instructional formats of both the split attention (conventional manual and computer) and redundancy (integrated manual and computer) group that was responsible for performance decrements experienced by these groups (Chandler and Sweller, 1996).

It is important to note that the low element interactivity material used by Sweller and Chandler (1994) and Chandler and Sweller (1996) did not consistent of trivial concepts, it merely did not impose a high intrinsic cognitive load since the elements did not interact. Skills such as moving a cursor in different denomination steps, using the grid lock or using the function menus of computer aided design software are difficult to learn because there are many separate skills to remember. Each skill can be learnt individually, however, since this learning requires only one or two information elements to be processed simultaneously. As a result, the intrinsic cognitive load imposed by material of this type is minimal. In order to undertake high element interactivity functions such moving between two absolute coordinate positions in computer aided design software, however, novice users of

the software are required to hold as many as nine elements of information concurrently. As a consequence, the intrinsic cognitive load imposed by material of this nature is much greater than in the material described previously.

The results discussed in the previous paragraphs indicate that instructional conditions that reduce extraneous cognitive load in situations where the intrinsic load imposed by the instructional material is high can have benefits for learning. Similarly, the results of several experiments indicate that instructional conditions that expand working memory resources can also benefit to learning in this situation. For instance, Tindall-Ford et al. (1997) compared the performance of a group that received information in a visual only format (text and diagram) to a group that received diagrammatic visual information and an equivalent auditory version of the textual information (dual modality group) using instructional material that imposed either a high or a low intrinsic load. The low element interactivity material consisted of a list of electrical symbols and their meaning (see Figure 2.5). Although there were many symbols to be learnt, the information was of low element interactivity since each symbol could be learnt and understood in isolation to the other symbols. As each symbol could be processed serially, the working memory load imposed by this material was minimal. In contrast, the high element interactivity material consisted of several electrical circuits (see Figure 2.5). To understand an electrical circuit, all parts of the circuit must be simultaneously processed since each part of the circuit is related to each other part. The requirement of simultaneous processing has the effect of imposing a high processing burden on working memory.

LOW ELEMENT
INTERACTIVITY MATERIAL

HIGH ELEMENT
INTERACTIVITY MATERIAL

Figure 2.5: An example of some of the low element interactivity material and the high element interactivity material used in Tindall-Ford et al. (1997).

For low element interactivity material, no difference was found between the visual only and the dual mode instructional technique in terms of test performance or subjective mental load. The efficiency of the instructional conditions for this material could not be differentiated. In contrast, for high element interactivity material, the students in the dual mode condition showed superior test performance in terms of scores and times as well as rating their mental effort significantly lower. The researchers also eliminated the possibility that these results may have been due to the students finding listening an easier task than reading. The efficiency of the instructional conditions was assessed and the results indicated that the dual mode condition was a significantly more efficient instructional technique in a situation where the learner was required to understand high element interactivity material. These results provide support to the cognitive load interpretation of a dual mode instructional technique: dual modality instruction expands working memory capacity and results in greater processing capacity. This is only beneficial to learning in situations when the intrinsic load of the instructional material is high enough to jeopardise working memory's limited processing ability.

The results of these studies that have focussed on information of differing element interactivity indicate that information can be difficult to learn for several reasons. Information may be difficult to learn because there is a large amount of information to be learnt or because there is a large amount of information to be processed concurrently (Tindall-Ford et al., 1997). Cognitive load concerns arise in the latter case and it is in situations such as these that the instructional technique used becomes a critical factor in learning. Instructional formats that reduce extraneous cognitive load or increase the ability of working memory to process information benefit learners when the information to be learnt imposes a high intrinsic cognitive load.

2.5 Cognitive load theory: a summary

Cognitive load theory applies knowledge about the structure of human cognitive architecture to the design of instructional material. The theory contends features of cognitive architecture and development of expertise such as the limited capacity of working memory, the virtually limitless storage available in long term memory and the acquisition and automation of schemas need to be considered when designing instruction. Further, the theory aims to develop instructional designs that facilitate learning by providing conditions which aid the processes of schema acquisition and automation. Since information imposes its own intrinsic load on working memory, in order to ensure sufficient processing resources are available for learning, instructions need to be designed so that they do not incorporate features that are extraneous to learning, yet necessitate the use of cognitive resources.

Cognitive load theory has assessed many commonly used instructional techniques and found that they impose a high cognitive burden that "interferes with the very learning intended by the instructional procedure" (Sweller, 1999: 55). These instructional procedures are cognitively demanding because the instructional designs themselves incorporate features that are extraneous to learning and, therefore, "interfere with rather than assist in schema acquisition and automation" (Sweller, 1999: 55). Cognitive load theory has been used to generate alternative instructional techniques which, while serving the same instructional purpose as the conventional procedures, either reduce the extraneous load imposed by the instructional design or expand the capacity of working memory. These techniques include goal free problems, increased exposure to worked examples, physical integration of split sources of information, elimination of redundant information and the use of several modalities in instruction.

Goal free problems prevent learners using means end analysis, a problem solving strategy that imposes a high cognitive load and directs the learner's attention to aspects of the problem that do not facilitate the development of schemas. Increasing the amount of worked examples a learner studies also prevents the use of means end analysis; in addition, this technique directs the learner's attention to aspects of the problem that need to be incorporated into a problem solving schema. The physical integration of mutually referring sources of information eliminates the split attention effect; this occurs when attention must be split between two sources of mutually referring information and the learner is forced to mentally integrate the information sources in order to understand the concept being presented. If the need to process redundant material incorporated into instructions is eliminated, more processing resources become available for learning. If several modalities are used to present instructional material, the partially independent modality processors existing in working memory are utilised; this has the effect of expanding the capacity of working memory to process information. The instructional techniques outlined above are of most use when the intrinsic cognitive load, caused by the instructional material itself, is high.

The focus of cognitive load theory research on developing instructional designs that reduce extraneous load or increase working memory capacity is, in part, due to the previously held belief that the intrinsic load of instructional material is unalterable (Sweller, 1993; Sweller and Chandler, 1994; Chandler and Sweller, 1996; Sweller, van Merrienboer and Paas, 1998). The next chapter of this thesis will discuss findings that indicate that this belief may be false. In other words, the intrinsic cognitive load of instructional material may be changeable.

Chapter 3 Pathways to meaningful learning

3.1 Types of learning

Historically in the field of education, a debate has raged over the necessity of understanding as a component of learning. Earlier this century, there was strong support for learning without understanding, that is rote learning: " for pure practical efficacy it is better that the rationale should not at the actual time of working be thought of at all. The pupil should contain his efforts to a rigid application of the rule" (Ballard, 1915: 12). By the middle of the century, there was a shift of support in the debate towards meaningful learning, that is learning with understanding. Experimental evidence increasingly supported this view; for instance, in a large study involving 4 cities, 41 classes and 1400 students, Brownell & Moser (1949) compared students who were taught to borrow in subtraction by either a meaningful or a mechanical (rote) method. The evidence overwhelming supported a meaningful learning approach.

Ausubel (1968a) contended that rote learnt material and meaningfully learnt material are represented, organised and stored differently as well being retained and forgotten at different rates. Rote learning produces "discrete, self contained entities organizationally isolated" from existing stored knowledge (Ausubel, 1968a: 109). In contrast, meaningful learning is learning that is anchored to existing knowledge in a non-arbitrary, substantive fashion.

Several factors must be present in order for meaningful learning to occur: the learner must be disposed to relate the new learning in a meaningful way to his or her existing knowledge; the relationship between the new concept and existing knowledge must be plausible (non-arbitrary) and not dependent upon a single set of words or phrases (non-verbatim); and most importantly, it must be possible for the learning task to be meaningfully related to existing knowledge (Ausubel, 1968a,b). For this final criterion to be achieved, the learner's cognitive structure must already contain an appropriate and relevant body of knowledge (Ausubel, 1977).

Subsequent research has reinforced Ausubel's contention about the importance of prior knowledge for meaningful learning; for example, Dochy, Segers & Buehl (1999: 173) concluded after their thorough review of the literature that "prior knowledge is indeed an effective aid for learning new knowledge". They also suggest that students with little or no prior knowledge in an area have difficulty in learning and structuring new information because they lack the necessary knowledge frameworks. Yates and Chandler (1991) label this phenomena the Matthew Effect: that is, the (knowledge) rich get richer. This idea is reminiscent of Ausubel's explanations and can also be understood from a schema perspective: if the learner does not possess a schema or it is not activated, comprehension and recall are reduced at the time of learning and beyond (e.g., see Bransford and Johnson, 1972).

Despite the more onerous conditions that must be fulfilled in order to achieve meaningful learning, this type of learning is more desirable than rote learning since knowledge learnt in this manner is more easily developed, it is retained better and is forgotten after a greater interval than knowledge gained through rote learning (Ausubel, 1968a; 1968b; 1977). Meaningful learning possesses these advantages over rote learning because of the meaningful relationship that is developed between the new knowledge and the learner's existing knowledge hierarchy (Ausubel, 1968b). If the learner does not possess prior knowledge relevant to the learning task, however, rote learning may be

unavoidable since "(i)f some existing … set of ideas cannot be utilised for assimilative purposes, the only alternative is rote learning" (Ausubel, 1968a: 131).

3.2 An alternative path to understanding

There is evidence in the literature that learning in the absence of full comprehension can ultimately benefit understanding; for example, King & Russell (1966) compared academically matched students given the task of reading a text from psychology with instructions to either aim to learn the text *word for word* or to focus on learning the *main ideas*. Both at immediate recall and at recall after a second review of the text, the *main ideas* group performed significantly better on measures of recall, such as number of letters, number of content words identical to the original text and number of sentences recalled. The groups did not differ statistically in terms of their memory for the meaning of the passage, identified in terms of their scores on measures such as number of idea units and sequences of words. Interestingly, by the second recall (undertaken after a second reading of the text) the trend of the data had reversed magnitude suggested that the *word for word* group was outperforming the *main ideas* group in terms of understanding the meaning of the passage. The authors were puzzled by the results, suggesting that the distinction between rote learning and meaningful learning is not clear cut. The results of this experiment can be reinterpreted in light of Hoosain's (1983) work.

Hoosain (1983) was interested in exploring if there is a cognitive foundation to the traditional Chinese practice of rote learning classical literary passages. Classical literary passages are complex pieces of text, different to both modern written Chinese and modern spoken Chinese because of their remote subject matter, use of archaic words and phrasal structures and their terse, non-repetitive style. In order to test the influence of rote learning, groups of female high school students were

instructed to either learn a text by rote, without trying to understand the ideas, or to try to understand the ideas in the text without focusing on the exact words. Four text passages were used in the experiment, differing in complexity of the content (difficult or easy) and the writing style (traditional literary or modern colloquial style). While the colloquial texts were approximately twice as long as the literary texts, the ideas presented in each set of texts were identical since colloquial texts were translations of the literary texts. The difference in length is testimony to the more descriptive, verbose style of the colloquial writing style as compared to the literary texts. To summarise, the experiment compared rote learning to main ideas learning using texts that differed in complexity and writing style.

The eight experimental groups were tested on their verbatim recall and their comprehension of the particular text allocated to their group. As might be expected, the rote memorisation group performed significantly better on the verbatim recall measure regardless of type of text. Comprehension was assessed through multiple choice questions based upon the content of the passage. Surprisingly, the comprehension performance of the rote and main ideas groups across all texttypes could not be distinguished. However, comprehension performance main effects for literary style and passage difficulty were significant: superior performance was recorded for groups studying colloquial texts in preference to literary texts and for groups studying the easier passage. When the raw data for comprehension was analysed, it showed that the rote memorisation group outperformed the ideasgroup on both the difficult literary and the difficult colloquial texts. In order to investigate this effect further, Hoosain replicated the experiment using male high school students. In this experiment only two groups were compared: a rote memorisation group and a main ideas group. The groups were compared on their verbatim recall and comprehension of the difficult literary text used in the previous experiment. A similar pattern of results to the previous experiment was recorded. The rote memorisation group recalled significantly more in terms of verbatim reproduction yet

their comprehension test performance could not be differentiated from the main ideas group.

Hoosain concluded the traditional practice of memorising classical Chinese texts does have a cognitive basis. He suggested "(r)ote memorisation, even with only partial understanding, could serve to retain the text in memory while waiting for a future opportunity to unravel meanings" (1983, p.196). Thus, understanding was able to continue after the initial learning because the student was still able to access a large part of the text in memory. Access to this material in memory may have given the rote memorisation students an advantage in the test because the questions may have directed them to parts of the text which they could then reflect upon. In contrast, students in the main ideas group who did not understand the meaning of parts of the text during their original learning had no opportunity to retrospectively acquire this understanding. This conclusion can also provide an explanation for King and Russell's (1966) results.

Hoosain's results and conclusions can be viewed from a cognitive load perspective. Learning a complex text by rote allowed working memory to process some of the necessary basic elements that constitute a schema for the information to develop. This schema helped learners in their subsequent comprehension test performance. This interpretation is supported by the finding that the rote memorisation group performed most strongly on the most difficult texts (the difficult literary and the difficult colloquial text). A cognitive load interpretation would predict that the most obvious difference between learners who have acquired a schema for the information and learners who have not would occur when the material to be learnt is at its most difficult. This is because the difficulty of a text is caused by a high degree of interactivity between the elements forming the text. As the element interactivity of learning material increases, the likelihood of working memory being overwhelmed also increases. If this overload occurs, a schema for the information cannot develop.

A lack of a schema further compounds the learner's ability to process the information and leads to an inability to comprehend the meaning of the text.

The fact that in both Hoosain's (1983) and King and Russell's (1966) research the groups instructed to learn without necessarily understanding the material displayed superior performance when tested for comprehension and understanding raises a question worthy of further investigation: would initially learning some elements of information, even if comprehension was not possible, be ultimately beneficial to learning? This question will be explored further in the next section.

Chapter 4 Introduction to the empirical investigations

4.1 A summary of the research pertinent to this thesis

The previous chapters have provided a detailed discussion of the theoretical assumptions and research relevant to the current thesis. These assumptions and research provide a theoretical framework for the current empirical investigations. This chapter will summarise this theoretical framework and briefly outline the experiments which follow.

The aspects of human cognitive architecture relevant to higher intellectual processes such as learning include the initial processing of information by working memory and storage of information in long term memory in schemas. Working memory is defined by its capacity limitations; it can process only a few elements of information at any one time (Simon, 1974; Miller, 1956a). This contrasts to long term memory which has a seemingly unlimited storage capacity (Newell & Simon, 1972). Knowledge is stored in long term memory in cognitive constructs called schemas. Schemas hold vast amounts of information; they are hierarchically organised. Schemas not only provide the structure for long term memory but also form the basis of our intellectual capacity (de Groot, 1966; Sweller and Cooper, 1985) as they reduce the processing burden placed on working memory by enabling information from different contexts to be recognised and categorised accordingly.

Schema acquisition and automation of the knowledge held in schemas are the critical factors involved in human learning, understanding and the development of expertise. Evidence has now emerged to strongly suggest that the differences between experts and novices is their domain specific knowledge stored in schemas. Expertise is determined by the extent of schema acquisition (de Groot, 1965; Chase and Simon, 1973), as well as the degree to which these schemas contain specific information on domain typical problem solutions (Chi et al., 1982; Larkin et al, 1980 ; Sweller, Mawer and Ward , 1983). With expertise, interacting elements that would otherwise overwhelm working memory can be incorporated in a schema that acts as a single element and imposes a minimal working memory load. Given these findings, the aim of instruction should be to build knowledge and expertise through the acquisition and automation of schemas.

Cognitive load theory (see Sweller, 1999 for a recent summary) has its foundations in the cognitive architecture described above. It uses some aspects of human cognitive architecture and of the structure of information to provide instructional designs that are conducive to the acquisition of schemas and their automation. The theory contends that the amount of mental load placed on working memory by an instructional task will be the critical factor in determining how effective learning has been. Working memory load plays such an essential role in learning because if the capacity of working memory is exceeded, schema acquisition and automation will be impeded.

Cognitive load theory makes a distinction between two sources of cognitive load. Extraneous cognitive load is generated by the manner in which information is presented to learners and is under the control of instructional designers. The intrinsic cognitive load of information is determined by the information itself; more specifically, it is determined by the extent to which various elements interact. While all information will impose some degree of intrinsic cognitive load, difficult to

understand material is difficult because its information elements interact extensively and this imposes a heavy load on working memory because these elements must be processed simultaneously (Halford, Wilson & Phillips, 1998; Chandler & Sweller, 1996; Marcus, Cooper & Sweller, 1996; Sweller, 1994).

It has been stated previously that the intrinsic load of instructional material is unalterable (Sweller, 1993; Sweller and Chandler, 1994; Chandler and Sweller, 1996; Sweller, van Merrienboer and Paas, 1998). This is true if immediate understanding is the goal of the task; however, it will be indicated in the following paragraphs that intrinsic load can be altered if a learning situation is created where understanding is temporarily not an objective.

The results of several experiments have suggested that presenting all the elements required for understanding does not benefit learning (Hoosain, 1983; King and Russell, 1966). In addition, research into learning hierarchies has suggested lower level skills are prerequisites to higher level skills (Gagne, 1973, 1970; Bloom, 1956) and that learning *part* of a *whole* skill in isolation to the *whole* will ultimately benefit understanding of the *whole* if this relationship between these entities is meaningful and it is made explicit (Schmidt, 1991; Singley & Anderson, 1989).

4.2 Learning complex information: a paradox

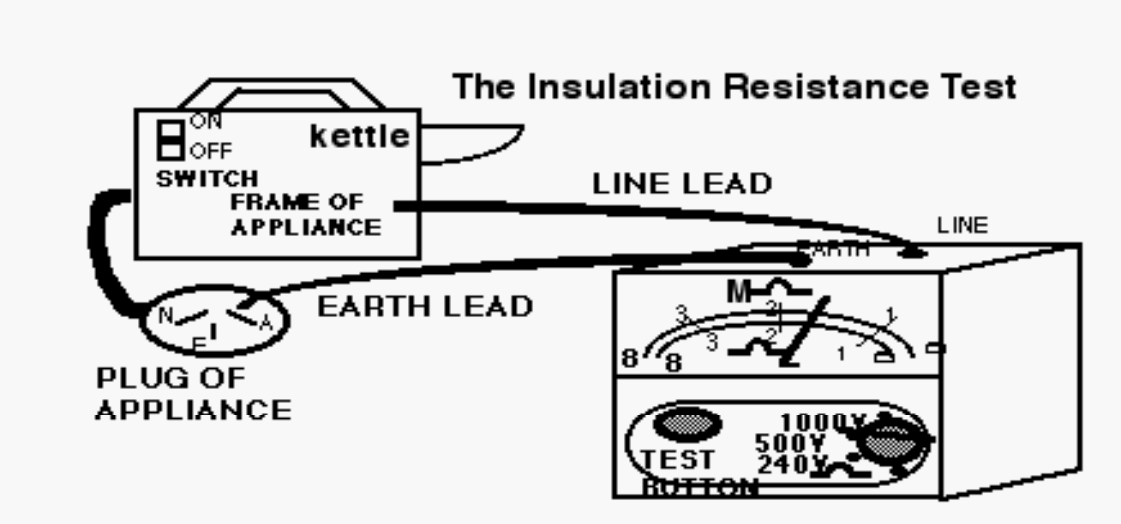
This combination of cognitive architecture and information structures constitutes the core of cognitive load theory; the theory can be used to design instruction intended to reduce extraneous cognitive load and facilitate learning through more rapid and effective schema construction. Nevertheless, as it stands, there is a gap in the theory. Very complex, high element interactivity material cannot be understood without processing all of the elements simultaneously in working memory. As indicated previously, human cognitive architecture overcomes this problem by incorporating the elements in a schema or limited number of schemas that can be processed easily in working memory. Once constructed and stored in long-term memory, a system of schemas overcomes the problem of a limited working memory. The difficulty arises when one considers how such schemas can be constructed in the first instance, if the learner possesses no or few *a priori* schemas in the domain. If the material is very complex, consisting of a large number of elements that are high in element interactivity, it will not be possible to process all of the elements in working memory simultaneously. But they must be processed simultaneously in working memory in order to be understood. The material cannot be understood if only some of the elements are processed in working memory. Furthermore, for a schema to be constructed, all of the elements must be processed in working memory. Thus, we have a paradox. Material can be understood once a schema has been constructed allowing all the elements to be processed in working memory simultaneously but until that point has been reached, the elements cannot be processed simultaneously in working memory and so cannot be understood. It seems to follow that for novices learning very complex material, the initial stages of schema construction must occur in an environment in which the material cannot be understood.

Most instructional design is intended to facilitate understanding and accordingly, instructional designers present their material in a manner that potentially permits understanding. That means incorporating all of the elements required for understanding in the instructions. As indicated above, for very complex material with many interacting elements, attempts by learners to assimilate all of the elements will inevitably fail due to working memory limitations. Understanding is impossible until after appropriate schemas have been acquired and attempts to facilitate understanding by presenting all the required elements simultaneously, will, according to cognitive load theory, also fail. Alternative modes of presentation may be required.

Cumulatively, the strands of research described in the previous section suggest a possible alternative to solve the paradox of learning complex information. This alternative instructional technique is the isolated-interacting elements approach.

4.3 An isolated–interacting elements instructional technique

Consider the instructional material of Figure 4.1 intended to teach electrical engineering students how to conduct a test of insulation resistance. In order to understand this material, the learner must simultaneously consider the following elements: the aim of the test, the required setting of the volt meter, the required setting of the appliance switch, the location of the earth lead and the line lead during both phases of the test and the criteria by which to judge the voltmeter readings as safe. For novices, these elements far exceed working memory limits. While an expert electrical engineer can process all of these elements simultaneously in working memory via previously acquired schemas, a relative novice (e.g., first year apprentice or trainee), not possessing the requisite schemas, can only hold and process a few of the elements simultaneously in working memory. If only a few of the elements are processed in working memory simultaneously, the material cannot be understood because the elements interact. Accordingly, this material, designed to facilitate understanding, must inevitably fail in its aim with both understanding and learning inhibited. This type of instructional procedure will be referred to as the "interacting elements" procedure. It is characterised by the inclusion of all elements required for understanding but at the cost of an impossibly high working memory load.



This test aims to check that there is no "leak" in the flow of current from either the active or neutral wires of the appliance to the frame of the appliance, which would cause the frame of the appliance to become electricaly active.

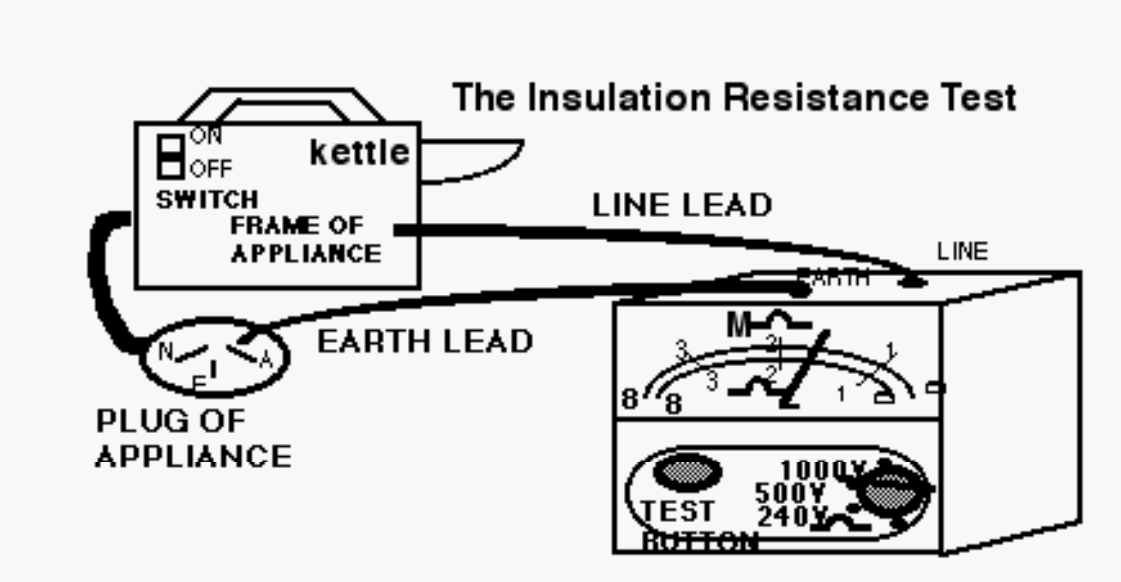
To achieve this two circuits are set up; the first including the frame of the appliance and the active of the appliance and a second including the frame of the appliance and the neutral of the appliance. Very large measures of insulation resistance (ideally "infinite" resistance) indicates that there are no leaks in the flow of current to each circuit. A megger meter measures the resistance of the circuit.

1. By setting the meter to read 500 volts, a larger than normal voltage will be introduced into each circuit in order to test the appliance under a heavy load. **Set the meter to read 500 volts**.
2. The appliance is switched "on" to allow current generated from the megger meter to flow into each circuit. **Make sure the switch is "on".**
3. To measure of the resistance of the first circuit the megger meter is connected into the circuit. **Place the line lead on the frame of the appliance.**
4. **Place the earth lead on the active pin of the appliance's plug.**
5. The test button of the megger meter introduces 500 volts into the circuit. **Press the test button.**
6. A measurement of insulation resistance of the circuit can be taken from the needle of the meter, with a reading of at least 1 Megohm (1,000,000 ohms) being sufficiently large to indicate that the active-frame circuit possesses an acceptable level of insulation. This ensures that no current leaks to the appliance's frame. **Read the resistance from the meter. The required result is a reading of at least one Megohm.**
7. To complete the test of insulation resistance the resistance of the second circuit must be measured. The second circuit is set up between the neutral point of the appliance and the appliance frame. **Remove the earth lead from the active pin and place it on the neutral pin.**
8. A reading of the resistance of the second circuit can be taken by pressing the test button of the megger meter and introducing 500 volts into the neutral-frame circuit. **Press the test button again.**

To pass the test, the second circuit is also required to have a safe level of insulation resistance which indicates no current has leaked from this circuit. **Read the resistance. A reading of at least one Megohm is again required.**

Figure 4.1: An example of interacting elements instructional material.

In contrast, consider the alternative instructional material consisting solely of the figure and text of Figure 4.2. Note that this text is the same as the text in bold type in Figure 4.1. These materials are intended to be learned element by element without a high degree of understanding. These instructions include the isolated elements of information that form the concept yet none of the interactions between the elements that allow the learner to understand the concept holisticly. They will be referred to as the "isolated elements" version of the instructional material. Unlike the interacting elements procedure, students who studied this set of instructions would easily be able to hold and process the isolated elements in working memory. The cost is reduced understanding but the potential benefit is an increased probability of learning because the reduced working memory load permits necessary elements to be held and processed in working memory. Once these materials are learned (albeit with limited understanding), it may be possible for learners to process the interacting elements materials in working memory allowing full understanding to occur. Previous study of these materials will have the effect of reducing the working memory burden on the learner as information about isolated elements included in the interacting elements instructions will already have been retained in a partial schema for the concept. The acquisition of this should allow the learner to be in a position to understand the full concept when he or she is exposed to the interacting elements instructions in the second phase of learning. It is predicted that studying isolated elements materials followed by interacting elements materials will be of more benefit to learning than repeated exposure to the interacting elements materials.



1. Set the meter to read 500 volts.
2. Make sure the switch is "on".
3. Place line lead on the frame of the appliance.
4. Place the earth lead on the active pin of the appliance's plug.
5. Press the test button.
6. Read the resistance from the meter. The required result is a reading of at least one Megohm.
7. Remove the earth lead from the active pin and place it on the neutral pin.
8. Press the test button again.
9. Read the resistance. A reading of at least one Megohm is again required.

Figure 4.2: An example of isolated elements instructional material.

This result is predicted despite the fact that there is too little information for understanding to occur with isolated elements but understanding potentially can occur with the presentation of interacting elements. It is predicted that these isolated elements of information will act as a body of knowledge that further learning can anchor to and be subsumed by. At the second learning phase, when the instructional material contains both isolated and interacting elements of information, meaningful learning can be achieved because Ausubel's conditions for this type of learning have been met: the new learning can be related to previously

acquired knowledge in a non-arbitrary and substantive manner. In effect, it is suggested that for complex materials, our cognitive architecture may dictate that sometimes, a degree of learning must occur before understanding, and that our instructional designs should reflect this imperative.

To summarise, the proposed instructional design aims to fill a gap that currently exists within cognitive load theory. This theory cannot explain how schemas and understanding of very complex concepts are developed by novices. These processes require all the elements of the concept to be processed simultaneously in working memory, yet, working memory's limitations preclude this from occurring in the case of a concept with high interactivity between the elements. The proposed instructional design aims to facilitate learning in this situation by attempting to show that initially learning about a concept without full comprehension will ultimately aid the student's ability to comprehend the concept by providing him or her with a knowledge framework to which the new learning can be meaningfully related.

The predicted benefits of an isolated-interacting elements instructional approach follows only under circumstances where the knowledge base of the learners is such that they are unable to process all of the interacting elements in working memory. A learner who has the ability to process the interacting elements simultaneously possesses this ability as a result of their previously acquired domain schemas; thus, a learner in this situation will gain little if any benefit from an instructional procedure whose aim is to facilitate the development of rudimentary domain schemas.

4.4 An introduction to the experiments conducted in this thesis

This thesis reports upon five experiments. Overall, these experiments seek to examine the usefulness of an isolated elements-interacting elements instructional technique. The experiments have used comprehensive instructional material from several domains. The knowledge of the learners after instruction has been assessed in multiple ways through both written and practical tests. In addition, the experiments seek to provide support for a cognitive load hypothesis. To this end, measures of mental load have been collected and analysed.

Experiment 1 investigated whether studying isolated elements materials followed by interacting elements materials will be of more benefit to learning than repeated exposure to the interacting elements materials when learning about complex information. This experiment used electrical engineering materials that were high in element interactivity.

Experiment 2 aimed to further define the conditions under which an isolated-interacting elements instructional technique is beneficial for learning. If learners have acquired sufficient schemas to process the required elements and their interactions simultaneously in working memory, the need for an isolated elements stage in the instruction process may be unnecessary. Experiment 2 replicated Experiment 1 exactly except for the expertise level of the participants. Using more experienced learners, it aimed to investigate whether the isolated–interacting elements instructional technique would benefit learners who already possess domain relevant schemas.

Experiments 3 and 4 were designed to replicate and extend the findings of Experiments 1 and 2. Novice learners (Experiment 3) and relatively more expert learners (Experiment 4) were instructed about a highly complex electrical circuit. This material used system as opposed to procedurally based information and as such aimed to extend the generality of the instructional technique.

Experiment 5 was designed to replicate and generalise the findings of Experiment 1 and 3. It also aimed to investigate the theoretical underpinnings of the isolated-interacting elements approach. The participants of this experiment were inexperienced learners in the domain of pneumatics. The students of Experiment 5, however, had previously participated in either Experiment 2 or 4, which had used instructional content from electrical engineering, as expert learners.

In the following chapters, each of these experiments will be described and discussed in full. These empirical details will be preceded by appropriate theoretical reasoning and specific hypotheses. Each experiment will be followed by a discussion of results in terms of cognitive load theory and other findings.

Chapter 5 Assimilating complex information: learning isolated elements then interacting elements

5.1 Experiment 1

The first experiment used two groups. The isolated-interacting elements group studied isolated elements instructions at phase one followed by instructions involving interacting elements at phase two. The initial learning phase was intended to provide the opportunity to develop a database of knowledge, albeit of discrete, organisationally isolated units. It was assumed that at this point, learners have sufficient memory capacity to be able to learn these isolated element steps but insufficient capacity to understand the reasoning behind the tests. The second learning phase, when the interacting version of the materials was studied, was intended to provide the forum to decipher meanings; to understand the relations present between elements and to establish an anchorage between the more complex information presented during this phase to the existing knowledge elements. It was hypothesised that the initial period of learning using the isolated elements instructions would allow the isolated-interacting elements group students to develop rudimentary schemas that would assist them in learning the interacting elements instructions during the next instruction phase.

The isolated-interacting elements group was compared to a more conventional method of instruction for promoting understanding, the interacting elements only group where instructions containing interacting elements were presented at both phases of instruction. It was expected that the interacting elements only group would perform more poorly at both phases. Compared to the isolated-interacting elements group, the working memory resources of the interacting elements only group would be overwhelmed because of the many elements they were required to mentally integrate due to the high element interactivity of the material.

To summarise, we predicted that first providing inexperienced students with isolated elements instructions (e.g., instructions of Figure 4.2) *then* presenting them with interacting elements instructions (e.g., instructions of Figure 4.1) would lead students to gain a superior understanding of the information than the traditional approach of immediately presenting students with the complete version of material explaining a complex concept.

Method

Participants

Twenty-two first year industrial trade students enrolled in a training course at a Sydney company participated in the experiment. The students were from a varied educational background but all had completed at least Year 10 of High School. They were enrolled in either a one month basic industrial skills course (9 students) or a six month manufacturing and engineering trainee-ship course (13 students). Students were tested within their first month of training so the electrical material covered at this point in both of the courses, was very similar. They understood electrical principles at a basic level, such as how to locate the neutral, active and earth pins of an electrical appliance. However, none of the participants had directly studied the electrical safety tests used in the experiment. Students were excluded

from the study if they had completed further studies in the field of electrical engineering; if English was not their first language; or if more than forty eight hours elapsed between the two learning sessions of the study (e.g., due to illness of a student).

Materials

The instructional materials used in the experiment consisted of information on three electrical safety tests that are performed on 240 volt electrical appliances using a volt meter. The three tests check various electrical circuits of the appliance to ensure that it is operating safely. In this experiment the electrical applianceused was an electrical kettle. The three tests were to check the: insulation between the frame and the active and neutral points of the appliance (the Insulation Resistance test); insulation between the frame and the earth point of the appliance (the Earth Continuity test) and the insulation of electrical extension leads (the Insulation Resistance of a Flexible Lead test).

For a novice learner, understanding why and how to do each of the electrical safety tests was a complex task as the material is high in element interactivity; for example, to understand the tests of Insulation Resistance of a Flexible Lead, the learner must simultaneously consider what the test aims to achieve, the required setting of the volt meter, the on/off status of the appliance, the placement position of both leads of the volt meter, the criteria set for safe and unsafe readings, the consequences of an unsafe reading and finally, the changes made to the procedure to test the second circuit and thus complete the test. To understand fully the concept of insulation resistance of a flexible lead, all the elements of the test, their relations and implications have to be considered simultaneously. The element interactivity of the material is high and so the intrinsic cognitive load of the concepts is also high.

Two sets of introductory instructional material were developed: one set presented isolated elements that had to be learned without emphasis on the interactions between them (isolated elements condition) while the other placed an emphasis on the interactions between elements (interacting elements condition). The instructional materials were designed in line with Mayer and colleagues' cognitive theory of multimedia learning; for example, the instructional material contained a combination of non-redundant verbal and visual material presented simultaneously (Mayer & Moreno, in press; Mayer, Steinhoff, Bowler & Mars, 1995). Each set of instructional materials consisted of a three page booklet. Each page of the booklet described a different electrical safety test and contained a numbered diagram with a separate section of correspondingly numbered text points below the diagram. Both sets of instructions were designed to introduce and demonstrate the equipment and procedures used in the three electrical safety tests described above. A copy of the isolated elements instructions has been included in Appendix 1; a copy of the interacting elements instructions has been included in Appendix2.

The isolated elements version of instructions explained the procedural steps of the electrical tests; for example, referring back to the instructional material illustrated in Figure 4.1, the step by step procedure for performing an insulation resistance test of an electrical appliance is displayed in the red text. The isolated elements instructions included no text other than the bold text. Each step of the test procedure can be learned in isolation without reference to previous steps. The interacting elements instructions replicated the isolated elements version of the information (red text in Figure 4.1), but also included other information relevant to the tests such as the aim of the test and the logic, in terms of electrical circuit theory, behind each step of the test (see normal text in Figure 4.1). The interacting elements instructions therefore provided all the information for gaining a meaningful understanding of the electrical test as a whole.

An analysis of the element interactivity of each of the electrical safety tests has been included in Appendix 3. The analysis estimates the number of elements that would need to be simultaneously processed for a first year electrical trainee with only a very limited electrical knowledge to fully understand the test. Encoding information from working memory should be much easier with the isolated elements version of the instructional materials than with the interacting elements approach because the isolated elements instructions have fewer interacting elements than the interacting elements instructions and so should impose a lower working memory load. The difference between the two conditions is illustrated clearly in the documents included in Appendix 3, where an estimation of the number of elements that interact, in both the isolated and interacting elements versions of the instructional material for the Insulation Resistance test, have been included.

Although it is easier to hold the information from the isolated elements version in working memory, it does not follow that it is possible to process all the information about a test simultaneously, even with the isolated elements version of the instructions. It may not be possible, given the nature of this material, to assimilate all the elements of the isolated elements version in one reading. (It is certainly not possible for the interacting elements version.) However, it is more likely that information from the isolated elements version could be more readily encoded into long term memory due to the reduced number of elements.

Cognitive load was measured in two ways: through the time required for instruction and through subjective ratings of task difficulty (see Paas & van Merrienboer, 1993; 1994). Subjective ratings were used as they "provide a powerful …(measure of) the subjective experience of workload" (Gopher & Braune, 1984: 529) since students have little difficulty assigning a numerical value to the imposed mental workload. In addition, subjective ratings are easy to implement, are non-intrusive on the primary task (i.e., learning the training materials) and correlate highly with more objective

measures of mental load (Moray, 1982; O'Donell & Eggemeier, 1986). The subjective mental load rating measure used was a seven point Likert type scale. Above the scale was the instruction "How easy or difficult was it to learn and understand the electrical tests from the instructions you were given? Place across on the rating scale below". Each number on the scale was labelled with a description ranging from extremely easy (1) to extremely hard (7). A copy of the subjective mental load rating scale used in the experiment has been included in Appendix 4.

The test material consisted of test items and equipment for both written and practical tests. Each test item was designed to be objective and was marked as either correct or incorrect. The written test consisted of twenty three items. Some problems had several parts so the test was marked out of a total of thirty-one points. (It was possible to score a total of eleven marks on items relating to the Earth Continuity test; nine marks from items relating to the Insulation Resistance Test; eight marks from items relating to the Flexible Lead test and three marks from items testing information that was provided in both the Insulation Resistance test and the Flexible Lead test). A copy of the written test used in the experiment has been included in Appendix 5.

There were two categories of questions in the written test: those tapping low element interactivity knowledge about the test procedures and those tapping high element interactivity knowledge of the electrical testing procedures and electrical circuitry. Low element interactivity questions were factually based, asking about a specific part of, or reason behind the procedure of a specified electrical test, for example: "What do you set the volt meter to in the Earth Continuity test?" or "How is power introduced to the circuit when testing for Insulation Resistance?". This category of questions was low in element interactivity because they required only the recall of factual information which could be processed as single element or a

few elements of information. This category contained eighteen questions. Each correct answer to these items was worth one mark.

High element interactivity knowledge questions required the students to practically apply their factual knowledge of the electrical tests. The application of knowledge required the student to be able to hold the whole concept in mind simultaneously, thus these questions could be considered high in element interactivity. Questions involved tasks such as nominating the appropriate test to be used, judging the appropriateness of a given test result and deciding what consequences may arise or what action was required to be taken due to this test result. Some examples of applied knowledge questions include: "An extension cord is tested for insulation resistance. The result is Neutral-Earth circuit = 0.5 megohms; Active-Earth circuit = 3 megohms. Would you pass this lead as safe? Explain your answer." or "A hair dryer is tested and the resistance reading is 2.9 ohms. A fault occurs. Given this information, where will the excess electrical current, from the power surge, go? Why does the excess current go to this point?". There were several diagrammatical questions where information about the situation had to be inferred from the diagrams. Alternatively, responses such as the placement of the meter leads or the expected test reading, also had to be physically represented in the correct locations on diagrams. This section also included two questions which required the students to apply their knowledge of the electrical testing procedure to an electrical appliance different from the one they had studied. Specifically, students were required to apply knowledge of safety testing an electrical kettle to testing an iron and a hair dryer. This category of questions required a holistic understanding of each electrical test because the components of the test needed to be understood and applied in their entirety to a new scenario. This category contained five questions, some with several parts. Each correct answer to each item or part was worth one mark; total marks possible for this category of questions was thirteen.

The practical test consisted of two items: both items required transfer of knowledge to a novel situation. Specifically, students were required to apply knowledge of an electrical kettle to safety tests of a fluorescent light. The students were required to perform an Earth Continuity test and an Insulation Resistance test to a fluorescent light. The equipment provided for the test was a fluorescent light and a volt meter. Key parts of the light were labelled, e.g. Earth (E), Active (A) and Neutral (N). The volt meter was set to 1000 Volts, a setting not required by either test: therefore in order to correctly perform the test the student had to reset the meter correctly. The performance of each electrical test was marked either correct or incorrect, so the practical test was scored out of a total of two marks. The element interactivity of knowledge tapped by the practical tests was high as students had to remember the entire test procedure and apply it to a novel situation.

Procedure

All the students were randomly assigned to either the isolated-interacting elements instruction or the interacting elements only group with 11 students in each group. They were tested individually, in a quiet room. The students were informed that they were to be given some electrical safety testing instructional material which would be followed by a written and practical test and that this session was to be repeated within 48 hours. Students were able to ask questions at any time during the experiment.

In the first experimental session, the students in the isolated-interacting elements instruction group were given the isolated elements instructional booklet while the students in the interacting elements only group were given the interacting elements instructional booklet. An unlimited time was allowed to read through the instructions and this time was recorded by the experimenter. All students were provided with a metal electric kettle and a volt meter. They were instructed to perform the tests with this equipment as they read through the instructions. At the

completion of the study phase, the students were provided with a subjective mental load rating scale, the format of which was explained to both groups. They were asked to rate the mental effort involved in understanding all of the electrical tests described in their training booklet on the scale described above.

The test section of the experiment followed. The students were asked to complete the written test, described in the materials section, which was common to both groups. They were allowed an unlimited time to complete the test. Students were prohibited from reattempting any question once they had attempted it. The instructional material was not available during the test period. The practical test followed. Students were required to attempt both parts of the practical test in a set order (Earth Continuity and then Insulation Resistance).

The second phase of the experiment was held within forty-eight hours of the first phase. The procedure of the second phase of the experiment was identical to the first, apart from two differences. 1) In the practical section, the order in which the two electrical safety tests were performed by the students was reversed (that is Insulation Resistance test followed by Earth Continuity test). 2) All students studied the interacting elements instructions. Therefore to summarise, over both periods of instruction the isolated-interacting elements instruction group studied the isolated elements instructional material followed by the interacting elements instructional material while the interacting elements only group was presented the interacting elements instructional material twice.

Results and Discussion

The test score variables under analyses were written test scores, divided into low element interactivity knowledge scores and high element interactivity knowledge scores, and practical test scores. Means and Standard Deviations for these variables are provided in Table 5.1. Other variables analysed included mental load ratings and instruction times. Means and Standard Deviations for these variables are also provided in Table 5.1.

Phase 1	Max score	Isolated-interacting elements group (n=11)		Interacting elements only group (n=11)	
		Mean	Std Dev	Mean	Std Dev
Low element interactivity score	100%	36.37	15.19	31.82	14.51
High element interactivity score	100%	17.73	8.95	12.27	11.93
Practical test score	2	0.18	0.41	0.10*	0.32*
Mental Load	7	2.91	1.38	4.09	0.94
Instruction Time	-	645.27	274.05	917.27	263.01
Phase 2					
Low element interactivity score	100%	54.04	19.26	41.92	11.21
High element interactivity score	100%	35.00	20.22	17.58	8.44
Practical test score	2	0.36	0.51	0.00	0.00
Mental Load	7	3.27	1.27	4.00	1.18
Instruction Time	-	603.46	328.11	676.09	240.94

Note: * = sample size reduced to 10 students due to equipment failure

Table 5.1: Test Scores, Mental Load Ratings and Instruction Time (in seconds) for Experiment 1.

Low and high element interactivity knowledge scores were analysed separately by 2 (Instruction Condition) x 2 (Test Phases) ANOVAs with repeated measures on the second factor. The results for low element interactivity knowledge question

scores indicated there was no difference between the instructional groups, $F(1,20) = 2.65$, $MSe = 288.82$. (The .05 level of significance is used throughout this paper.) The test phases main effect indicated a significant difference, $F(1,20) = 11.77$, $MSe = 180.10$, demonstrating an improvement over the two instructional phases. There was no significant interaction, $F(1,20) = 0.87$, $MSe = 180.10$. The results for high element interactivity question scores indicated there was a significant difference between the instructional conditions in the expected direction, $F(1,20) = 7.06$, $MSe = 203.87$, with the isolated-interacting elements instruction group performing at a superior level. The test phases main effect was also significant, $F(1,20) = 9.50$, $MSe = 147.53$, demonstrating an improvement over time. There was no significant interaction, $F(1,20) = 2.67$, $MSe = 147.53$.

There were 2 practical tests. Each test was scored correct or incorrect and the results of the two tests were added together to give a practical score out of two points. These scores were assessed by a 2 (Instruction Condition) x 2 (Test Phases) ANOVA with repeated measures on the last factor. Results indicated a possible difference between the instructional conditions, $F(1, 20) = 4.03$, $MSe = 0.14$, $p = 0.058$ favouring the isolated-interacting elements group. There was no improvement over time, $F(1, 20) = 0.20$, $MSe = 0.11$, nor was there a significant interaction, $F(1, 20) = 1.80$, $MSe = 0.11$.

These results suggest that as the application of knowledge was increased, through the need to apply knowledge to either entire situations with their interacting elements (high element interactivity knowledge question scores) or in application of the knowledge to a realistic, practical transfer task, differences in performance emerged, suggesting superiority of performance by the isolated-interacting elements instruction group.

The time taken by the participants for instruction were assessed by a 2 (Instruction Condition) x 2 (Test Phases) ANOVA with repeated measures on the last factor.

Results indicated a possible difference between instructional conditions in the time taken for instruction over both phases, $F(1, 20) = 3.33$, $MSe = 98161.07$, $p = 0.08$. An inspection of the means indicated that trend was toward the isolated-interacting elements group requiring less instructional time in total than the interacting elements only group. This measure also showed a possible difference in the time taken over the two instruction phases, $F(1, 20) = 3.88$, $MSe = 56831.73$, $p = 0.06$. There was no significant interaction, $F(1, 20) = 1.92$, $MSe = 56831.73$.

Mental load ratings showed a significant instruction condition main effect in the expected direction, $F(1, 20) = 4.74$, $MSe = 2.11$, suggesting that the isolated-interacting elements instruction group found their instructions less demanding than the interacting elements only group over both phases. In phase one this difference was to be expected given the disparity in the amount of information in the different sets of instructions. At phase 2 however, when both groups received the same interacting elements version of the instructions, (the interacting elements only group for the second time), a difference is still maintained (see Table 1). In fact, the mean of the isolated-interacting elements group at Phase 2 is still well below that of the interacting elements only group at Phase 1. There was no improvement over time, $F(1, 20) = 0.26$, $MSe = 0.77$, nor was there a significant interaction, $F(1, 20) = 0.72$, $MSe = 0.77$. The lack of a significant interaction also suggests that the difference in mental load ratings between groups was similar through both instructional phases.

Paas and van Merrienboer (1993;1994) devised a method of combining performance measures with subjective mental load ratings to investigate the relative efficiency of instruction conditions. Instructional efficiency increases when learners demonstrate either a superior performance than would be expected from their level of mental effort or require less mental effort to achieve a given performance score. Paas & van Merrienboer suggested:

> combined measures (of performance and mental effort) may provide a better insight into the cognitive consequences of training and task environments and thus improve attempts to design optimal environments (1993: 742).

Performance can be measured by an objective test score relevant to the learning task; subjective mental effort ratings require the learner to translate the perceived mental effort required by the task into a numerical value. Efficiency (E) is calculated by transforming both mental effort (M) and performance scores (P) to standardised z scores. These z scores for each instruction condition are then combined in the following manner to give a single efficiency score for each condition:

$$E = \frac{/M-P/}{\sqrt{2}}$$

The sign of the relative condition efficiency depends on the relation between metal load and performance scores. It is calculated according to these rules: If M-P<0, then E is positive; if M-P>0, then E is negative. The greater the value of E, the more efficient the instruction condition.

It was hypothesised that the isolated-interacting elements instruction group would be more efficient than the interacting elements only group. In other words, it was expected that the isolated-interacting elements instruction group would achieve a greater performance for less mental effort invested. The mean and standard deviations for normalised (Z) mental effort and performance scores and the condition efficiency scores for each group are shown in Table 5.2.

	Isolated-interacting elements group				Interacting elements only group			
Phase 1	Z		Efficiency		Z		Efficiency	
	X	S.D.	X	S.D.	X	S.D.	X	S.D.
Low element interactivity score	-.27	0.89	.17	1.00	-.54	.89	-.68	.87
High element interactivity score	-.19	.57	.24	.86	-.54	.77	-.68	.51
Practical Score	.06	1.16	.41	1.26	-.18	.81*	-.42	.65*
Mental Load	-.52	1.09	-	-	0.41	.75	-	-
Phase 2								
Low element interactivity score	.77	1.13	.71	1.20	0.05	.66	-.20	.93
High element interactivity score	.93	1.31	.82	1.33	-.20	.55	-.38	.59
Practical Score	.55	1.36	.57	1.45	-.43	0.00	-.55	0.66
Mental Load	-.23	1.01	-	-	.34	.94	-	-

* = sample size reduced to 10 students due to equipment failure

Note: The efficiency scores are based on the normalised mental effort scores given in Table 5.1 for each group for each learning phase.

Table 5.2: Normalised (Z) scores and Condition Efficiency Scores for Performance and Mental Load for Experiment 1.

Efficiency measures for both high and low element interactivity questions were calculated and analysed using 2 (Instruction Condition) x2 (Test Phases) ANOVAs with repeated measures on the last factor. Results for low element interactivity question efficiency scores indicated there was a significant instruction condition main effect favouring the isolated-interacting elements group, $F(1, 20) = 6.75$, $MSe = 1.26$. There was no improvement over time, $F(1, 20) = 3.59$, $MSe = 0.77$. There was no significant interaction between the factors, $F(1, 20) = 0.01$, $MSe = 0.77$. The results for high element interactivity question efficiency scores also indicated a significant difference between the instructional conditions favouring the isolated-interacting elements instruction group, $F(1, 20) = 12.64$, $MSe = 0.97$. There was no improvement over time, $F(1, 20) = 3.63$, $MSe = 0.59$. There was no significant interaction between the factors, $F(1, 20) = 0.40$, $MSe = 0.59$. These results are represented in Figure 5.1.

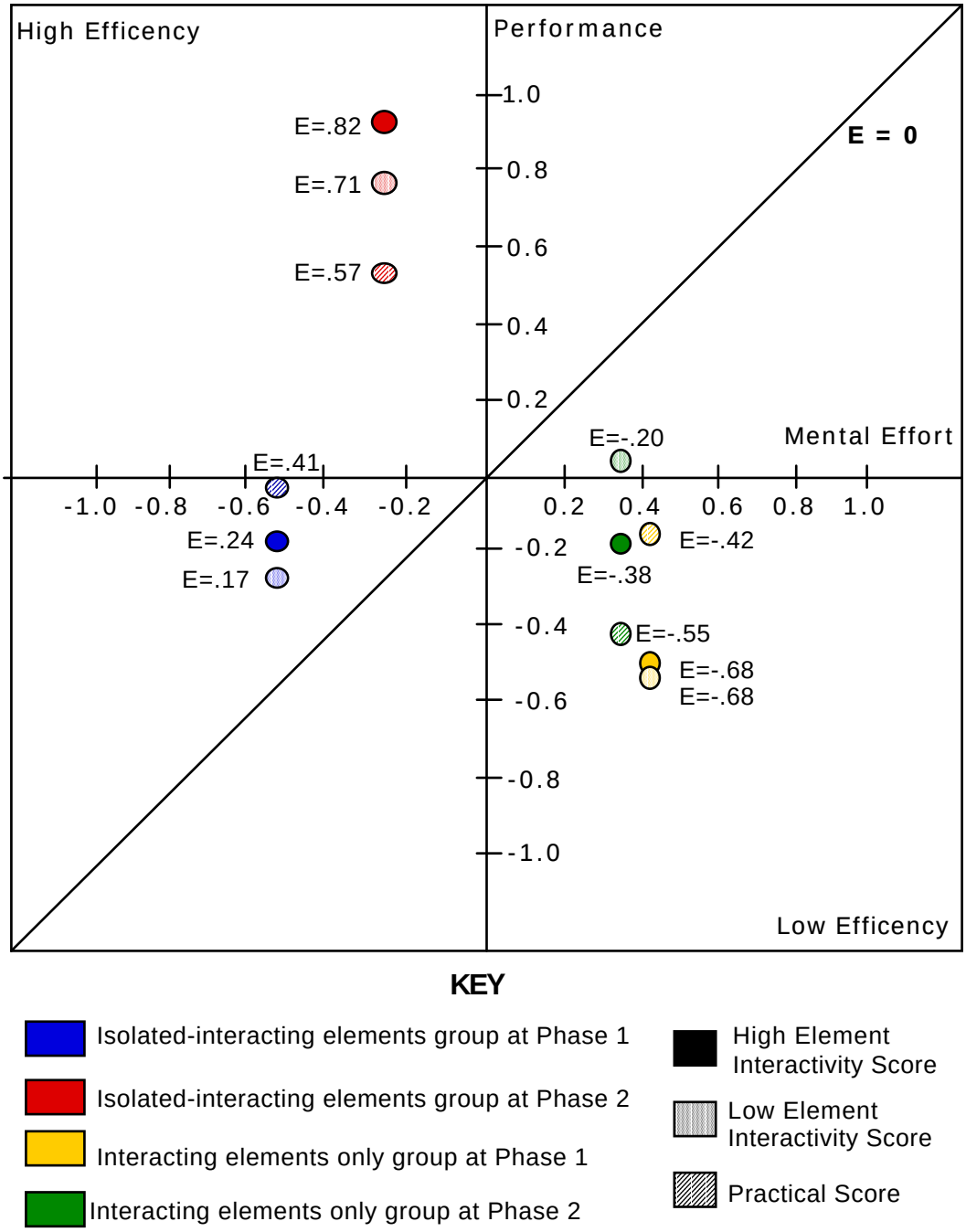


Figure 5.1: Relative instruction condition efficiency as a function of performance and mental effort ratings.

Practical score efficiency measures were analysed in a 2 (Instruction Condition) x2 (Test Phases) ANOVA with repeated measures on the second factor. Table 2 shows the means and standard deviations for standardised (Z) mental effort and practical performance scores and the condition efficiency scores for each group. Results indicated there was a significant instruction condition main effect favouring the isolated-interacting elements instruction group, $F(1, 20) = 5.64$, $MSe = 1.83$. There was no improvement over time, $F(1, 20) = 0.00$, $MSe = 0.44$, nor was there an interaction effect, $F(1, 20) = 0.45$, $MSe = 0.44$. These results are also represented in Figure 5.1 above.

The general methodology introduced by Paas and van Merrienboer (1993; 1994) can be adapted to indicate instructional efficiency using any two measures that can be related in a coherent fashion. For example, it is reasonable to propose that an instructional procedure that requires little student time but results in high performance scores is more efficient that one that requires more time but results in lower performance scores. Performance can be measured by a test score relevant to the learning task; learning times can be measured by the amount of time students spent learning from the materials. Learning times have also been used as a indirect measure of mental load related to a training task (see Chandler & Sweller, 1991). Learning time and performance score are standardised and efficiency (E) then can be calculated using a version of the Paas and van Merrienboer formula:

$$E = \frac{/I\text{-}P/}{\text{square root (2)}}$$

where E = efficiency, I = instruction time and P = performance score on a test.

Using this procedure, it was hypothesised that the isolated-interacting elements instruction group would be more efficient than the interacting elements only group. In other words, it was expected that the isolated-interacting elements instruction

group would achieve a greater performance for less instruction time invested. The mean and standard deviations for normalised (Z) mental effort and performance scores and the condition efficiency scores for each group are shown in Table 5.3.

	Isolated-interacting elements group				Interacting elements only group			
Phase 1	Z		Efficiency		Z		Efficiency	
	X	S.D.	X	S.D.	X	S.D.	X	S.D.
Low element interactivity score	-.27	.89	-.04	1.05	-.54	.89	-.88	.78
High element interactivity score	-.19	.57	.02	.74	-.54	.77	-.88	.80
Practical score	.06	1.16	.20	1.16	-.18	.81*	-.63*	.75
Instruction Time	-.22	.93	-	-	.70	.89	-	-
Phase 2								
Low element interactivity score	.77	1.13	.80	.82	.05	.66	.12	.47
High element interactivity score	.93	1.31	.91	1.36	-.20	.55	-.06	.58
Practical score	.55	1.36	.65	1.18	-.43	0.00	-.22	.58
Instruction Time	-.36	1.11	-	-	-.12	.82	-	-

* = sample size reduced to 10 students due to equipment failure

Note: The efficiency scores are based on the normalised instructional time scores given in Table 5.1 for each group for each learning phase.

Table 5.3: Normalised (Z) scores and Condition Efficiency Scores for Performance and Instructional Time (in seconds) for Experiment 1.

Efficiency measures for both high and low element interactivity knowledge questions were calculated using the learning time/performance methodology and analysed separately in 2 (Instructional Condition) x 2 (Test Phases) ANOVAs with repeated measures on the last factor. Results for low element interactivityquestion efficiency scores indicated there was a significant instruction condition main effect favouring the isolated-interacting elements group, $F(1, 20) = 6.94$, $MSe = 0.91$. There was a significant main effect for test phases, $F(1, 20) = 23.67$, $MSe = 0.39$, indicating an improvement over time. There was no significant interaction between the factors, $F(1, 20) = 0.18$, $MSe = 0.39$. These results are represented in Figure 5.2.

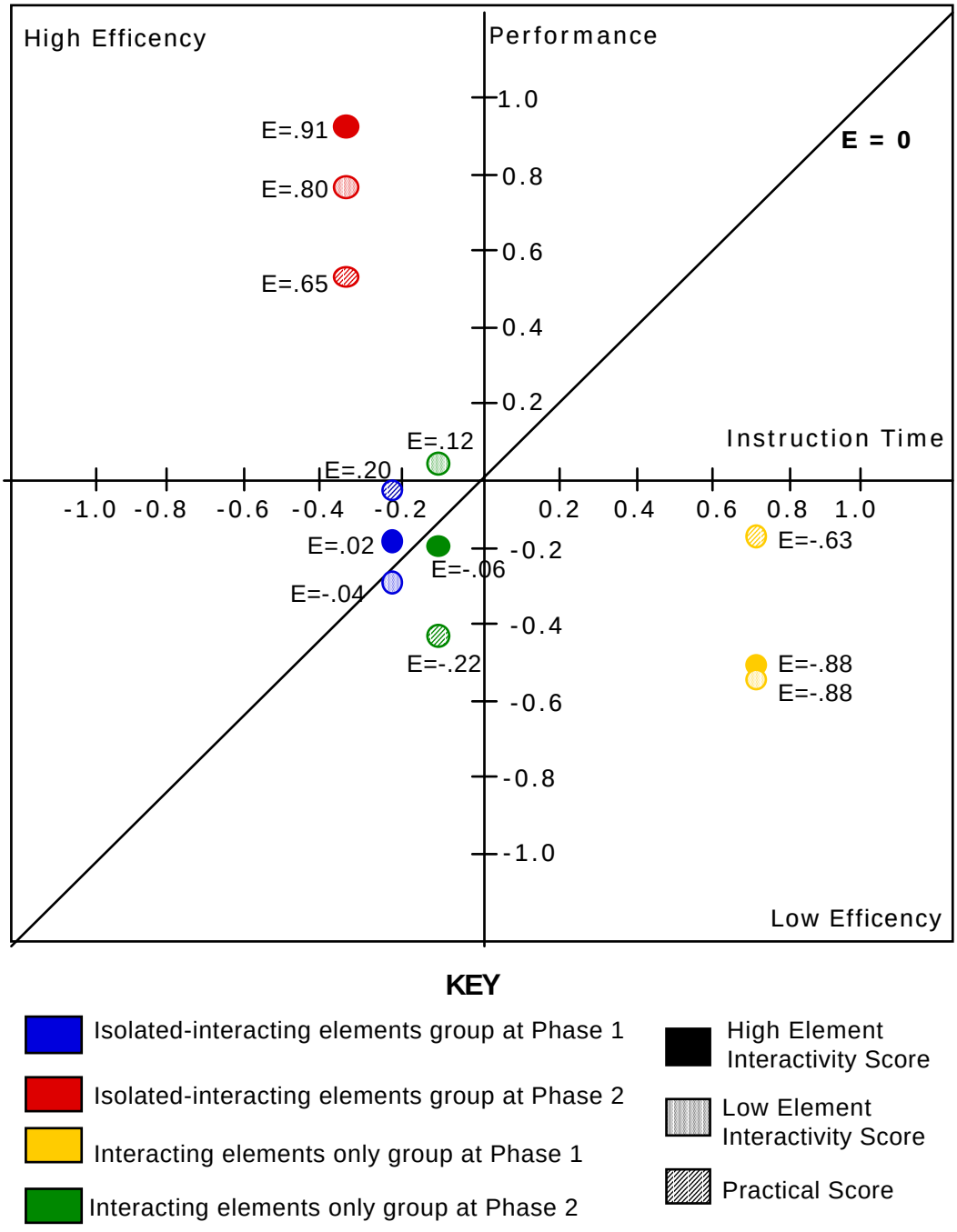


Figure 5.2: Relative instruction condition efficiency as a function of performance and instruction time for Experiment 1.

The results for high element interactivity question efficiency scores also indicated a significant difference between the instructional conditions favouring the isolated-interacting elements instruction group, $F(1, 20) = 9.09$, $MSe = 1.06$. The test phases main effect was also significant, $F(1, 20) = 12.97$, $MSe = 0.62$, demonstrating an improvement over time. There was no significant interaction between the factors, $F(1, 20) = 0.02$, $MSe = 0.62$. These results are also represented in Figure 5.2 above.

The results for practical question efficiency scores also indicated a significant difference between the instructional conditions favouring the isolated-interacting elements instruction group, $F(1, 20) = 6.56$, $MSe = 1.20$. No difference was suggested over the test phases, $F(1, 20) = 3.24$, $MSe = 0.61$. There was no significant interaction between the factors, $F(1, 20) = 0.01$, $MSe = 0.61$. These results are also represented in Figure 5.2 above.

The results from measuring instructional condition efficiency through both mental effort and performance and instructional time and performance indicate that the isolated-interacting elements method of instruction was superior in terms of its efficiency on all measures of performance compared to the interacting elements only instructional technique. In other words, the isolated-interacting elements method of instruction has resulted in a much greater performance per unit of mental effort or per unit of instruction time for all measures of test performance using this group of students.

The results from the written test and the subjective mental load measures support the hypothesis that the isolated-interacting elements instruction group performed at a superior level to the interacting elements only group because this method of instruction imposed a lower cognitive load. We assume that in the initial learning phase, the isolated-interacting elements group of students did not fully understand

the complex concept of electrical safety testing, but they acquired the rudimentary schemas. By obviating the need to process all interacting elements required for understanding in working memory, students may have been able to acquire at least some of the necessary elements, resolving the paradox of learning without processing all interacting elements required for understanding in working memory. Subsequently, the interactions between these elements could be learned in the second phase allowing a more complete understanding of the material. In contrast, the interacting elements only group may have acquired few elements in either the first or the second phase because the interacting elements may have exceeded working memory capacity.

Students in the isolated-interacting elements method of instruction gained a greater comprehension of the topic at the most meaningful level of understanding by the end of training. Their superior performance was most evident where application of their knowledge to a novel setting was required, a task requiring holistic understanding of the information. This learning was achieved despite a single exposure to the interacting elements explanation compared to the repeated exposure of the interacting elements only group. The isolated-interacting elements method students required less mental effort in order to attain this level of understanding. In addition, this group achieved this level of performance after less time studying the instructions than the interacting elements only instructional technique.

These results can be explained from a cognitive load perspective. Novice students often have difficulty acquiring complex concepts because their working memory is overwhelmed. In order to allow novices to process the interacting elements of complex material, initially they may need to be presented with information on the topic that has had the intrinsic cognitive load or difficulty of the material, artificially reduced. The student is able to isolate elements that constitute the topic, forming a rudimentary schema without full understanding. At the next learning stage, when

presented with a complete picture of the complex topic, students who have acquired this partial schema are much better able to process in working memory all of the elements and construct the more complete and complex schemas required for a full understanding of the topic.

5.2 Experiment 2

The results of Experiment 1 pointed to the success of the isolated-interacting elements method of instruction over interacting elements only instructions for inexperienced learners. Cognitive load theory suggests the performance of the novice learners was improved by having an initial stage of learning using isolated elements instructions, thus artificially reducing the intrinsic load of the material. This procedure allowed the formation of rudimentary schemas. At phase 2, learners were able to more easily process the interacting elements information presented because the partial schemas they had developed in phase 1 reduced their working memory load.

If schema formation was the key to the success of the isolated-interacting elements method of instruction with inexperienced learners, that method of instruction may be of little or no benefit to more experienced or expert learners. These learners, by definition of their expertise, already possess more sophisticated schemas in the given area and so may be able to process the interacting elements in working memory without undue difficulty. The purpose of Experiment 2 was to investigate the success of the isolated-interacting elements method of instruction compared with the interacting elements only method, over two learning periods using participants with relatively more expertise than those of Experiment 1. Specifically, Experiment 2 attempted to further define the conditions under which the isolated-interacting elements method of instruction is successful. Therefore, Experiment 1 was replicated in all respects, except, in the expertise of the participants.

Experiment 2 used students who were considerably more expert than those of Experiment 1. They were electrical and mechanical apprentices who had completed almost half of their first year electrical training. At this point of their training, all apprentices had demonstrated competency in many aspects of their

domestic and industrial electrical wiring courses. Thus, they had a sound knowledge of all basic electrical concepts and the general area of electrical testing, but they had not previously been directly exposed to the instructional material used in the study. In accord with our earlier theorising, the results of Experiment 1 indicated inexperienced learners benefited from isolated elements instructions followed by the interacting elements instructional format, rather than receiving only the interacting elements format. For the more experienced learners of Experiment 2, it was proposed that there would be no advantage in providing isolated elements instructions followed by the interacting elements format. It was hypothesised that these more expert learners would already possess many of the rudimentary electrical schemas that isolated elements instruction helps to form. Therefore, the benefit gained from this method of instruction may be negated using more experienced learners.

Method

Participants

Twenty-five first year electrical and mechanical apprentices taking an apprenticeship training course at a Sydney company participated in the experiment. All of the students had completed at least Year 10 of high school and were enrolled in various first year courses at trade colleges. At the time of testing, all apprentices were well into their eight month training course in industrial and domestic electricity (students had completed a mean of 8.32 units from a 21 unit course). As stated, at this stage of their course all apprentices had demonstrated their competency in all basic aspects of electrical testing. They were excluded from the study if they had completed further studies in the field of electrical engineering; if English was not their first language; or if more than forty eight hours had elapsed between the two learning sessions.

Materials

The instructional material used in this experiment was identical to that used in Experiment 1.

Procedure

The procedure used in this experiment was almost identical to that used in Experiment 1. The only difference was in the allocation of participants to instructional groups. Participants were matched for apprenticeship type (i.e. electrical, mechanical, dual apprenticeship, fitter-turner, tool maker) before being randomly assigned to either the isolated-interacting elements instruction group or the interacting elements only group.

Results and Discussion

The test score variables under analysis were high element interactivity knowledge score, low element interactivity knowledge score and practical test score. Means and Standard Deviations for these variables are provided in Table 4. Other variables analysed included mental load rating and instruction time. Means and Standard Deviations for these variables are provided in Table 5.4.

Phase 1	Max score	Isolated-interacting elements group (n=12)		Interacting elements only group (n=13)	
		Mean	Std Dev	Mean	Std Dev
Low element interactivity score	100%	60.76	13.47	58.65	18.36
High element interactivity score	100%	31.81	25.32	33.85	19.20
Practical test score	2	0.67	0.65	0.54	0.66
Mental Load	7	2.17	0.94	2.85	0.80
Instruction Time	-	392.92	105.47	739.31	186.18
Phase 2					
Low element interactivity score	100%	72.22	17.97	71.15	17.05
High element interactivity score	100%	48.33	18.95	46.28	19.52
Practical test score	2	0.50	0.52	0.62	0.77
Mental Load	7	2.42	1.38	2.46	0.97
Instruction Time	-	522.17	109.32	458.31	208.25

Table 5.4: Test Scores, Mental Load Ratings and Instruction Time (in seconds) for Experiment 2.

Low and high element interactivity knowledge questions were analysed separately using a 2 (Instruction Condition) x 2 (Test Phases) ANOVA with repeated measures on the second factor. The results for low element interactivity knowledge

question scores indicated there was no difference between the instructional groups, $F(1,23) = 0.07$, $MSe = 462.92$. The test phases main effect indicated a significant difference, $F(1,23) = 16.94$, $MSe = 105.74$, demonstrating an improvement over time. There was no significant interaction, $F(1,23) = 0.03$, $MSe = 105.74$. The results for high element interactivity question scores did not indicate a difference between the instructional conditions, $F(1,23) = 0.00$, $MSe = 682.07$. The test phases main effect indicated a significant difference, $F(1,23) = 13.97$, $MSe = 187.42$, demonstrating an improvement over time. There was no significant interaction, $F(1,23) = 0.28$, $MSe = 187.42$. Practical scores were assessed by a 2 (Instruction Condition) x 2 (Test Phases) ANOVA with repeated measures on the last factor. These results suggested there was no difference between the instructional conditions, $F(1,23) = 0.01$, $MSe = 0.55$. There was no improvement over time, $F(1, 23) = 0.06$, $MSe = 0.32$, nor was there a significant interaction, $F(1, 23) = 0.58$, $MSe = 0.32$.

The results from the written and practical tests suggest there are no differences in test performance between the two instructional conditions. Thus, regardless of the type of test, no performance differences were observed between the groups, although the test of both groups tended to improve over time.

Subjective mental effort rating results did not indicate a difference between the instructional groups, $F(1,23) = 0.97$, $MSe = 1.69$. There was no improvement over time, $F(1, 23) = 0.17$, $MSe = 0.46$, nor was there a significant interaction, $F(1,23) = 2.71$, $MSe = 0.46$. These results suggest that these participants required the same degree of mental effort to learn, regardless of their instructional condition. Of interest, was the finding that the degree of mental effort required for learning did not alter over time.

The only significant differences to be found related to instruction time. There was a significant difference between groups in instruction time, $F(1,23) = 5.79$, $MSe =$

43048.96, in time taken over the two instructional phases, $F(1,23) = 10.158$, $MSe = 8699.05$ and a significant interaction between the factors, $F(1,23) = 60.36$, $MSe = 8699.05$. In order to assess the interaction effect, a test of simple effects was carried out at each phase. At Phase 1, a significant difference was found between the instructional groups, $t(23) = -5.66$ with the isolated-interacting elements group requiring less learning time. At Phase 2, when the groups studied the same version of the experimental material, no difference was found between the groups, $t(23) = 0.95$. These differences between groups were due entirely to the isolated-interacting elements group requiring less time to read their instructions in Phase 1.

Instruction condition efficiency measures were calculated by combining mental load and performance scores according to the method devised by Paas and Van Merrienboer (1993;1994). The performance variables used were high element interactivity knowledge questions and low element interactivity knowledge questions. Means and Standard Deviations for normalised (Z) mental effort ratings and performance scores as well as instructional efficiency scores are provided in Table 5.5 shows the mean and standard deviations for normalised (Z) mental effort and performance scores and the condition efficiency scores for each group.

	Isolated-interacting elements group				Interacting elements only group			
Phase 1	Z		Efficiency		Z		Efficiency	
	X	S.D.	X	S.D.	X	S.D.	X	S.D.
Low element interactivity score	-.26	.77	.01	.91	-.40	1.05	-.53	1.09
High element interactivity score	-.36	1.13	-.05	1.18	-.29	.89	-.45	.90
Practical test score	.12	.97	.29	.88	-.06	1.04	-.30	1.08
Mental Load	-.28	.87	-	-	.35	.77	-	-
Phase 2								
Low element interactivity score	.35	.99	.29	1.29	.32	.98	.24	1.07
High element interactivity score	.36	.85	.29	1.33	.29	.91	.22	1.12
Practical test score	-.12	.78	-.04	1.20	.06	1.21	.05	1.10
Mental Load	-.06	1.28	-	-	-.02	.93	-	-

Note: The efficiency scores are based on the normalised mental effort scores given in Table 5.4 for each group for each learning phase.

Table 5.5: Normalised (Z) scores and Condition Efficiency Scores for Performance and Mental Load for Experiment 2.

Efficiency measures for each of the performance scores were calculated and analysed using a 2 (Instruction Condition) x2 (Test Phases) ANOVA with repeated

measures on the last factor. The results for low element interactivity question efficiency scores indicated no difference between the instructional conditions, $F(1, 24) = 0.58$, $MSe = 2.02$. The test phases main effect was significant, $F(1, 24) = 8.67$, $MSe = 0.41$, demonstrating an improvement over time. There was no significant interaction between the factors, $F(1, 24) = 1.99$, $MSe = 0.41$. These results are represented in Figure 5.3.

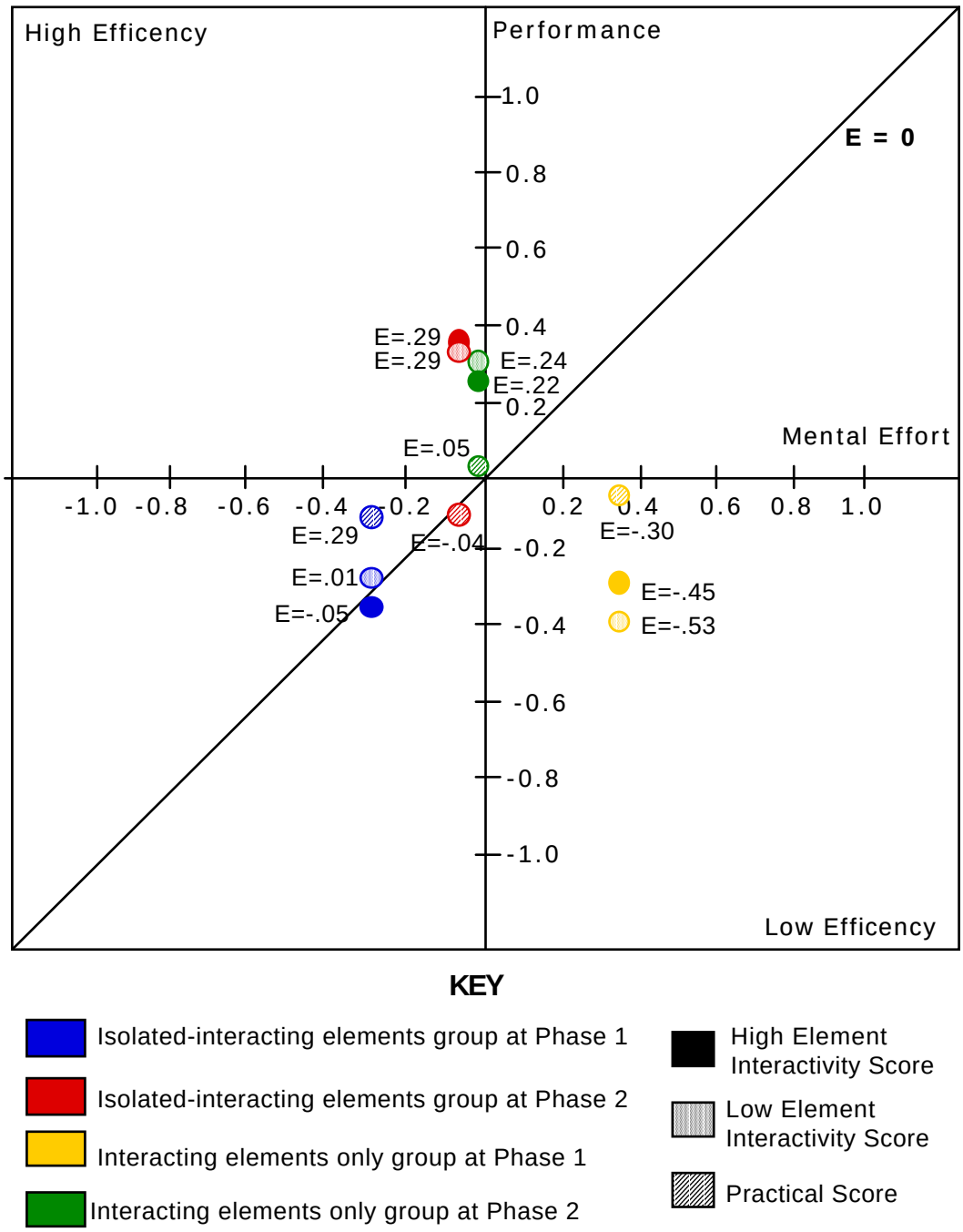


Figure 5.3: Relative instruction condition efficiency as a function of performance and mental effort ratings in Experiment 2.

Results for high element interactivity question efficiency scores indicated no difference between the instructional conditions in terms of performance efficiency, $F(1, 24) = 0.32$, $MSe = 2.29$. A significant difference between the test phases was indicated, $F(1, 24) = 10.46$, $MSe = 0.32$, demonstrating an improvement over time for both groups. There was no significant interaction between the factors, $F(1, 24) = 1.09$, $MSe = 0.32$. These results are also represented in Figure 5.3 above.

Practical score efficiency measures were analysed in a 2 (Instruction Condition) x2 (Test Phases) ANOVA with repeated measures on the second factor. Results indicated there was no difference between the instructional conditions in terms of performance efficiency, $F(1, 24) = 0.46$, $MSe = 1.71$. There was no improvement over time, $F(1, 24) = 0.00$, $MSe = 0.49$, nor was there a significant interaction between the factors, $F(1, 24) = 3.03$, $MSe = 0.49$. These results are also represented in Figure 5.3 above.

Instruction condition efficiency measures as a factor of study time were calculated by combining instruction time and performance scores in the manner devised by the author from the methodology of Paas and Van Merrienboer (1993;1994). The performance variables used were high element interactivity knowledge questions and low element interactivity knowledge questions. Means and Standard Deviations for normalised (Z) mental effort ratings and performance scores as well as instructional efficiency scores are provided in Table 5.6 shows the mean and standard deviations for normalised (Z) mental effort and performance scores and the condition efficiency scores for each group.

	Isolated-interacting elements group				Interacting elements only group			
Phase 1	Z		Efficiency		Z		Efficiency	
	X	S.D.	X	S.D.	X	S.D.	X	S.D.
Low element interactivity score	-.26	.77	.26	.72	-.40	1.05	-1.00	1.04
High element interactivity score	-.36	1.13	.19	.95	-.29	.89	-.92	1.07
Practical test score	.12	.97	.53	.71	-.06	1.04	-.77	1.03
Instruction Time	-.62	.53	-	-	1.02	.91	-	-
Phase 2								
Low element interactivity score	.35	.99	.27	.86	.32	.98	.47	1.25
High element interactivity score	.36	.85	.28	.85	.29	.91	.46	1.04
Practical test score	-.12	.78	-.05	.64	.06	1.21	.29	.98
Instruction Time	-.04	.51	-	-	-.36	1.02	-	-

Note: The efficiency scores are based on the normalised instructional time scores given in Table 5.4 for each group for each learning phase.

Table 5.6: Normalised (Z) scores and Condition Efficiency Scores for Performance and Instructional Time (in seconds) for Experiment 2.

Efficiency measures for each type of score were calculated and analysed using a 2 (Instruction Condition) x 2 (Test Phases) ANOVA with repeated measures on the last factor. The results for low element interactivity question efficiency scores

indicated no difference between the instructional conditions, $F(1, 24) = 2.17$, $MSe = 1.69$. A significant difference between the test phases was indicated, $F(1, 24) = 27.40$, $MSe = 0.26$, demonstrating an improvement over time. A significant interaction was indicated between the factors, $F(1, 24) = 26.19$, $MSe = 0.26$. Tests of simple effects were used to explore the interaction. At phase 1, a significant difference was indicated between the instructional groups, $t(24) = 3.61$, with the isolated-interacting elements group showing greater performance efficiency for low element interactivity questions. At phase 2, no difference was indicated between the instructional conditions, $t(24) = -0.47$. These results are represented in Figure 5.4.

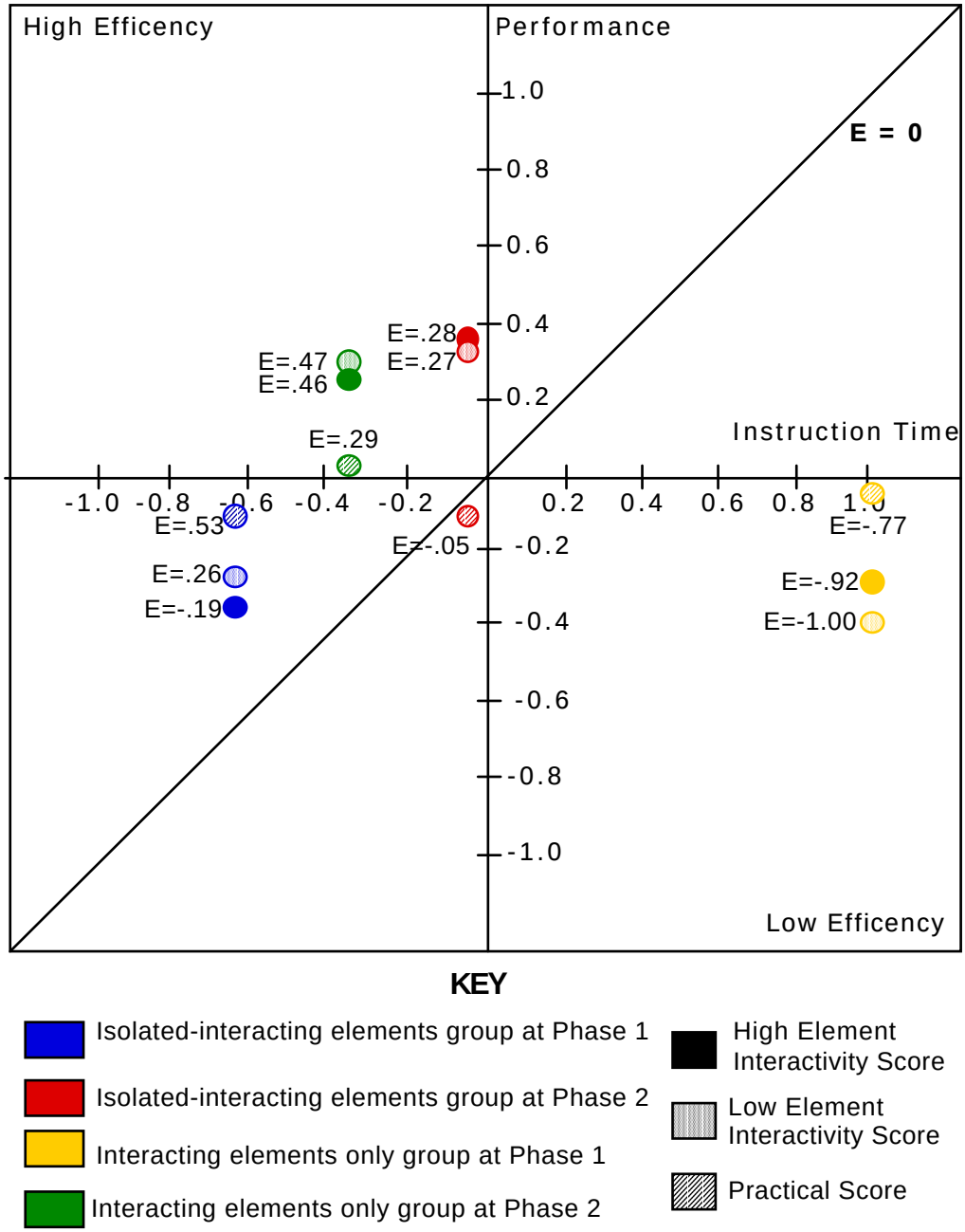


Figure 5.4: Relative instruction condition efficiency as a function of performance and instruction time in Experiment 2.

Results for high element interactivity question efficiency scores indicated no difference between the instructional conditions in terms of performance efficiency, $F(1, 24) = 1.73$, $MSe = 1.65$. A significant difference between the test phases was indicated, $F(1, 24) = 25.48$, $MSe = 0.28$, demonstrating an improvement over time for both groups. A significant interaction was indicated between the factors, $F(1, 24) = 19.64$, $MSe = 0.28$. Tests of simple effects were used to explore the interaction. At Phase 1, a significant difference was found between the instructional groups, $t(24) = 2.80$ with the isolated-interacting elements group showing greater performance efficiency for high element interactivity questions. At phase 2, no difference was indicated between the instructional groups, $t(24) = -0.47$. These results are also represented in Figure 5.4 above.

Results for practical question efficiency scores indicated no difference between the instructional conditions in terms of performance efficiency, $F(1, 24) = 2.86$, $MSe = 1.03$. There was no improvement over time, $F(1, 24) = 1.67$, $MSe = 0.44$. A significant interaction was indicated between the factors, $F(1, 24) = 19.95$, $MSe = 0.44$. Tests of simple effects were used to explore the interaction. At Phase 1, a significant difference was found between the instructional groups, $t(24) = 3.74$ with the isolated-interacting elements group showing greater performance efficiency for the practical questions. At phase 2, no difference was indicated between the instructional groups, $t(24) = -1.059$. These results are also represented in Figure 5.4 above.

As noted previously, the calculation of relative efficiency allows two instruction conditions to be compared. These results indicate when instruction condition efficiency is measured in terms of performance per unit of mental effort, the teaching techniques cannot be distinguished from each other. When efficiency is measured in terms of performance per unit of instruction time, an overall difference does not exist; however, at phase one students in the isolated-interacting elements

condition do show greater performance per unit of instruction time. These results can be explained by the smaller amount of time required for instruction by the Isolated-interacting elements group when studying the isolated version of the materials in phase one.

The results from Experiment 2 contrast sharply with those of Experiment 1, where instruction by the isolated-interacting elements method significantly improved test performance. The level of expertise of the participants was the only identifiable difference between the two experiments. This observation lends credence to the hypothesis that the isolated-interacting elements method of instruction benefits learners through assisting the process of schema construction. This hypothesis explains why the isolated-interacting elements method of instruction showed distinct performance advantages for novice learners (Experiment 1), who previously had only very primitive schemas in the area. The more expert learners in Experiment 2, already in possession of schemas for the general area of electrical circuitry, learnt equally effectively, regardless of the instruction method used.

Chapter 6 Assimilating complex information: a replication

6.1 Introduction to Experiments 3 and 4

Experiments 1 and 2 investigated the viability of the isolated-interacting elements instruction condition, a method of instruction that aims to increase understanding by reducing intrinsic cognitive load due to element interactivity. Experiment 1 demonstrated that students in the isolated-interacting elements instruction condition achieved superior performance and required less mental effort to understand the material than the interacting elements only instruction group receiving the interacting elements instructions at both phases. Experiment 2 examined learners who were relatively more expert and the results suggested no performance benefits were to be gained from the isolated-interacting elements method of instruction. We suggest that the isolated-interacting elements method of instruction is more effective for novice learners because isolated elements instruction at phase 1 allows critical elements to be processed in working memory and so promotes the development of partial schemas that enhance knowledge acquisition and more comprehensive schema development at the second phase of instruction. These results were not obtained in Experiment 2 because students had sufficiently comprehensive schemas to allow them to process all of the interacting elements from the interacting elements instructions in working memory. They did not need to have the instructions presented as isolated elements.

While the differing results of Experiments 1 and 2 were theoretically coherent, they could have been due to the particular procedures and methods used. There was a heavy emphasis on procedural learning in Experiments 1 and 2 and it could be argued that the sequential and directive nature of this type of information may have influenced the results. It is important to replicate the results of Experiments 1 and 2 using materials that place a greater emphasis on conceptualisation rather than on procedures. Accordingly, Experiments 3 and 4 used materials requiring a higher degree of conceptualisation than Experiments 1 and 2. Identical materials and procedures were used for both experiments but the students of Experiment 3 had considerably less background knowledge than those of Experiment 4. We predicted that the isolated-interacting elements instruction condition would be superior to the interacting elements instruction condition in Experiment 3 but that this difference would disappear in Experiment 4.

6.2 Experiment 3

Experiment 3 compared the isolated-interacting elements and interacting elements only conditions using apprentices who had only a limited background in electrical principles. As in Experiment 1, it was predicted that the isolated-interacting elements condition would be superior to the interacting elements only condition. The experiment used material depicting an electrical circuit used to power an industrial oven. This material was very complex, containing more elements of information and more interactions between the elements than the materials employed previously in Experiments 1 and 2. It can be regarded as particularly high in element interactivity because all of the elements that form a circuit relate to and in turn effect, other circuit elements.

Method

Participants

Eighteen first year industrial trade students enrolled in a training course at a Sydney company participated in the experiment. All of the students had completed at least Year 10 of High School. They were selected from students enrolled in one of the following courses: a six month manufacturing and engineering trainee-ship course (13 students) or the practical section of a Year 11 school electronics course (5 students). Students were tested within their first fortnight of training so both groups had little electrical knowledge and what had been covered, with respect to simple electrical concepts, was very similar for both courses. Both groups had been introduced to very basic electrical principles (e.g., A represents active, N neutral and current flows from active to neutral) but none of the participants had ever studied the electrical circuit or any similar circuits to that used in the experiment. Students were excluded from the study if they had completed further studies in the field of electrical engineering; if English was not their first language; or if there were more than forty eight hours between the two learning sessions of the study.

Materials

The participant information sheet recorded demographic information such as age, years of schooling, qualification being studied and stage in current course. Participants were also asked details of any previous experience with electrical training.

The instructional materials used in the experiment consisted of information describing the electrical circuit used to power an industrial oven. The oven circuit consisted of two interconnecting circuits: the power circuit and the control circuit. The power circuit is formed by switches and fuses which connect and power a three phase motor. The power circuit is connected into the main circuit: the control

circuit. This circuit controls the function and sequence of events required to bake goods in the oven. The function and workings of the oven circuit were impossible to understand without reference to the whole circuit; for example, in order to understand the purpose and function of the oven's timer, the students were required to consider the interplay between the timer, the conveyor, the motor and various switches. The material is high in element interactivity because simultaneous processing of many elements is required to understand the oven baking process and electrical system.

Two sets of instructional materials were developed to introduce the electrical circuit: an isolated elements version and an interacting elements version.. Both sets of instructions were designed to introduce and demonstrate the function and sequence of events in an industrial oven baking process. The instructional materials were designed in line with Mayer and colleagues' cognitive theory of multimedia learning (Mayer & Moreno, in press; Mayer, et al., 1995); for example, the instructional material contained a combination of non-redundant verbal and visual material presented simultaneously. Each set of instructions provided the student with a diagram and text describing the electrical circuit that operated the oven. The isolated elements version of the instructional materials aimed to artificially reduce the element interactivity of the material by focusing only on individual elements of information, not on the interconnections between these elements. Each element of the circuit was represented as an autonomous unit in the sequence of events, that together formed the oven cycle.

In contrast to isolated elements version, the interacting elements version of the instructional materials presented material at its original level of element interactivity, therefore, the interactions between elements were emphasised. This version of the oven circuit also described the oven process and so text was incorporated which was virtually identical to that used in the isolated elements version. The difference between the two versions lay in the emphasis the

interacting elements version placed upon the cause and effect relationships that drove the system; for example, these instructions highlighted how a switch closing allows a coil to energise which then causes the forward contact switch in the power circuit to close, allowing the motor to start and move the conveyor in a forward direction so the goods are taken toward the heating elements for baking. Theoretical information on electrical principles and practise are also included in the interacting elements version of the materials. An analysis of the element interactivity of the electrical circuit has been included in Appendix 6.

In a similar manner to the instructional materials of Experiments 1 and 2, the isolated elements instructions included fewer elements than the interacting elements instructions. Nevertheless, the instructional material is very complex so it may be the case that there are too many elements to be processed simultaneously in working memory even for the isolated elements condition. The materials only can be processed by forming partial schemas in long-term memory to reduce working memory load. This process should be easier for the isolated elements instruction condition because of the reduction in the number of elements compared to the interacting elements instruction condition.

The isolated elements version was presented on a single A4 (21 X 29.5 cm) sized page. Numbered text points were integrated into the circuit diagram. The interacting elements version was presented in an identical format to the isolated elements version except that it was on an A3 size page (42 X 29.5 cm). This larger diagram was necessary because of the difference in the amount of text contained in the interacting elements version. A copy of the isolated elements instructions has been included in Appendix 7; a copy of the interacting elements instructions has been included in Appendix 8.

Both groups also received a single, A4 (21 X 29.5 cm) sized page that depicted a diagram representing the oven baking process. The diagram included key features

of the oven such as the motor, the heating elements, the conveyor and the oven control panel realistically rather than as electrical circuitry. The diagram used these features to show how the electrical circuit runs the oven baking process. The diagram was accompanied by this text, "This circuit enables goods to be baked in an oven. The goods are on a conveyor, which is driven by a 3 phase motor. The conveyor transports goods to the oven's heating elements where it stops and cooking occurs for a prescribed time. At the end of the set cooking time, the conveyor is reactivated and the goods are moved away from the cooking elements back to their original position". The aim of this diagram was to provide the students with holistic background information on the oven process in order to enable them to gain an understanding of how the electrical circuitry combined to enable the oven baking process to occur. A copy of this diagram has been included in Appendix 9.

Another diagram, depicting only the oven electrical circuit, was available to students during testing. Switches and symbols were labelled on the diagram but no textual description was included. It was presented on an A4 (21 X 29.5 cm) sized page. A copy of this diagram has been included in Appendix 10.

The subjective mental load rating measure used a nine point Likert type scale. Above the scale was the instruction "How easy or difficult was it to learn and understand the electrical circuit from the instructions you were given? Place a cross on the rating scale below". Each number on the scale was labelled with a description ranging from extremely easy (1) to extremely difficult (9). The number of points on the mental effort Likert scale were increased from seven in Experiment 1 and 2, to nine in this experiment, in order to give the students a greater range of options to describe their subjective mental load when learning. A copy of the subjective mental load rating scale used in the experiment has been included in Appendix 11.

There were two categories of questions in the written test: those tapping low and high element interactivity knowledge. Both types of questions included textual and diagrammatical test items. Low element interactivity questions asked about a specific part or fact of the electrical circuit. For example: "How can the operation of the circuit be stopped manually?" or "What stops Coil R from overheating?". The questions were low in element interactivity because the questions involved only a single section of the circuit and the information required to answer the questions could be learnt in isolation from the rest of the circuit. Answering the question required only at most one or two elements of information to be processed simultaneously. This section of questions tapping low element interactivity knowledge included four test items with the total marks possible for this category of questions being eight. Some questions contained sub-parts; for example, "What three actions need to be performed by someone using the oven for the baking process to begin?". This question required three responses, each response was worth one mark.

High element interactivity questions dealt with issues such as the progression of events in the circuit or the structure and function of the oven circuit. These questions required students to have a holistic understanding of the circuit and its processes. In order to answer the questions, the students had to simultaneously refer to many elements of the circuit.. For example: "Circle the option that completes the sentence correctly. When the goods initially reach the heating elements, Limit switch 1 is signalled. The top of double pole switch 1.....?" or "For the conveyor to start moving the goods away from the heating elements, what has to change in this circuit? Place a cross on the diagram on all the item/s in the diagram that need to be different. Indicate what change is required". The information required to answer each question involved the simultaneous application at least four elements of information. There were eight test items in the high element interactivity category of questions and the total marks possible for this category of questions was twenty-seven. Some questions contained sub-parts; for

example, "When the goods are baking, which switches are open?". A correct answer to this question would identify seven open switches. Each correct response was awarded one mark. A copy of the written test used in the experiment has been included in Appendix 12.

Procedure

All the students were randomly assigned to either the isolated-interacting elements instruction group or the interacting elements only group. All students were tested individually, in a quiet room. There were nine students in each group. The students were informed that they were to be given instructional material regarding an electrical circuit which would be followed by a written test and that this session was to be repeated within forty-eight hours. Students filled in an information sheet at this stage that asked for biographical and educational details. Any students considered to have prior knowledge (see criteria outlined in the "participants" section) were eliminated at this stage. Students were able to ask questions at any time during the experiment.

In the first experimental session, the students in the isolated-interacting elements instruction group studied the isolated elements instructional booklet while the students in the interacting elements only group were given the interacting elements instructional booklet. Both groups also received the schematic diagram of the oven baking process. Students were allowed a maximum of ten minutes to read through the instructional material. If a student finished before this time, they were asked to review the information for the remaining time. This procedure was used in order to provide additional control by equalising study times rather than allow unlimited times as occurred in Experiments 1 and 2. At the completion of the study phase, the students were provided with a subjective mental load rating scale, the format of which was explained to both groups. They were asked to rate the mental effort expended in understanding the oven circuit and to translate this to a numerical value.

The test section of the experiment followed. The students were asked to complete the written test, which was common to both groups. The instructional materials were not available to the students during this test period. The circuit diagram (labelled circuit elements with no text description) was available to the students at this stage but was removed for diagrammatical questions that required the students to find faults in the circuit. At all stages of the test period, students were prohibited from reattempting questions after they had attempted them and from reviewing previously completed test answers.

The second phase of the experiment was held within forty-eight hours of the first phase. The procedure of the second phase of the experiment was identical to the first, except that all students studied the interacting elements instructions.

Results and Discussion

The test score variables under analysis were low and high element interactivity knowledge scores and mental load rating. Means and Standard Deviations for these variables are provided in Table 6.1.

Phase 1	Max score	Isolated-interacting elements group (n=9)		Interacting elements only group (n=9)	
		Mean	Std Dev	Mean	Std Dev
Low element interactivity score	100%	50.00	25.00	26.39	17.05
High element interactivity score	100%	25.51	10.55	19.75	15.16
Mental Load	9	5.67	1.00	6.00	1.12
Phase 2					
Low element interactivity score	100%	55.56	25.85	38.89	26.10
High element interactivity score	100%	40.33	14.87	23.46	14.58
Mental Load	9	5.22	1.09	6.00	1.73

Table 6.1: Test Scores and Mental Load Ratings for Experiment 3

Low and high element interactivity knowledge scores were analysed separately by 2 (Instruction Condition) x 2 (Test Phases) ANOVAs with repeated measures on the second factor. The results for low element interactivity knowledge question scores indicated a possible difference between the instructional groups favouring the isolated-interacting elements instruction group, $F(1,16) = 3.86$, $MSe = 6.06$, $p = 0.07$. The test phases main effect also indicated a possible difference over time, $F(1,16) = 3.93$, $MSe = 1.19$, $p = 0.07$. There was no significant interaction, $F(1,16) = 0.58$, $MSe = 1.19$.

The results for high element interactivity question scores indicated a possible difference favouring the isolated-interacting elements instruction group, $F(1,16) = 3.18$, $MSe = 26.42$, $p = 0.09$. The test phases main effect indicated a significant difference, $F(1,16) = 31.03$, $MSe = 1.81$, demonstrating an improvement over time. A significant interaction was displayed between the factors, $F(1,16) = 11.17$, $MSe = 1.81$. Tests of simple effects were used to explore the interaction. No difference was found between the instructional conditions at phase 1, $t(16) = 0.936$. At phase 2, a significant difference was indicated between the groups, $t(16) = 2.43$, with the isolated-interacting elements group performing at a superior level.

Mental load ratings were analysed in 2 (Instructional Condition) x 2 (Test Phases) ANOVA with repeated measures on the last factor. Results indicated no differences between the instructional conditions, $F(1, 16) = 1.19$, $MSe = 2.34$. There was no improvement over time, $F(1, 16) = 0.50$, $MSe = 0.88$, nor was therea significant interaction, $F(1, 16) = 0.50$, $MSe = 0.88$.

Instructional condition efficiency measures were calculated according to the method devised by Paas and van Merrienboer (1993;1994). The performance variables used were high element interactivity knowledge questions and low element interactivity knowledge questions. The means and standard deviations of the normalised (Z) mental effort and performance scores and the condition efficiency scores for each group are shown in Table 6.2.

	Isolated-interacting elements group				Interacting elements only group			
Phase 1	Z		Efficiency		Z		Efficiency	
	X	S.D.	X	S.D.	X	S.D.	X	S.D.
Low element interactivity score	.29	.98	.23	1.05	-.64	.67	-.61	.85
High element interactivity score	-.11	.68	-.05	.86	-.49	.98	-.50	.77
Mental Load	-.04	.80	-	-	.22	.89	-	-
Phase 2								
Low element interactivity score	.51	1.02	.64	1.13	-.15	1.03	-.26	1.42
High element interactivity score	.84	.96	.88	1.21	-.25	.94	-.33	1.22
Mental Load	-.40	.87	-	-	.22	1.38	-	-

Note: The efficiency scores are based on the normalised mental effort scores given in Table 6.1 for each group for each learning phase.

Table 6.2: Normalised (Z) scores and Condition Efficiency Scores for Performance and Mental Load for Experiment 3.

Efficiency measures for each type of score were calculated and analysed using a 2 (Instruction Condition) x 2 (Test Phases) ANOVA with repeated measures on the last factor. The results for low element interactivity question efficiency scores indicated a possible difference between the instructional conditions favouring the isolated-interacting elements instruction group, $F(1, 16) = 3.42$, $MSe = 2.01$, $p = 0.08$. No difference was indicated between the test phases, $F(1, 16) = 2.32$, $MSe =$

0.55. There was no significant interaction between the factors, $F(1, 16) = 0.01$, $MSe = 0.55$. These results are represented in Figure 6.1.

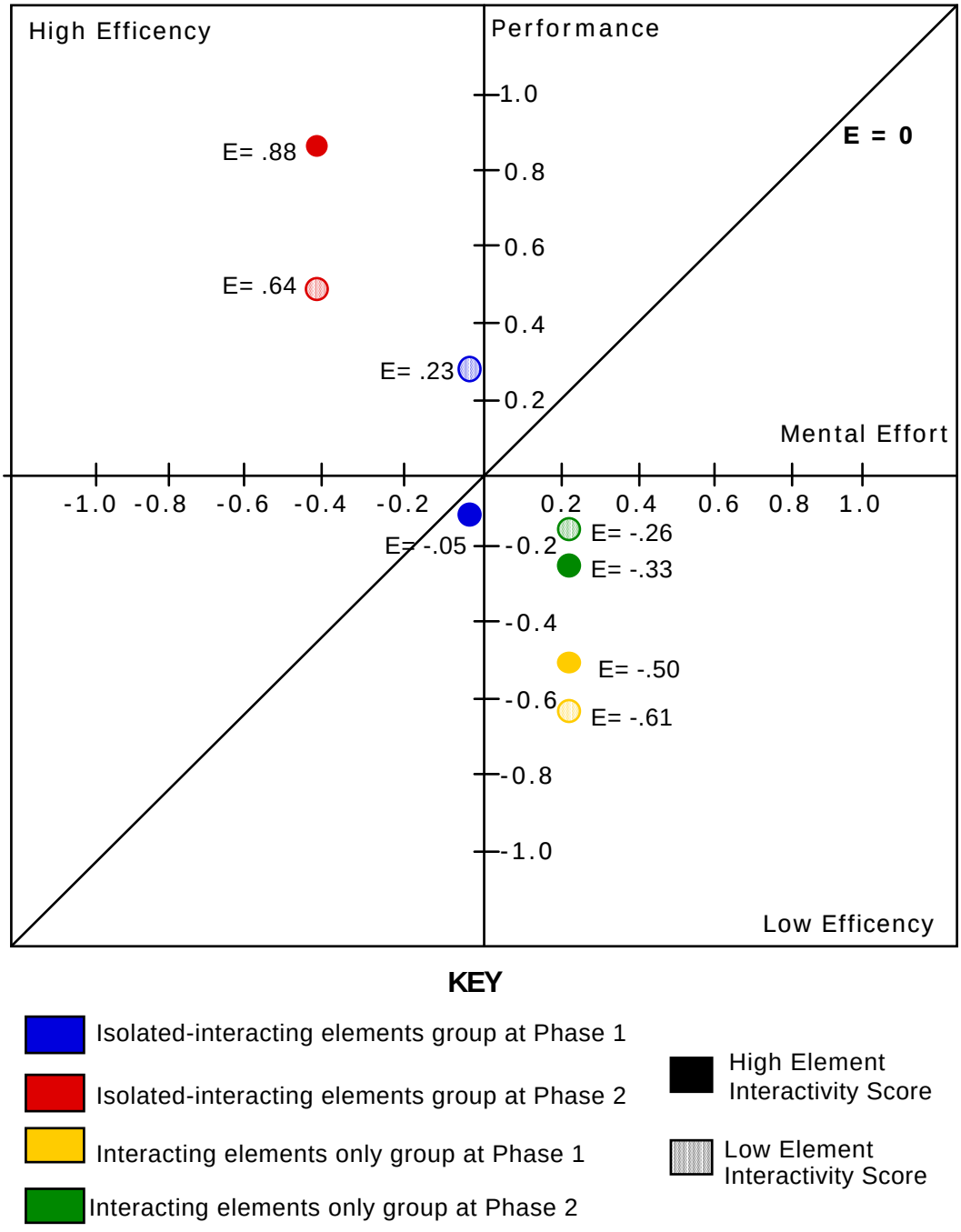


Figure 6.1: Relative instruction condition efficiency as a function of performance and mental effort ratings in Experiment 3.

Results for high element interactivity question efficiency scores indicated a possible difference favouring the isolated-interacting elements instruction group with respect to performance efficiency, $F(1, 16) = 3.34$, $MSe = 1.86$, $p = 0.09$. A significant difference was indicated between the test phases, $F(1, 16) = 9.52$, $MSe = 0.28$, demonstrating an improvement over time for both groups. A significant interaction was indicated between the factors, $F(1, 16) = 4.55$, $MSe = 0.28$. Tests of simple effects was used to explore the interaction. At phase 1, no difference was indicated between the instructional conditions, $t(16) = 1.17$. At phase 2, a significant difference existed between the instructional groups, $t(16) = 2.11$, with the isolated-interacting elements group displaying superior instruction condition efficiency by phase 2 of the learning process. These results are also represented in Figure 6.1 above.

As predicted, the results of Experiment 3 do replicate and support the findings of the previous experiment with novice learners (Experiment 1). The major difference favouring the isolated-interacting elements instruction condition occurred during phase 2 using questions testing for high element interactivity knowledge. In a seeming paradox, for these students, the "understanding" or interacting elements instructions of phase 2 were only effective after studying material during phase 1 that did not emphasise understanding. Interacting elements instructions in phase 1 reduced rather than facilitated the effectiveness of Interacting elements instructions in phase 2.

The isolated-interacting elements instructional condition has now been established as a superior means of information delivery using both procedural and more conceptually complex material. The data support the theoretical mechanism proposed to explain the success of the isolated-interacting elements instructional condition. Learning by the isolated-interacting elements instructional condition reduces the working memory load required to process a complex concept.

Learning isolated elements of information at Phase 1 frees memory capacity at Phase 2, providing the learner with the ability to focus on understanding the complex relations between the elements of a concept.

6.3 Experiment 4

The results of Experiments 1 and 3 indicated the superiority of the isolated-interacting elements method of instruction over the interacting elements only instructional technique for inexperienced learners. The theoretical explanation generated from a cognitive load theory perspective suggests the isolated-interacting elements instruction condition reduces working memory load during the isolated elements phase resulting in the acquisition of partial schemas that permit a more complete understanding during the subsequent high cognitive load, high element interactivity phase. Experiment 4 used more experienced learners who could be expected to have previously acquired the partial schemas that reduce working memory load sufficiently to allow understanding. If interacting elements instructions can be assimilated we can expect that isolated elements play no role when using more experienced learners thus eliminating an advantage for a isolated-interacting elements instruction condition.

Experiment 4 replicated the design of the previous experiments using the materials and procedure of Experiment 3 but with a sample of more experienced learners. The participants involved in this study were industrial apprentices, experienced in the area to be studied and so it was predicted that no performance differences would be obtained between the two instructional conditions. It was expected that these more expert learners would not benefit from the isolated-interacting elements method of instruction as the interacting elements instructions should not overburden working memory.

Method

Participants

Eighteen first year electrical and mechanical apprentices enrolled in an apprenticeship training course at a Sydney company participated in the experiment. All of the students had completed at least Year 10 of High School and were enrolled in various first year courses at trade colleges. The participants were tested more than three quarters of the way through their eight month training course in industrial and domestic electricity (students had completed a mean of 28.55 units from a 34 unit course). Apprentices were excluded from the study if they had completed further studies in the field of electrical engineering, if English was not their first language, or if there were more than forty eight hours between the two learning sessions.

Materials

The instructional materials used in this experiment was identical to that used in Experiment 3.

Procedure

The procedure used in this experiment was virtually identical to that used in Experiment 3. The only difference was in the allocation of participants to instructional groups. Participants were matched for apprenticeship type (i.e. electrical, mechanical, dual apprenticeship, fitter-turner, tool maker) before being randomly assigned to either the isolated-interacting elements instruction group or the interacting elements only group.

Results and Discussion

The test score variables under analysis were low and high element interactivity knowledge scores and mental load rating. Means and standard deviations for these variables are provided in Table 6.3.

Phase 1	Max score	Isolated-interacting elements group (n=9)		Interacting elements only group (n=9)	
		Mean	Std Dev	Mean	Std Dev
Low element interactivity score	100%	65.59	20.52	69.44	15.45
High element interactivity score	100%	32.51	11.97	39.51	20.37
Mental Load	9	5.44	1.13	4.67	1.87
Phase 2					
Low element interactivity score	100%	70.83	27.95	76.39	17.05
High element interactivity score	100%	46.50	14.12	58.44	21.26
Mental Load	9	4.22	1.39	3.44	1.51

Table 6.3: Test Scores and Mental Load Ratings for Experiment 4.

Low and high element interactivity knowledge scores were analysed separately by 2 (Instruction Condition) x 2 (Test Phases) ANOVAs with repeated measures on the second factor. The results for low element interactivity knowledge question scores did not indicate a difference between the instructional groups, $F(1,16) = 0.27$, $MSe = 775.82$. The test phases main effect indicated a possible difference,

F(1,16) = 3.90, MSe = 90.06, p = 0.07, with performance improving over time. There was no significant interaction, F(1,16) = 0.05, MSe = 90.06.

The results for high element interactivity question scores did not indicate a difference between the instructional conditions, F(1,16) = 1.46, MSe = 552.22. The test phases main effect indicated a significant difference, F(1,16) = 46.46, MSe = 52.49, demonstrating an overall improvement over time. No interaction between the factors was indicated, F(1,16) = 1.05, MSe = 52.49.

The results for subjective mental load ratings indicated no difference between the instructional conditions, F(1, 16) = 1.33, MSe = 4.09. The test phases main effect indicated a significant difference, F(1,16) = 32.81, MSe = 0.41, demonstrating an improvement over time. There was no interaction between the factors, F(1,16) = 0.00, MSe = 0.41.

Instructional condition efficiency measures were calculated according to the method devised by Paas and Van Merrienboer (1993;1994). The performance variables used were high element interactivity knowledge questions and low element interactivity knowledge questions. The means and standard deviations of the normalised (Z) mental effort and performance scores and the condition efficiency scores for each group are provided in Table 6.4.

	Isolated-interacting elements group				Interacting elements only group			
Phase 1	Z		Efficiency		Z		Efficiency	
	X	S.D.	X	S.D.	X	S.D.	X	S.D.
Low element interactivity score	-.26	1.01	-.62	1.02	-.05	.76	-.13	1.27
High element interactivity score	-.61	.62	-.87	.82	-.25	1.06	-.27	1.41
Mental Load	.62	.70	-	-	.14	1.16	-	-
Phase 2								
Low element interactivity score	.02	1.38	.11	1.52	.29	.84	.64	.93
High element interactivity score	.12	.73	.18	1.07	.74	1.10	.96	1.37
Mental Load	-.14	.87	-	-	-.62	.94	-	-

Note: The efficiency scores are based on the normalised mental effort scores given in Table 6.3 for each group for each learning phase.

Table 6.4: Normalised (Z) scores and Condition Efficiency Scores for Performance and Mental Load for Experiment 4.

Efficiency measures for each type of score were calculated and analysed using a 2 (Instruction Condition) x 2 (Test Phases) ANOVA with repeated measures on the last factor. The results for low element interactivity question efficiency scores

indicate no difference between the instructional conditions, $F(1, 16) = 0.882$, $MSe = 2.66$. A significant difference was indicated between the test phases, $F(1, 16) = 21.70$, $MSe = 0.24$, demonstrating an improvement over time. There was no significant interaction between the factors, $F(1, 16) = 0.02$, $MSe = 0.24$. These results are represented in Figure 6.2.

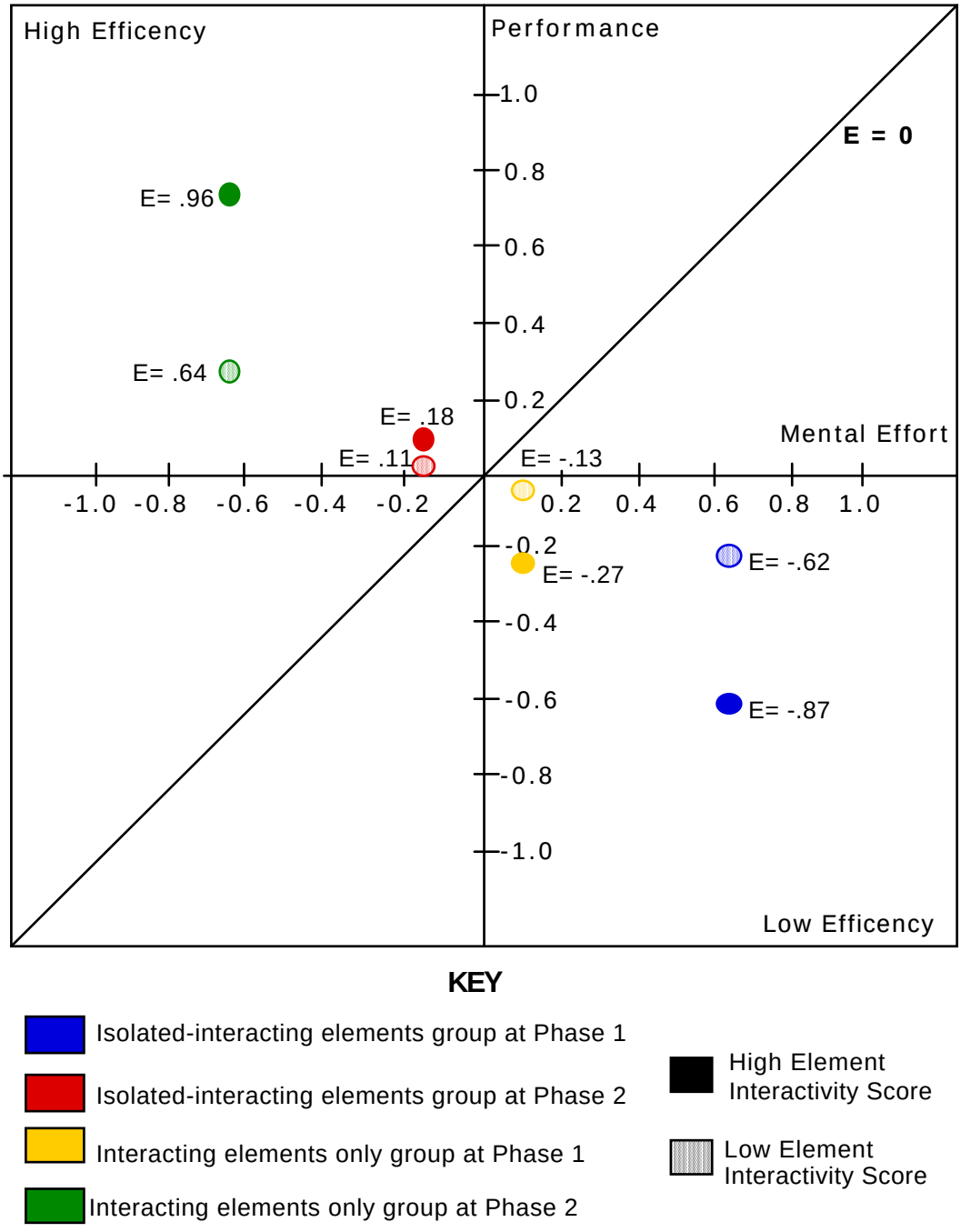


Figure 6.2: Relative instruction condition efficiency as a function of performance and mental effort ratings for Experiment 4.

Results for high element interactivity question efficiency scores indicated no difference between the instructional conditions, $F(1, 16) = 1.63$, $MSe = 2.62$. A significant difference was displayed between the test phases, $F(1, 16) = 56.75$, $MSe = 0.21$, demonstrating an improvement over time. There was no significant interaction between the factors, $F(1, 16) = 0.36$, $MSe = 0.21$. These results are also represented in Figure 6.2 above.

In a pattern of results similar to those observed in Experiment 2 in which more experienced learners also participated, there were no significant differences between groups on either performance, subjective mental ratings measures or instructional efficiency. These results are in line with our predictions. It can be concluded that there is no performance benefit for expert learners using the isolated-interacting elements method of instruction. This experiment has also generalised the lack of effect gained by using the isolated-interacting elements method of instruction for more expert learners because the same pattern of results has occurred using different and arguably, more complex material. The lack of a isolated-interacting elements instruction condition effect for expert learners has now been established using both procedural and more heavily conceptually based material.

We suggest that expert learners will not gain any advantage from an instructional condition intended to more readily permit processing in working memory. The processing and working memory capacity advantages that novice learners gain from an initial exposure to instructional material which had a reduced intrinsic load had no effect on this participant sample because the learners were already capable of processing the complex concepts in their entirety.

Chapter 7 Assimilating complex information: theoretical support

7.1 Introduction to Experiment 5

Experiments 1 and 2 used identical experimental material with vastly different results. This pattern of results was replicated in Experiments 3 and 4. In Experiments 1 and 3, performance benefits were associated with studying isolated elements instructions followed by interacting elements instructions. This instructional technique was compared to a more conventional method of instruction for promoting understanding where instructions containing interacting elements were presented repeatedly. In contrast, in Experiments 2 and 4 when the same two instructional techniques were compared, no differences between the two instructional conditions were apparent. It is our contention that the disparity in the results that emerged from the experiments was due to a difference in the expertise or prior knowledge level of the two participant populations. Experiments 1 and 3 used two groups of students who were new to the concepts of electrical engineering, the topic area employed in the experiments. The samples used in Experiments 2 and 4 however, had completed considerably more domain specific electrical training than their counterparts in Experiments 1 and 3 respectively.

Experiment 5 aimed to directly explore the hypothesis that the critical factor in the success of the isolated-interacting elements method of instruction is the expertise

level of the students used. To this end, Experiment 5 aimed to partially replicate the findings of Experiments 1 and 3 which had used samples of learners who were inexperienced in the knowledge domain investigated. Therefore, Experiment 5 used a sample of students with little experience in the material to be studied; however, the sample used consisted of the same students who had participated in Experiments 2 and 4 as expert learners. This apparent contradiction is explained by three factors: the domain specific nature of expertise, the different learning materials employed by each of the experiments and the participants' previous experience with these subject topics.

As stated previously in chapter 1, *expertise* is not generic. It is developed only in a specific knowledge area with experience and practise (De Groot, 1965; Chase & Simon, 1973a,b; Chi, 1978). In Experiments 2 and 4, the materials were drawn from the area of electrical engineering, in particular, electrical appliance safety testing and electrical circuitry. In contrast, Experiment 5 used materials from mechanical engineering, namely pneumatic or air powered mechanical systems. In Experiments 2 and 4, the participants could be regarded as experienced learners because of their previous exposure to the electrical material used for instruction in the experiments. In Experiment 5, these same students could be considered novice learners because they lacked knowledge about the pneumatic concepts used in the learning material.

The results from Experiments 2 and 4 suggested that the participants in these experiments did not benefit from the isolated-interacting elements method of instruction. Experiment 5, by using different instructional material, aimed to demonstrate that the isolated-interacting elements method could be effective with these same students, provided they were inexperienced in the knowledge area under investigation. Theoretically, this contrast can be explained by the contention that the isolated-interacting elements instruction method benefits the inexperienced

learner because the process of schema construction is facilitated through an initial reduction of the cognitive load of the learning material.

7.2 Experiment 5

It was hypothesised that Experiment 5 would replicate the findings of Experiments 1 and 3 and show a superior performance by the group taught by the isolated-interacting elements method of instruction. It was expected that these students would achieve this performance using less mental effort and would require less learning time.

Method

Participants

Forty-two first year electrical and mechanical apprentices involved in a training course at a Sydney company participated in the experiment. All of the participants had completed at least Year 10 of High School and were enrolled in various first year trade courses at trade colleges. The majority of the participants had received no exposure to the area of pneumatics but ten students had received limited exposure to pneumatics; all of these students had received the same training (which did not directly deal with the pneumatic system used in this experiment) held approximately five months prior to the experiment. These participants were distributed evenly between the experimental groups. Half of the students in Experiment 5 had participated in Experiment 2 of this thesis. A large majority of the remaining students had participated in Experiment 4.

Students were excluded from the study if they had completed further recognised studies in the field of pneumatics other than stated above; if English was not their first language or if more than forty eight hours had elapsed in between the two learning sessions. Students were also asked details of any previous experience

with the subject area of pneumatics. If they indicated previous knowledge of pneumatics they were asked to provide details of their experience and self rate their pneumatic knowledge as high, medium or low.

Materials

The instructional material used in the experiment described an indirectly controlled pneumatic circuit for a single acting cylinder. For novice students, the pneumatic circuit was complex because the material was very high in element interactivity. The information is high in element interactivity because in order to understand the system as a whole, one has to consider how the parts of the system interact; for example, in order to fully understand the concept of indirect pneumatic control of a single acting cylinder, the student had to simultaneously consider the role of the pushbutton and how this interacted with the 3/2 way valve, the air pilot line, the air supply sources, the control valve and the position of the cylinder. A student would be unable to understand how the system worked by merely considering a single element, such as the cylinder. An analysis of the element interactivity of the pneumatic circuit has been included in Appendix 13. The analysis estimates the number of elements involved in the system for a first year apprentice with limited pneumatic knowledge.

Two sets of instructions were developed: an isolated elements version and an interacting elements version. Each set of instructions provided information about an indirectly controlled pneumatic circuit. The instructional materials were designed in line with Mayer and colleagues' cognitive theory of multimedia learning (Mayer & Moreno, in press; Mayer, et al., 1995); for example, the instructional material contained a combination of non-redundant verbal and visual material presented simultaneously. Each version of the instructional material consisted of a booklet containing two diagrams on separate pages. The first diagram illustrated the pneumatic circuit in the *off* position. In the *off* position there is no air pressure in the circuit and the cylinder's piston is retracted. The second diagram showed the circuit

in the *on* position. In this position the cylinder's piston is extended because of the air pressure in the circuit. Each diagram contained textual information describing the circuit which was integrated into the diagram. These text points were numbered in the sequence the student was to follow. The diagrams indicated the presence or absence of air pressure through colour. A colour legend was included to allow the students to decipher this information. A copy of the isolated elements instructions has been included in Appendix 14; a copy of the interacting elements instructions has been included in Appendix 15.

The isolated elements version of the instructions provided the student with a rudimentary outline of the purpose and sequence of events in the pneumatic circuit. The instructions contained a basic explanation of the components of the pneumatic circuit and an outline of the events occurring in the circuit; for example, information such as: "An air pilot line connects the control valve and the Pushbutton valve" provides an explanation of the circuit components; while "Air flows into the chamber of the single acting cylinder from the control valve's air source. The cylinder piston extends" provides information on the sequence of events occurring in the circuit. The order of events in the circuit was also indicated by the presentation of the explanatory text in chronological point form.

The interacting elements version of the instructions contained a detailed explanation of the circuit components and a description of the cause and effect relations involved in the system. In comparison to the isolated elements version of the instructions, the majority of the textual points in the interacting elements version dealt with several elements of the circuit (including details of their complex structure) and their relations to each other; for example, in reference to the control valve, the isolated version of the instructions explains, "The control valve rests in the *Off* position so no air flows through the valve". In contrast, the interacting elements version states "In the *Off* position, Port 2(A) bleeds air from the cylinder chamber through the valve to port 3(R) where it is expelled to the atmosphere". It is

clear from this example the complexity of explanation is increased in the interacting elements version through an increase in the number of elements described in each text point (the internal structure of the valve is also referred too) but also because this version describes how the elements (the cylinder and the valve) relate in the pneumatic system. Some points of the explanation, however, were identical; for example, an initial statement: "The pneumatic circuit is in the initial rest position with the cylinder piston rod retracted" was the same in both versions.

Cognitive load was measured in two ways: through the time required for instruction and through subjective ratings of task difficulty (see Paas & van Merrienboer, 1993: 1994). The subjective mental load rating measure used was a nine point Likert type scale. Above the scale was the instruction "How easy or difficult was it to learn and understand the pneumatic circuit from the instructions you were given?". Each number on the scale was labelled with a description ranging from extremely easy (1) to extremely difficult (9). A copy of the subjective mental load rating scale used in the experiment has been included in Appendix 16.

The test material consisted of a fault finding task. The task consisted of a diagram of a pneumatic circuit similar to the one used in the instructional phase of the experiment. The major difference between these circuits, however, was the presence of eight faults within the pneumatic system used in the fault finding task. The students were asked to identify the system faults, thus, the task aimed to identify the students who held a complete knowledge of how the pneumatic circuit functioned and was connected. The students were asked to mark the faults they had identified in the circuit on the diagram and to provide an explanation why each of the components they had identified was faulty.

There were two categories of faults in the pneumatic circuit used in the fault finding task: those tapping low element interactivity knowledge about the circuit and those tapping high element interactivity knowledge about the circuit and pneumatic

systems. A low element interactivity fault indicated a fault which, in the circuit's current state of work, did not impact upon the ability of the circuit to function; for example, the four low element interactivity faults involved misplaced internal valve and cylinder components. High element interactivity faults in the fault finding task were more directly concerned with the flow of air through the system. The high element faults involved more fundamental changes to the system that would effect the ability of the pneumatic circuit to function in its current state; for example, the faults involved aspects of the system such as: an incorrect connection point between the cylinder and the rest of the system; an extra air source; an incorrect connection between a valve and the source of compressed air and an incorrect connection between two valves. These types of faults would not allow the circuit to run normally.

Each of the test items in the fault finding task was designed to be objective and could be marked as either correct or incorrect. There were four faults in each of the categories. Each fault was worth one mark. Therefore, both the low element interactivity fault category and the high element interactivity fault category were scored out of a total of four marks. A copy of the fault finding task used in the experiment has been included in Appendix 17.

Procedure

Students were matched for apprenticeship type (i.e. electrical, mechanical, dual apprenticeship, fitter-turner, tool maker) and experience with pneumatics before being randomly assigned to either the isolated-interacting elements instruction group or the interacting elements only group. There were twenty students in the interacting elements only group and twenty two students in the isolated-interacting elements instruction group.

The students were informed that they were to be given instructional material in regard to a pneumatic circuit which would be followed by a written test and that this session was to be repeated within forty - eight hours. All students were tested individually, in a quiet room. Students filled in an information sheet. Any students who did not meet the student criteria (see criteria outlined in the participants section) were eliminated at this stage. Students were able to ask questions at any time during the experiment.

In the first experimental session, the students in the isolated-interacting elements instruction group studied the isolated elements instructional booklet while the students in the interacting elements only group were given the interacting elements instructional booklet. Students were allowed an unlimited time to read through the instructional material; this time was recorded by the experimenter. At the completion of the study phase, the students were provided with a subjective mental load rating scale, the format of which was explained to both groups. They were asked to rate the mental effort they had expended to understand the pneumatic circuit and to translate this to a numerical value.

The test section of the experiment followed. The students were asked to complete the written test, which was common to both groups. The instructional materials were not available to the students during this test period. At all stages of the test period students were prohibited from reattempting questions after they had answered them and from reviewing previously completed test answers.

The second phase of the experiment was held within forty-eight hours of the first phase. The procedure of the second phase of the experiment was identical to the first, except that all students studied the interacting elements instructions. To summarise, over both periods of instruction, the isolated-interacting elements group studied the isolated elements instructional material followed by the

interacting elements instructional material. The interacting elements only group was presented with the interacting elements instructional material twice.

Results and Discussion

The variables under analysis were: fault finding test score, divided into high element interactivity fault score and low element interactivity fault score; mental load rating and instruction time. Means and Standard Deviations for these variables are provided in Table 7.1.

Phase 1	Max score	Isolated-interacting elements group (n=22)		Interacting elements only group (n=20)	
		Mean	Std Dev	Mean	Std Dev
Low element fault score	100%	20.45	21.32	11.25	20.64
High element fault score	100%	39.77	33.33	41.25	23.33
Mental Load	9	4.50	1.57	4.75	1.52
Instruction Time	-	207.59	78.32	380.00	115.96
Phase 2					
Low element fault score	100%	19.32	23.06	17.50	24.47
High element fault score	100%	61.36	25.27	47.50	24.20
Mental Load	9	4.00	1.69	3.90	1.65
Instruction Time	-	277.86	97.13	295.50	109.36

Table 7.1: Test Score, Instruction Time (in seconds) and Mental Load Ratings for Experiment 5.

Low and high element interactivity fault scores were analysed separately by a 2 (Instructional Condition) x 2 (Test Phases) ANOVA with repeated measures on the second factor. The results for low element fault finding score suggested there was no difference between the instructional conditions, $F(1,40) = 0.85$, $MSe = 1.20$. There was no indication of a difference between the test phases, $F(1,40) = 0.54$,

MSe = 0.41. There was no significant interaction between the factors, F(1,40) = 1.12, MSe = 0.41.

The results for high element interactivity fault finding scores did not indicate a difference between the instructional groups, F(1,40) = 0.66, MSe = 1.95. A difference was found between the test phases however, F(1,40) = 17.13, MSe = 0.38, demonstrating an improvement in performance over time. The interaction between the factors was significant, F(1,40) = 5.20, MSe = 0.38. In order to assess the interaction effect, a test of simple effects was carried out at each phase. At phase 1, no difference was indicated between the instructional conditions, t(40) = 0.16. At Phase 2, however, results indicated a significant difference between the instructional groups, t(40) = 1.81, with the isolated-interacting elements instructional condition performing at a superior level.

Subjective mental effort scores were analysed in 2 (Instructional Condition) x 2 (Test Phases) ANOVA with repeated measures on the second factor. Results indicated no difference between the instructional groups, F(1,40) = 0.03, MSe = 4.78. The test phases main effect indicated a significant difference, F(1, 40) = 23.30, MSe = 0.40, demonstrating a reduction in the subjective mental effort rating of both groups over the two instructional phases. There was no significant interaction, F(1,40) = 1.60, MSe = 0.40.

The time taken by the students during instruction was analysed in 2 (Instructional Condition) x 2 (Test Phases) ANOVA with repeated measures on the second factor. Results indicated a significant difference between the instructional groups in the time taken for instruction over both phases, F(1,40) = 16.45, MSe = 11506.33. This result suggested that the isolated-interacting elements instruction group required significantly less total instruction time. The results for instruction time did not indicate a difference in test performance between the test phases, F(1,40) = 0.028, MSe = 8734.44. The interaction between both factors was significant,

F(1,40) = 14.36, MSe = 8734.44. A test of simple effects was carried out at each phase in order to assess the interaction. At Phase 1, a significant difference between the instruction groups was found, t(40) = 5.69; this result indicates that the isolated-interacting elements group took less time to study the instructions at phase 1 than the interacting elements only group. It should be noted that the groups were studying the different versions of the instructions during this phase. At Phase 2, when the groups studied the same instructions, no difference was indicated between the groups, t(40) = 0.56. An inspection of the means indicated, however, that the isolated-interacting elements group required less instruction time at both time phases (see Table 7.1). This result was achieved despite Phase 2 being the first exposure of the isolated-interacting elements group to the interacting elements version of the instructions.

Instruction condition efficiency measures were calculated according to the method devised by Paas and Van Merrienboer (1993; 1994). Variables investigated included the condition efficiency measures for high and low element interactivity fault finding question scores. Table 7.2 shows the mean and standard deviations for normalised (Z) mental effort and performance scores and the condition efficiency scores for each group.

	Isolated-interacting elements group				Interacting elements only group			
Phase 1	Z		Efficiency		Z		Efficiency	
	X	S.D.	X	S.D.	X	S.D.	X	S.D.
Low element fault score	0.14	0.96	0.01	1.00	-.25	0.88	-.36	1.04
High element fault score	-.28	1.20	-.29	1.14	-.21	0.80	-.33	1.06
Mental Load	0.13	0.97	-	-	0.26	0.90	-	-
Phase 2								
Low element fault score	0.09	1.04	0.19	1.03	0.01	1.04	0.16	1.17
High element fault score	0.49	0.91	0.47	1.07	0.00	0.83	0.15	1.08
Mental Load	-.18	1.05	-	-	-.22	0.97	-	-

Note: The efficiency scores are based on the normalised mental effort scores given in Table 7.1 for each group for each learning phase.

Table 7.2: Normalised (Z) scores and Condition Efficiency Scores for Performance and Mental Load for Experiment 5.

Instruction condition efficiency scores for each question type were calculated and analysed separately in 2 (Instructional Condition) x 2 (Test Phases) ANOVA with repeated measures on the second factor. Results for low element interactivity fault finding efficiency scores indicated no difference between the instructional conditions in terms of performance efficiency, $F(1, 42) = 0.45$, $MSe = 1.86$. Results

indicated a significant difference between the test phases, $F(1, 42) = 6.80$, $MSe = 0.39$, suggesting an improvement in performance efficiency over time for both groups. There was no interaction between the factors, $F(1, 42) = 1.61$, $MSe = 0.39$. These results are represented in Figure 7.1.

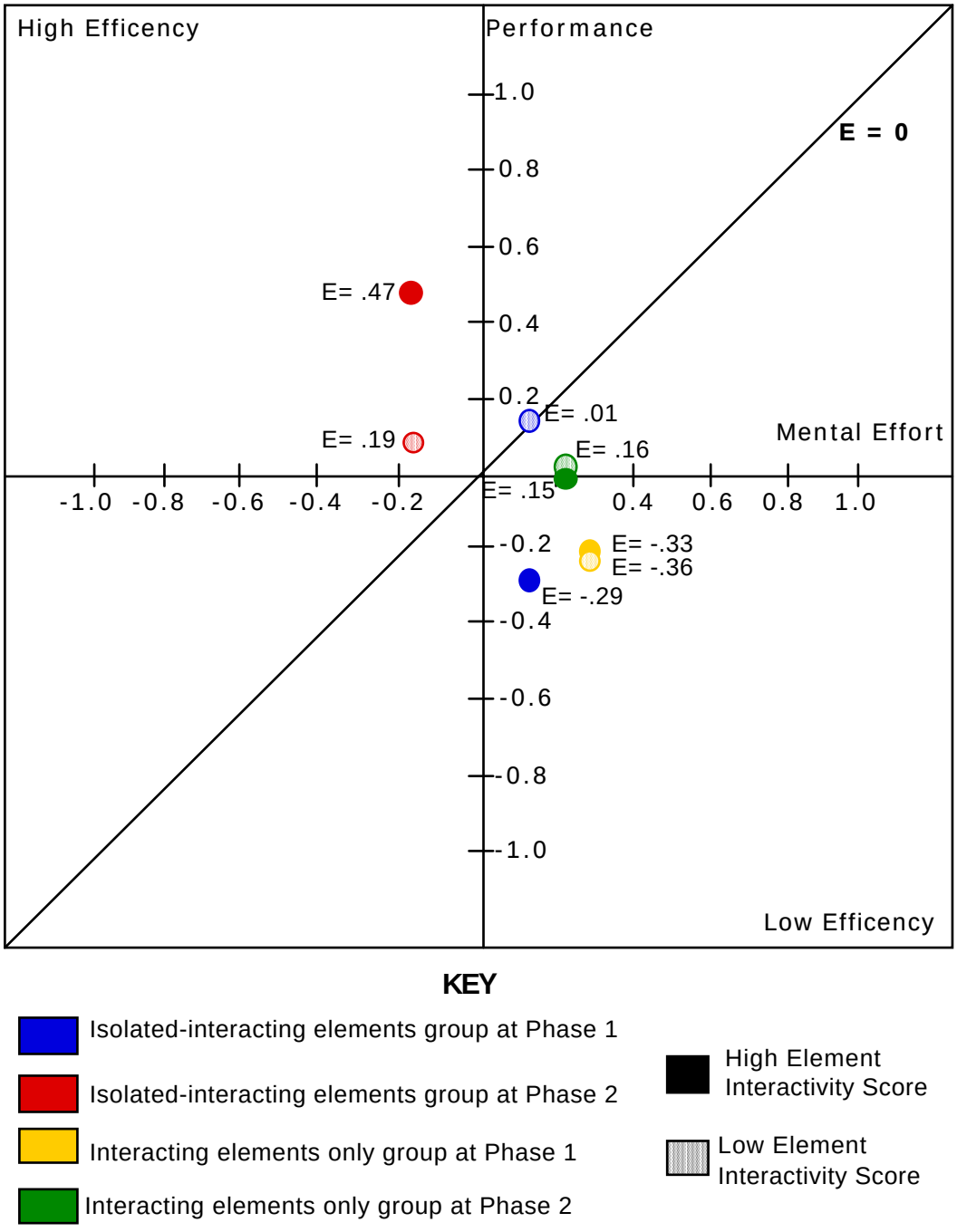


Figure 7.1: Relative instruction condition efficiency as a function of performance and mental effort ratings for Experiment 5.

The results for high element interactivity fault finding efficiency scores indicated no difference between the instructional conditions, $F(1, 42) = 0.34$, $MSe = 2.13$. A significant difference was indicated between the test phases indicating an improvement in high element interactivity fault finding performance efficiency over time, $F(1, 42) = 36.85$, $MSe = 0.23$. There was no significant interaction between the factors, $F(1, 42) = 1.91$, $MSe = 0.23.2$. These results are also represented in Figure 7.1 above.

Instruction condition efficiency measures were calculated by combining instruction time and performance scores in the manner devised by the author which is derived from the methodology of Paas and Van Merrienboer (1993; 1994). Efficiency measures for both low and high element interactivity fault finding questions were calculated using the instruction time/performance methodology and analysed separately in 2 (Instructional Condition) x 2 (Test Phases) ANOVAs with repeated measures on the last factor. Table 7.3 shows the mean and standard deviations for normalised (Z) instruction time and performance scores and the condition efficiency scores for each group.

	Isolated-interacting elements group				Interacting elements only group			
Phase 1	Z		Efficiency		Z		Efficiency	
	X	S.D.	X	S.D.	X	S.D.	X	S.D.
Low element fault score	0.14	0.96	0.59	0.85	-.25	0.88	-.68	0.88
High element fault score	-.28	1.20	0.29	1.07	-.21	0.80	-.66	0.65
Instruction Time	-.69	0.67	-	-	0.72	0.98	-	-
Phase 2								
Low element fault score	0.09	1.04	0.12	1.08	0.01	1.04	-.03	1.07
High element fault score	0.49	0.91	0.41	0.87	0.00	0.83	-.04	0.95
Instruction Time	-.09	0.83	-	-	0.06	0.89	-	-

Note: The efficiency scores are based on the normalised instructional time scores given in Table 7.1 for each group for each learning phase.

Table 7.3: Normalised (Z) scores and Condition Efficiency Scores for Performance and Instructional Time (in seconds) for Experiment 5.

The results for high element interactivity fault finding time performance efficiency scores indicated a significant difference between the instructional conditions, $F(1, 42) = 8.98$, $MSe = 1.20$. An improvement in high element interactivity fault finding performance efficiency was found over time, $F(1, 42) = 7.29$, $MSe = 0.40$. A

possible interaction between the factors was suggested, $F(1, 42) = 3.27$, $MSe = 0.40$, $p=0.078$. These results are represented in Figure 7.2.

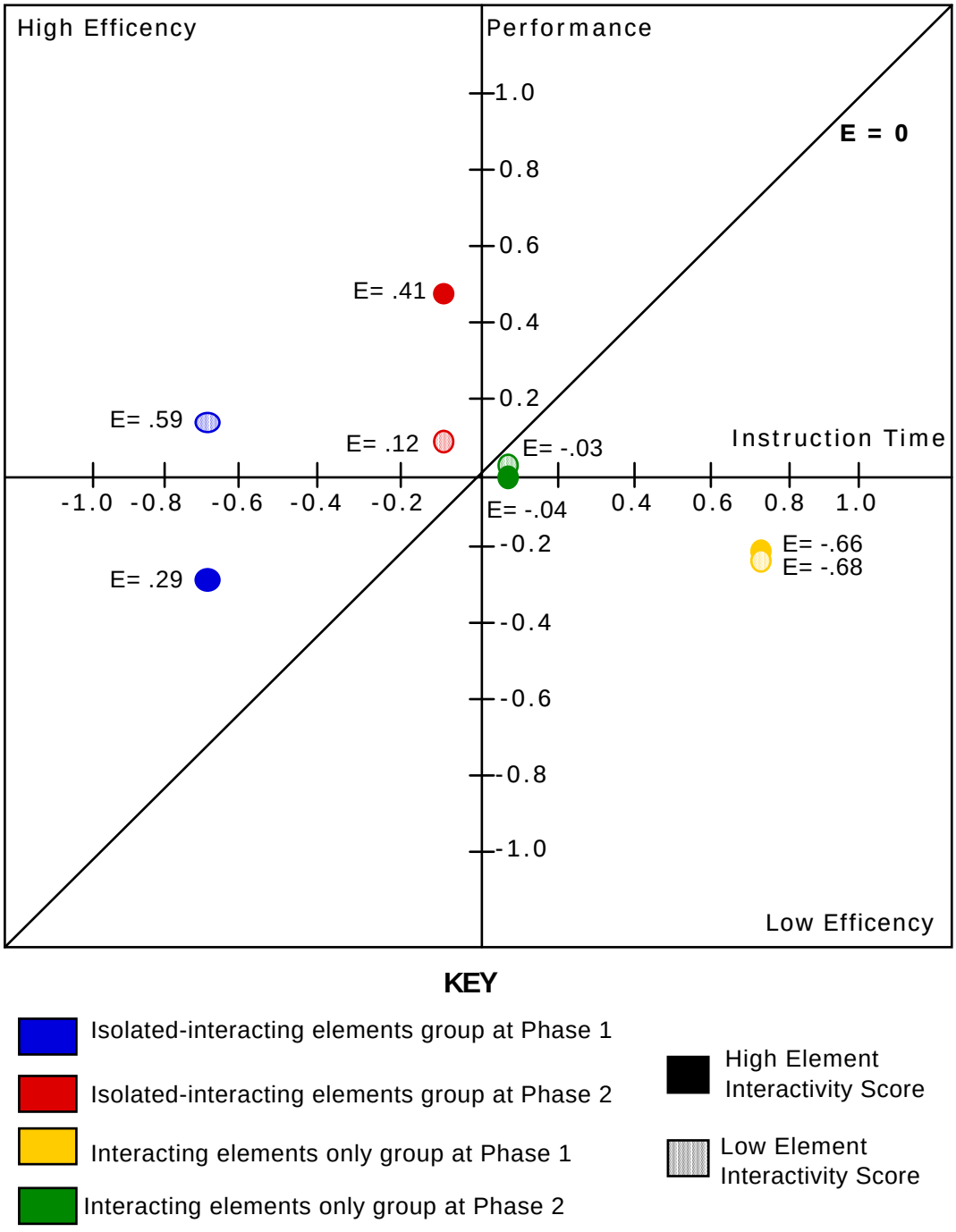


Figure 7.2: Relative instruction condition efficiency as a function of performance and instruction time for Experiment 5.

Results for low element interactivity fault finding time performance efficiencyscores suggested a significant difference between the instructional conditions in terms of performance efficiency, $F(1, 42) = 8.56$, $MSe = 1.31$, favouring the isolated-interacting elements instruction group. The test phases main effect did not indicate a difference, $F(1, 42) = 0.30$, $MSe = 0.59$. A significant interaction was indicated between the factors, $F(1, 42) = 11.49$, $MSe = 0.59$. In order to assess the interaction effect, a test of simple effects was carried out at each phase. At Phase 1, a significant difference was found between the instructional groups, $t(42)= 4.86$ with the isolated-interacting elements group showing superior performance efficiency. At Phase 2, no difference was indicated between the groups, $t(42)= -0.49$. These results are also represented in Figure 7.2 above.

As predicted, the results of Experiment 5 do replicate and support the findings of Experiments 1 and 3. By the end of training, students in the isolated-interacting elements method of instruction displayed a greater comprehension of the system at the most meaningful level of understanding. This superior performance was achieved using significantly less total training time and despite only a single exposure to the interacting elements explanation of the system. In comparison, the interacting elements only group had repeated exposure to the full, complex concept through the interacting elements version, yet, this reduced rather than increased their performance on the most difficult questions. A comparison of the groups in terms of performance efficiency when this measure is assessed as a factor of performance per unit of study time indicated the isolated-interacting elements instructional condition was a superior instructional technique.

The importance of Experiment 5's results, however, go beyond a confirmation of the "isolated-interacting elements instruction method" effect. The majority of the students included in this experiment had participated in either Experiment 2 or 4. Results of Experiments 2 and 4 suggested these particular students had gained no

benefit from the use of the isolated-interacting elements method of instruction; yet the results of this experiment, using the same learners, suggested the opposite.

The difference in the success of the isolated-interacting elements method of instruction across these experiments creates direct evidence for the importance of the level of prior knowledge in the success of the isolated-interacting elements instruction method. This method of instruction benefits learners with little experience in an area because it permits students who have not acquired the critical schemas needed to permit processing of material in working memory, to begin to do so. Participants in Experiments 2 and 4 did not benefit from this type of instruction because in the subject areas chosen for the experiment the students already possessed sufficient schematic knowledge to process the material in working memory. The key difference in Experiment 5 was that the learners were inexperienced in the selected field and so required an isolated elements presentation in order to be able to process the material in working memory. The consequence was improved performance when instructed using the isolated-interacting elements method of instruction.

Chapter 8 General discussion

8.1 The paradox of learning complex concepts

This thesis has investigated what seems to be a paradox of learning: some material which is very high in element interactivity cannot be processed in working memory with understanding until after it has been stored in schematic form in long term memory. The thesis has investigated this paradox using a cognitive load perspective. Cognitive load theory is based on the assumption that features of human cognitive architecture have important implications for how we learn and understand: notably, a limited capacity working memory or processing centre; an ability to store immense amounts of schematically organised knowledge in long term memory, which allows vast quantities of information to act as a single information element, alleviating the processing burden on working memory. Expertise develops as schemas are constructed and automated. The theoretical assumptions underlying the experiments discussed in this paper were derived from this framework.

The structure of the information to be learnt also has critical implications for learning. Information is comprised of a number of elements which the student must learn and understand. If the information elements do not interact, they can be processed separately in working memory with little cognitive burden. However, if the elements forming the information interact extensively, working memory must process the elements simultaneously. Under some circumstances, such as when material is complex and so high in element interactivity, this task may be

problematic given that working memory can only deal with a small number of elements at a time. This difficulty may be mitigated if the learner already possesses a schema for at least some of the information. While understanding of high element interactivity material can follow schema construction because all relevant elements are incorporated in the schema and so can be processed in working memory as a single element, a paradox was evident because it was not clear how the elements that constitute a complex schema could be processed in working memory *before* the schema had been constructed. This paper undertook to investigate an instructional technique, the isolated-interacting elements instructional technique, developed to combat this paradoxical situation.

It was proposed that initially, the element interactivity of complex material had to be artificially reduced in order to enable a schema or partial schema for the information to be developed. That reduction could be accomplished by presenting the material as isolated elements that could be processed in working memory. The consequence of the reduction in element interactivity of material may result in an initial decrease in the student's capacity for understanding. Over the longer term however, it was hypothesised that the promotion of schema construction would lead to an increase in the learner's understanding. The isolated-interacting elements instructional technique had two stages: firstly, students studied instructions detailing only the individual information elements forming a concept; in stage two, learners studied a version of the instruction that expounded all the information elements and how they interact. In comparison, the interacting elements only instruction condition took a more conventional approach to fostering full comprehension by presented all the information for understanding at both phases.

8.2 Summary of results

The experiments reported in this paper provided powerful evidence that for certain groups of learners, information is better learnt through an instructional method that presents isolated elements instruction followed by interacting elements instruction. The first experiment demonstrated that when novice students are required to learn complex information, the isolated-interacting elements method of instruction was superior to the conventional method. Students in the isolated-interacting elements instructional group demonstrated superior performance to the conventional learning group on a knowledge test and practical task. In addition, the isolated-interacting elements group rated their perceptions of the mental effort required to achieve this performance as lower. When the relative efficiencies of the instructional conditions were calculated, the results suggested the isolated-interacting instruction condition was significantly more efficient than the interacting elements only condition both in terms of performance per unit of mental effort and performance per unit of instructional time required.

The finding of performance superiority of students studying in the isolated-interacting elements group was replicated in Experiment 3 using different students who were also novices to the content domain. This experiment used different, more complex content material from the domain of electrical engineering. At phase 2, after both instructional groups had studied the interacting elements instructions in the final training phase, the isolated- interacting group performed better on the most difficult test items that demanded a full understanding of the complex relationships governing the electrical system the students had studied.

In contrast to the results of Experiments 1 and 3, the results of Experiments 2 and 4 seemed to contradict the success of the isolated-interacting elements instructional technique. In Experiments 2 and 4, (using the experimental material

from Experiments 1 and 3 respectively), learners using the conventional full interacting elements only method of instruction performed as well as the isolated-interacting elements instructional group. The contrast of these results to those of Experiments 1 and 3 can be explained by the differing levels of expertise of the learners who had participated in each of the experiments. The learners in Experiments 2 and 4 had more experience in the relevant knowledge domains than the learners from Experiments 1 and 3. It was expected that the isolated-interacting elements method of instruction, which allows schema acquisition in order to facilitate learning and understanding, would be of little use to learners who already possess rudimentary schemas and so may not experience a heavy cognitive load. While an instruction time measure favoured the isolated-interacting elements group due to the reduced amount of instructional material this group had to study in phase 1, the subjective mental effort ratings and performance efficiency measure of the instructional conditions did not differ, providing credence to the explanation this thesis has provided.

The results of Experiment 5 provide significant theoretical support for the claim that expertise is the key factor implicated in the beneficial effects associated with the isolated-interacting elements instructional technique. Participants from Experiments 2 and 4 who had previously not demonstrated any differences between isolated only and isolated-interacting elements instruction conditions, were given materials to study in Experiment 5 from the field of pneumatics. This domain was quite unfamiliar to these learners. In Experiment 5, it was found that students performed better if they had studied the isolated elements version of the instructions followed by the interacting elements version in comparison to repeated exposure to the interacting elements instructions. The achievement of this performance superiority also required less total learning time. The results of Experiment 5 replicated those found in Experiments 1 and 3. The results provide strong theoretical support to the claims of this thesis since the same participants who did not benefit from isolated-interacting elements instruction in a domain in

which their expertise had been identified *a priori* did benefit from this same approach in a domain in which they were novice learners.

The cumulative evidence of the five experiments reported in this thesis suggests an alternative instructional technique can be used to circumvent the paradox of learning about complex concepts. That is, when novice learners are required to learn about complex concepts that are high in element interactivity, they will benefit from isolated elements instruction followed by interacting elements instruction. This method of instruction facilitates the construction of schemas by initially artificially reducing the intrinsic cognitive load of the material to be learnt. Although this restricts and perhaps prohibits understanding from occurring initially, aspects of human cognitive architecture seem to dictate that when the concept to be learnt is high in element interactivity and the learner possesses few if any schemas in a domain then a degree of learning must occur before understanding. Further, the results of Experiments 1, 3 and 5 strongly indicate that by the end of training, students who have studied isolated-interacting elements instruction actually understand more about the complex concept to be learnt than students who have been exposed to all the elements of information repeatedly. The students have gained this knowledge using less mental effort and less total instruction time. The students who studied only the interacting elements managed to understand very little about the concept because their working memories were overwhelmed causing an obstruction to schema acquisition.

It has been further shown in Experiments 2 and 4 that the benefits associated with the isolated-interacting elements instructional technique are limited to novices in a particular content domain. Experts, who by definition already possess schemas in a particular domain, can already hold the interacting elements of the concept in working memory simultaneously and so do not unduly benefit from an instructional technique that facilitates schema development.

8.3 Theoretical and methodological issues arising from the research

8.3.1 Intrinsic cognitive load

It has been previously stated that intrinsic cognitive load is unalterable (Sweller and Chandler, 1994; Sweller, van Merrienboer and Paas, 1998) because this type of cognitive load reflects the internal structure of the material itself, that is, the number of information elements and the degree of interaction between them. This proposition is true, keeping in mind that understanding is an essential concomitant of learning. Nevertheless, while understanding must be one of the ultimate goals of learning, it may not be necessary or indeed, possible, as an intermediate goal. The isolated followed by interacting elements approach suggests that when dealing with very complex information, in order to allow novice students to process interacting elements, the intrinsic cognitive load of the material should initially be artificially reduced. Technically, this is achieved by eliminating the interactions between the information elements. A large degree of interaction between the information elements is the foundation of a concept's intrinsic cognitive load since this interaction requires information elements to be processed simultaneously. If the interactivity between the elements is reduced, discrete information units are essentially created. Serial rather than simultaneous processing of information elements can then be undertaken, substantially reducing working memory load. Students can study the concept at the original level of intrinsic load in a subsequent learning phase of the isolated-interacting elements instructional condition, at which point full understanding becomes possible. The success of the isolated-interacting elements instructional condition is assessed by testing students' understanding of this version of the material.

8.3.2 Relationship to the redundancy effect

The experimental design used in the current experiments bears a close similarity to that used to demonstrate the redundancy effect (Chandler & Sweller, 1991; 1996; Kalyuga, Chandler & Sweller, 1998; Mayer, Bove, Bryman, Mars & Tapangco, 1996; Reder & Anderson, 1980; 1982; Sweller & Chandler, 1994; Yeung, Jin & Sweller, 1998; Kalyuga, Chandler & Sweller, 1998; 1999) which occurs when students presented additional, redundant information perform more poorly than students presented less information. The similarities between the redundancy experiments and the experiments described in this thesis are superficial, however, due to substantial differences in the nature and construction of the experimental material. In the case of redundancy experiments, both the material containing the redundant information and the material with redundancy removed, are fully understandable in their own right. Frequently, the redundant material simply consists of a repetition of the basic information in a different form; for example, a fully intelligible diagram might be redescribed in words (Kalyuga, et al., 1998; Chandler & Sweller, 1991) or a full text may supply detail and background information that a summary condition eliminates (Mayer et al., 1996; Reder and Anderson, 1980; 1982). In the current experiments, the isolated elements information was difficult or impossible to understand because critical relations integral to gaining a holistic understanding of the concept were omitted in order that the information could be processed in working memory. Notwithstanding this distinction, it needs to be born in mind that both the redundancy effect and the results of the current experiments can be explained by assuming that a set of reduced materials can facilitate learning by reducing cognitive load.

8.4 Application of the research & future directions

The results of the research presented in this thesis have significance for instructional design in the traditional classroom or paper based format as well as for more recent educational technologies. Anecdotal evidence suggests that an informal isolated–interacting elements approach to instruction for novices has been utilised in classrooms for many years. The approach described by the experiments presented in this thesis formalises this procedure, yet, it also has a much wider application for the field of education given the rise of multimedia and computer based instruction in this field.

The technology underlying computer based instruction supports and facilitates the use of an isolated-elements instructional design due to the instructional designer's ability to control the sequence of instruction in this medium. Computer designed and presented instruction can easily provide and enforce multiple phases of instruction. During these multiple phases of instruction, the element interactivity of the to-be-learnt concept can be manipulated; for instance, initially the element interactivity of information could be restricted, then the interacting elements version of the instruction could be presented at a point designated either by the instructor or by the learner. The learning outcomes of user control in comparison to instructor control over the ability to access more complex instructional information poses an interesting question for future research. In addition, the ideal sequencing of information from isolated elements to interacting elements would also be worthy of further exploration; for instance, it may be the case that some middle stage of instruction between the isolated elements and the interacting elements versions of the instructions would be beneficial to learners' understanding of the concept.

The results presented in this thesis have further application to instruction presented and designed on computer. The results suggested that students with some knowledge in a domain are able to cope with the interacting elements version of the instructions immediately. Computer based instruction has the ability to discriminate between learners through posing some type of knowledge assessment before a concept is presented. Different paths of instruction could be made available depending on the learner's results on the assessment. One version could be based on the interacting elements only instructional technique reported in this thesis. The concept to be learnt about would be presented in its full complexity. This path would be for the relatively more expert learners in the domain. Another learning pathway could be provided for inexperienced learners; this would be based upon the isolated-interacting instructional technique presented in this thesis and would contain multiple phases of instruction where information presented about a concept is increased in complexity across the phases.

Future research could also investigate other potential measures of cognitive load. In the current experiments, cognitive load was assessed through subjective ratings of mental effort and time taken for instruction. It has been suggested (Bannert, 2000, pers. comm.) that a record of the number and content of students' questions may be another fruitful method of assessing the cognitive load imposed during instruction. Although the procedure of the experiments described in this thesis allowed for and responded to students' questions during instruction, no formal data of this measure were collected. It would certainly be of interest in the future to assess whether, as a cognitive load explanation would hypothesise, students in the interacting elements only condition required more support in order to comprehend the instructions than the isolated-interacting group.

8.5 Concluding comments

The conclusion that understanding and learning of very complex information can be facilitated by presenting the information in a manner that initially precludes full understanding is likely to be seen as highly controversial by some. Nevertheless, from a theoretical perspective, there is a degree of inevitability about this conclusion. We know that: (a) working memory is very limited; (b) complex material involves more interacting elements than can be processed simultaneously in working memory unless they have been incorporated into a schema; (c) understanding requires all interacting elements to be processed simultaneously in working memory. If these three propositions are valid, understanding cannot occur until after appropriate schemas have been constructed. Instructional designs intended to encourage students to understand very complex material prior to the construction of appropriate schemas will fail. The experiments of this paper provide empirical support for these conclusions.

Reference List

Adelson, B. (1981) Problem Solving and the development of abstract categories in programming languages. *Memory and Cognition*, 9, 422-433.

Akin, O. (1980). *Models of architectural knowledge.* London: Pion.

Allport, A.D., Antonis, B., & Reynolds,P. (1972). On the division of attention: A disproof of the single channel hypothesis. *Quarterly Journal of Experimental Psychology*, 24, 225-235.

Anderson, J. R. (1985) *Cognitive psychology and its implications*. (2nd edition) New York : Freeman.

Anderson, J.R. (1980). *Cognitive psychology and its implications*. San Francisco: Freeman.

Anderson, J.R. (1983). *The architecture of cognition*. Cambridge, M.A: Harvard University Press.

Anderson, R. C., Reynolds, R. E., Schallert, D. L. and Goetz, E. T. (1977) Frameworks for Comprehending Discourse. *American Educational Research Journal*, 14(4), 367-381.

Anzai, Y. and Simon, H. A. (1979) The theory of learning by doing. *Psychological Review*, 86, 124-180.

Atkinson, R. C., & Shiffrin, R. M. (1968). Human memory: A proposed system and its control processes. In K. W. Spence & J. T. Spence (Eds.) *Advances in the psychology of learning and motivation research theory*, Vol. 2. New York: Academic Press.

Atkinson, R.K., Derry, S.J. Renkl, A. & Wortham, D. (in press) Learning from examples: Instructional principles from the worked examples research. *Review of Educational Research*.

Ausubel, D.P. (1968a). *Educational Psychology: A Cognitive View.* New York: Holt, Rinehart and Winston, Inc.

Ausubel, D.P. (1968b) A subsumption theory of meaningful verbal learning and retention. From R.G. Kuhlen (ed.) *Studies in Educational Psychology* (pp.167-174). Boston: Blaisdell Publishing Co.

Ausubel, D.P. (1977) The facilitation of meaningful verbal learning in the classroom. *Educational Psychologist*, 12, 162-178.

Ayres, P. (1993) Why goal free problems can facilitate learning. *Contemporary Educational Psychology*, 18, 376-381.

Baddeley, A. (1994). The magical number seven: Still magic after all these years. *Psychological Review*, 101(2), 353-356.

Baddeley, A. (1992) Working Memory. *Science* ; 255, 556-559.

Baddeley, A.D. (1990) *Working memory: Human memory, theory and practice.* United Kingdom. Lawrence Erlbaum Associates.

Baddeley, A. (1981) The concept of working memory: A view of its current state and probable future development. *Cognition*, 10, 17-23.

Baggett, P. (1975) Memory for explicit and implicit information in picture stories. *Journal of Verbal Learning and Verbal Behaviour*, 14, 538-548.

Ballard, P.B. (1915) Norms of performance in the fundamental processes of arithmetic. *Journal of Experimental Pedagogy*, III, 9-20.

Bartlett, F. C. (1932/1995) *Remembering: A study in experimental and social psychology.* Cambridge: Cambridge University Press.

Bloom, B. (1956). (Ed.) *Taxonomy of Educational Objectives: Cognitive Domain.* NY: McKay.

Bower, G. H., Black, J. B. and Turner, T. J. (1979) Scripts in memory for text. *Cognitive Psychology*, 11, 177-220.

Bransford, J. D. and Franks, J. J. (1971) The abstraction of linguistic ideas. *Cognitive Psychology*, 2, 331-350.

Bransford, J. D. and Johnson, M K. (1972) Contextual Prerequisites for Understanding: Some investigations of comprehension and recall. *Journal of Verbal Learning and Behaviour*, 11, 717-726.

Bransford, J. D., Barclay, J. R. and Franks, J. J. (1972) Sentence memory: A constructive versus interpretive approach. *Cognitive Psychology*, 3, 193-209.

Brownell, W.A & Moser, H.E. (1949) *Meaningful vs. Mechanical learning: A study in grade III subtraction*. Durham, North Carolina: Duke University Press.

Bruner, J. S., Goodnow, J. J. and Austin, G. A. (1956) *A Study of Thinking*. New York: John Wiley.

Bruning, R. H., Schraw, G. J., and Ronning, R. R. (1995) *Cognitive Psychology and Instruction* (2nd Ed). Englewood Cliffs, N. J.: Merrill.

Butler, B. E. (1974) The limits of selective attention in tachistospcoic recognition. *Canadian Journal of Psychology*, 28, 199-213.

Carmichael, L. L., Hogan, H. P., and Walter, A. A. (1932) An experimental study of the effect of language on the reproduction of visually perceived form. *Journal of Experimental Psychology*, 15, 73-86.

Cerpa, N., Chandler, P., & Sweller, J. (1996). Some conditions under which integrated computer based training software can facilitate learning. *Journal of Educational Computing Research* ,15, 345-367.

Chandler, P., & Sweller, J. (1996). Cognitive load while learning to use a computer program. *Applied Cognitive Psychology*, 10,151-170.

Chandler, P., & Sweller, J. (1992) The split attention effect as a factor in the design of instruction. *British Journal of Education Psychology*, 62, 233-246.

Chandler, P., & Sweller, J. (1991) Cognitive load theory and the format of instruction. *Cognition and Instruction*, 8, 293-332.

Charness, N. (1976). Memory for chess positions: Resistance to interference. *Journal of Experimental Psychology: Human Learning and Memory,* 2, 641-653.

Charness, N. (1979) Components of skill in bridge. *Canadian Journal of Psychology*, 33, 1-16.

Charness, N. (1992) The impact of chess on cognitive science. *Psychological Research,* 54(4), 4-9.

Chase, W.G., & Simon, H.A. (1973a). Perception in chess. *Cognitive Psychology,* 4, 55-81.

Chase, W.G., & Simon, H.A. (1973b). The mind eye in chess. In W.G. Chase (Ed.), *Visual Information Processing*. New York : Academic Press.

Chi, M. (1978). Knowledge structures and memory development. In R. S. Seigler (Ed.), *Children's Thinking: What Develops?* Hillsdale, New Jersey. Lawrence Erlbaum Associates.

Chi, M. T. H., Feltovich, P. J., and Glaser, R. (1981) *Categorization and representation of physics problems by experts and novices. Technical Report No. 4 (18th May, 1981).* Pittsburgh: University of Pittsburgh, Learning Research and Development Center.

Chi, M., Glaser, R., & Rees, E. (1982). Expertise in problem solving. In R. Stenberg (Ed.), *Advances in psychology of human intelligence*. Hillsdale, New Jersey: Lawrence Erlbaum Associates.

Christensen, E. E., Murry, R. C., Holland, K., Reynolds, J., Landy, M. J.& Moore, J. G. (1981) The effect of search time on perception. *Radiology*, 138, 361-365.

Clement, J. (1991) Nonformal reasoning in experts and science students: The use of analogies, extreme cases, and physical intuition. In J. F. Voss, D. N. Perkins and J. Segal (Eds.), *Informal Reasoning and Education* (pp. 345-362). Hillsdale, N. J.: Lawrence Erlbaum Associates Inc.

Cook, L. E. and Mayer, R. E. (1988). Teaching readers about the structure of Scientific Text. *Journal of Educational Psychology*, 80(4), 448-456.

Cooper, G., & Sweller, J. (1987). Effects of schema acquisition and rule automation on mathematical problem solving transfer. *Journal of Educational Psychology*, 79, 347-362.

Craik, F. I. M. and Lockhart, R. S. (1972) Levels of Processing: A framework for memory research. *Journal of Verbal Learning and Verbal Behaviour*, 11, 671-684.

Darwin, C. J., Turvey, M. T. and Crowder, R. G. (1972) An auditory analogue of the Sperling partial report procedure: Evidence for brief auditory storage. *Cognitive Psychology*, 3, 255-267.

DeGroot, A. D. (1965). *Thought and Choice in Chess*. The Hague : Mouton.

DeGroot, AD. (1966). Perception and memory versus thought : Some old ideas and recent findings. In B. Kleinmuntz (Ed.) *Problem Solving*. New York: Wiley.

Detweiler, M. and Schneider, W. (1991) Modelling the acquisition of dual task skill in a connectionist/control architecture. In D. L. Dumas (Ed.), *Multiple-Task Performance.* London: Taylor & Francis.

Dochy, F., Segers, M. & Buehl, M.M. (1999) The relation between assessment practices and outcomes of studies: the case of research on prior knowledge. *Review of Educational Research*, 69, 145-186.

Egan, D. E., & Greeno, J. G. (1974) Theories of rule induction: knowledge acquired in concept learning, serial pattern learning and problem solving. In W. G. Gregg (Ed.). *Knowledge and Cognition*. Hillsdale, New Jersey: Lawrence Erlbaum Associates.

Egan, D. E., & Schwartz, B. J. (1979). Chunking in recall of symbolic drawings. *Memory and Cognition*, 7*,* 149-158.

Ellis, H. C. and Hunt, R. R. (1989) *Fundamentals of Human Memory and Cognition (4th ed)*. Dubuque, Iowa: Wm. C. Brown.

Ericsson, K. A. and Kintsch, W. (1995) Long Term Working Memory. *Psychological Review*, 102 (2), 211-245.

Ericsson, K. A., Chase, W. G. and Faloon, S. (1980) Acquisition of a memory skill. *Science*, 208, 1181-1182.

Ericsson, K.A. and Polson, P.G. (1988) An experimental analysis of the mechanisms of memory skill. *Journal of Experimental Psychology: Learning, Memory and Cognition,* 14(2), 305-316.

Eysenck, M. W. and Keane, M. T. (1990) Cognitive psychology: A Student's handbook. Hillsdale, New Jersey: Lawrence Erlbaum Associates.

Fisk, A., & Schneider,W. (1984). Memory as a function of attention, level of processing and automatization. *Journal of Experimental Psychology : Learning, Memory and Cognition*, 10, 181-197.

Frey, P. W. and Adesman, P. (1976) Recall memory for visually presented chess positions. *Memory and Cognition*, 4, 541-547.

Freyhof, H., Gruber, H. and Ziegler, A. (1992) Expertise and hierarchal knowledge representation in chess. *Psychological Research*, 54, 32-37.

Frick, R. (1984) Using an auditory and a visual short-term store to increase digit span. *Memory and Cognition*, 12, 507-514.

Gabrys, G., Weiner, A., and Lesgold, A. (1993) Learning by problem solving in a coached apprenticeship system. In M. Rabinowitz, (Ed.), *Cognitive Science: Foundations of Instruction*. Hillsdale, N.J.: Lawrence Erlbaum Associates Inc.

Gagne, R. (1970). *The Conditions of Learning* (2nd ed.). NY: Holt, Rinehart & Winston.

Gagne, R.M. (1973). Learning and Instructional Sequence. *In* Fred N. Kerlinger (Ed) *Review of Research in Education* (Vol. 1, pp. 3-33). Itasca, Illinois: F.E. Peacock (A publication of the American Educational Research Association).

Gick, M. L. (1986) Problem-Solving Strategies. *Educational Psychologist*, 21, 99-120.

Gick, M. L., & Holyoak, K. J. (1983). Schema induction & analogical transfer. *Cognitive Psychology*, 15,1-38.

Glaser, R. and Chi, M. T. H. (1988). Overview. In M. T. H. Chi, R. Glaser, M. J. Farr (Eds), *The Nature of Expertise* (pp. xv - xxviii). Hillsdale, N. J.: Lawrence Erlbaum Associates Inc.

Gobet, F. and Simon, H. A. (1996) Templates in Chess memory: A Mechanism for recalling several boards. *Cognitive Psychology*, 31, 1-40.

Godden, D. R. & Baddeley, A. D.(1980) When does context influence recognition memory? *British Journal of Psychology*, 71, 99-104.

Gopher, D. & Braune, R. (1984) On the psychophysics of workload: Why bother with subjective measures? *Human Factors*, 26, 519-532.

Green, A. J. K. and Gilhooly, K. J. (1992) Empirical advances in expertise research. in M. T. Keane and K. J. Gilhooly (Eds.), *Advances in the Psychology of Thinking (Vol. 1)*. (pp45-70) New York: Harvester Wheatsheaf.

Greeno, J. G. (1973) The structure of memory and the process of solving problems. In R.L. Solso (ed) Contemporary Issues in Cognitive Psychology: The Loyola Symposium (pp103-133). Washington, DC: V. H. Winston and Sons.

Greeno, J. G. (1978) Natures of problem Solving Abilities. In W. K. Estes (Ed.) *Handbook of learning and cognitive processes (Vol. 5)*. Hillsdale, New Jersey: Lawrence Erlbaum Associates.

Groen, G. J. & Patel, V. L. (1988) The relationship between comprehension and reasoning in medical expertise. In M. T. H. Chi, R. Glaser, M. J. Farr (Eds) *The Nature of Expertise.* (pp. 287-310) Hillsdale, N. J.: Lawrence Erlbaum Associates Inc.

Haberlandt, K. (1994) *Cognitive Psychology*. Boston: Allyn and Bacon.

Halford, G., Maybery, M., and Bain, J. (1986). Capacity limitations in children's reasoning: A dual task approach. *Child Development,* 57*,* 616-627.

Halford, G.S., Wilson, W.H. and Phillips, S. (1998) Processing capacity defined by relational complexity: Implications for comparative, developmental, and cognitive psychology. *Behavioural and Brain Sciences*, 21, 803-864.

Hardiman, P. T., Defresne, R. and Mestre, J. P. (1989) The relation between problem categorization and problem solving among experts and novices. *Memory and Cognition*, 17(5), 627-638.

Hastie, R. (1981) Schematic principles in human memory. In E. T. Higgins, C. D. Herman and M. P. Zanna (Eds.) *Social Cognition: the Ontario Symposium.* Hillsdale, N. J.: Lawrence Erlbaum Associates Inc.

Hinsley, D. A., Hayes, J. R. and Simon, H. A. (1977) From Words to Equations: Meaning and Representation in Algebra Word Problems. In M. A. Just and P. A. Carpenter (Eds.), *Cognitive Processes in Comprehension* (pp33-62). Hillsdale, New Jersey: Lawrence Erlbaum Associates.

Hoosain, R. (1983). Memorization of Classical Chinese. *Psychologia: An International Journal of Psychology in the Orient*, *26*(3), 193-197.

Howard R.W. (1987) *Concepts and Schemata: An Introduction.* London: Cassell.

Jacoby, L. L. and Witherspoon, D. (1982). Remembering without awareness. Canadian Journal of Psychology, 36, 300-324.

Jefferies, R., Turner, A. T., Polson, P. G., & Atwood, M. E. (1981) Processes involved in designing software. In J. R. Anderson (Ed.), *Cognitive skills and their acquisition.* Hillsdale, N.J. : Lawrence Erlbaum Associates.

Jelsma, O. and van Merrienboer, J.G. (1989) Contextual interference: Interactions with reflection-impulsivity. *Perceptual and Motor Skills*, 68, 1055-1064.

Jeung, H., Chandler, P. & Sweller, J. (1997). The role of visual indicators in dual sensory mode instruction. *Educational Psychology,* 17, 329 -343.

Johnson, M. K., Bransford, J. D. and Solomon, S. (1973) Memory for tacit implications of sentences. *Journal of Experimental Psychology*, 98, 203-205.

Kalyuga, S., Chandler, P., & Sweller. J. (1998) Levels of expertise and instructional design. *Human Factors*, 40, 1-17.

Kalyuga, S., Chandler, P., & Sweller. J. (1999). Managing split attention and redundancy in multi media instruction. *Applied Cognitive Psychology,* 13, 351-371.

King, D.J. & Russell,G.W. (1966). A Comparison of Rote and Meaningful Learning of Connected Meaningful Material. *Journal of Verbal Learning and Verbal Behaviour*, *5*, 478-483.

Kintsch, W. (1977) On comprehending stories. In M. A. Just and P. A. Carpenter (Eds.), *Cognitive Processes in Comprehension* (pp33-62). Hillsdale, New Jersey: Lawrence Erlbaum Associates.

Kintsch, W. (1995) Introduction. To F. C. Bartlett (1932/1995) *Remembering: A study in experimental and social psychology.* Cambridge: Cambridge University Press.

Kotovsky, K., Hayes, J. R., & Simon, H. A. (1985). Why are some problems hard? Evidence from Tower of Hanoi. *Cognitive Psychology,* 17*,* 248-294.

Kundel, H. L. and Nodine, C. F. (1975) Interpreting chest radiographs without visual search. *Radiology*, 116, 527-532.

Laird, J. E., Newell, A., Rosenbloom, P. S. & (1987) SOAR: An architecture for general intelligence. *Artificial Intelligence*, 33, 1-64.

Larkin, J., (1983). The role of problem representation in physics. In D. Genter and A. Stevens (Eds.) *Mental Models* (pp. 75-98) Hillsdale, N. J.: Lawrence Erlbaum Associates Inc.

Larkin, J., (1981) Enriching formal knowledge: A model for learning to solve model textbook physics problems. In J. R. Anderson (Ed.), Cognitive Skills and their acquisition. Hillsdale, N. J.: Lawrence Erlbaum Associates Inc.

Larkin, J., (1979) Information processing models and science instruction. In J. Lochhead and J. Clement (Eds.), *Cognitive Process Instruction*. Philadelphia: Franklin Institute Press.

Larkin, J., McDermott, J., Simon, D. P., & Simon, H. A. (1980a). Expert and Novice Performance in solving physics problems. *Science,* 208 (20th June, 1980), 1335-1342.

Larkin, J., McDermott, J., Simon, D. P, & Simon, H. A. (1980b). Models of competence in solving physics problems. *Cognitive Science,* 11,65-99.

Lesgold, A, Rubinson, H., Feltovich, P., Glaser, R., Klopfer, D., and Wang, Y. (1988) Expertise in a complex Skill: Diagnosing X-Ray pictures. In M. T. H. Chi, R. Glaser, M. J. Farr (Eds.) *The Nature of Expertise.* (pp. 311-342). Hillsdale, N. J.: Lawrence Erlbaum Associates Inc.

Lim, E.L., Dixon, R.S. and Moore, D.W. (1996) Worked examples versus non-goal-specific problems: A test of schema development in geometry. *Educational Psychology*, 16, 421-431.

Logan, G. D. (1988a) Towards an instance theory of automatization. *Psychological Review*, 95(4), 492-527.

Logan, G. D. (1988b) Automaticity, resources, and memory: Theoretical controversies and practical implications. *Human Factors*, 30(5), 583-598.

Low, R., & Over, R. (1992). Hierarchal ordering of schematic knowledge relating to area-of-rectangle problems. *Journal of Educational Psychology,* 84, 62-69.

Luchins, A. S. (1942) Mechanization in problem solving. *Psychological Monographs*, 54, (6, Whole No. 248).

Mandler, J. M. (1984) *Stories, scripts, and scenes: Aspects of schema theory.* Hillsdale, New Jersey : Lawrence Erlbaum Associates.

Mantyla, T. (1986) Optimising cue effectiveness: Recall of 500 and 600 incidentally learned words. *Journal of Experimental Psychology: Learning, Memory and Cognition,* 12, 66-71.

Marcus, N., Cooper, M., & Sweller, J. (1996). Understanding Instructions. *Journal of Educational Psychology*, 88, 49-63.

Mawer, R., & Sweller, J. (1982). The effects of subgoal density and location on learning during problem solving. *Journal of Experimental Psychology: Learning, Memory and Cognition*, 8, 252-259.

Mayberry, M., Bain, G. & Halford, J. (1986). Information processing demands of transitive inference. *Journal of Experimental Psychology: Learning, Memory and Cognition*, 12, 600-613.

Mayer, R.E. (1992) *Thinking, Problem Solving & Cognition.* (2nd Edition). New York : W.H. Freeman and Company.

Mayer, R.E., & Moreno, R. (in press) Aids to computer-based multimedia learning. *Learning and Instruction*.

Mayer, R. E., Bove, W., Bryman, A., Mars, R., & Tapangco, L. (1996) When less is more: Meaningful learning from visual and verbal summaries of science textbook lessons. *Journal of Educational Psychology,* 88, 64-73.

Mayer, R.E., Steinhoff, K., Bowler, G. and Mars, R. (1995) A generative theory of textbook design: Using annotated illustrations to foster meaningful learning of science text. *Educational Technology Research & Development*, 43, 31-43.

McCelland, J. L., Rumelhart, D. E., and Hinton, G. E. (1986) The appeal of parallel distributed processing. In J. L. McCelland, D. E. Rumelhart and the PDP Research Group (Eds.), *Parallel Distributed Processing, Vol 1, Foundations.* Cambridge, Mass: MIT Press.

McKeithen, K. B., Reitman, J. S., Reuter, H. H., & Hirtle, S. (1981). Knowledge organisation and skilled differences in computer programmers. *Cognitive Psychology*, 13, 307-325.

Medin, D. L., Wattenmaker, W. D. & Hampson, S. E. (1987) Family resemblance, conceptual cohesiveness and category construction. *Cognitive Psychology*, 19, 242- 278.

Miller, G.A. (1956a) The magical number seven, plus or minus two: Some limits on our capacity for processing information. *Psychological Review,* 63 ,81-97.

Miller, G.A. (1956b) Information and Memory. *Scientific American,* 195, 42-46.

Miller. W. (1937) The picture crutch in reading. *Elementary English Review,* 14, 263-264.

Moray, N. (1982). Subjective mental workload. *Human Factors*, 24, 25-40.

Mousavi, S., Low, R., & Sweller, J. (1995) Reducing cognitive load by mixing auditory and visual presentation modes. *Journal of Educational Psychology,* 87, 319-334.

Newell, A. (1990) *Unified theories of cognition*. Cambridge, Mass: Harvard University Press.

Newell, A, & Simon, H. (1972). *Human Problem Solving*. Englewood Cliffs, New Jersey: Prentice Hall.

Newell, A., Simon, H.A, & Shaw, J. C. (1965) Elements of a Theory of Problem Solving. In R. C. Anderson and D. P. Ausubel (Eds.) *Readings in the Psychology of Cognition*. New York: Holt, Rinehart and Winston Inc.

Norman, D.A (1970). Introduction: Models of Human Memory. In D. A. Norman (Ed.), *Models of Human Memory* (pp. 1-15). New York: Academic Press.

Norman, D. A., & Rumelhart, D. E. (1970). A System for Perception and Memory. In D. A. Norman (Ed.), *Models of Human Memory* (pp. 19-64). New York: Academic Press.

O'Donnell, R.D., & Eggemeir, F.T. (1986). Workload assessment methodology. In K.R. Boff, L. Kaufman, & J.P. Thomas (Eds.), *Handbook of perception and human performance* (Vol. 2, Chapter 42, pp. 1-49). New York: Wiley.

Overton, D. A. (1985) Contextual stimulus effects of drugs and internal states. In P.D. Balsam and A. Tomie (Eds.), *Context and Learning*. (pp 357-384) Hillsdale, NJ: Lawrence Erlbaum Associates.

Owen, E., & Sweller, J. (1985). What do students learn while solving mathematics problems. *Journal of Educational Psychology*, 77, 272-284.

Paas, F.G.W.C. (1992) Training strategies for obtaining transfer in statistics: A cognitive load approach. *Journal of Educational Psychology*, 84, 429-434.

Paas, F., & Van Merrienboer, J. (1994). Variability of worked examples and transfer of geometric problem-solving skills: A cognitive load approach. *Journal of Educational Psychology*, 86, 122-133.

Paas, F., & Van Merrienboer, J. (1993). The efficiency of instructional conditions: An approach to combine mental-effort and performance measures. *Human Factors*, 35, 737-743.

Patel, V. L. and Groen, G. J. (1986) Knowledge-based solution strategies in medical reasoning. *Cognitive Science*, 10, 91-116.

Pavio, A. (1990) *Mental representations : A dual coding approach.* New York: Oxford University Press.

Perkins, D. N., and Salomon, G. (1989). Are Cognitive Skills Context-Bound? *Educational Researcher*, 18(1), 16-25.

Peterson, L.R., and Peterson, M.J. (1959) Short term retention of individual verbal items. *Journal of Experimental Psychology.* 58, 193-198.

Pichert, J. W. and Anderson, R. C. (1977) Taking different perspectives on a story. *Journal of Educational Psychology*, 69, 309-315.

Pierce, K.A., Duncan, M.K., Gholson, B., Ray, G.E. and Kamhi, A.G. (1993) Cognitive load, schema acquisition, and procedural adaptation in nonisomorphic analogical transfer. *Journal of Educational Psychology*, 85, 66-74.

Pinchus, R. H., Geller, E. B., and Stover, E. M. (1986) A technique for using the story schema as a transition to understanding and summarising event based magazine articles. *Journal of Reading,* 30, 152-158.

Polya, G. (1957). *How to solve it: A new aspect of mathematical method.* Garden City, New York: Doubleday.

Posner, M. I. (1988). Introduction: What is it to be an expert. In M. T. H. Chi, R. Glaser, M. J. Farr (Eds.), *The Nature of Expertise.* (pp. xxix - xxxvi) Hillsdale, N. J.: Lawrence Erlbaum Associates Inc.

Posner, M. I. and Rossman, E. (1965) Effect of size and location of informational transforms upon short term retention. *Journal of Experimental Psychology*, 70, 496-505.

Rabinowitz, M. and Mandler, J. M. (1983) Organisation and information retrieval. *Journal of Experimental Psychology: Learning, Memory and Cognition*, 9, 430 - 439.

Reder, L.M., & Anderson, J.R. (1980) A comparison of text and their summaries: Memorial consequences. *Journal of Verbal Learning and Verbal Behaviour*, 19, 121-134.

Reder, L.M., & Anderson, J.R. (1982) Effects of spacing and embellishment on memory for main points of text. *Memory and Cognition,* 10, 97-102.

Reed, S. K., Ernst, G. W., & Banerji, R. (1974) The role of analogy in transfer between similar problem states. *Cognitive Psychology*, 6, 436-450.

Reif, F. (1995). Millikan Lecture 1994: Understanding and teaching important scientific thought processes. *American Journal of Physics,* 63(1), 17-32.

Reitman, J. S. (1976) Skilled perception in GO: Deducing memory structures from inter response times. *Cognitive Psychology*, 8, 336-356.

Reitman, W. R. (1965) *Cognition and Thought*. New York: John Wiley.

Renkl, A., (1999) Learning mathematics from worked-out examples: Analyzing and fostering self-explanations. *European Journal of Psychology of Education*, 14,477-488.

Renkl, A. (1997) Learning from worked-out examples: A study on individual differences. *Cognitive Science*, 21, 1-29.

Renkl, A., Atkinson, R.K. & Maier, U.H. (2000) From studying examples to solving problems: Fading worked-out solutions steps helps learning. In L. Gleitman & AK. Joshi (Eds.), *Proceedings of the 22nd Annual Conference of the Cognitive Science Society* (p.393-398). Mahwah, NJ: Erlbaum.

Renkl, A., Stark, R., Gruber, H. & Mandl, H. (1998) Learning from worked-out examples: The effects of example variability and elicited self-explanations. *Contemporary Educational Psychology*, 23, 90-108.

Rich, E. (1983) *Artificial Intelligence*. New York: McGraw Hill.

Rosch, E. (1978) Principles of Categorization. In E. Rosch and B. B. Lloyd (Eds) *Cognition and Categorization* (pp. 28-48). Hillsdale, NJ: Lawrence Erlbaum Associates.

Rumelhart D. E. (1984) Schemata and the cognitive system. The effect of pictures on the acquisition of a small vocabulary of similar sight words. *British Journal of Educational Psychology*, 54, 265-275.

Rummelhart, D. E. (1981). Schemata: The building Blocks of Cognition. In R. T. Guthrie (Ed.), *Comprehension and teaching: Research Reviews*. Newark, DE: International Reading Association.

Rumelhart D. E. (1980) Schemata: The Building Blocks of Cognition. In R. C. Anderson, R. J. Spiro, & W. E. Montague (Eds.) *Theoretical Issues in Reading Comprehension* (pp. 33-58). Hillsdale, N. J.: Lawrence Erlbaum Associates.

Rummelhart, D. E., & Norman, D. A. (1978). Accretion, tuning and restructuring. In J. W. Cotton & R. L. Klatzky (Eds.), *Semantic factors in cognition*. Hillsdale, New Jersey: Lawrence Erlbaum Associates.

Rummelhart, D. E., & Ortony, A. (1977). The representation of knowledge in memory. In R. C. Anderson, R. J. Spiro, & W. E. Montague (Eds.), *Schooling and the acquisition of knowledge*. Hillsdale, New Jersey: Lawrence Erlbaum Associates.

Samuels, S. J., Termyson, R., Sax, L., Mulcahy, P., Schermer, N., and Hajovy, H. (1988) Adult's use of a scientific text structure in the recall of a scientific journal article. *Journal of Educational Research*, 81, 171-174.

Schank, R. C., & Abelson, R. (1977). *Scripts, plans, goals and understanding.* Hillsdale, New Jersey : Lawrence Erlbaum Associates.

Schmidt, R.A. (1991) *Motor Learning & Performance: From Principles to Practice.* Champaign, Illinois: Human Kinetics Books.

Schneider, W. (1993) Varieties of working memory as seen in biology and connectionist/control architectures. *Memory and Cognition*, 21, 184-192.

Schneider, W. and Detweiler, M. (1987) A connnectionist/control architecture for working memory. In G.H. Bower (Ed) *The Psychology of Learning and Motivation* (vol 21, pp 54-119). New York: Academic Press.

Schneider, W., & Shiffrin, R. (1977). Controlled and automatic human information processing: I. Detection, search and attention. *Psychological Review*, 84, 1-66.

Schoenfeld, A. H. (1979). Explicit heuristic training as a variable in problem solving performance. *Journal for Research in Mathematics Education,* 10, 173-187.

Schoenfeld, A. H., & Herrmann, D. J. (1982). Problem perception and knowledge structure in expert and novice mathematical problem solvers. *Journal of Experimental Psychology; Learning, Memory & Cognition,* 8,484-494.

Shanteau, J. (1992). How Much Information does an Expert Use? *Acta Psychologia*, 81(1), 75-86.

Shiffrin, R. M.(1993) Short Term memory: A brief commentary. *Memory and Cognition* ; 21(2):193-197.

Shiffrin, R. M. (1988) Attention. In R. C. Atkinson, R. J. Herrnstein, G. Lindzey, R. Duncan Luce (Eds.), *Steven's Handbook of Experimental Psychology: Volume 2: Learning and Cognition* (2nd ed) (pp739-811). New York: John Wiley.

Shiffrin, R. M., Dumais, S. T., Schneider, W. (1981) Characteristics of automation. In J. Long and A. Baddeley (Eds.), *Attention and Performance IX* (pp. 223-238). Hillsdale, N.J.: Lawrence Erlbaum and Associates.

Shiffrin, R. & Schneider, W. (1977). Controlled and automatic human information processing: II. Perceptual learning, automatic attending, and a general theory. *Psychological Review,* 84, 127-190.

Simon, H.A. (1986). The parameters of human memory. In F. Klix & H. Hagendorf (Eds.), *Human memory and cognitive capabilities : Mechanisms and performances.* (Part A) Amsterdam : North Holland.

Simon , H.A. (1975). The functional equivalence of problem solving skills. *Cognitive Psychology*, 7, 268-288.

Simon, H. A. (1974) How Big Is a Chunk? *Science*, 183 (8/2), 482-488.

Simon, H. A. and Gilmartin, K. (1973) A simulation of memory for chess positions. *Cognitive Psychology*, 5, 29-46.

Simon, D. P. and Simon, H. A. (1978) Individual differences in solving physics problems. In R. Sielgler (ED.), *Children's Thinking: What Develops?* Hillsdale, N. J.: Lawrence Erlbaum Associates Inc.

Singley, M.K. & Anderson, J.R. (1989) *The Transfer of Cognitive Skill.* Cambridge, Massachutsus: Harvard University Press.

Smith, M. U. (1992) Expertise and the organisation of knowledge: unexpected differences among genetic counsellors, faculty, and students on problem categorisation tasks. *Journal of Research in Science Teaching*, 29(2), 179-205.

Soloway, E., Adelson, B., and Ehrlich, K. (1988) Knowledge and Processes in the Comprehension of Computer programs. In M. T. H. Chi, R. Glaser, M. J. Farr (Eds) *The Nature of Expertise*. (pp. 129-152). Hillsdale, N. J.: Lawrence Erlbaum Associates Inc.

Sperling, G. (1960) The information available in brief visual presentations. *Psychological Monographs*., 74, 1-29.

Stark, R., Mandl, H., Gruber, H., Renkl, A. (in press) Conditions and effects of example elaboration. *Learning and Instruction.*

Sweller, J. (1999) *Instructional Design.* Melbourne: ACER.

Sweller, J. (1994). Cognitive load theory, learning difficulty and instructional design. *Learning and Instruction,* 4, 295-312.

Sweller, J. (1993). Some cognitive processes and their consequences for the organisation and presentation of information. *Australian Journal of Psychology,* 45, 1-8.

Sweller, J. (1989) Cognitive technology: Some procedures for facilitating learning and problem solving in mathematics and science. *Journal of Educational Psychology*, 81, 457-466.

Sweller, J. (1988) Cognitive load during problem solving: Effects on learning. *Cognitive Science*, 12, 257-285.

Sweller, J., & Chandler, P. (1994) Why some material is difficult to learn. *Cognition and Instruction*, 12, 185-233.

Sweller, J., & Chandler, P. (1991) Evidence for cognitive load theory. *Cognition and Instruction*, 8, 351-362.

Sweller, J., & Cooper, G. A. (1985). The Use of Worked Examples as a Substitute for Problem Solving in Learning Algebra. *Cognition and Instruction*, 2 , 59-89.

Sweller, J., & Levine, M. (1982) Effects of sub-goal density on means-ends analysis and learning. *Journal of Experimental Psychology: Human Learning and Memory*, 8, 463-474.

Sweller, J., Chandler, P., Tierney, P., & Cooper, M. (1990) Cognitive load in the structuring of technical material. *Journal of Experimental Psychology: General*, 119, 176-192.

Sweller, J., Mawer, R., & Howe, W. (1982) Concequences of history-cued and means-ends strategies in problem solving. *American Journal of Psychology*, 95, 455-483.

Sweller, J., Mawer, R. & Ward, M. (1983). Development of Expertise in Mathematical Problem Solving. *Journal of Experimental Psychology: General*, 112(4), 639-661.

Sweller, J., Van Merrienboer, J. & Paas, F. (1998) Cognitive Architecture and Instructional Design. *Educational Psychology Review*, 10, 251-296.

Tarmizi, R., & Sweller, J. (1988) Guidance during Mathematical Problem Solving. *Journal of Educational Psychology*, 80, 424-436.

Thorndyke, P. W. and Hayes-Roth, B. (1977) The Use of Schemata in the Acquisition and Transfer of Knowledge. *Cognitive Psychology,* 11, 82-106.

Tindall-Ford, S., Chandler, P. & Sweller, J. (1997). When two sensory modes are better than one. *Journal of Experimental Psychology: Applied*, 3, 257-287.

Treisman, A.M. (1964) Verbal cues, language, and meaning in selective attention. *American Journal of Psychology*, 77, 206-219.

Tulving, E. (1983) *Elements of episodic memory*. Oxford: Oxford University Press.

Tulving, E. (1972) Episodic and semantic memory. In E. Tulving and W. Donaldson (eds.), *Organisation of Memory.* London: Academic Press.

Tulving, E., and Osler, S. (1968) Effectiveness of retrieval cues in memory for words. *Journal of Experimental Psychology*, 77, 593-601.

Tversky, A. (1977) Features of similarity. *Psychological Review*, 84, 327-352.

Van Bruggen, J.M., Kirschner, P.A. & Jochems, W. (in press) External representation of argumentation in CSCL and the management of cognitive load. *Learning and Instruction*.

Van Gerven, P.W.M., Van Merrienboer, J.J.G., Paas, F.G.W.C. & Schmidt, H.G. (in press) Cognitive load and Aging: Effects of worked examples on training efficiency. *Learning and Instruction*.

Van Merrienboer, J.J.G., Schuurman, J.G., de Croock, M.B.M., & Paas, F.G.W.C. (in press) Redirecting Learners' attention during training: Effects on cognitive load, transfer test performance and training efficiency. *Learning and Instruction*.

Voss, J. F. & Post, T. A. (1988) On the Solving of Ill-Structured Problems. In M. T. H. Chi, R. Glaser, M. J. Farr (Eds) *The Nature of Expertise.* (pp. 261-285) Hillsdale, N. J.: Lawrence Erlbaum Associates Inc.

Ward, M., & Sweller, J. (1990) Structuring effective worked examples. *Cognition & Instruction*, 7, 1-39.

Wessels, M. G. (1982). *Cognitive Psychology*. New York : Harper & Row.

Wiedenbeck, S. (1985) Novice/expert differences in programming skills. *International Journal of Man-Machine Studies*, 23, 383-390.

Wineburg, S. S. (1991) Historical problem solving: a study of the cognitive processes used in the evaluation of documentary and pictorial evidence. *Journal of Educational Psychology*, 83(1), 73-87.

Wu, H.M., & Solman, R.T. (1994). Effective use of pictures as extra stimulus prompts. *British Journal of Educational Psychology*, 63, 144-160.

Yates, G. and Chandler, M. (1991) The cognitive psychology of knowledge: Basic findings and educational implications. *Australian Journal of Education*, 35, 131-153.

Yates, G. and Chandler, M. (1994) Prior knowledge and how it influences classroom learning. *Research Information for Teachers* (Set no. 2, Item no. 6.) Wellington: New Zealand Council for Educational Research.

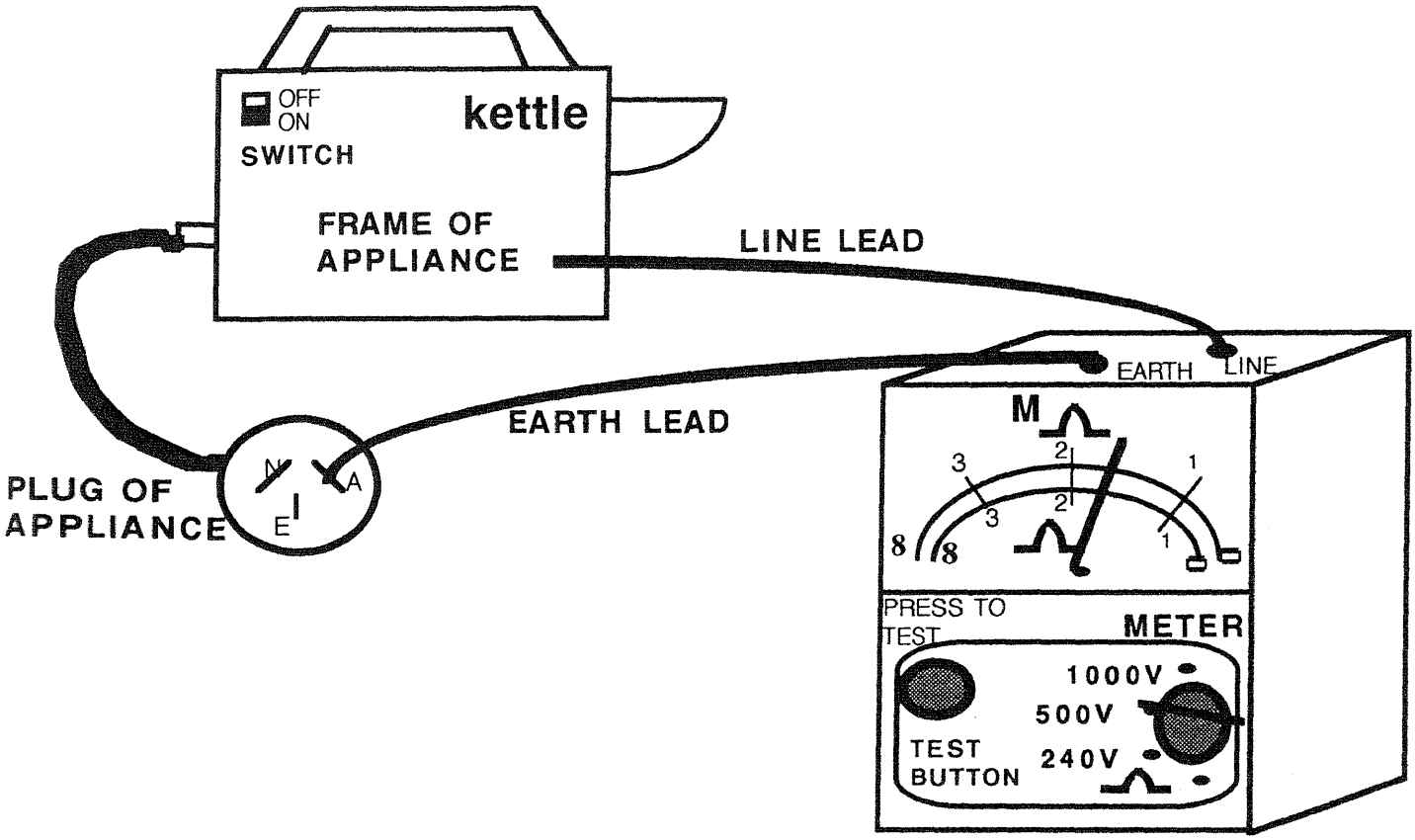
Yeung, A.S., Jin, P. & Sweller, J. (1998) Cognitive load and learner expertise: Split attention and redundancy effects in reading with explanatory notes. *Contemporary Educational Psychology*, 23, 1-21.

Zhu, X., & Simon, H.A. (1987) Learning mathematics by examples and doing. *Cognition and Instruction*, 4, 137-166.

Appendix 1:

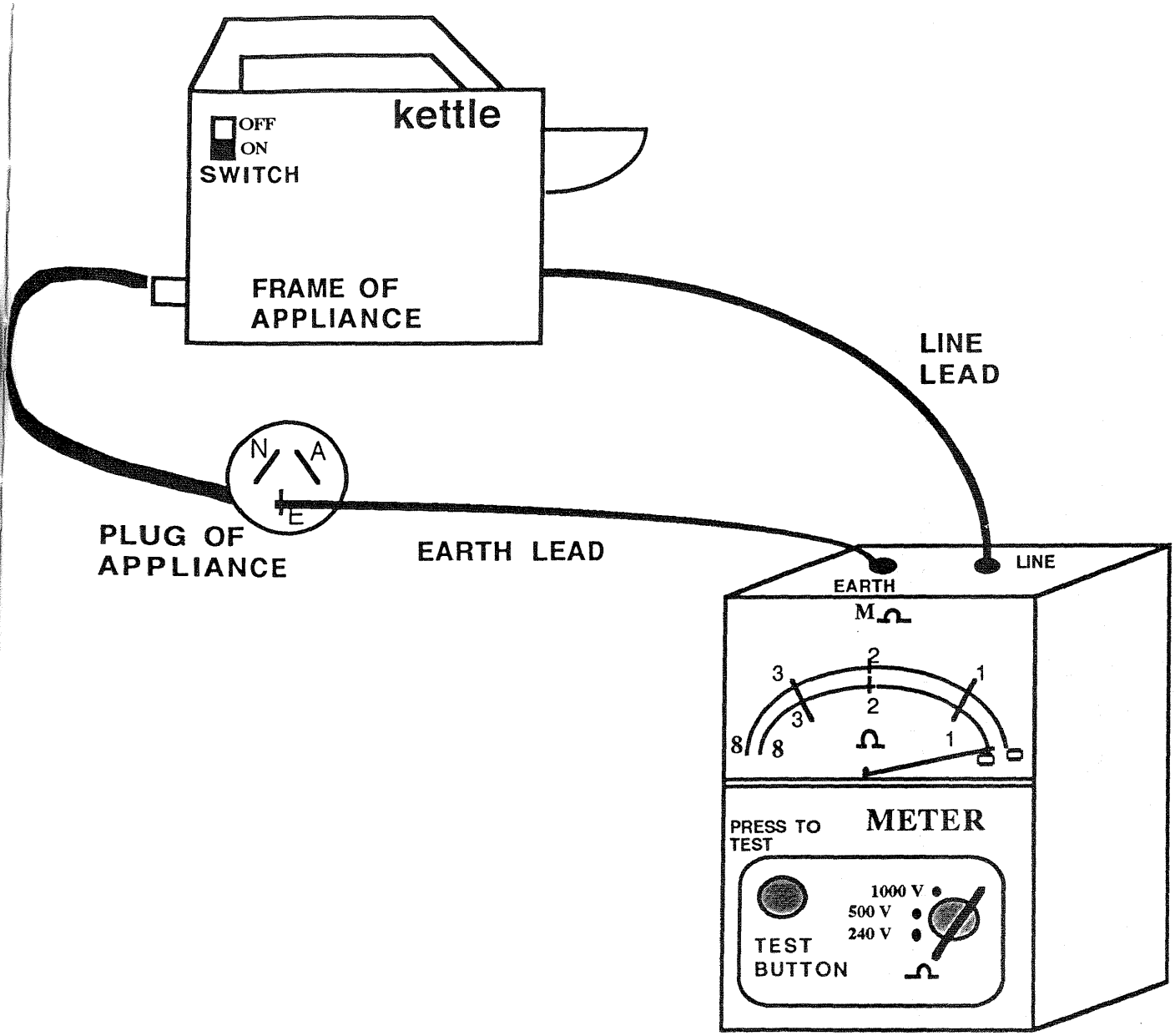
Isolated elements instructional material used in Experiments 1 & 2.

The Insulation Resistance Test



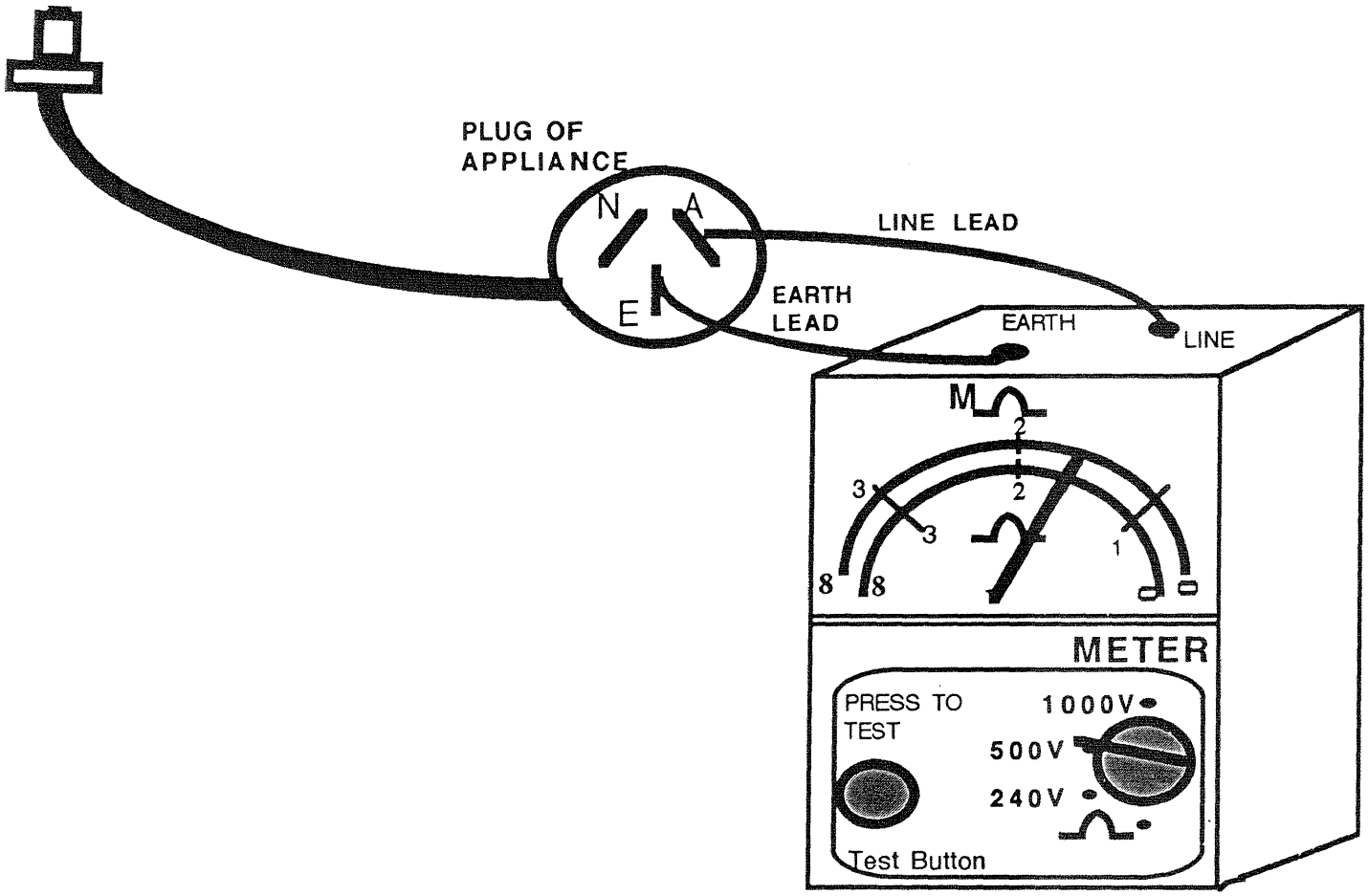
1. Set the meter to read 500 volts.
2. Make sure the switch is "on".
3. Place line lead on the frame of the appliance.
4. Place the earth lead on the active pin of the appliance's plug.
5. Press the test button.
6. Read the resistance from the meter. The required result is a reading of at least one Megohm.
7. Remove the earth lead from the active pin and place it on the neutral pin.
8. Press the test button again.
9. Read the resistance. A reading of at least one Megohm is again required.

The Earth Continuity Test



1. Set the meter to read ohms
2. Place the earth lead from the meter on the earth pin of the appliance's plug
3. Place the line lead on the frame of the appliance
4. Press the test button
5. Read the resistance from the meter. The required result is a reading of less than two ohms

The Insulation Resistance of A Flexible Lead Test

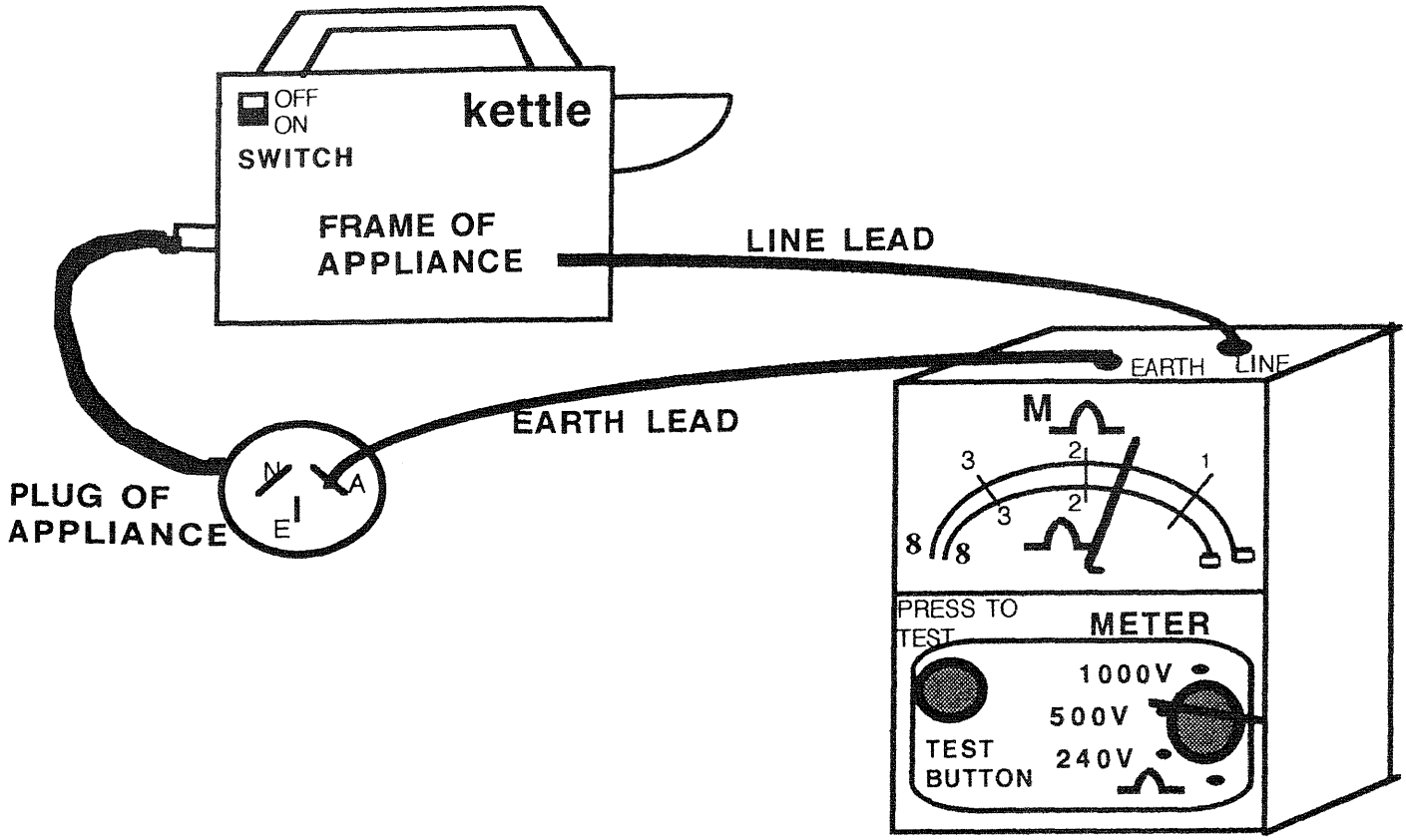


1. Set the meter to read 500 volts
2. Place the earth lead from the meter on the earth pin of the flexible lead
3. Place the line lead on the active pin of the flexible lead.
4. Press the test button
5. Read the resistance from the meter. The required result is a reading of at least one Megohm
6. Remove the line lead from the active pin and place it on the neutral pin
7. Press the test button again
8. Read the resistance. A reading of at least one Megohm is again required

Appendix 2:

Interacting elements instructional material used in Experiments 1 & 2.

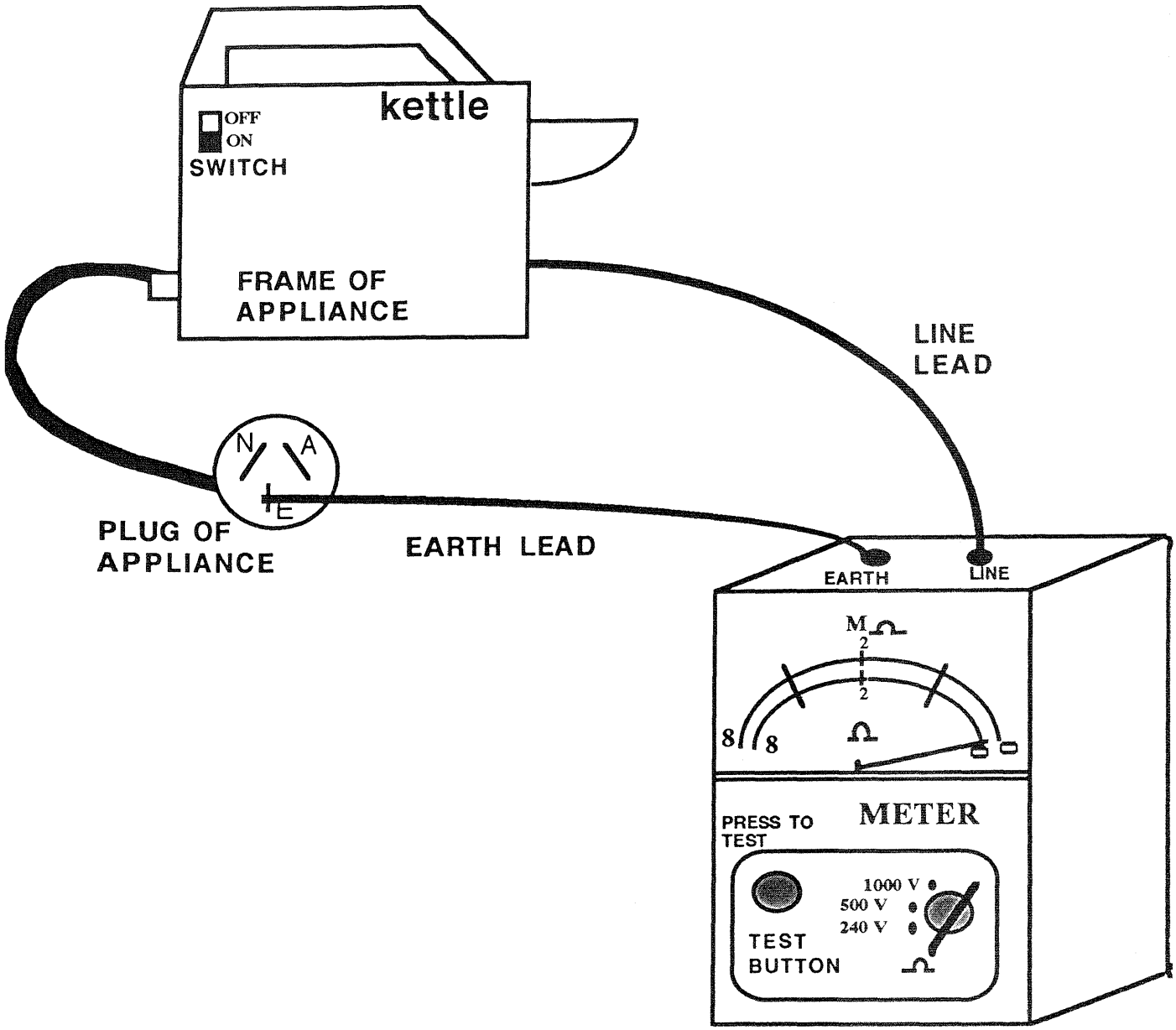
The Insulation Resistance Test



This test aims to check that there is no "leak" in the flow of current from either the active or neutral wires of the appliance to the frame of the appliance, which would cause the frame of the appliance to become electrically active.
To achieve this two circuits are set up; the first including the frame of the appliance and the active of the appliance and a second including the frame of the appliance and the neutral of the appliance. Very large measures of insulation resistance (ideally "infinite" resistance) indicates that there are no leaks in the flow of current to each circuit. A megger meter measures the resistance of the circuit.

1. By setting the meter to read 500 volts, a larger than normal voltage will be introduced into each circuit in order to test the appliance under a heavy load. Set the meter to read 500 volts.
2. The appliance is switched "on" to allow current generated from the megger meter to flow into each circuit. Make sure the switch is "on".
3. To measure the resistance of the first circuit the megger meter is connected into the circuit. Place the line lead on the frame of the appliance.
4. Place the earth lead on the active pin of the appliance's plug.
5. The test button of the megger meter introduces 500 volts into the circuit. Press the test button.
6. A measurement of insulation resistance of the circuit can be taken from the needle of the meter, with a reading of at least 1 Megohm (1,000,000 ohms) being sufficiently large to indicate that the active-frame circuit possesses an acceptable level of insulation. This ensures that no current can leak to the appliance's frame. Read the resistance from the meter. The required result is a reading of at least one Megohm.
7. To complete the test of insulation resistance the resistance of the second circuit must be measured. The second circuit is set up between the neutral point of the appliance and the appliance frame. Remove the earth lead from the active pin and place it on the neutral pin.
8. A reading of the resistance of the second circuit can be taken by pressing the test button of the megger meter and introducing 500 volts into the neutral-frame circuit. Press the test button again.
9. To pass the test, the second circuit is also required to have a safe level of insulation resistance which indicates current has not leaked from this circuit. Read the resistance. A reading of at least one Megohm is again required.

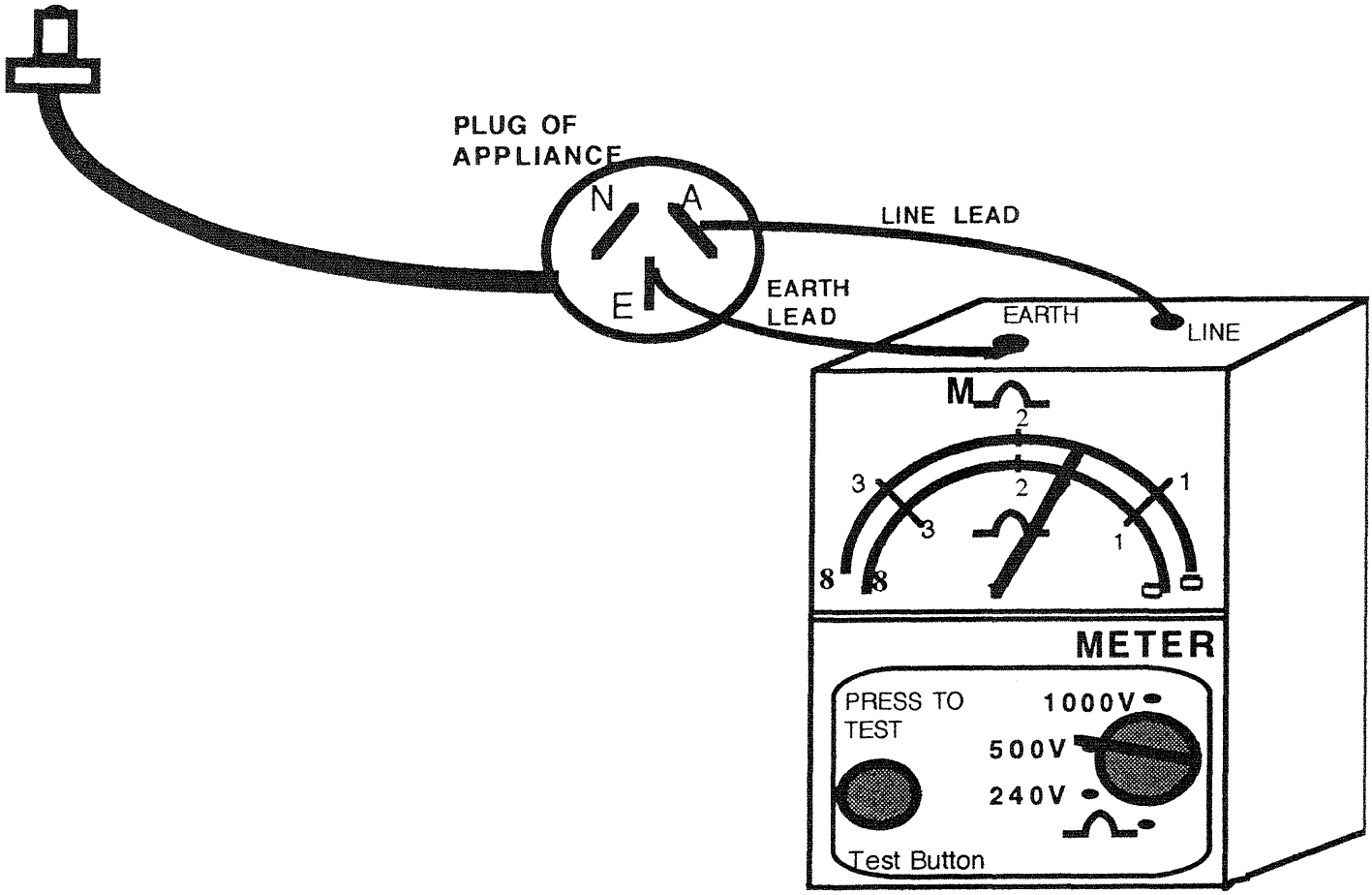
The Earth Continuity Test



This test aims to check that in the event of a fault, the frame of the appliance does not become electrically active from current which should be directed to the earth. This can be achieved by setting up a circuit including the frame of the appliance and its earth point and measuring the resistance between the two points. Resistance is a barrier which can keep current in the circuit. Ideally, this circuit should have zero resistance.

1. A megger meter is connected into each circuit to measure the resistance. The ohms setting is used to measure the very small resistances expected of this circuit. Set the meter to read ohms.
2. To set up to measure the resistance of the circuit, place the earth lead of the megger meter on the earth pin of the appliance's plug and then
3. Place the line lead on the frame of the appliance.
4. When the test button of the megger meter is pressed, a measurement of the resistance of the circuit can be taken from the needle of the meter. Press the test button.
5. A resistance reading of less than 2 ohms is sufficiently small to indicate that there is no dangerous barrier in the earth-frame circuit, blocking current from going to the earth point. Read the resistance from the meter. The required result is a reading of less than two ohms.

The Insulation Resistance of a Flexible Lead Test



This test aims to check that there is no "leak" in the flow of current from either the active or neutral wires to the earth wire of the flexible lead. This would cause current to flow through the earth wire back to the power source and could be potentially dangerous. To achieve this two circuits are set up; the first including the earth of the flexible lead and the active of the flexible lead and a second including the earth of the flexible lead and the neutral of the flexible lead. Very large measures of insulation resistance (ideally "infinite" resistance) indicates that there are no leaks in the flow of current to each circuit. A megger meter is connected into each circuit to measure the resistance.

1. By setting the meter to 500 volts, a larger than normal voltage will be introduced into each circuit in order to test the appliance under a heavy load. Set the meter to read 500 volts.
2. To measure the resistance of the first circuit the megger meter is connected into the circuit. Place the earth lead from the meter on the earth pin of the flexible lead and then
3. Place the line lead on the active pin of the flexible lead.
4. The test button of the megger meter introduces 500 volts into the circuit in order to test it. Press the test button.
5. A measurement of insulation resistance of the circuit can be taken from the needle of the meter, with a reading of at least 1 Megohm (1,000,000 ohms) being sufficiently large to indicate that the earth-active circuit possesses an acceptable level of insulation. This ensures there is no leak in the flow of current to earth. Read the resistance from the meter. The required result is a reading of at least one Megohm.
6. To complete the test of insulation resistance the resistance of the second circuit must be measured. The second circuit is set up between the neutral and the earth points of the lead therefore remove the line lead from the active pin and place it on the neutral pin.
7. A reading of the resistance of the second circuit can be taken by pressing the test button of the megger meter and introducing 500 volts into the neutral-earth circuit. Press the test button again.
8. To pass the test, the second circuit is also required to have a safe level of insulation resistance which indicates current has not leaked from this circuit. Read the resistance. A reading of at least one Megohm is again required.

Appendix 3:

An analysis of the element interactivity of the instructional material used in Experiments 1 & 2.

The elements of interacting information have been estimated for each of the electrical safety tests. These estimates represent what could reasonably be supposed to be the case for novices electricians.

Earth Continuity test-

Isolated Elements version of Instructions

1. The meter must be set to ohms.
2. The earth lead of the meter must be placed to the earth pin on the appliance plug.
3. The line lead of the meter must be placed to some point on the metallic frame of the appliance.
4. The test is performed by pressing the test button
5. The result is taken from the meter and checked against the criteria of less than two ohms.

Interacting Elements Version of Instructions

1. The test checks that, in the event of a fault, the appliance frame will not carry current which should go to earth.
2. The test measures resistance.
3. Resistance is a barrier which can stop current leaving a circuit.
4. This circuit should ideally have zero resistance.
5. Resistance is measured by a megger meter.
6. The meter must be set to ohms.
7. Ohms are units that are able to measure very small levels of resistance.
8. The earth lead of the meter must be placed to the earth pin on the appliance plug.
9. The line lead of the meter must be placed to some point on the metallic frame of the appliance.

10. The test is performed by pressing the test button.
11. A resistance reading can be taken from the meter and checked against the criteria of less than two ohms.
12. Although zero ohms is ideal, a result of less than two ohms is an acceptable level of resistance for the circuit to possess.

Insulation Resistance test

Isolated Elements version of Instructions

1. The meter is set to 500 volts.
2. The appliance is switched "on".
3. The line lead of the meter must be placed to some point on the metallic frame of the appliance.
4. The earth lead of the meter must be placed to the active pin on the appliance plug.
5. The test is performed by pressing the test button and then checking the result meets the criteria of at least one megohm.
6. Change the earth lead to the neutral pin of the appliance pin.
7. Test this circuit by pressing the test button again and checking the result is at least one megohm.

Interacting Elements Version of Instructions

1. The test checks that there is no leak in the flow of current through the appliance circuit that would cause the appliance frame to become electrically active.
2. To establish that there is no leaks, two circuits have to be tested: the circuit between the active point and the appliance frame and the circuit between the neutral point and the appliance frame.

3. The test measures the insulation resistance of the circuit. If the reading is high, it follows that current is not being leaked from the circuit.
4. Ideally a circuit should have infinite resistance but a safe criteria of at least 1 megohm has been set.
5. The megger meter is set to 500 volts.
6. This setting means the circuits are tested under a larger than normal voltage.
7. The appliance is switched "on".
8. This is so current from the megger meter can flow through each circuit.
9. To set up the first circuit, the line lead of the meter must be placed to some point on the metallic frame of the appliance.
10. The earth lead of the meter must be placed to the active pin on the appliance plug.
11. The test is performed by pressing the test button which introduces 500 volts to the circuit.
12. The result must meet the criteria of at least one megohm to ensure that no current is being leaked to the appliance frame.
13. One megohm is equal to 1 000 000 ohms.
14. The resistance of the second circuit must be measured to complete the test.
15. The earth lead is moved onto the neutral pin of the appliance plug.
16. Test the second circuit by pressing the test button again and checking the result is at least one megohm.

Insulation Resistance of a Flexible Lead test-

Isolated Elements version of Instructions

1. The meter is set to 500 volts.
2. The earth lead of the meter must be placed to the earth pin of the flexible lead.
3. The line lead of the meter must be placed active pin of the flexible lead.
4. The test is performed by pressing the test button and then checking the result meets the criteria of at least one megohm.
5. Change the line lead to the neutral pin of the flexible lead.
6. Test this by pressing the test button again and checking the result is at least one megohm.

Interacting Elements Version of Instructions

1. The test checks that there is no leak in the flow of current from either the active or neutral wires to the earth wire of the flexible lead.
2. A leak in the flow of current would cause power to flow back to the power source, a dangerous situation.
3. To establish that there is no leaks, two circuits have to be tested: the circuit between the active and earth wires of the flexible lead and the circuit between the neutral and earth wires of the flexible lead.
4. The test measures the insulation resistance of the circuits created. If the reading is high, it follows that current is not being leaked from the circuit.
5. Ideally a circuit should have infinite resistance but a safe criteria of at least 1 megohm has been set.
6. The megger meter is set to 500 volts.
7. This setting means the circuits are tested under a larger than normal voltage.
8. To set up the first circuit, the earth lead of the meter must be placed onto earth

pin of the flexible lead.

9. The line lead of the meter must be placed to the active pin of the flexible lead.

10. The test is performed by pressing the test button which introduces 500 volts to the circuit.

11. The result must meet the criteria of at least one megohm to ensure that no current is being leaked to earth.

12. One megohm is equal to 1 000 000 ohms.

13. The resistance of the second circuit must be measured to complete the test.

14. The line lead is moved onto the neutral pin of the flexible lead.

15. Test the second circuit by pressing the test button again and checking the result is at least one megohm.

Appendix 4:

Subjective mental load rating scale used in Experiments 1 & 2.

SUBJECT NUMBER: _____

LEARNING ELECTRICAL TESTS

How easy or difficult was it to learn and understand the electrical tests from the instructions you were given ?

Place a cross on the rating scale below

1	2	3	4	5	6	7
Extremely Easy	**Very** Easy	**Easy**	**Neither Easy nor Difficult**	**Difficult**	**Very Difficult**	**Extremely Difficult**

Appendix 5:

Written test used in Experiments 1 & 2.

High element interactivity test questions

Question 1

Question 7

Question 8

Question 14

Question 16

Low element interactivity test questions

Question 2

Question 3

Question 4

Question 5

Question 6

Question 9

Question 10

Question 11

Question 12

Question 13

Question 15

Question 17

Question 18

Question 19

Question 20

Question 21

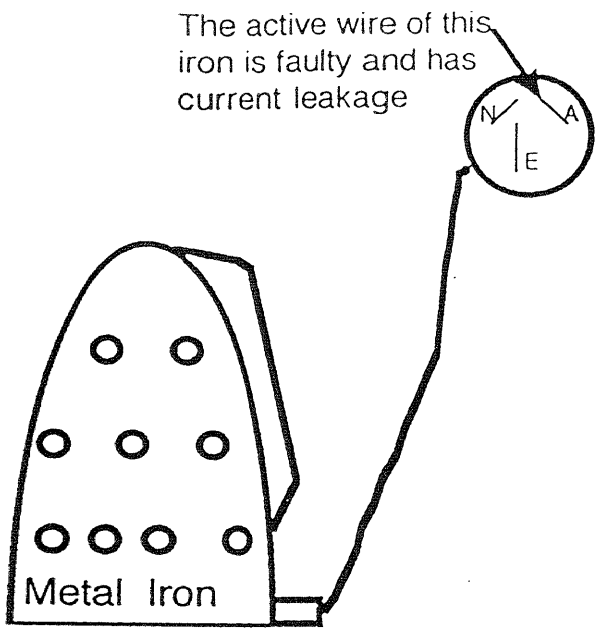
Question 22

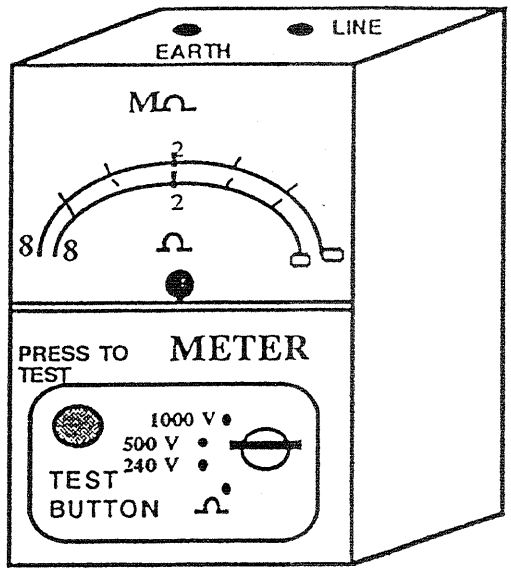
Question 23

APPLIANCE TESTING

SUBJECT NUMBER: _____

1.





A) Where will this leakage of electrical current go ? **(Indicate on diagram)**

B) Draw the leads of the megger meter onto the diagram to create the circuit which can indicate this fault

C) Draw a needle on the megger meter and colour the area above or below the needle to show where the resistance reading of this faulty wire would fall

Questions 2-5 refer to the EARTH CONTINUITY test

2. What do you set the megger meter to?

3. Where is the earth lead placed during the test?

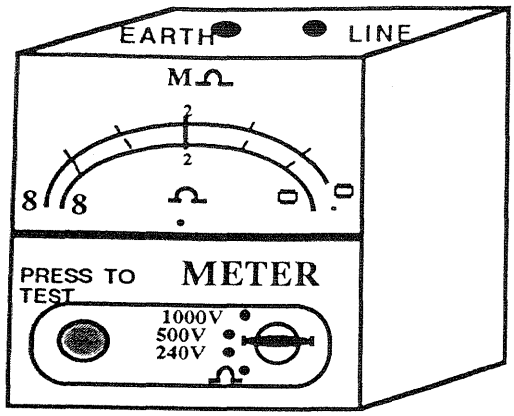
4. What is the required result of the test?

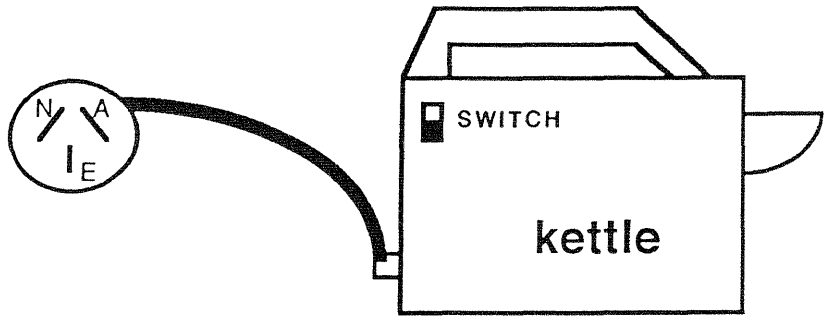
5. How many readings are required to be taken to complete the whole test?

6. How is power introduced to the circuit when testing for insulation resistance?

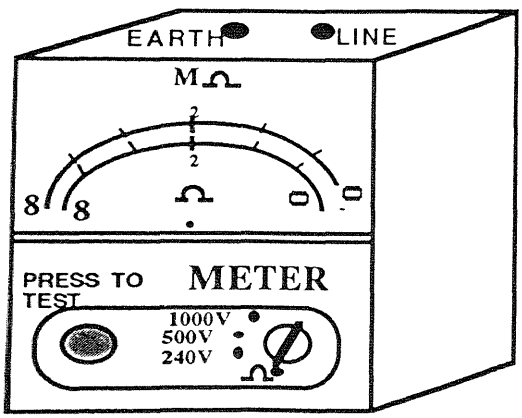
7. For each of the three tests shown below **draw on the diagram**
   i) Where the **leads** should be placed to perform the test
   ii) The **needle of the megger meter** if the kettle had passed the test

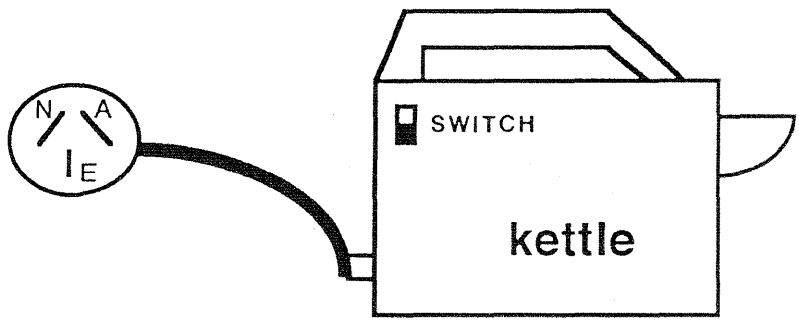
A) The insulation resistance test



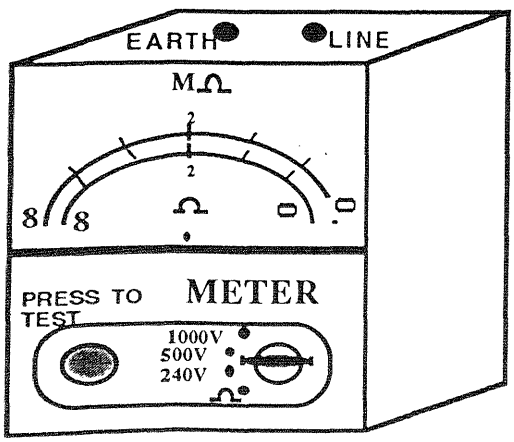


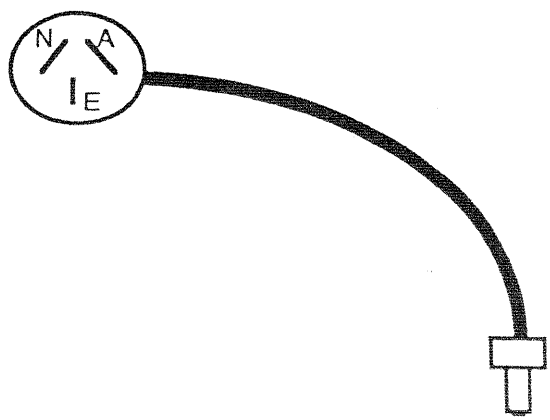
B) The earth continuity test





C) The insulation of a flexible lead test





8. An extension cord is tested for it's insulation resistance. The result is

A) Neutral - earth circuit = 0.5 megohms
Active-earth circuit = 3 megohms

Would you pass the extension lead as safe? Explain your answer.

B) Neutral - earth circuit = 1.1 megohms
Active-earth circuit = 1.1 megohms

Would you pass the extension lead as safe? Explain your answer.

9. A normal appliance requires 240 volts to operate. Why is 500 volts used to power the circuit when testing for insulation resistance?

Questions 10-13 refer to the INSULATION RESISTANCE test

10. Where is the line lead placed?

11. What is the required result of the test?

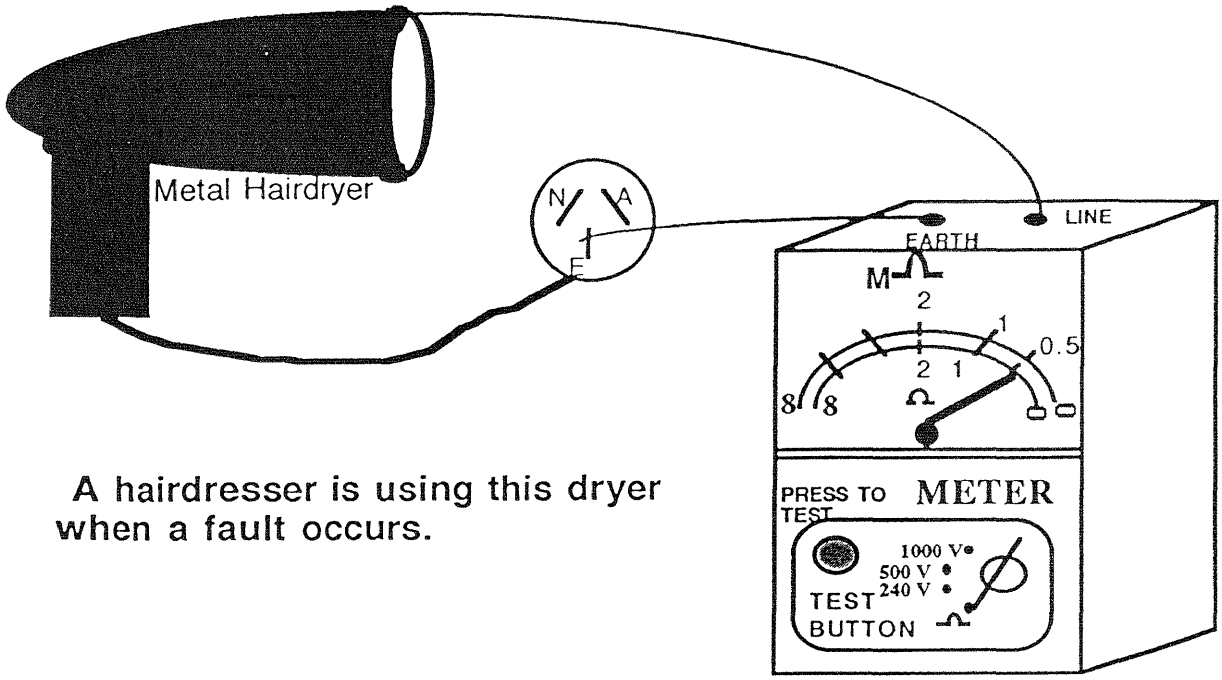
12. What do you set the megger meter to ?

13. How many times do you have to push the test button to complete the whole test?

14. A flexible lead has an insulation resistance of 0.5 Megohms. A fault occurs. What is the effect?

15. Why is the appliance switched **ON** in the insulation resistance test?

16.



A hairdresser is using this dryer when a fault occurs.

A) Will the hairdresser
(Circle the correct answer)

i) be electrocuted
ii) receive a mild shock
iii) not be affected

B) Explain why this occurs?

C) Another hair dryer is tested and the resistance reading is 2.9 ohms. Given this information

i) Where will the excess electric current, from the power surge go?
Place a cross (X) on diagram above

ii) Why does the excess current go to this point?

Questions 17-20 refer to the INSULATION RESISTANCE OF A FLEXIBLE LEAD test

17. What is the megger meter set to?

18. Where is the earth lead placed?

19. What units is this test measured in?

20. How many readings are required to be taken to complete the whole test?

21. What is the role of the megger meter in the circuit when performing the earth continuity test?

22. Why are two resistance readings required to complete the insulation resistance test?

23. Why is the ohms setting of the megger meter used in the earth continuity test?

Appendix 6:

An analysis of the element interactivity of the instructional material used in Experiments 3 & 4.

The elements of interacting information have been estimated for each versions of the oven circuit. These estimates represent what could reasonably be supposed to be the case for novices electricians.

Isolated Elements version of Instructions

1. There are two circuits for the oven: a power circuit & a control circuit
2. When oven door shuts, the oven door limit switch is closed.
3. Main isolator switch is connected in circuit with the motor.
4. The main isolator switch must be turned on.
5. The baking process begins when Start is pressed
6. Pressing start activates Coil F.
7. Switch F1 immediately closes.
8. Start button can now be released.
9. Coil F closes the Forward contact switch in the power circuit.
10. The motor starts, moving the conveyor and the goods to baked toward the heating elements.
11. Limit switch 1 is a double pole switch. The moving conveyor activates it.
12. The top of Double pole switch 1 opens while the bottom of Double pole switch 1 closes.
13. The forward contact switch opens.
14. The timer is activated.
15. At the end of the set baking time, switch T1 closes
16. Coil R is energised
17. Switch R1 closes
18. Switch R2 opens
19. Switch T1 opens
20. The Reverse Contact switch in the power circuit closes
21. The motor starts, moving the conveyor and the baked goods away from the heating elements.
22. Limit switch 2 opens when the conveyor reaches its original position.
23. The Reverse Contact switch in the power circuit opens
24. TOLs are line line with the coils and the motor to stop overheating.
25. Pressing the stop button will open the circuit.
26. Pressing the emergency stop push button will open the circuit.

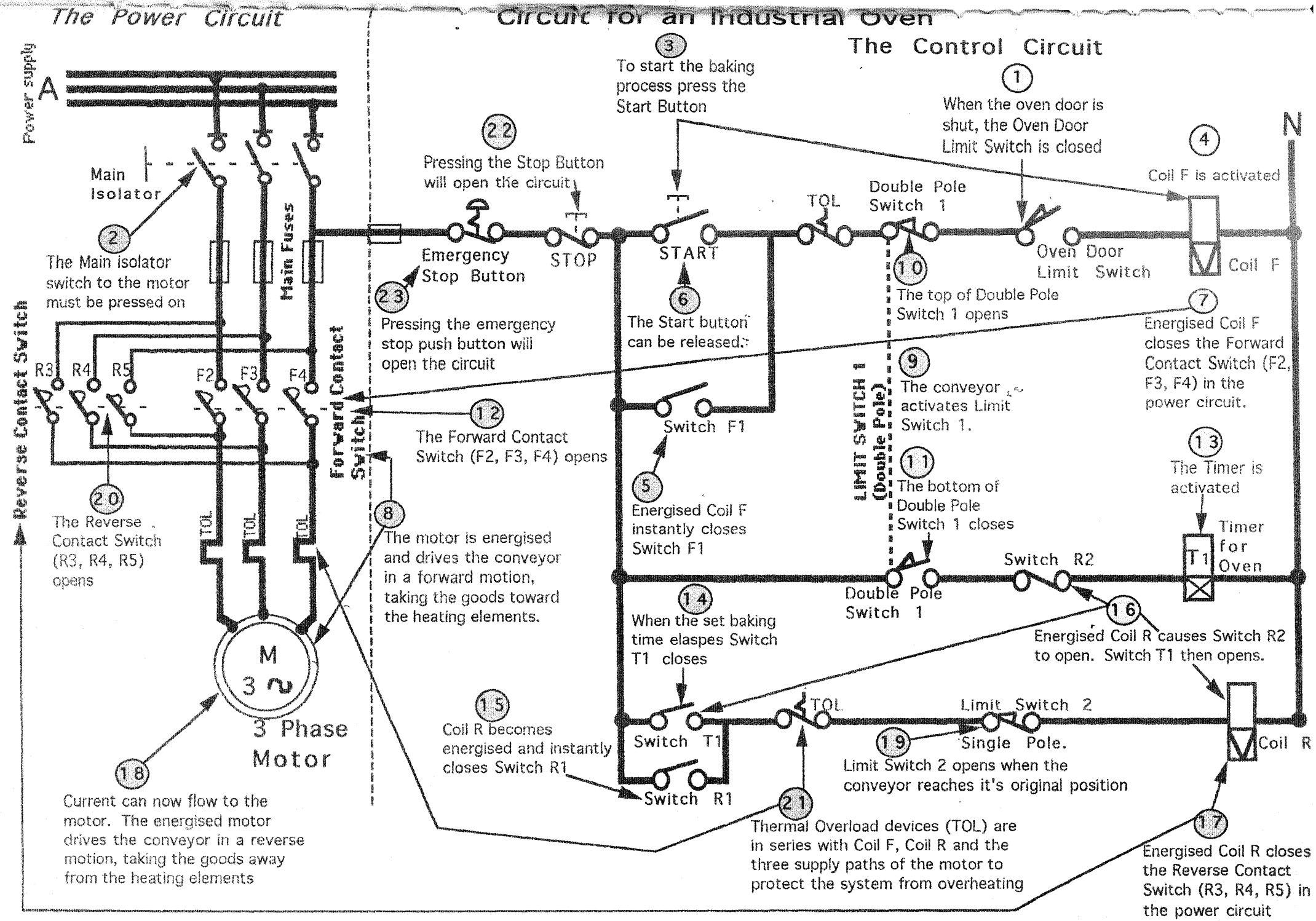
Interacting Elements Version of Instructions

1. Current flows from active to neutral and can only flow through a completely closed circuit
2.There are two circuits for the oven: a power circuit & a control circuit
3. When oven door shuts, the oven door limit switch is closed.
4. Main isolator switch is connected in circuit with the motor.
5.Main isolator switch controls the flow of current to the motor
6. The main isolator switch must be turned on to start the oven baking.
7. The power circuit needs a separate switch to the control circuit as the motor may be in a separate location to the oven and/or conveyor
8. The baking process begins when Start is pressed.
9. Depressing the start button creates a closed circuit
10. Current can flow through and activate Coil F.
11. Activated Coil F immediately closes Switch F1
12. This creates a holding contact .
13. Start button can now be released
14. Because current is supplied to coil F through he holding contact created by F1.
15. Coil F closes the Forward contact switch in the power circuit.
16.The closed Forward contact switch allows current to activate the motor
17. The motor starts, moving the conveyor and the goods to baked toward the heating elements.
18. Limit switch 1 is a double pole switch. The moving conveyor activates it.
19. The top of Double pole switch 1 opens stopping the supply of current to coil F.
20. This causes the Forward contact switch to open, stopping the motor and the conveyor
21. This switch opens as a safety measure so the motor is supplied with current to go forward and reverse at he same time
22. The bottom of Double pole switch 1 closes, supplying current to the Timer
23. The timer is activated. It has a pre set baking time to cook the goods.
24. At the end of the set baking time, switch T1 closes
25. Switch T1 is activated by the timer
26. The closure of T1 creates a closed circuit so current can flow to Coil R

27. Coil R is energised
28. Coil R immediately closes Switch R1
29. Switch R1 acts as a holding circuit, maintaining the supply of current to Coil R
30. Coil R opens Switch R2
31. Current is no longer supplied to the timer
32. The de-energised timer opens Switch T1
33. Coil R closes the Reverse Contact switch in the power circuit
34. The motor starts, moving the conveyor and the baked goods away from the heating elements.
35. Limit switch 2 opens when the conveyor reaches its original position.
36. This stops the supply of current to Coil R
37. This causes the Reverse Contact switch in the power circuit to open
38. This stops the supply of current to the motor, stopping the conveyor
39. TOLs are line line with the coils and the motor to stop overheating.
40. If overheating occurs, the TOLs open, stopping the supply of current
41. Pressing the stop button will open the circuit.
42. Pressing this button will stop the oven or the conveyor.
43. This button is accessed by the oven operator
44. Pressing the emergency stop push button will open the circuit.
45. This button will stop any part of the oven process as active cannot flow to neutral.
46. This button is probably not located at the actual oven but in a more general position in the factory

Appendix 7:

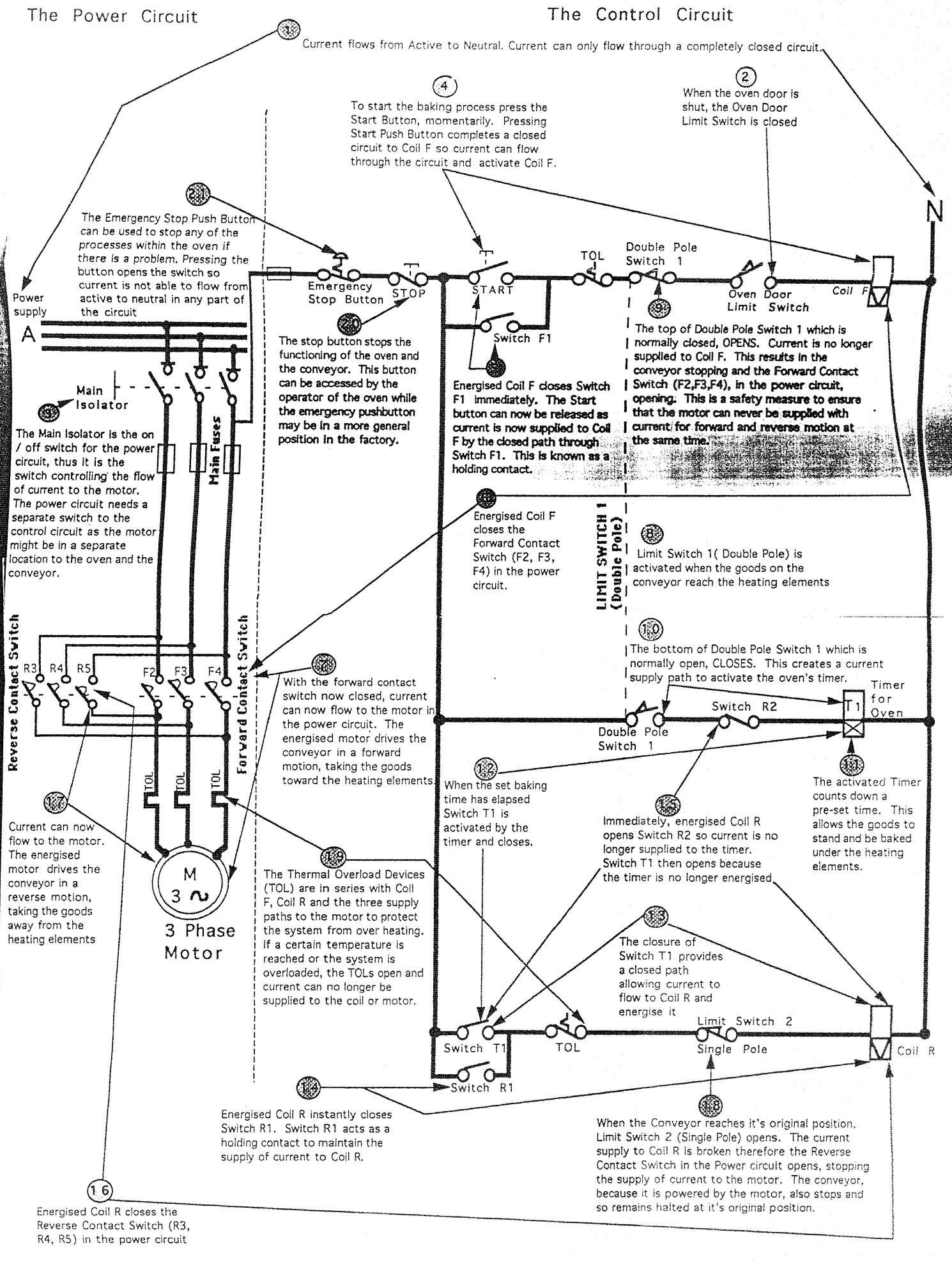
Isolated elements instructional material used in Experiments 3 & 4.


The Power Circuit
Circuit for an Industrial Oven
The Control Circuit
Power supply
A
N
Main Isolator
Main Fuses
Reverse Contact Switch
Forward Contact Switch
R3
R4
R5
F2
F3
F4
TOL
M
3 ~
3 Phase Motor
Emergency Stop Button
STOP
START
Switch F1
Double Pole Switch 1
Oven Door Limit Switch
Coil F
LIMIT SWITCH 1 (Double Pole)
Switch R2
Timer for Oven
T1
Switch T1
Switch R1
Limit Switch 2
Single Pole.
Coil R
1 When the oven door is shut, the Oven Door Limit Switch is closed
2 The Main isolator switch to the motor must be pressed on
3 To start the baking process press the Start Button
4 Coil F is activated
5 Energised Coil F instantly closes Switch F1
6 The Start button can be released.
7 Energised Coil F closes the Forward Contact Switch (F2, F3, F4) in the power circuit.
8 The motor is energised and drives the conveyor in a forward motion, taking the goods toward the heating elements.
9 The conveyor activates Limit Switch 1.
10 The top of Double Pole Switch 1 opens
11 The bottom of Double Pole Switch 1 closes
12 The Forward Contact Switch (F2, F3, F4) opens
13 The Timer is activated
14 When the set baking time elaspes Switch T1 closes
15 Coil R becomes energised and instantly closes Switch R1
16 Energised Coil R causes Switch R2 to open. Switch T1 then opens.
17 Energised Coil R closes the Reverse Contact Switch (R3, R4, R5) in the power circuit
18 Current can now flow to the motor. The energised motor drives the conveyor in a reverse motion, taking the goods away from the heating elements
19 Limit Switch 2 opens when the conveyor reaches it's original position
20 The Reverse Contact Switch (R3, R4, R5) opens
21 Thermal Overload devices (TOL) are in series with Coil F, Coil R and the three supply paths of the motor to protect the system from overheating
22 Pressing the Stop Button will open the circuit
23 Pressing the emergency stop push button will open the circuit

Appendix 8:

Interacting elements instructional material used in Experiments 3 & 4.

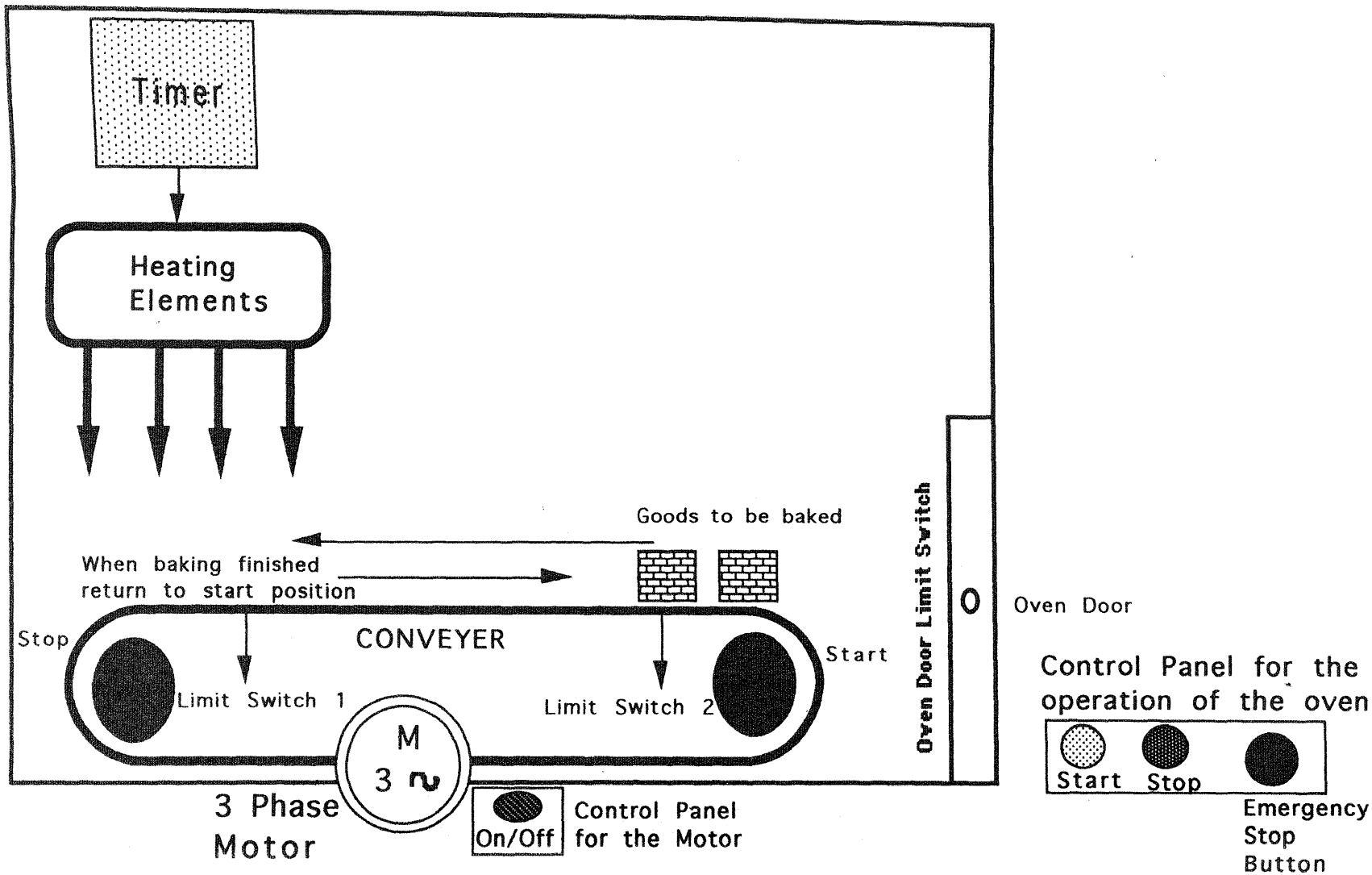
Circuit for an Industrial Oven



Appendix 9:

Diagram illustrating the oven baking process used in Experiments 3 & 4.

Oven

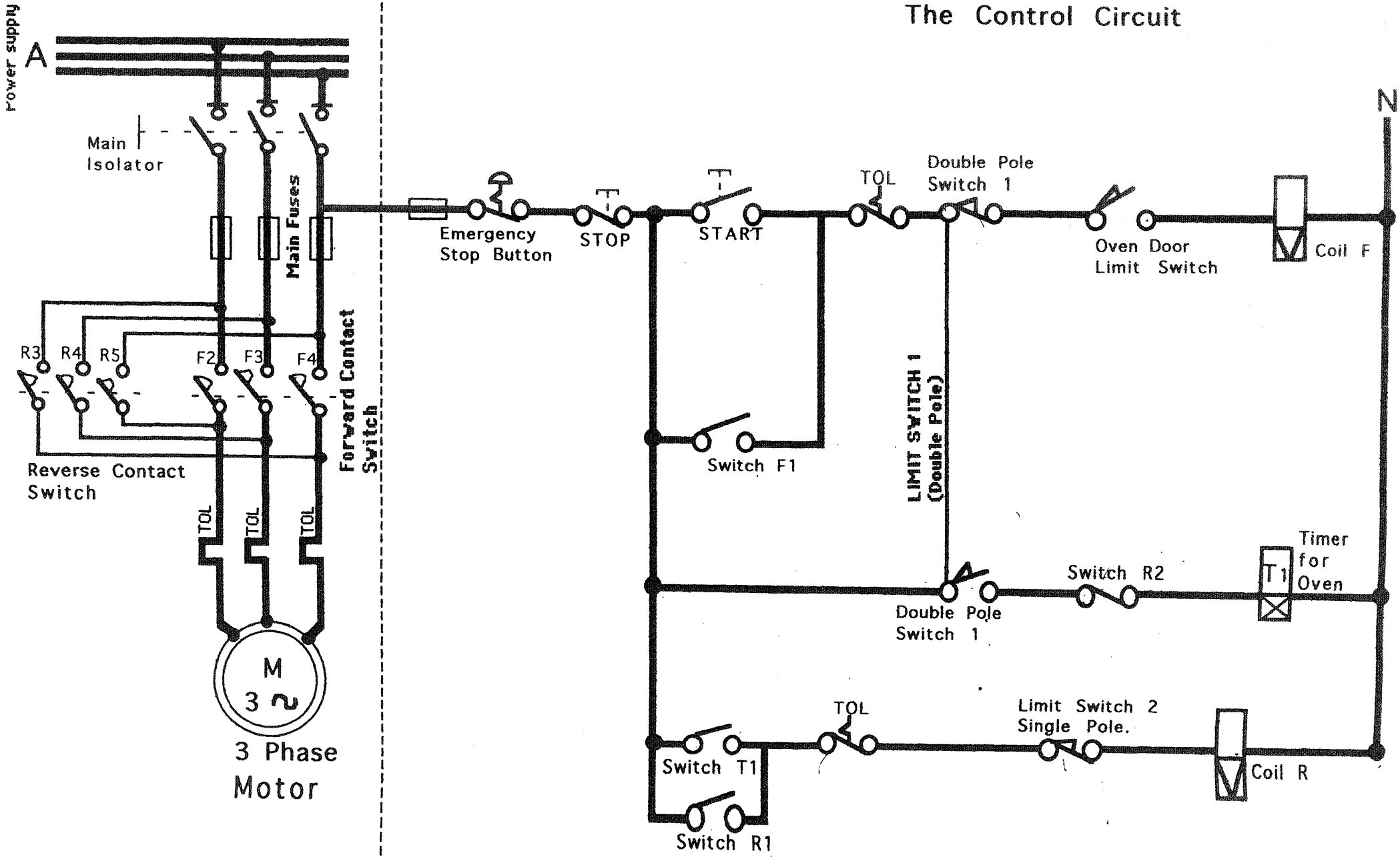


This circuit enables goods to be baked in an oven. The goods are on a conveyor, which is driven by a 3 phase motor. The conveyor transports the food to the oven's heating elements where it stops and cooking occurs for a prescribed time. At the end of the set cooking time, the conveyor is re-activated and the goods are moved away from the cooking elements back to their original position.

Appendix 10:

Diagram depicting the oven electrical circuit used in Experiments 3 & 4.

Circuit for an Industrial Oven



Appendix 11:

Subjective mental load rating scale used in Experiments 3 & 4.

Name:

Subject Number:

LEARNING AN ELECTRICAL CIRCUIT

How easy or difficult was it to learn and understand the electrical circuit from the instructions you were given?

Place a cross on the rating scale below

1	2	3	4	5	6	7	8	9
Extremely Easy	Very Easy	Easy	Slightly Easy	Neither Easy nor Difficult	Slightly Difficult	Difficult	Very Difficult	Extremely Difficult

Appendix 12:

Written test used in Experiments 3 & 4.

Low element interactivity test questions

Question 1

Question 2

Question 5

Question 11

High element interactivity test questions

Question 3

Question 4

Question 6

Question 7

Question 8

Question 9

Question 10

Question 12

Name:
Number:

Oven Circuit Test

1. What three actions need to be performed by someone using the oven for the baking process to begin?

2. How can the operation of the circuit be stopped manually?

3. What switches are closed when the conveyor starts moving the goods toward the oven's heating elements?

4. What causes Limit Switch 2 to open?

5. i) What stops coil R from overheating?

ii) How?

6. When the goods are baking, which switches are open?

7. What event causes Switch T1 to open?

8. Circle the option that completes the sentence correctly

When the goods initially reach the heating elements, Limit switch 1 is signalled. The top of double pole switch 1 .....

A opens, starting the conveyor, while the bottom of double pole switch 1 closes, stopping the timer .

B closes, stopping the conveyor, while the bottom of double pole switch 1 opens, starting the timer.

C opens, stopping the conveyor while the bottom of double pole switch 1 closes, starting the timer.

D closes, starting the conveyor, while the bottom of double pole switch 1 opens, stopping the timer.

9. Circle the option that completes the sentence correctly

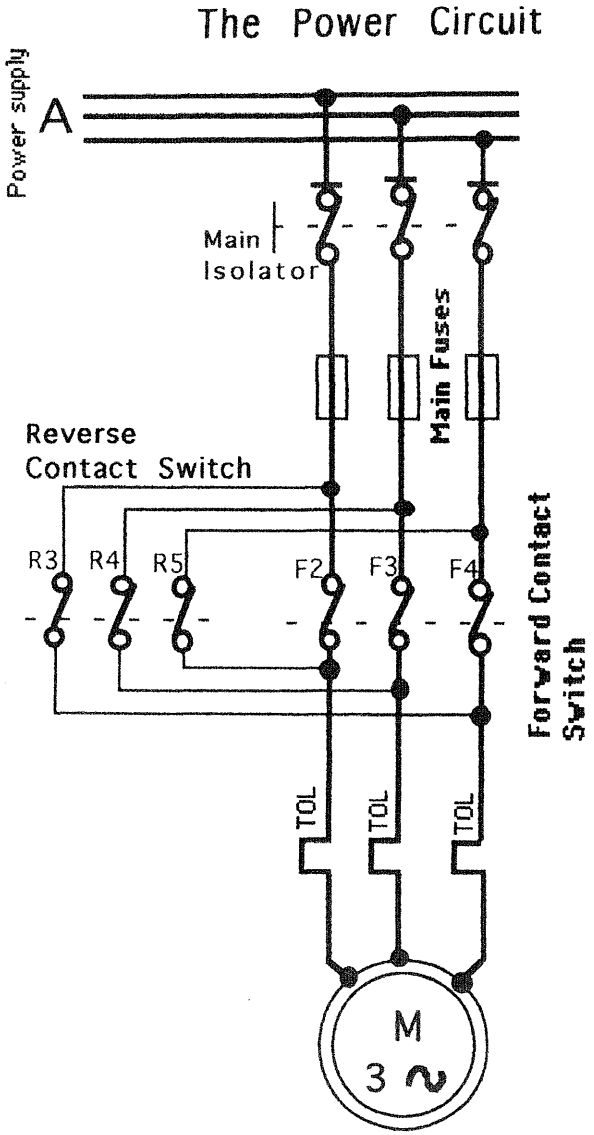
At the completion of baking .....

A Switch T1 opens, energising Coil R. Coil R opens Switch R2, stopping the timer.

B Switch T1 closes, energising Coil R. Coil R opens Switch R2, stopping the timer.

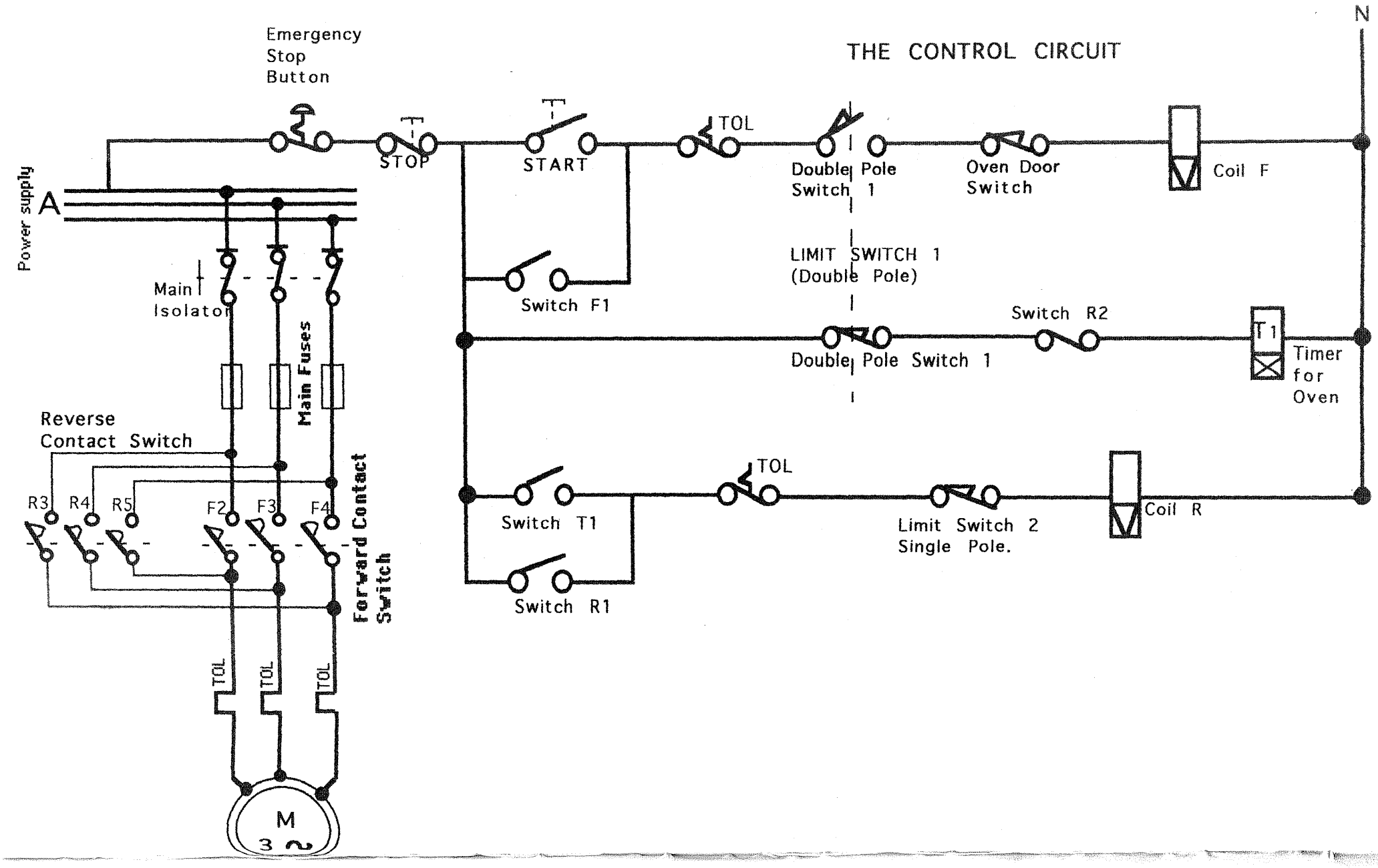
C Switch T1 closes, energising Coil R. Coil R closes Switch R2, stopping the timer.

D Switch T1 opens, energising Coil R. Coil R closes Switch R2, stopping the timer.



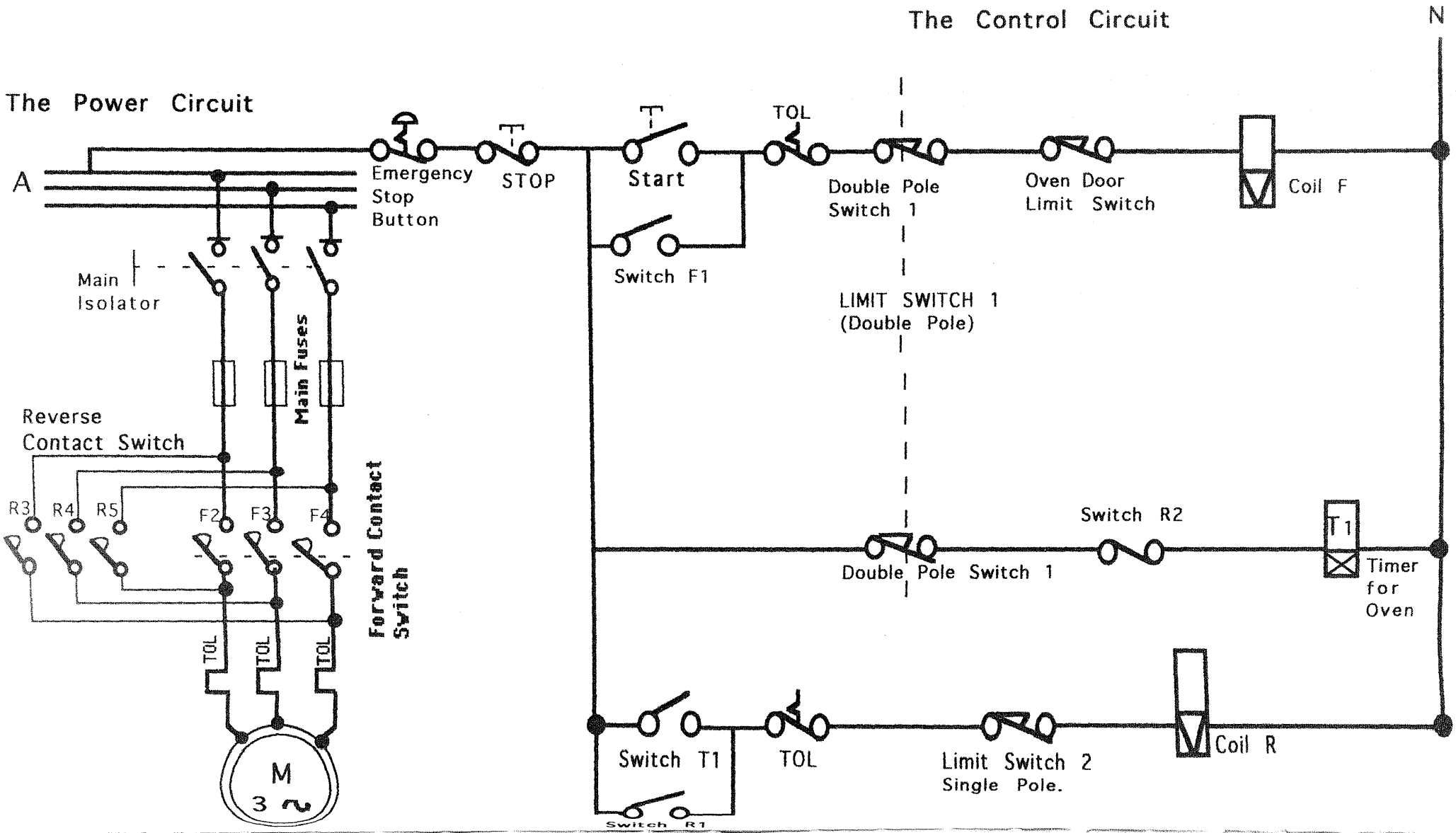
10) Identify any fault/s in this circuit and state the reasons for your answer

11) According to the diagram below, what item/s in the circuit are currently operating?



12) For the conveyor to start moving the goods away from the heating elements, what has to change in this circuit?

Place a cross on all the item/s in the diagram which need to be different. Indicate what change is required.



Appendix 13:

An analysis of the element interactivity of the instructional material used in Experiment 5.

Isolated Elements version of Instructions

1. A single acting cylinder is indirectly controlled because it operates at a high speed.
2. There are two sources of air supplying the circuit.
3. Valves control air supply. This circuit has two valves.
4. When the pneumatic circuit is at rest the cylinder's piston is retracted and no air flows through the circuit.
5. One valve is controlled by a pushbutton. Pushing this button moves the circuit into the On position.
6. When the button is pushed, air flows from the pushbutton valve to the control valve.
7. Air from the Control valve supply flows to the cylinder chamber,causing the piston to extend to date stamp boxes on a production line.
8. When the pushbutton is released the flow of air around the circuit stops, the cylinder's piston retracts and the state of the circuit reverts back to the Off position.

Interacting Elements Version of Instructions

1. A single acting cylinder is indirectly controlled because it operates at a high speed.
2. There are two sources of air supplying the circuit.
3. Valves control the air supply of the circuit through their switching positions. Each switching position is configured differently to channel the flow of air.
4. The 3/2 way push button valve has 3 openings or ports and 2 possible switching positions. Pushing the button puts in the valve into the On position.
5. When the pushbutton valve is in the Off switching position, the supply of air through port 1P is blocked and air from the circuit is expelled into the atmosphere through port 3R.
6. The circuit has a second valve, the control valve. It also has two switching positions.
7. Air supplied by the pushbutton valve through a pilot line (12Z) to the control valve directs the choice of switching position.
8. When the control valve is in the Off position, no air comes through the pilot line (12Z) and the control valve's air supply through port 1P is capped.
9. When the control valve is in the Off position, air is bled out from the cylinder to the atmosphere through the control valve's port 3R.
10 All these factors result in the retracted position of the cylinder's piston.
11. When the pushbutton valve is in the On position, air flows through port 1P into the valve and then into the pilot line (12Z) supplying the control valve, forcing it into the On position
12. When the Control valve is in the On position, it supplies air through port 1P, from its own source, to the cylinder.
13. This causes the cylinder's piston to extend and date stamp boxes on a production line.
14. When the pushbutton is released, the pushbutton valve moves back to the Off position, causing air supply through the pilot line to stop.
15. The air that forced the control valve into the On position is bled into the

atmosphere through port 3R in the pushbutton valve causing the control valve to move back into the Off position.

16. Air is no longer supplied to the cylinder from the control valve. Air from the cylinder is bled to the atmosphere through the Control Valve.

17. These factors cause the cylinder's piston to retract.

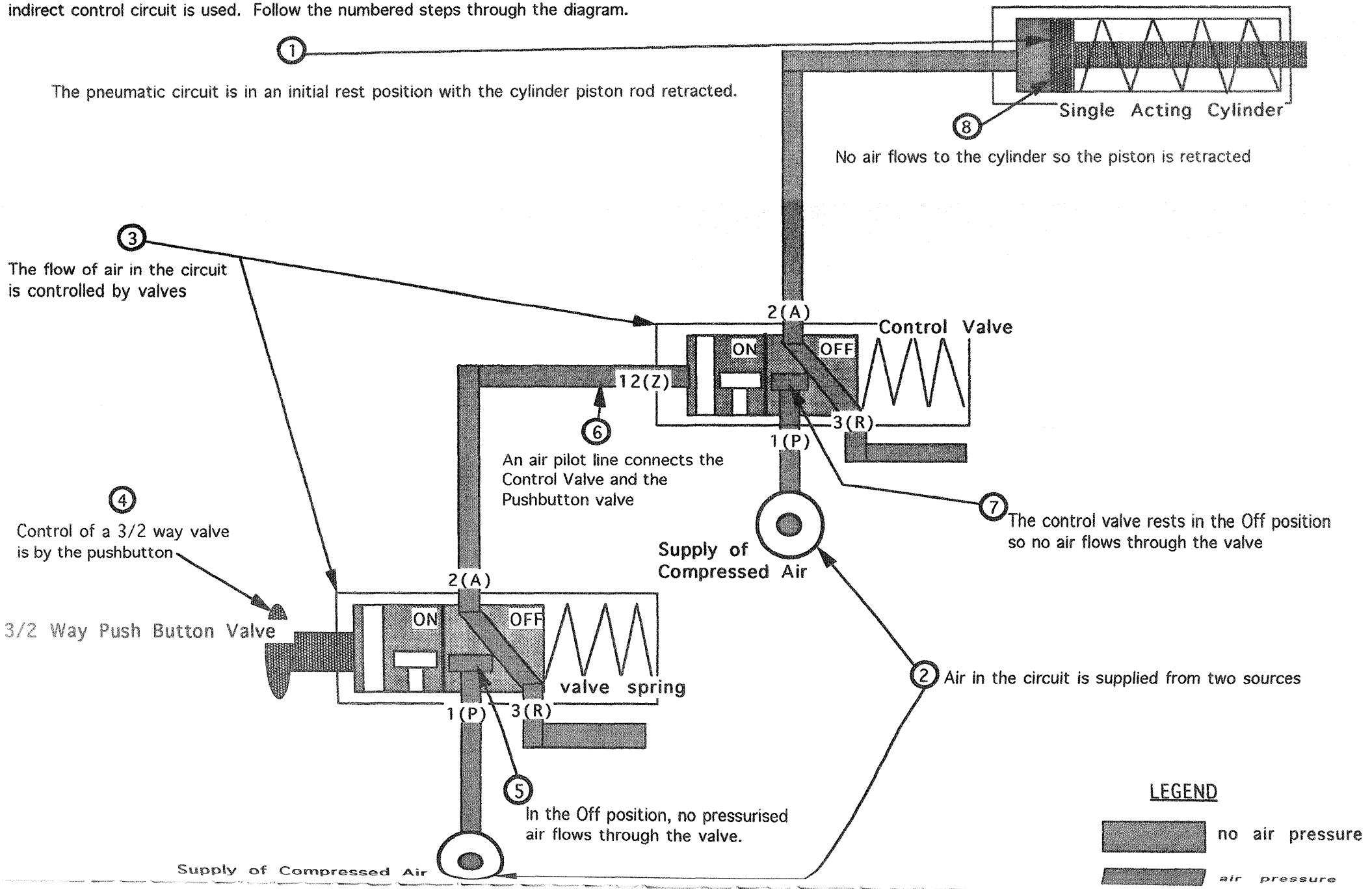
Appendix 14:

Isolated elements instructional material used in Experiment 5.

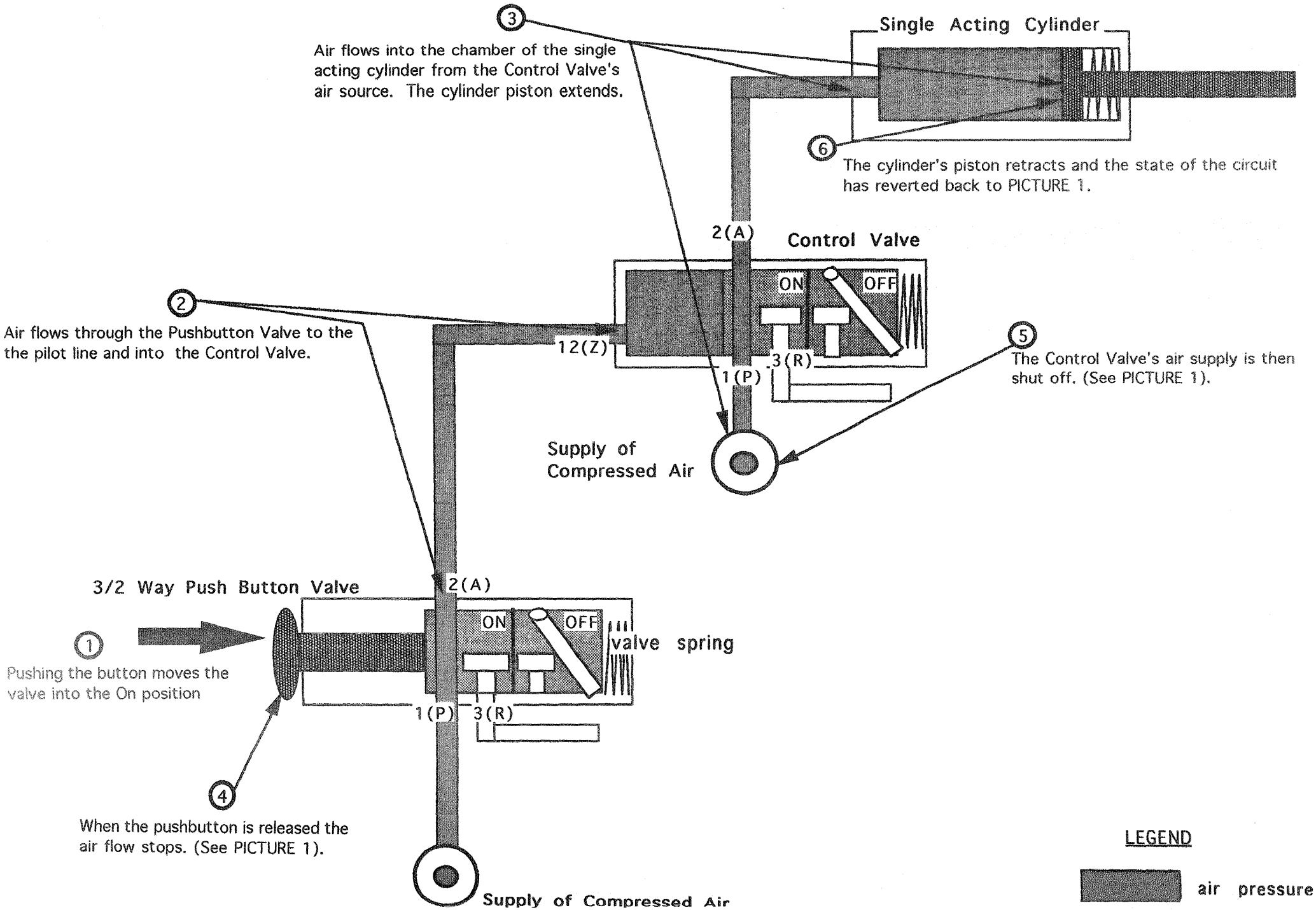
A PNEUMATIC CIRCUIT FOR INDIRECT CONTROL OF A SINGLE ACTING CYLINDER

PICTURE 1 - OFF POSITION

A single acting cylinder is designed to extend and date stamp boxes on a production line when a pushbutton is pressed. The cylinder operates at a high speed therefore an indirect control circuit is used. Follow the numbered steps through the diagram.



PICTURE 2 - ON POSITION



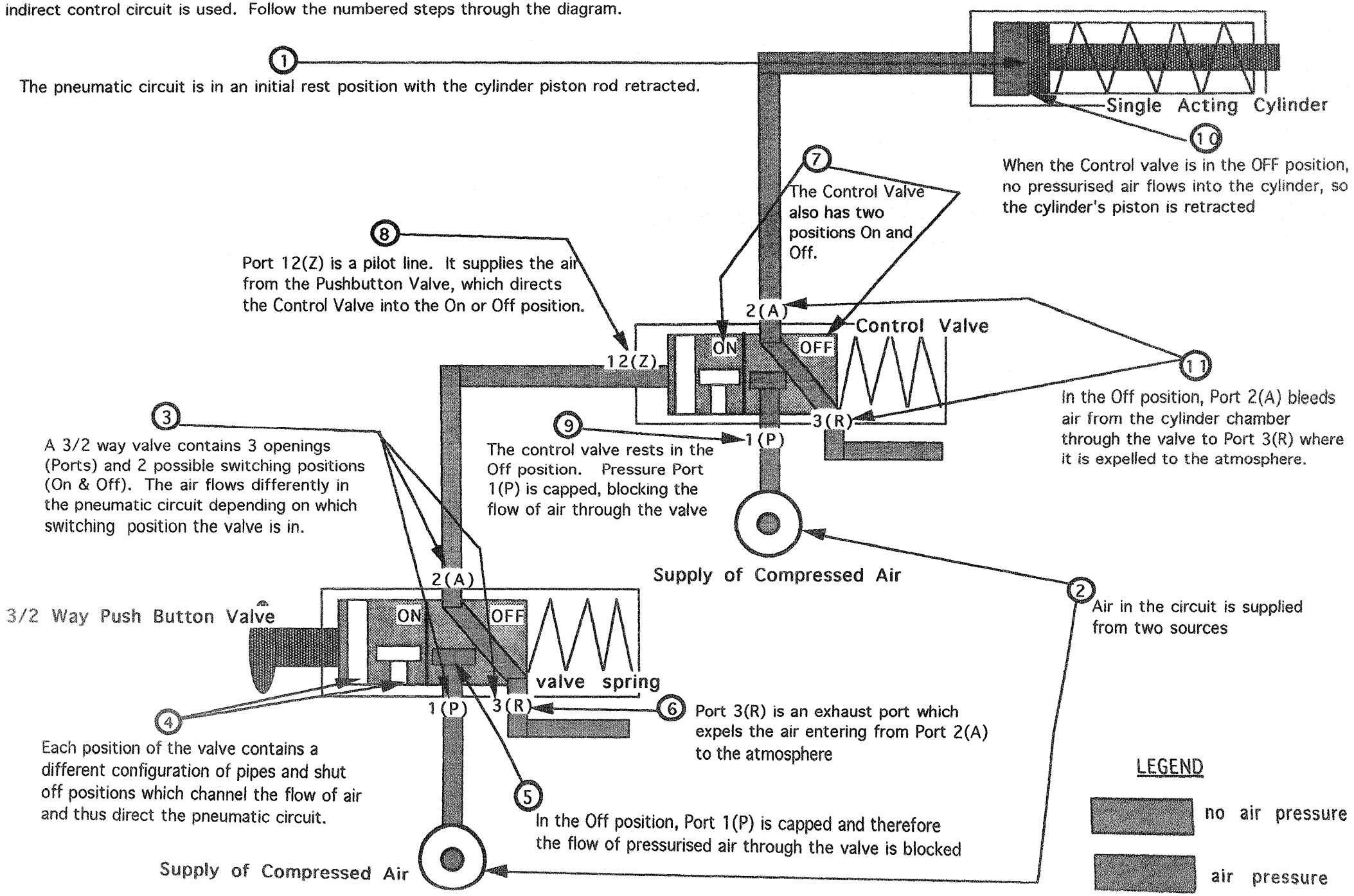
Appendix 15:

Interacting elements instructional material used in Experiment 5.

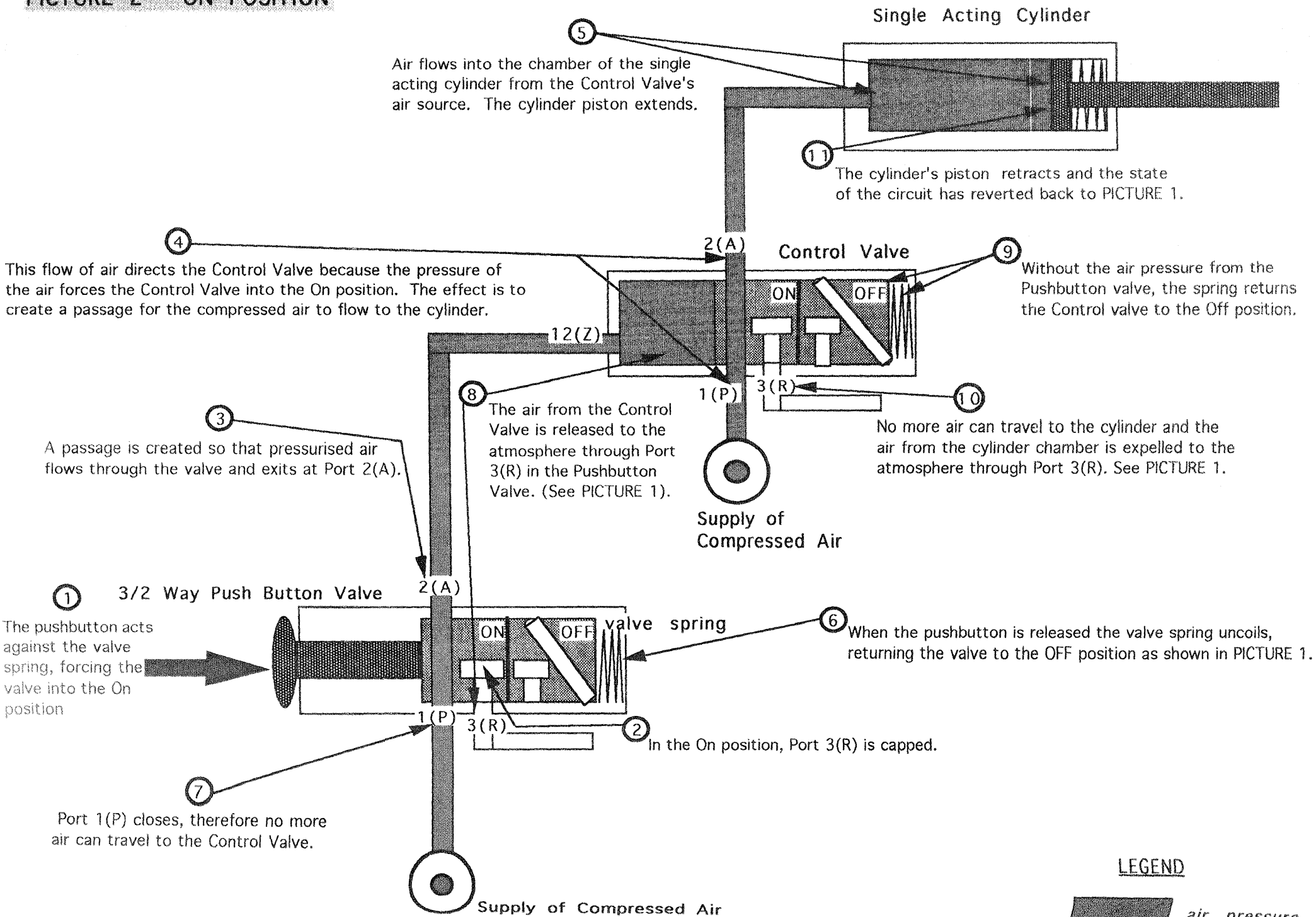
A PNEUMATIC CIRCUIT FOR INDIRECT CONTROL OF A SINGLE ACTING CYLINDER

PICTURE 1 - OFF POSITION

A single acting cylinder is designed to extend and date stamp boxes on a production line when a pushbutton is pressed. The cylinder operates at a high speed therefore an indirect control circuit is used. Follow the numbered steps through the diagram.



PICTURE 2 - ON POSITION



Appendix 16:

Subjective mental load rating scale used in Experiment 5.

Name:

Subject Number:

LEARNING A PNEUMATIC CIRCUIT

How easy or difficult was it to learn and understand the pneumatic circuit from the instructions you were given?

Place a cross on the rating scale below

1	2	3	4	5	6	7	8	9
Extremely Easy	Very Easy	Easy	Slightly Easy	Neither Easy nor Difficult	Slightly Difficult	Difficult	Very Difficult	Extremely Difficult

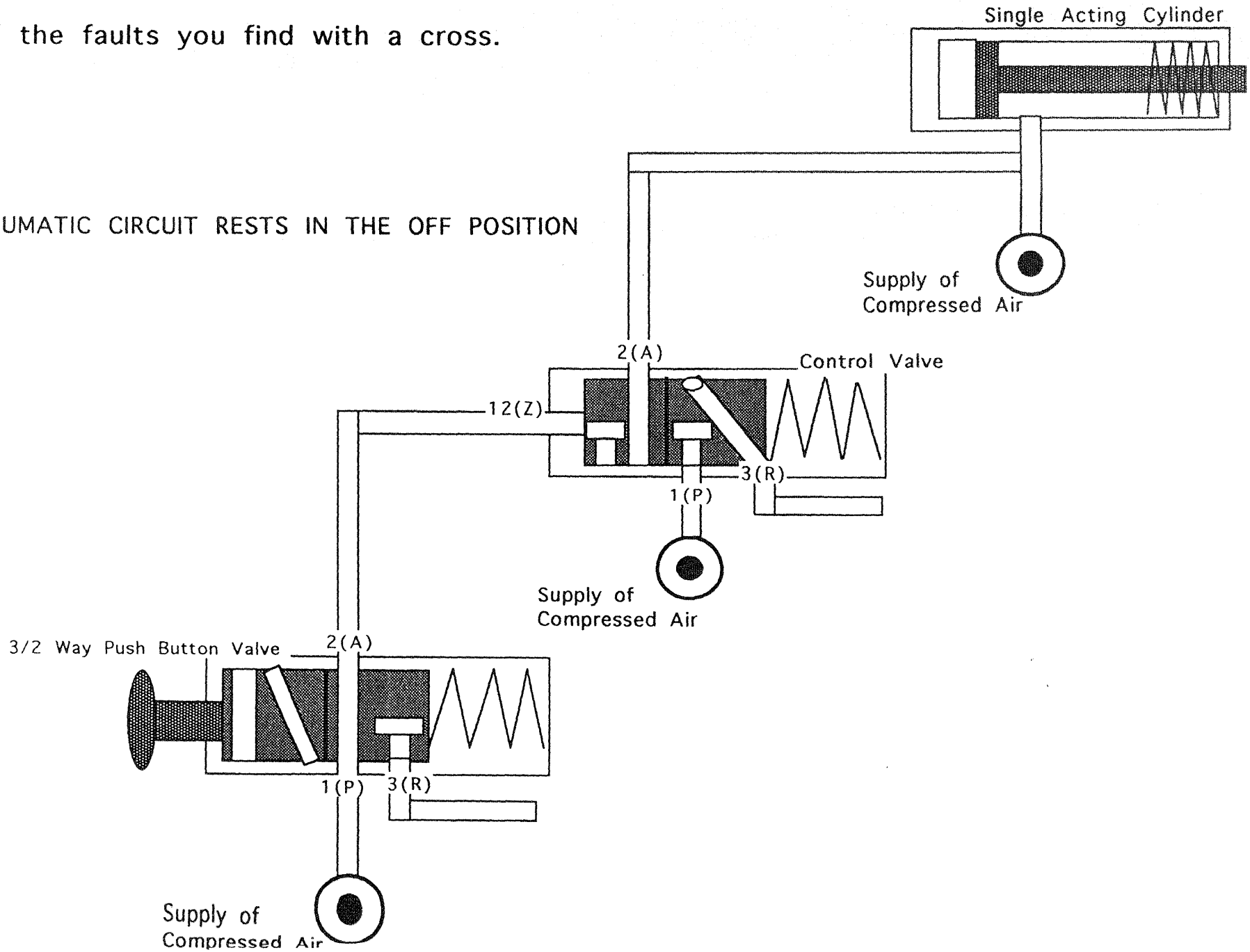
Appendix 17:

Fault finding test used in Experiment 5.

There are faults in this pneumatic circuit.

Mark all the faults you find with a cross.

THE PNEUMATIC CIRCUIT RESTS IN THE OFF POSITION



Appendix 18:

Pollock, E.; Chandler, P. & Sweller, J. (in press) Assimilating complex information. *Learning and Instruction.*

Assimilating complex information

Edwina Pollock, Paul Chandler and John Sweller
University of New South Wales

Keywords: learning; instruction; element interactivity; cognitive load; working memory; long-term memory; schemas

The work reported in this paper was supported by a grants from the Australian Research Council. The authors wish to thank Email Ltd and BHP Ltd for their ongoing support and commitment to our joint research programs. In particular, we wish to thank Bryan Jones, Richard Winter and the technical training team, trainees and apprentices of Email Training Services and Eddy Gosek, Brian Walker the training staff and apprentices of BHP Training Centre, Port Kembla.

Abstract

Methods of instruction which are intended to facilitate understanding tend to incorporate all of the information elements required for understanding in the instructions. Frequently, these types of instructions may overwhelm a learner's limited working memory and hinder learning. In four experiments, a two phase, isolated-interacting elements learning approach was developed in which in the first phase, the element interactivity of complex material was artificially reduced by presenting the material as isolated elements of information that could be processed serially, rather than simultaneously, in working memory. In the second phase, all the information for understanding was presented. The control group was simply presented with all the information for understanding in both Phase 1 and 2. The results provided powerful evidence that for certain groups of learners, information is better learnt through the isolated-interacting elements instructional method.

Consider a student faced with the task of understanding and learning very taxing material that will require considerable time, effort and thought. What mental processes does the student engage in and what instructional procedures and designs can best facilitate learning? Cognitive load theory is concerned with issues associated with cognitive processes and instructional design and may assist in answering these questions. Nevertheless, as will be seen below, when dealing with highly complex information, there is a gap in the theory as currently formulated. This paper considers theoretical points and provides data intended to address some of the issues associated with learning intellectually difficult material.

Cognitive load theory

Cognitive load theory (Sweller, 1999; Sweller, van Merrienboer & Paas, 1998) uses some aspects of human cognitive architecture and of the structure of information to provide instructional designs that facilitate understanding, learning and problem solving. The theory assumes; 1) a limited working memory that can process only a few elements of current information at any given time (Miller, 1956), 2) an effectively unlimited long-term memory holding knowledge that can be used to overcome the limitations of working memory, 3) schemas (Chi, Glaser & Rees, 1982; Larkin, McDermott, Simon & Simon, 1980), held in long-term memory and used to structure knowledge by organising elements of information consisting of lower order schemas into higher order schemas that require less working memory capacity and, 4) automation that allows schemas to be processed automatically rather than consciously in working memory thus reducing working memory load (Kotovsky, Hayes & Simon, 1985; Schneider & Shiffrin, 1977; Shiffrin & Schneider, 1977).

Cognitive load theory makes a distinction between two sources of cognitive load. Extraneous cognitive load is generated by the manner in which information is presented to learners and is under the control of instructional designers. A major assumption of cognitive load theory is that instruction should be structured to reduce unnecessary extraneous working memory load (see Sweller, 1999; Sweller, van Merrienboer & Paas, 1998, for a range of instructional techniques designed to

reduce cognitive load, such as the completion effect discussed by van Merriënboer, Schuurman, de Croock & Paas, this issue: the use of worked examples, Stark, Mandl, Gruber, & Renkl, this issue; van Gerven, Paas, van Merriënboer & Schmidt, this issue: split-attention, discussed by Mayer & Moreno, this issue; van Bruggen, Kirschner & Jochems, this issue: or the modality effect, Mayer & Moreno, this issue). In contrast, intrinsic load is imposed by the intellectual complexity of information. The intrinsic characteristics of some information is such that if understanding is to occur, it will impose a heavy working memory load irrespective of instructional design considerations. It is that complex information that is the subject of this paper.

The intrinsic cognitive load of information is determined by the extent to which various elements interact. An element is information that can be processed by a particular learner as a single unit in working memory. Some information can be understood and learned, individual element by individual element because the elements do not interact. Learning a new vocabulary provides an example. Each new vocabulary item can be learned without reference to any of the other items. For instance, the Spanish word for *bird* can be assimilated and learned independently of the word for *cat*. The information is low in element interactivity. It imposes a low intrinsic cognitive load because in order to understand and learn it, only a limited number of elements need to be processed in working memory at any given time. In contrast, high element interactivity material consists of elements that cannot be understood in isolation because they interact. We can learn vocabulary items individually but we cannot learn grammatical syntax without considering several vocabulary items and their relations: We can learn the names and perhaps even the functions of individual electrical components one at a time but we cannot understand an electrical circuit consisting of those components without simultaneously considering several components and their relations. Understanding syntax or an electrical circuit are high element interactivity endeavours for learners with little experience in these areas. As a consequence, they impose a high, intrinsic cognitive load because many elements must be processed in working

memory simultaneously. Difficult to understand material is difficult precisely because it is high in element interactivity and imposes a heavy load on working memory (Chandler & Sweller, 1996; Marcus, Cooper & Sweller, 1996; Sweller, 1994; see also Halford, Wilson & Phillips, 1998, for a similar formulation). The intrinsic cognitive load of instructional material is fixed if immediate understanding is the goal of the task. However, as will be indicated below, intrinsic load can be altered if a learning situation is created where understanding is temporarily not an objective.

Our cognitive architecture handles the problem of an excessively high working memory load by constructing schemas. A schema is a cognitive construct that organises elements of information categorically and stores them in long term memory. Bartlett (1932) provided an early conceptualisation of schema theory that forms the foundation of modern work. Schemas abstract knowledge, with the result that the infinite variety of information captured by our senses can be ignored, reducing the burden on working memory capacity and allowing knowledge represented in one context to be used in many contexts (Thorndyke and Hayes-Roth, 1977; Rumelhart, 1980). Expertise is determined by the extent of schema acquisition (de Groot, 1965; Chase and Simon, 1973), as well as the degree to which these schemas contain specific information on domain typical problem solutions (Chi et al., 1982; Larkin et al, 1980 ; Sweller, Mawer and Ward , 1983). With expertise, interacting elements that would otherwise overwhelm working memory can be incorporated in a schema that acts as a single element and imposes a minimal working memory load. Thus, expert English speakers have a schema that tells them that "the cat sat on the mat" is syntactically correct but "cat the on sat mat the" is not. Algebraic schemas tell domain experts that $a/b = c$ can be transformed into $a = cb$. Both equations and their transformation can be processed easily in working memory by anyone who is competent at elementary algebra despite the large number of elements. (Elements include relations between symbols as well as the symbols themselves.) Once acquired, the otherwise difficult task of processing all the interacting elements in working memory becomes a

relatively simple task because the elements are incorporated in a schema that itself acts as an element and can be processed readily in working memory.

A paradox

This combination of cognitive architecture and information structures constitutes the core of cognitive load theory and can be used to design instruction intended to reduce extraneous cognitive load and facilitate learning through more rapid and effective schema construction (see Sweller, 1999; Sweller, van Merrienboer & Paas, 1998). Nevertheless, as it stands, there is a gap in the theory. Very complex, high element interactivity material cannot be understood without processing all of the elements simultaneously in working memory. As indicated above, our cognitive architecture overcomes this problem by incorporating the elements in a schema or limited number of schemas that can be processed easily in working memory. The difficulty arises when one considers how such schemas can be constructed in the first instance, if the learner possesses no or few *a priori* schemas in the domain. If the material is very complex, consisting of a large number of elements that are high in element interactivity, it will not be possible to process all of the elements in working memory simultaneously. But they must be processed simultaneously in working memory in order to be understood and for a schema to be constructed. Thus, we have a paradox. Material can be understood once a schema has been constructed allowing all the elements to be processed in working memory simultaneously but until that point has been reached, the elements cannot be processed simultaneously in working memory and so cannot be understood. It seems to follow that for novices learning very complex material, the initial stages of schema construction must occur in an environment in which the material cannot be understood.

Most instructional design is intended to facilitate understanding and accordingly, instructional designers present their material in a manner that potentially permits understanding. That means incorporating all of the elements required for understanding in the instructions. As indicated above, for very complex material with many interacting elements, attempts by learners to assimilate all of the

elements will inevitably fail due to working memory limitations. Alternative modes of presentation may be required.

There is evidence in the literature that presenting all the elements of information required for understanding does not benefit learning. King & Russell (1966) compared academically matched students, given the task of listening to an English passage with instructions to either aim to learn the passage 'word for word' or to focus on learning the main ideas. By the second recall (undertaken after a second session of listening to the passage) the trend of the data suggested that the word for word group was outperforming the main ideas group in terms of understanding the meaning of the passage.

Hoosain (1983) was interested in exploring if a cognitive basis existed for the traditional Chinese practise of rote learning classical literary passages. Groups were instructed to either learn the passages verbatim, without trying to understand the ideas, or to aim to understand the ideas in the passage without focusing on the exact words. As might be expected, the rote group performed significantly better on verbatim recall of the passage but surprisingly, the performance of the rote and main ideas groups in terms of passage comprehension could not be distinguished. These results provide credence to our hypothesis that initially learning some elements of information, even if comprehension is not possible, may ultimately increase a student's understanding of a topic.

Consider the instructional material of Figure 1 intended to teach electrical engineering students how to conduct a test of insulation resistance. In order to understand this material, the learner must simultaneously consider the following elements: the aim of the test, the required setting of the volt meter, the required setting of the appliance switch, the location of the earth lead and the line lead during both phases of the test and the criteria by which to judge the voltmeter readings as safe. For novices, these elements far exceed working memory limits. While an expert electrical engineer can process all of these elements simultaneously in working memory via previously acquired schemas, a relative novice (e.g., first year apprentice or trainee), not possessing the requisite

schemas, can only hold and process a few of the elements simultaneously in working memory. If only a few of the elements are processed in working memory simultaneously, the material cannot be understood because the elements interact. Accordingly, this material, designed to facilitate understanding, must inevitably fail in its aim with both understanding and learning inhibited. This type of instructional procedure will be referred to as the "interacting elements" procedure. It is characterised by the inclusion of all elements required for understanding but at the cost of an impossibly high working memory load.

Insert Figure 1 about here

In contrast, consider the alternative instructional material consisting solely of the figure and bold type text of Figure 1. These materials are intended to be learned element by element without a high degree of understanding. They will be referred to as the "isolated elements" procedure. Unlike the interacting elements procedure, isolated elements can be easily held and processed in working memory. The cost is reduced understanding but the potential benefit is an increased probability of learning because the reduced working memory load permits necessary elements to be held and processed in working memory. Once these materials are learned (albeit with limited understanding), it may be possible for learners to process the interacting elements materials in working memory allowing full understanding to occur. We predict that studying isolated elements materials followed by interacting elements materials will be of more benefit to learning than repeated exposure to the interacting elements materials. This prediction is supported by research into learning hierarchies which suggests that lower level skills are prerequisites to higher order skills (Bloom, 1956 and Gagne, 1970). This prediction follows only under circumstances where the knowledge base of the learners is such that they are unable to process all of the interacting elements in working memory. Experiments 1 and 3 test this prediction. In contrast, if learners have acquired sufficient schemas to process the required elements and their interactions simultaneously in working memory, the need for an isolated elements approach may be unnecessary. Experiments 2 and 4 test this prediction.

Experiment 1

The first experiment used two groups. The isolated-interacting elements group studied isolated elements instructions at phase one followed by instructions involving interacting elements at phase two. The isolated-interacting elements group was compared to a more conventional method of instruction for promoting understanding, the interacting elements only group where instructions containing interacting elements were presented at both phases of instruction. It was expected that the interacting elements only group would perform more poorly at both phases. Compared to the isolated-interacting elements group, the working memory resources of the interacting elements only group would be overwhelmed because of the many elements they were required to mentally integrate due to the high element interactivity of the material.

Method

Participants

Twenty-two first year industrial trade students enrolled in a training course at a Sydney company participated in the experiment. The students were from a varied educational background but all had completed at least Year 10 of High School. They were enrolled in either a one month basic industrial skills course (9 students) or a six month manufacturing and engineering trainee-ship course (13 students). Students were tested within their first month of training so the electrical material covered at this point in both of the courses, was very similar. They understood electrical principles at a basic level, such as how to locate the neutral, active and earth pins of an electrical appliance. However, none of the participants had directly studied the electrical safety tests used in the experiment. Students were excluded from the study if they had completed further studies in the field of electrical engineering; if English was not their first language; or if more than forty eight hours elapsed between the two learning sessions of the study (e.g., due to illness of a student).

Materials

The instructional materials used in the experiment consisted of information on three electrical safety tests that are performed on 240 volt electrical appliances using a volt meter. The three tests check various electrical circuits of the appliance to ensure that it is operating safely. In this experiment the electrical appliance used was an electrical kettle. The three tests were to check the: insulation between the frame and the active and neutral points of the appliance (the Insulation Resistance test); insulation between the frame and the earth point of the appliance (the Earth Continuity test) and the insulation of electrical extension leads (the Insulation Resistance of a Flexible Lead test). For a novice learner, understanding what and why to do each of the tests was a complex task as the material is high in element interactivity. For example, to understand the tests of Insulation Resistance of a Flexible Lead, the learner must simultaneously consider what the test aims to achieve, the required setting of the volt meter, the on/off status of the appliance, the placement position of both leads of the volt meter, the criteria set for safe and unsafe readings, the consequences of an unsafe reading and finally, the changes made to the procedure to test the second circuit and thus complete the test.

Two sets of introductory instructional material were developed: one set presented isolated elements that had to be learned without emphasis on the interactions between them (isolated elements condition) while the other placed an emphasis on the interactions between elements (interacting elements condition). Each set of instructional materials consisted of a three page booklet. Each page of the booklet described a different electrical safety test and contained a numbered diagram with a separate section of correspondingly numbered text points below the diagram. Both sets of instructions were designed to introduce and demonstrate the equipment and procedures used in the three electrical safety tests described above.

The isolated elements version of instructions explained the procedural steps of the electrical tests. For example, in the bold text of Figure 1, the step by step procedure for performing an insulation resistance test of an electrical appliance is displayed. The isolated elements instructions included no text other than the bold

text. The interacting elements instructions replicated the isolated elements version of the information (bold text in Figure 1), but also included other information relevant to the tests such as the aim of the test and the logic, in terms of electrical circuit theory, behind each step of the test (see normal text in Figure 1). The interacting elements instructions therefore provided all the information for gaining a meaningful understanding of the electrical test as a whole.

An analysis of the element interactivity of the Insulation Resistance test has been included in the Appendix. The analysis estimates the number of elements that would need to be simultaneously processed for a first year electrical trainee with only a very limited electrical knowledge to fully understand the test.

Subjective ratings of task difficulty were used as a measure of cognitive load (see Paas & van Merrienboer, 1993: 1994). The subjective mental load rating measure used was a seven point Likert type scale. Above the scale was the instruction "How easy or difficult was it to learn and understand the electrical tests from the instructions you were given? Place a cross on the rating scale below". Each number on the scale was labelled with a description ranging from extremely easy (1) to extremely hard (7).

The test material consisted of test items and equipment for both written and practical tests. Each test item was designed to be objective and was marked as either correct or incorrect. The written test consisted of twenty three items. Some problems had several parts so the test was marked out of a total of thirty-one points.

There were two categories of questions in the written test: those tapping low element interactivity knowledge about the test procedures and those tapping high element interactivity knowledge of the electrical testing procedures and electrical circuitry. Low element interactivity questions were factually based, asking about a specific part of, or reason behind the procedure of a specified electrical test. This category contained eighteen questions. Each correct answer to these items was worth one mark.

High element interactivity knowledge questions required the students to practically apply their factual knowledge of the electrical tests. This category contained five questions, some with several parts. Each correct answer to each item or part was worth one mark; total marks possible for this category of questions was thirteen.

The practical test consisted of two items: both items required transfer of knowledge to a novel situation. Specifically, students were required to apply knowledge of an electrical kettle to safety tests of a fluorescent light. The performance of each electrical test was marked either correct or incorrect, so the practical test was scored out of a total of two marks. The element interactivity of knowledge tapped by the practical tests was high as students had to remember the entire test procedure and apply it to a novel situation.

Procedure

All the students were randomly assigned to either the isolated-interacting elements instruction or the interacting elements only group with 11 students in each group. They were tested individually, in a quiet room. The students were informed that they were to be given some electrical safety testing instructional material which would be followed by a written and practical test and that this session was to be repeated within 48 hours. In the first experimental session, the students in the isolated-interacting elements instruction group were given the isolated elements instructional booklet while the students in the interacting elements only group were given the interacting elements instructional booklet. An unlimited time was allowed to read through the instructions and this time was recorded by the experimenter. All students were provided with a metal electric kettle and a volt meter. They were instructed to perform the tests with this equipment as they read through the instructions. At the completion of the study phase, the students were provided with a subjective mental load rating scale, the format of which was explained to both groups. They were asked to rate the mental effort involved in understanding all of the electrical tests described in their training booklet on the scale described above. The test section of the experiment followed. The students were asked to complete the written test, described in the materials section, which was common to both

groups. They were allowed an unlimited time to complete the test. Students were prohibited from reattempting any question once they had attempted it. The instructional material was not available during the test period. The practical test followed. Students were required to attempt both parts of the practical test in a set order (Earth Continuity and then Insulation Resistance).

The second phase of the experiment was held within forty-eight hours of the first phase. The procedure of the second phase of the experiment was identical to the first, apart from two differences. 1) In the practical section, the order in which the two electrical safety tests were performed by the students was reversed (that is Insulation Resistance test followed by Earth Continuity test). 2) All students studied the interacting elements instructions.

Results and Discussion

The test score variables under analyses were written test scores, divided into low element interactivity knowledge scores and high element interactivity knowledge scores, practical test scores and mental load ratings. Means and Standard Deviations for these variables are provided in Table 1.

Insert Table 1 about here

Low and high element interactivity knowledge scores were analysed separately by 2 (Instruction Condition) x 2 (Test Phases) ANOVAs with repeated measures on the second factor. The results for low element interactivity knowledge question scores indicated there was no difference between the instructional groups, $F(1,20) = 2.65$, $MSe = 288.82$. (The .05 level of significance is used throughout this paper.) The test phases main effect indicated a significant difference, $F(1,20) = 11.77$, $MSe = 180.10$, demonstrating an improvement over the two instructional phases. There was no significant interaction, $F(1,20) = 0.87$, $MSe = 180.10$. The results for high element interactivity question scores indicated there was a significant difference between the instructional conditions in the expected direction, $F(1,20) = 7.06$, $MSe = 203.87$, with the isolated-interacting elements instruction group performing at a superior level. The test phases main effect was also significant,

$F(1,20) = 9.50$, $MSe = 147.53$, demonstrating an improvement over time. There was no significant interaction, $F(1,20) = 2.67$, $MSe = 147.53$.

There were 2 practical tests. Each test was scored correct or incorrect and the results of the two tests were added together to give a practical score out of two points. These scores were assessed by a 2 (Instruction Condition) x 2 (Test Phases) ANOVA with repeated measures on the last factor. Results indicated a possible difference between the instructional conditions, $F(1, 20) = 4.03$, $MSe = 0.14$, $p = 0.058$ favouring the isolated-interacting elements group. There was no improvement over time, $F(1, 20) = 0.20$, $MSe = 0.11$, nor was there a significant interaction, $F(1, 20) = 1.80$, $MSe = 0.11$.

The time taken by the participants for instruction were assessed by a 2 (Instruction Condition) x 2 (Test Phases) ANOVA with repeated measures on the last factor. Results indicated a possible difference between instructional conditions in the time taken for instruction over both phases, $F(1, 20) = 3.33$, $MSe = 98161.07$, $p = 0.08$. An inspection of the means indicated that trend was toward the isolated-interacting elements group requiring less instructional time in total than the interacting elements only group. This measure also showed a possible difference in the time taken over the two instruction phases, $F(1, 20) = 3.88$, $MSe = 56831.73$, $p = 0.06$. There was no significant interaction, $F(1, 20) = 1.92$, $MSe = 56831.73$.

Mental load ratings showed a significant instruction condition main effect in the expected direction, $F(1, 20) = 4.74$, $MSe = 2.11$, suggesting that the isolated-interacting elements instruction group found their instructions less demanding than the interacting elements only group over both phases. In phase one this difference was to be expected given the disparity in the amount of information in the different sets of instructions. At phase 2 however, when both groups received the same interacting elements version of the instructions, (the interacting elements only group for the second time), a difference is still maintained (see Table 1). In fact, the mean of the isolated-interacting elements group at Phase 2 is still well below that of the interacting elements only group at Phase 1. There was no improvement over time, $F(1, 20) = 0.26$, $MSe = 0.77$, nor was there a significant interaction, $F(1, 20) =$

0.72, MSe = 0.77. The lack of a significant interaction also suggests that the difference in mental load ratings between groups was similar through both instructional phases.

Paas and van Merrienboer (1993;1994) devised a method of combining performance measures with subjective mental load ratings to investigate the relative efficiency of instruction conditions. Instructional efficiency increases when learners demonstrate either a superior performance than would be expected from their level of mental effort or require less mental effort to achieve a given performance score. Paas & van Merrienboer suggested that "combined measures (of performance and mental effort) may provide a better insight into the cognitive consequences of training and task environments and thus improve attempts to design optimal environments" (1993, p.742). Performance can be measured by an objective test score relevant to the learning task; mental effort ratings require the learner to translate the perceived mental effort required by the task into a numerical value. Efficiency (E) is calculated by transforming both mental effort (M) and performance scores (P) to standardised z scores. These z scores for each instruction condition are then combined in the following manner to give a single efficiency score for each condition:

$$E = \frac{/M-P/}{\text{square root}(2)}$$

The sign of the relative condition efficiency depends on the relation between metal load and performance scores. It is calculated according to these rules: If M-P<0, then E is positive; if M-P>0, then E is negative. The greater the value of E, the more efficient the instruction condition. It was hypothesised that the isolated-interacting elements instruction group would be more efficient than the interacting elements only group. In other words, it was expected that the isolated-interacting elements instruction group would achieve a greater performance for less mental effort invested.

Efficiency measures for both high and low element interactivity questions were calculated and analysed using 2 (Instruction Condition) x 2 (Test Phases) ANOVAs

with repeated measures on the last factor. Table 2 shows the mean and standard deviations for normalised (Z) mental effort and performance scores and the condition efficiency scores for each group. Results for low element interactivity question efficiency scores indicated there was a significant instruction condition main effect favouring the isolated-interacting elements group, $F(1, 20) = 6.75$, $MSe = 1.26$. There was no improvement over time, $F(1, 20) = 3.59$, $MSe = 0.77$. There was no significant interaction between the factors, $F(1, 20) = 0.01$, $MSe = 0.77$. The results for high element interactivity question efficiency scores also indicated a significant difference between the instructional conditions favouring the isolated-interacting elements instruction group, $F(1, 20) = 12.64$, $MSe = 0.97$. There was no improvement over time, $F(1, 20) = 3.63$, $MSe = 0.59$. There was no significant interaction between the factors, $F(1, 20) = 0.40$, $MSe = 0.59$.

Insert Table 2 about here

Practical score efficiency measures were analysed in a 2 (Instruction Condition) x 2 (Test Phases) ANOVA with repeated measures on the second factor. Table 2 shows the means and standard deviations for standardised (Z) mental effort and practical performance scores and the condition efficiency scores for each group. Results indicated there was a significant instruction condition main effect favouring the isolated-interacting elements instruction group, $F(1, 20) = 5.64$, $MSe = 1.83$. There was no improvement over time, $F(1, 20) = 0.00$, $MSe = 0.44$, nor was there an interaction effect, $F(1, 20) = 0.45$, $MSe = 0.44$.

The results from the written test and the subjective mental load measures support the hypothesis that the isolated-interacting elements instruction group performed at a superior level to the interacting elements only group because this method of instruction imposed a lower cognitive load. We assume that in the initial learning phase, the isolated-interacting elements group of students did not fully understand the complex concept of electrical safety testing, but they acquired rudimentary schemas. By obviating the need to process all interacting elements required for understanding in working memory, students may have been able to acquire at least some of the necessary elements, resolving the paradox of learning without

processing all interacting elements required for understanding in working memory. Subsequently, the interactions between these elements could be learned in the second phase allowing a more complete understanding of the material. In contrast, the interacting elements only group may have acquired few elements in either the first or the second phase because the interacting elements may have exceeded working memory capacity.

Experiment 2

If schema formation was the key to the success of the isolated-interacting elements of instruction with inexperienced learners in Experiment 1, that method of instruction may be of little or no benefit to more experienced or expert learners. These learners, by definition of their expertise, already possess more sophisticated schemas in the given area and so may be able to process the interacting elements in working memory without undue difficulty. The purpose of Experiment 2 was to investigate the success of the isolated-interacting elements method of instruction compared with the interacting elements only method, over two learning periods using participants with relatively more expertise than those of Experiment 1. Experiment 1 was replicated in all respects, except, in the expertise of the participants.

Method

Participants

Twenty-five first year electrical and mechanical apprentices taking an apprenticeship training course at a Sydney company participated in the experiment. All of the students had completed at least Year 10 of high school and were enrolled in various first year courses at trade colleges. At the time of testing, all apprentices were well into their eight month training course in industrial and domestic electricity (students had completed a mean of 8.32 units from a 21 unit course). At this stage of their course all apprentices had demonstrated their competency in all basic aspects of electrical testing.

Materials and Procedure

The instructional material used in this experiment was identical to that used in Experiment 1.

The procedure was almost identical to that used in Experiment 1. The only difference was in the allocation of participants to instructional groups. Participants were matched for apprenticeship type (i.e. electrical, mechanical, dual apprenticeship, fitter-turner, tool maker) before being randomly assigned to either the isolated-interacting elements instruction group or the interacting elements only group.

Results and Discussion

The test score variables under analysis were high element interactivity knowledge score, low element interactivity knowledge score and practical test score. Means and Standard Deviations for these variables are provided in Table 3. Other variables analysed included mental load rating and instruction time. Means and Standard Deviations for these variables are provided in Table 3.

Insert Table 3 about here

There were no significant differences between the groups found for either low or high element interactivity knowledge questions or for the practical scores. Neither were significant effects found for subjective mental ratings. As might be expected given these results, there were no significant differences between groups in instructional efficiency using subjective mental load ratings and any of the performance scores. Means and Standard Deviations for normalised (Z) mental effort ratings and performance scores as well as instructional efficiency scores are provided in Table 4.

Insert Table 4 about here

The only significant differences to be found related to instruction time. There was a significant difference between groups in instruction time, $F(1,23) = 5.79$, $MSe = 43048.96$, in time taken over the two instructional phases, $F(1,23) = 10.158$, $MSe = 8699.05$ and a significant interaction between the factors, $F(1,23) = 60.36$, $MSe = 8699.05$. In order to assess the interaction effect, a test of simple effects was carried out at each phase. At Phase 1, a significant difference was found between

the instructional groups, $t(23) = 5.66$ with the isolated-interacting elements group requiring less learning time. At Phase 2, when the groups studied the same version of the experimental material, no difference was found between the groups, $t(23)= 0.95$. These differences between groups were due entirely to the isolated-interacting elements group requiring less time to read their instructions in Phase 1.

The results from Experiment 2 contrast sharply with those of Experiment 1, where instruction by the isolated-interacting elements method significantly improved test performance. The level of expertise of the participants was the only identifiable difference between the two experiments. This observation lends credence to the hypothesis that the isolated-interacting elements method of instruction benefits learners through assisting the process of schema construction. This hypothesis explains why the isolated-interacting elements method of instruction showed distinct performance advantages for novice learners (Experiment 1), who previously had only very primitive schemas (e.g., A represents active) in the area. The more expert learners in Experiment 2, already in possession of schemas for the general area of electrical circuitry, learnt equally effectively, regardless of the instruction method used.

Introduction to Experiments 3 and 4

While the differing results of Experiments 1 and 2 were theoretically coherent, they could have been due to the particular procedures and methods used. There was a heavy emphasis on procedural learning in Experiments 1 and 2 and it could be argued that the sequential and directive nature of this type of information may have influenced the results. It is important to replicate the results of Experiments 1 and 2 using materials that place a greater emphasis on conceptualisation rather than on procedures. Accordingly, Experiments 3 and 4 used materials requiring a higher degree of conceptualisation than Experiments 1 and 2. Identical materials and procedures were used for both experiments but the students of Experiment 3 had considerably less background knowledge than those of Experiment 4. We predicted that the isolated-interacting elements instruction condition would be

superior to the interacting elements only instructions condition in Experiment 3 but that this difference would disappear in Experiment 4.

Experiment 3

Experiment 3 compared the isolated-interacting elements and interacting elements only conditions using apprentices who had only a limited background in electrical principles. As in Experiment 1, it was predicted that the isolated-interacting elements condition would be superior to the interacting elements only condition. The experiment used material depicting an electrical circuit used to power an industrial oven. This material was very complex, containing more elements of information and more interactions between the elements than the materials employed previously in Experiments 1 and 2. It can be regarded as particularly high in element interactivity because all of the elements that form a circuit relate to and in turn effect, other circuit elements.

Method

Participants

Eighteen first year industrial trade students enrolled in a training course at a Sydney company participated in the experiment. All of the students had completed at least Year 10 of High School. They were selected from students enrolled in one of the following courses: a six month manufacturing and engineering trainee-ship course (13 students) or the practical section of a Year 11 school electronics course (5 students). Students were tested within their first fortnight of training so both groups had little electrical knowledge and what had been covered, with respect to simple electrical concepts, was very similar for both courses. Both groups had been introduced to very basic electrical principles (e.g., A represents active, N neutral and current flows from active to neutral) but none of the participants had ever studied the electrical circuit or any similar circuits to that used in the experiment.

Materials

The participant information sheet recorded demographic information such as age, years of schooling, qualification being studied and stage in current course.

Participants were also asked details of any previous experience with electrical training.

The instructional materials used in the experiment consisted of information describing the electrical circuit used to power an industrial oven. The function and workings of the oven circuit were impossible to understand without reference to the whole circuit.

Two sets of instructional materials were developed to introduce the electrical circuit: an isolated elements version and an interacting elements version. The difference between the two versions lay in the emphasis the interacting elements version placed upon the cause and effect relationships that drove the system.

The isolated elements version was presented on a single A4 (21 X 29.5 cm) sized page. Numbered text points were integrated into the circuit diagram. The interacting elements version was presented in an identical format to the isolated elements version except that it was on an A3 size page (42 X 29.5 cm). This larger diagram was necessary because of the difference in the amount of text contained in the interacting elements version. (The size of the instructional diagram made it impossible to include as a figure in this paper. It may be obtained from the authors.)

Both groups also received a single, A4 (21 X 29.5 cm) sized page that depicted a diagram representing the oven baking process. The diagram included key features of the oven such as the motor, the heating elements, the conveyor and the oven control panel realistically rather than as electrical circuitry. Another diagram, depicting only the oven electrical circuit was available to students during testing. Switches and symbols were labelled on the diagram but no textual description was included. It was presented on an A4 (21 X 29.5 cm) sized page.

The subjective mental load rating measure used a nine point Likert type scale. The number of points on the mental effort Likert scale were increased from seven in Experiment 1 and 2, to nine in this experiment, in order to give the students a greater range of options to describe their subjective mental load when learning.

There were two categories of questions in the written test: those tapping low and high element interactivity knowledge. Both types of questions included textual and diagrammatical test items. Low element interactivity questions asked about a specific part or fact of the electrical circuit. This section of questions included four test items with the total marks possible for this category of questions being eight. High element interactivity questions dealt with issues such as the progression of events in the circuit or the structure and function of the oven circuit. These questions required students to have a holistic understanding of the circuit and its processes. There were eight test items in the high element interactivity category of questions and the total marks possible for this category of questions was twenty-seven because some questions contained subparts.

Procedure

All the students were randomly assigned to either the isolated-interacting elements instruction group or the interacting elements only group. All students were tested individually, in a quiet room. There were nine students in each group. The students were informed that they were to be given instructional material regarding an electrical circuit which would be followed by a written test and that this session was to be repeated within forty-eight hours. Students filled in an information sheet at this stage that asked for biographical and educational details. Any students considered to have prior knowledge (see criteria outlined in the "participants" section) were eliminated at this stage. In the first experimental session, the students in the isolated-interacting elements instruction group studied the isolated elements instructional booklet while the students in the interacting elements only group were given the interacting elements instructional booklet. Both groups also received the schematic diagram of the oven baking process. Students were allowed a maximum of ten minutes to read through the instructional material. If a student finished before this time, they were asked to review the information for the remaining time. This procedure was used in order to provide additional control by equalising study times rather than allow unlimited times as occurred in Experiments 1 and 2. At the completion of the study phase, the students were

provided with a subjective mental load rating scale, the format of which was explained to both groups. They were asked to rate the mental effort expended in understanding the oven circuit and to translate this to a numerical value. The test section of the experiment followed. The students were asked to complete the written test, which was common to both groups. The instructional materials were not available to the students during this test period. The circuit diagram (labelled circuit elements with no text description) was available to the students at this stage but was removed for diagrammatical questions that required the students to find faults in the circuit. At all stages of the test period, students were prohibited from reattempting questions after they had attempted them and from reviewing previously completed test answers.

The second phase of the experiment was held within forty-eight hours of the first phase. The procedure of the second phase of the experiment was identical to the first, except that all students studied the interacting elements instructions.

Results and Discussion

The test score variables under analysis were low and high element interactivity knowledge scores and mental load rating. Means and Standard Deviations for these variables are provided in Table 5.

Insert Table 5 about here

Low and high element interactivity knowledge scores were analysed separately by 2 (Instruction Condition) x 2 (Test Phases) ANOVAs with repeated measures on the second factor. The results for low element interactivity knowledge question scores indicated a possible difference between the instructional groups favouring the isolated-interacting elements instruction group, $F(1,16) = 3.86$, $MSe = 6.06$, $p = 0.07$. The test phases main effect also indicated a possible difference over time, $F(1,16) = 3.93$, $MSe = 1.19$, $p = 0.07$. There was no significant interaction, $F(1,16) = 0.58$, $MSe = 1.19$. The results for high element interactivity question scores indicated a possible difference favouring the isolated-interacting elements instruction group, $F(1,16) = 3.18$, $MSe = 26.42$, $p = 0.09$. The test phases main effect indicated a significant difference, $F(1,16) = 31.03$, $MSe = 1.81$,

demonstrating an improvement over time. A significant interaction was displayed between the factors, $F(1,16) = 11.17$, $MSe = 1.81$. Tests of simple effects were used to explore the interaction. No difference was found between the instructional conditions at phase 1, $t(16) = 0.936$. At phase 2, a significant difference was indicated between the groups, $t\ (16) = 2.43$, with the isolated-interacting elements group performing at a superior level.

Mental load ratings were analysed using 2 (Instructional Condition) x 2 (Test Phases) ANOVA with repeated measures on the last factor. Results indicated no differences between the instructional conditions, $F(1, 16) = 1.19$, $MSe = 2.34$. There was no improvement over time, $F(1, 16) = 0.50$, $MSe = 0.88$, nor was there a significant interaction, $F(1, 16) = 0.50$, $MSe = 0.88$.

Instructional condition efficiency measures were calculated according to the method devised by Paas and van Merrienboer (1993;1994). Table 6 shows the means and standard deviations of the normalised (Z) mental effort and performance scores and the condition efficiency scores for each group.

Insert Table 6 about here

Efficiency measures for each type of score were calculated and analysed using a 2 (Instruction Condition) x 2 (Test Phases) ANOVA with repeated measures on the last factor. The results for low element interactivity question efficiency scores indicated a possible difference between the instructional conditions favouring the isolated-interacting elements instruction group, $F(1, 16) = 3.42$, $MSe = 2.01$, $p = 0.08$. No difference was indicated between the test phases, $F(1, 16) = 2.32$, $MSe = 0.55$. There was no significant interaction between the factors, $F(1, 16) = 0.01$, $MSe = 0.55$. Results for high element interactivity question efficiency scores indicated a possible difference favouring the isolated-interacting elements instruction group with respect to performance efficiency, $F(1, 16) = 3.34$, $MSe = 1.86$, $p = 0.09$. A significant difference was indicated between the test phases, $F(1, 16) = 9.52$, $MSe = 0.28$, demonstrating an improvement over time for both groups. A significant interaction was indicated between the factors, $F(1, 16) = 4.55$, $MSe = 0.28$. Tests of simple effects were used to explore the interaction. At phase 1, no

difference was indicated between the instructional conditions, $t(16) = 1.17$. At phase 2, a significant difference existed between the instructional groups, $t(16) = 2.11$, with the isolated-interacting elements group displaying superior instruction condition efficiency by phase 2 of the learning process.

As predicted, the results of Experiment 3 do replicate and support the findings of the previous experiment with novice learners (Experiment 1). The major difference favouring the isolated-interacting elements instruction condition occurred during phase 2 using questions testing for high element interactivity knowledge. In a seeming paradox, for these students, the "understanding" instructions of phase 2 were only effective after studying material during phase 1 that did not emphasise understanding. Understanding instructions in phase 1 reduced rather than facilitated the effectiveness of understanding instructions in phase 2.

Experiment 4

Experiment 4 replicated the design of the previous experiments using the materials and procedure of Experiment 3 but with a sample of more experienced learners. The participants involved in this study were industrial apprentices, experienced in the area to be studied and so it was predicted that no performance differences would be obtained between the two instructional conditions. It was expected that these more expert learners would not benefit from the isolated-interacting elements method of instruction as the understanding instructions should not overburden working memory.

Method

Participants

Eighteen first year electrical and mechanical apprentices enrolled in an apprenticeship training course at a Sydney company participated in the experiment. All of the students had completed at least Year 10 of High School and were enrolled in various first year courses at trade colleges. The participants were tested more than three quarters of the way through their eight month training course in industrial and domestic electricity (students had completed a mean of 28.55 units from a 34 unit course).

Materials and Procedure

The instructional materials used in this experiment was identical to that used in Experiment 3. The procedure used in this experiment was virtually identical to that used in Experiment 3. The only difference was in the allocation of participants to instructional groups. Participants were matched for apprenticeship type (i.e. electrical, mechanical, dual apprenticeship, fitter-turner, tool maker) before being randomly assigned to either the isolated-interacting elements instruction group or the interacting elements only group.

Results and Discussion

The test score variables under analysis were low and high element interactivity knowledge scores and mental load rating. Means and standard deviations for these variables are provided in Table 7.

Insert Table 7 about here

As was the case for Experiment 2 when using more experienced learners, there were no significant differences between groups on either performance or mental ratings measures. These results are in line with our predictions. As might be expected given these results, there were no significant differences between groups in instructional efficiency using subjective mental load ratings and any of the performance scores. Means and Standard Deviations for normalised (Z), mental effort ratings and performance scores as well as instructional efficiency scores are provided in Table 8. It can be concluded that there is no performance benefit for expert learners using the isolated-interacting elements method of instruction.

Insert Table 8 about here

General Discussion

Cognitive load theory is based on the assumption that certain features of the structure of information and of human cognitive architecture interact and this interaction has important implications for how we learn and understand. The paradox of learning seems to be that some material which is very high in element interactivity cannot be processed simultaneously in working memory with understanding until after it has been stored in schematic form in long-term memory.

While understanding of high element interactivity material follows schema construction because all relevant elements are incorporated in the schema and so can be processed in working memory as a single element, it was not clear how the elements that constitute a complex schema could be processed in working memory *before* the schema had been constructed. This paper undertook to investigate an instructional technique to combat this paradoxical situation.

It was proposed that initially, the element interactivity of complex material had to be artificially reduced in order to enable a schema or partial schema for the information to be developed. That reduction could be accomplished by presenting the material as isolated elements that could be processed in working memory. The consequence of the reduction in element interactivity of material may result in an initial decrease in the student's capacity for understanding. Over the longer term however, it was hypothesised that the promotion of schema construction would lead to an increase in the learner's understanding. The isolated-interacting elements method instructional technique had two stages: firstly students studied instructions detailing only the individual information elements forming a concept; in stage two, learners studied a version of the instruction that expounded all the information elements and how they interact. In comparison, the interacting elements only method of instruction presented all the information for understanding at both phases.

The experiments reported in this paper provided powerful evidence that for certain groups of learners, information is better learnt through the isolated-interacting elements instructional method. Experiments 1 and 3 demonstrated that when novice students are required to learn complex information, the isolated-interacting elements method of instruction was superior to the interacting elements only method. In contrast, the results of Experiments 2 and 4 seemed to contradict the success of the isolated-interacting elements instructional technique. In Experiments 2 and 4, learners using the interacting elements only method where full understanding was required in both phases performed as well as the isolated-interacting elements instructional group. Differing levels of expertise could be used

to explain the contrasting results. The learners in Experiments 2 and 4 had experience in the relevant knowledge domains. It was expected that the isolated-interacting elements method of instruction, which allows schema acquisition in order to facilitate learning and understanding, would be of little use to learners who already possess rudimentary schemas and so may not experience a heavy cognitive load. In addition, the subjective mental effort ratings and performance efficiency measure of the instructional conditions did not differ, giving further credence to this explanation.

It has been previously stated that intrinsic cognitive load of information is unalterable (Sweller and Chandler, 1994; Sweller, van Merrienboer and Paas, 1998) because this type of cognitive load reflects the internal structure of the material itself, that is, the number of information elements and the degree of interaction between them. This proposition is true, keeping in mind that understanding is an essential concomitant of learning. Nevertheless, while understanding must be one of the ultimate goals of learning, it may not be necessary or indeed, possible, as an intermediate goal. The isolated followed by interacting elements approach suggests that when dealing with very complex information, in order to allow novice students to process interacting elements, the intrinsic cognitive load of the material should initially be artificially reduced. Technically, this is achieved by eliminating the interactions between the information elements. A large degree of interaction between the information elements is the foundation of a concept's intrinsic cognitive load since this interaction requires information elements to be processed simultaneously. If the interactivity between the elements is reduced, discrete information units are essentially created. Serial rather than simultaneous processing of information elements can then be undertaken, substantially reducing working memory load. Students can study the concept at the original level of intrinsic load in a subsequent learning phase of the isolated-interacting elements instructional condition. The success of the isolated-interacting elements instructional condition is assessed by testing students' understanding of this version of the material.

The conclusion that understanding and learning of very complex information can be facilitated by presenting the information in a manner that initially precludes full understanding is likely to be seen as highly controversial by some. Nevertheless, from a theoretical perspective, there is a degree of inevitability about this conclusion. We know that: (a) working memory is very limited; (b) complex material involves more interacting elements than can be processed simultaneously in working memory unless they have been incorporated into a schema; (c) understanding requires all interacting elements to be processed simultaneously in working memory. If these three propositions are valid, understanding cannot occur until after appropriate schemas have been constructed. Instructional designs intended to encourage students to understand very complex material prior to the construction of appropriate schemas will fail. The experiments of this paper provide empirical support for these conclusions.

References

Bartlett, F. C. (1932). *Remembering a study in experimental and social psychology.* London. Cambridge University Press.

Bloom, B. (1956). (Ed.) *Taxonomy of Educational Objectives: Cognitive Domain.* NY: McKay.

Chandler, P., & Sweller, J. (1996). Cognitive load while learning to use a computer program. *Applied Cognitive Psychology*. 10, 151-170.

Chase, W. G., and Simon, H. A. (1973). Perception in chess. *Cognitive Psychology*, 4, 55-81.

Chi, M. T. H, Glaser, R., & Rees, E. (1982). Expertise in problem solving. In R. Stenberg (Ed.), *Advances in psychology of human intelligence.* Hillsdale, New Jersey : Erlbaum.

De Groot, A. D. (1965). *Thought and Choice in Chess*. The Hague : Mouton.

Gagne, R. (1970). *The Conditions of Learning* (2nd ed.). NY: Holt, Rinehart & Winston.

Halford, G. S., Wilson, W. H. and Phillips, S. (1998) Processing capacity defined by relational complexity: Implications for comparative, developmental, and cognitive psychology. *Behavioural and Brain Sciences*, 21 (6), 803-864.

Hoosain, R. (1983). Memorization of classical Chinese. *Psychologia: An International Journal of Psychology in the Orient*, 26(3), 193-197.

King, D. J., and Russell, G. W. (1966). A comparison of rote and meaningful learning of connected meaningful material. *Journal of Verbal Learning and Verbal Behaviour*, 5, 478-483.

Kotovsky, K., Hayes, J. R., & Simon, H. A. (1985). Why are some problems hard? Evidence from Tower of Hanoi. *Cognitive Psychology*, 17, 248-294.

Larkin, H., McDermott, J., Simon, D., & Simon, H. (1980). Models of competence in solving physics problems. *Cognitive Science*, 11,65-99.

Marcus, N., Cooper, M., & Sweller, J. (1996). Understanding Instructions. *Journal of Educational Psychology,* 88(1), 49-63.

Mayer, R. & Moreno, R. (this issue). Aids to computer-based multimedia learning. *Learning & Instruction.*

Miller, G. A. (1956). The magical number seven plus or minus two : Some limits on our capacity for processing information. *Psychological Review*, 63, 81-97.

Paas, F., & van Merrienboer, J. (1993). The efficiency of instructional conditions: An approach to combine mental-effort and performance measures. *Human Factors*, 35, 737-743.

Paas, F., & van Merrienboer, J. (1994). Variability of worked examples and transfer of geometric problem-solving skills: A cognitive load approach. *Journal of Educational Psychology*, 86, 122-133.

Rumelhart D.E. (1980) Schemata: The Building Blocks of Cognition. In R.C. Anderson, R.J. Spiro, & W.E. Montague (Eds.) *Theoretical Issues in Reading Comprehension* (pp 33-58). Hillsdale, N. J.: Lawrence Erlbaum Associates.

Schneider, W., & Shiffrin, R. (1977). Controlled and automatic human information processing: I. Detection, search and attention. *Psychological Review*, 84, 1-66.

Shiffrin, R. & Schneider, W. (1977). Controlled and automatic human information processing: II. Perceptual learning, automatic attending, and a general theory. *Psychological Review*, 84, 127-190.

Stark, R., Mandl, H., Gruber, H. & Renkl, A. (this issue). Conditions and effects of example elaboration. *Learning & Instruction.*

Sweller, J. (1994). Cognitive load theory, learning difficulty and instructional design. *Learning and Instruction*, 4, 295-312.

Sweller, J. (1999). *Instructional Design.* Melbourne: ACER.

Sweller, J., & Chandler, P. (1994). Why some material is difficult to learn. *Cognition and Instruction*, 12 (3), 185-233.

Sweller, J., Mawer, R., Ward, M. (1983). Development of Expertise in Mathematical Problem Solving. *Journal of Experimental Psychology: General*, 112(4), 639-661.

Sweller, J., van Merrienboer, J. J. G., & Paas, F. G. W. C. (1998). Cognitive architecture and instructional design. *Educational Psychology Review*, 10(3), 251-296.

Thorndyke, P. W. and Hayes-Roth, B. (1977) The Use of Schemata in the Acquisition and Transfer of Knowledge. *Cognitive Psychology*, 11, 82-106.

Van Bruggen, J., Kirschner, P., & Jochems, W. (this issue). External Representation of Argumentation in CSCL and the Management of Cognitive Load. *Learning & Instruction.*

Van Gerven, P., Paas, F., van Merriënboer, J., & Schmidt, H. (this issue). Cognitive load theory and aging: effects of worked examples on training efficiency. *Learning & Instruction.*

Van Merriënboer, J., Schuurman, J., de Croock, M. & Paas, F. (this issue). Redirecting learners' attention during training: effects on cognitive load, transfer test performance and training efficiency. *Learning & Instruction.*

List of Figures

Figure 1: Insulation Resistance Test: Interacting elements version of instructions

Note: Normal text - Interacting elements version; Bold text - isolated elements version

Table 1 Test Score, Instruction Time (in seconds) and Mental Load Ratings for Experiment 1

		Group			
		Isolated-interacting (n=11)		**Interacting only (n=11)**	
	Max. score	Mean	Std Dev	Mean	Std Dev
Phase 1					
Low element interactivity	(100%)	36.37	15.19	31.82	14.51
High element interactivity	(100%)	17.73	8.95	12.27	11.93
Practical test	(2)	0.18	0.41	0.10*	0.32*
Mental Load	(7)	2.91	1.38	4.09	0.94
Instruction Time	(-)	645.27	274.05	917.27	263.01
Phase 2					
Low element interactivity	(100%)	54.04	19.26	41.92	11.21
High element interactivity	(100%)	35.00	20.22	17.58	8.44
Practical test	(2)	0.36	0.51	0.00	0.00
Mental Load	(7)	3.27	1.27	4.00	1.18
Instruction Time	(-)	603.46	328.11	676.09	240.94

* Sample size reduced to 10 students due to equipment failure

Table 2 Normalised (Z) scores and Condition Efficiency Scores for Performance and Mental Load for Experiment 1

	Group			
	Isolated-interacting (n=11)		**Interacting only (n=11)**	
	Z	**Efficiency**	**Z**	**Efficiency**
	Mean (S.D.)	Mean (S.D.)	Mean (S.D.)	Mean (S.D.)
Phase 1				
Low element interactivity	-0.27 (0.89)	0.17 (1.00)	-0.54 (0.89)	-0.68 (0.87)
High element interactivity	-0.19 (0.57)	0.24 (0.86)	-0.54 (0.77)	-0.68 (0.51)
Practical Score	0.06 (1.16)	0.41 (1.26)	-0.18 (0.81)*	-0.42 (0.65)*
Mental Load	-0.52 (1.09)	-	0.41 (0.75)	-
Phase 2				
Low element interactivity	0.77 (1.13)	0.71 (1.20)	0.05 (0.66)	-0.20 (0.93)
High element interactivity	0.93 (1.31)	0.82 (1.33)	-0.20 (0.55)	-0.38 (0.59)
Practical Score	0.55 (1.36)	0.57 (1.45)	-0.43 (0.00)	-0.55 (0.66)
Mental Load	-0.23 (1.01)	-	0.34 (0.94)	-

Note: The efficiency scores are based on the normalised mental load scores given in Table 1 for each instructional group for each learning phase.

* Sample size reduced to 10 students due to equipment failure

Table 3 Test Score, Instruction Time (in seconds) and Mental Load Ratings for Experiment 2

		Group			
		Isolated-interacting (n=12)		**Interacting only (n=13)**	
	Max. score	Mean	Std Dev	Mean	Std Dev
Phase 1					
Low element interactivity	(100%)	60.76	13.47	58.65	18.36
High element interactivity	(100%)	31.81	25.32	33.85	19.20
Practical Score	(2)	0.67	0.65	0.54	0.66
Mental Load	(7)	2.17	0.94	2.85	0.80
Instruction Time	(-)	392.92	105.47	739.31	186.18
Phase 2					
Low element interactivity	(100%)	72.22	17.97	71.15	17.05
High element interactivity	(100%)	48.33	18.95	46.28	19.52
Practical Score	(2)	0.50	0.52	0.62	0.77
Mental Load	(7)	2.42	1.38	2.46	0.97
Instruction Time	(-)	522.17	109.32	458.31	208.25

Table 4 Normalised (Z) scores and Condition Efficiency Scores for Performance and Mental Load for Experiment 2

	Group			
	Isolated-interacting (n=12)		**Interacting only (n=13)**	
	Z	**Efficiency**	**Z**	**Efficiency**
	Mean (S.D.)	Mean (S.D.)	Mean (S.D.)	Mean (S.D.)
Phase 1				
Low element interactivity	-0.26 (0.77)	0.01 (0.91)	-0.40 (1.05)	-0.53 (1.09)
High element interactivity	-0.36 (1.13)	-0.05 (1.18)	-0.29 (0.89)	-0.45 (0.90)
Practical Score	0.12 (0.97)	0.29 (0.88)	-0.06 (1.04)	-0.30 (1.08)
Mental Load	-0.28 (0.87)	-	0.35 (0.77)	-
Phase 2				
Low element interactivity	0.35 (0.99)	0.29 (1.29)	0.32 (0.98)	0.24 (1.07)
High element interactivity	0.36 (0.85)	0.29 (1.33)	0.29 (0.91)	0.22 (1.12)
Practical Score	-0.12 (0.78)	-0.04 (1.20)	0.06 (1.21)	0.05 (1.10)
Mental Load	-0.06 (1.28)	-	-0.02 (0.93)	-

Note: The efficiency scores are based on the normalised mental load scores given in Table 3 for each instructional group for each learning phase.

Table 5 Test Scores and Mental Load Ratings for Experiment 3

		Group			
		Isolated-interacting (n=9)		**Interacting only (n=9)**	
	Max. score	Mean	Std Dev	Mean	Std Dev
Phase 1					
Low element interactivity	(100%)	50.00	25.00	26.39	17.05
High element interactivity	(100%)	25.51	10.55	19.75	15.16
Mental Load	(9)	5.67	1.00	6.00	1.12
Phase 2					
Low element interactivity	(100%)	55.56	25.85	38.89	26.10
High element interactivity	(100%)	40.33	14.87	23.46	14.58
Mental Load	(9)	5.22	1.09	6.00	1.73

Table 6 Normalised (Z) scores and Condition Efficiency Scores for Performance and Mental Load for Experiment 3

	Group			
	Isolated-interacting (n=9)		**Interacting only (n=9)**	
	Z	**Efficiency**	**Z**	**Efficiency**
	X(S.D.)	**X(S.D.)**	**X(S.D.)**	**X(S.D.)**
Phase 1				
Low element interactivity	0.29 (0.98)	0.23 (1.05)	-0.64 (0.67)	-0.61 (0.85)
High element interactivity	-0.11 (0.68)	-0.05 (0.86)	-0.49 (0.98)	-0.50 (0.77)
Mental Load	-0.04 (0.80)	-	0.22 (0.89)	-
Phase 2				
Low element interactivity	0.51 (1.02)	0.64 (1.13)	-0.15 (1.03)	-0.26 (1.42)
High element interactivity	0.84 (0.96)	0.88 (1.21)	-0.25 (0.94)	-0.33 (1.22)
Mental Load	-0.40 (0.87)	-	0.22 (1.38)	-

Note: The efficiency scores are based on the normalised mental load scores given in Table 5 for each instructional group for each learning phase.

Table 7 Test Scores and Mental Load Ratings for Experiment 4

		Group			
	Isolated-interacting (n=9)			**Interacting only (n=9)**	
	Max. score	Mean	Std Dev	Mean	Std Dev
Phase 1					
Low element interactivity	(100%)	65.29	20.52	69.44	15.45
High element interactivity	(100%)	32.51	11.97	39.51	20.37
Mental Load	(9)	5.44	1.13	4.67	1.87
Phase 2					
Low element interactivity	(100%)	70.83	27.95	76.39	17.05
High element interactivity	(100%)	46.50	14.12	58.44	21.26
Mental Load	(9)	4.22	1.39	3.44	1.51

Table 8 Normalised (Z) scores and Condition Efficiency Scores for Performance and Mental Load for Experiment 4

	Group			
	Isolated-interacting (n=9)		**Interacting only (n=9)**	
	Z	**Efficiency**	**Z**	**Efficiency**
	Mean (S.D.)	Mean (S.D.)	Mean (S.D.)	Mean (S.D.)
Phase 1				
Low element interactivity	-0.26 (1.01)	-0.62 (1.02)	-0.05 (0.76)	-0.13 (1.27)
High element interactivity	-0.61 (0.62)	-0.87 (0.82)	-0.25 (1.06)	-0.27 (1.41)
Mental Load	0.62 (0.70)	-	0.14 (1.16)	-
Phase 2				
Low element interactivity	0.02 (1.38)	0.11 (1.52)	0.29 (0.84)	0.64 (0.93)
High element interactivity	0.12 (0.73)	0.18 (1.07)	0.74 (1.10)	0.96 (1.37)
Mental Load	-0.14 (0.87)	-	-0.62 (0. 94)	-

Note: The efficiency scores are based on the normalised mental effort scores given in Table 7 for each instructional group for each learning phase.

APPENDIX

Estimation of the element interactivity of the Insulation Resistance test instructional material used in Experiments 1 and 2

The elements of interacting information have been estimated for the Insulation Resistance electrical safety test. These estimates represent what could reasonably be supposed to be the case for novice electricians. The estimates are based on the instructional material provided in Figure 1.

The Insulation Resistance test

Each point refers to one element of information.

Isolated Elements version of Instructions

The volt meter is set to 500 volts.
The appliance is switched "on".
The line lead of the volt meter must be placed on some point on the metallic frame of the appliance.
The earth lead of the volt meter must be placed on the active pin on the appliance plug.
The test is performed by pressing the test button and then checking the result meets the criteria of at least one Megohm.
Change the earth lead to the neutral pin of the appliance pin.
Test this circuit by pressing the test button again and checking the result is at least one Megohm.

Interacting Elements Version of Instructions

The test checks that there is no leak in the flow of current through the appliance circuit that would cause the appliance frame to become electrically active.

To establish that there are no leaks, two circuits have to be tested: the circuit between the active point and the appliance frame and the circuit between the neutral point and the appliance frame.

The test measures the insulation resistance of the circuit. If the reading is high, it follows that current is not being leaked from the circuit.

Ideally a circuit should have infinite resistance but a safe criteria of at least 1 Megohm has been set.

The volt meter is set to 500 volts.

This setting means the circuits are tested under a larger than normal voltage.

The appliance is switched "on".

This is so current from the volt meter can flow through each circuit.

To set up the first circuit, the line lead of the volt meter must be placed to some point on the metallic frame of the appliance.

The earth lead of the volt meter must be placed to the active pin on the appliance plug.

The test is performed by pressing the test button which introduces 500 volts to the circuit.

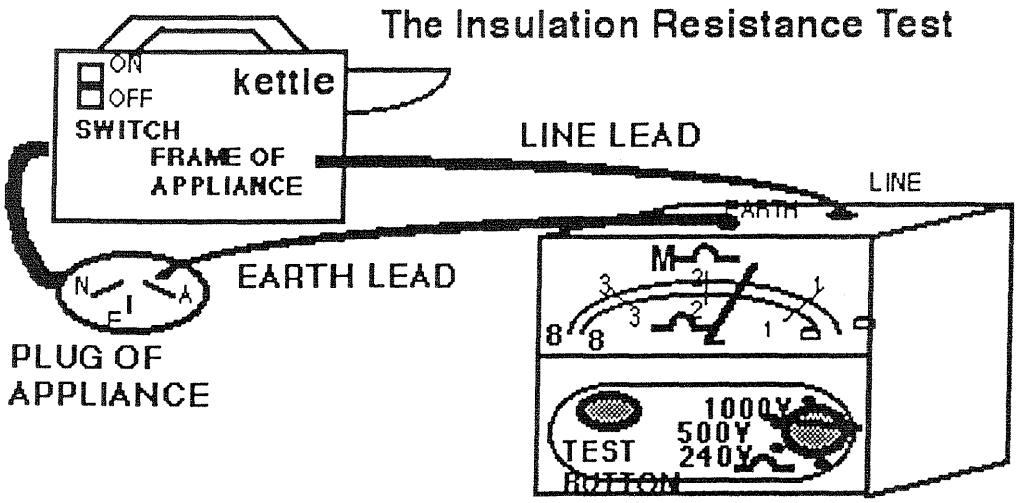
The result must meet the criteria of at least one Megohm to ensure that no current is being leaked to the appliance frame.

One Megohm is equal to 1 000 000 ohms.

The resistance of the second circuit must be measured to complete the test.

The earth lead is moved onto the neutral pin of the appliance plug.

Test the second circuit by pressing the test button again and checking the result is at least one Megohm.



This test aims to check that there is no "leak" in the flow of current from either the active or neutral wires of the appliance to the frame of the appliance, which would cause the frame of the appliance to become electrically active.

To achieve this two circuits are set up; the first including the frame of the appliance and the active of the appliance and a second including the frame of the appliance and the neutral of the appliance. Very large measures of insulation resistance (ideally "infinite" resistance) indicates that there are no leaks in the flow of current to each circuit. A megger meter measures the resistance of the circuit.

1. By setting the meter to read 500 volts, a larger than normal voltage will be introduced into each circuit in order to test the appliance under a heavy load. **Set the meter to read 500 volts.**
2. The appliance is switched "on" to allow current generated from the megger meter to flow into each circuit. **Make sure the switch is "on".**
3. To measure of the resistance of the first circuit the megger meter is connected into the circuit. **Place the line lead on the frame of the appliance.**
4. **Place the earth lead on the active pin of the appliance's plug.**
5. The test button of the megger meter introduces 500 volts into the circuit. **Press the test button.**
6. A measurement of insulation resistance of the circuit can be taken from the needle of the meter, with a reading of at least 1 Megohm (1,000,000 ohms) being sufficiently large to indicate that the active-frame circuit possesses an acceptable level of insulation. This ensures that no current leaks to the appliance's frame. **Read the resistance from the meter. The required result is a reading of at least one Megohm.**
7. To complete the test of insulation resistance the resistance of the second circuit must be measured. The second circuit is set up between the neutral point of the appliance and the appliance frame. **Remove the earth lead from the active pin and place it on the neutral pin.**
8. A reading of the resistance of the second circuit can be taken by pressing the test button of the megger meter and introducing 500 volts into the neutral-frame circuit. **Press the test button again.**

To pass the test, the second circuit is also required to have a safe level of insulation resistance which indicates no current has leaked from this circuit. **Read the resistance. A reading of at least one Megohm is again required.**